

Hyper-V documentation

Hyper-V in Windows Server enables you to run Windows and Linux virtual machines at scale, providing flexibility and maximizing resources.

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Hyper-V virtualization in Windows Server and Windows

08/05/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V is Microsoft's enterprise-grade hypervisor technology built into Windows Server and Windows. It provides hardware virtualization capabilities that enable organizations to create, manage, and run virtual machines at scale. As a type-1 hypervisor, Hyper-V runs directly on computing hardware, delivering near-native performance and robust isolation for virtualized workloads.

Virtualizing your workloads with Hyper-V enables you to maximize hardware utilization, streamline IT operations, and enhance business continuity. It supports a wide range of operating systems for guest virtual machines, including many versions of Windows, Linux, and FreeBSD, making it a versatile solution for diverse IT environments.

Whether you're consolidating servers, spanning on-premises and hyperscale cloud providers, or enabling flexible development and testing environments, Hyper-V offers scalable solutions for modern IT needs.

Hyper-V in Windows Server and Windows has different capabilities and features, but both provide a solid foundation for virtualization. Hyper-V in Windows Server is designed for enterprise deployments with advanced features like live migration, high availability, and disaster recovery. Hyper-V in Windows provides IT professionals and developers a lightweight solution suitable for development and testing scenarios.

This overview explains Hyper-V's benefits, key features, and common use cases, and provides guidance on getting started.

Benefits of Hyper-V

Hyper-V delivers significant organizational benefits across many areas, including:

- **Cost optimization:** reduce hardware acquisition and maintenance costs through server consolidation, while lowering datacenter space, power, and cooling requirements. Organizations can achieve high consolidation ratios, dramatically reducing total cost of ownership. Hyper-V is included with Windows Server and Windows, eliminating additional hypervisor licensing costs.

- **Operational efficiency:** streamline IT operations with centralized management, automated provisioning, and simplified backup and disaster recovery processes. Virtual machine templates and PowerShell automation reduce deployment time from hours to minutes.
- **Business agility:** rapidly provision new environments to support business initiatives, scale resources up or down based on demand, and quickly adapt to changing market conditions without lengthy hardware procurement cycles.
- **Enhanced security:** implement strong workload isolation, use shielded virtual machines for sensitive data, and maintain secure environments with features like Secure Boot and TPM 2.0 support.
- **Improved resilience:** minimize downtime through high availability features, implement comprehensive disaster recovery strategies with Hyper-V Replica, and ensure business continuity with live migration capabilities.
- **Development acceleration:** enable rapid creation of development and testing environments, support DevOps practices with consistent infrastructure, and reduce time-to-market for new applications and services.
- **Simplified compliance:** meet regulatory requirements more easily through consistent configurations, centralized monitoring, and comprehensive audit trails across virtualized environments.

Technical capabilities of Hyper-V

Here's a video that provides a comprehensive technical overview of Hyper-V, covering its hypervisor architecture, performance characteristics, and scalability features. It demonstrates how Hyper-V delivers enterprise-grade virtualization capabilities, discusses high availability and clustering features, and showcases real-world deployment scenarios for organizations looking to implement virtualization solutions.

<https://www.youtube-nocookie.com/embed/iTtYp-Qdvl4> ↗

The following sections provide an overview into some of the key enterprise-grade features available in Hyper-V that address critical business requirements around security, availability, management, and cost optimization.

Extensibility and integration

An important aspect of Hyper-V is its extensibility and integration with other Windows Server features and other Microsoft and third-party products and services to create a comprehensive virtualization solution ready for enterprise deployments.

For example, Hyper-V works seamlessly with Windows Server Failover Clustering to provide high availability, connects to a range of storage providers for flexible storage options, supports Azure integration for hybrid cloud scenarios, and integrates with Windows Admin Center or System Center Virtual Machine Manager for advanced management.

Hyper-V also is a key component of [Azure Local](#), enabling organizations to run Azure services on-premises with consistent management and development experiences. This integration allows for hybrid cloud scenarios where workloads can be seamlessly moved between on-premises environments and Azure, providing flexibility and scalability.

Security and isolation

Robust security features help protect virtual machines and their data, ensuring that workloads run in isolated environments:

- **Shielded Virtual Machines** provide comprehensive protection for sensitive workloads through BitLocker encryption, Secure Boot verification, and TPM 2.0 attestation. Host Guardian Service (HGS) ensures only authorized hosts can run shielded VMs, protecting against compromised administrators and malicious insiders.
- **Generation 2 virtual machines** deliver enhanced security with UEFI firmware, Secure Boot capabilities, and improved malware resistance. Virtual TPM 2.0 support enables BitLocker encryption within guest operating systems and supports Windows Hello for Business authentication.
- **Network isolation** through Virtual LANs (VLANs), private virtual switches, and software-defined networking (SDN) creates secure network segments that prevent unauthorized lateral movement between workloads.

Storage flexibility

Flexible storage options meet diverse organizational requirements, from local disks to cost-effective hyperconverged solutions to existing SAN deployments:

- **Storage Spaces Direct** creates hyperconverged infrastructure by pooling local storage across cluster nodes into a single, highly available storage namespace. This software-defined storage solution eliminates traditional SAN requirements while delivering enterprise-class features including fault tolerance, automatic repair, and performance optimization. Organizations achieve lower storage costs compared to traditional SAN

solutions while gaining simplified management and improved performance through NVMe and persistent memory support.

- **Storage Spaces** enables disaggregated storage architectures by pooling Just a Bunch of Disks (JBOD) enclosures into resilient storage pools with configurable redundancy levels. This approach provides flexibility to scale storage independently from compute resources while maintaining high availability through mirror, parity, or dual-parity configurations.
- **Local storage** options support various deployment scenarios from development environments using single drives to production workloads with NVMe arrays. Direct-attached storage provides optimal performance for I/O-intensive applications while reducing complexity for smaller deployments or edge computing scenarios.
- **SAN connectivity** supports both iSCSI and Fibre Channel protocols for integration with existing enterprise storage infrastructure. Multipath I/O (MPIO) ensures high availability and load balancing across multiple storage paths, while support for SMB 3.0 and NFS enables file-based storage scenarios with features like transparent failover and scale-out file server clustering.

High availability and business continuity

Comprehensive high availability and business continuity features help ensure mission-critical workloads remain operational:

- **Failover Clustering** with Cluster Shared Volumes (CSV) enables automatic failover of virtual machines between cluster nodes, providing valuable uptime for mission-critical applications. Live migration enables planned maintenance without service interruption. CSV Cache improves performance by caching frequently accessed data in memory, while CSV Block Cache uses system memory on storage-rich nodes to accelerate read operations.
- **Hyper-V Replica** delivers cost-effective disaster recovery by asynchronously replicating virtual machines to secondary sites over standard network connections, eliminating expensive SAN replication requirements. Recovery Point Objectives (RPO) can be as low as 30 seconds to ensure minimal data loss.
- **Shared nothing live migration** enables virtual machines to be moved between hosts without shared storage, enabling flexible resource allocation and load balancing across the cluster. This feature supports scenarios where hosts have different storage configurations or when migrating workloads between geographically distributed sites.
- **Storage Migration** enables moving virtual machine storage between different storage arrays without downtime, facilitating storage upgrades and load balancing across the

infrastructure.

- **Azure Site Recovery** extends Hyper-V disaster recovery capabilities to the cloud, providing automated replication, failover, and failback for virtual machines. This integration enables organizations to use Azure as a secondary site for disaster recovery without complex configuration or extra hardware.

Performance and scalability

High performance and scalability help meet demanding enterprise workloads:

- **Dynamic Memory** automatically adjusts memory allocation based on actual workload demands, maximizing host utilization while maintaining performance. Smart paging and memory overcommit ratios enable higher consolidation ratios.
- **SR-IOV and RDMA support** delivers near-native network performance by bypassing the hypervisor for high-bandwidth, low-latency networking scenarios.
- **GPU-Partitioning** (GPU-P) enables multiple virtual machines to share GPU resources for AI and graphics workloads.
- **Persistent Memory** integration provides ultra-low latency storage access for databases and in-memory computing scenarios.

Advanced management capabilities

A range of management tools and automation capabilities simplify operations and enhance productivity:

- **Windows Admin Center** provides modern web-based management with hyperconverged infrastructure monitoring, performance analytics, and streamlined troubleshooting workflows accessible from any device.
- **Hyper-V Manager** provides intuitive graphical management for day-to-day operations, while the Hyper-V module for Windows PowerShell enables advanced scripting and automation scenarios.
- **System Center Virtual Machine Manager** delivers enterprise-scale management with automated provisioning, compliance monitoring, capacity planning, and self-service portals for delegated administration.
- **PowerShell Direct** enables secure management of virtual machines without network connectivity, simplifying automation and troubleshooting. Extensive PowerShell cmdlet library supports Infrastructure as Code practices.

- **Seamless Active Directory integration** as Hyper-V uses existing Windows authentication infrastructure without extra configuration. Domain-joined hosts inherit existing Group Policy settings and security configurations automatically.

Cost optimization features

Optimize costs associated with virtualization:

- **Existing Windows Server licensing** enables you to run virtual machine without extra hypervisor licensing costs. Hyper-V is included with Windows Server and Windows, with the Datacenter edition of Windows Server providing unlimited virtual machine rights, meaning you can run Windows Server virtual machines at no extra cost. This approach simplifies licensing calculations and reduces overall costs.
- **Deduplication and compression** greatly reduce storage requirements in scenarios such as virtual desktop infrastructure (VDI) and data backup.
- **Nested virtualization** enables running hypervisors within virtual machines, supporting development scenarios and multitenant cloud environments without extra hardware.

Benefits for developers and IT professionals

Hyper-V in both Windows Server and Windows provides developers and IT professionals with powerful virtualization capabilities that accelerate development workflows and streamline IT operations for Windows and Linux environments. Developers can rapidly create isolated development and testing environments that mirror production configurations, enabling them to test applications across multiple operating systems and versions without requiring dedicated hardware. IT professionals benefit from the ability to quickly provision sandbox environments for testing patches, updates, and new configurations before deploying to production systems.

The ability to snapshot virtual machines before making changes provides a safety net for experimentation and rollback scenarios, while virtual machine templates and PowerShell automation enable consistent environment provisioning in minutes rather than hours. This capability is valuable for both development teams creating standardized build environments and IT teams maintaining consistent infrastructure configurations.

For Windows 11 users, Hyper-V offers Quick Create for streamlined virtual machine setup, making it easy to spin up development environments for different projects or create isolated testing environments for IT evaluation scenarios. The integration with Visual Studio and development tools, combined with features like nested virtualization, enables developers to create complex multi-tier applications and test containerized workloads locally, while IT professionals can evaluate new technologies and solutions in isolated environments.

Get started with Hyper-V

Hyper-V is readily available as a server role across all editions of Windows Server 2025, with multiple installation options to meet different organizational needs. Organizations can choose between Server with Desktop Experience for full GUI management or Server Core for reduced attack surface and resource usage.

For Windows 11, Hyper-V is included in the Pro, Enterprise, and Education editions, enabling developers and IT professionals to use virtualization capabilities directly on their desktops.

[Install Hyper-V](#)

Related content

Here are some additional resources to help you learn more about Hyper-V and its capabilities:

- [Hardware requirements for Hyper-V.](#)
- [Plan for Hyper-V networking.](#)
- [Storage architectures for Hyper-V.](#)
- [Virtual machine generations.](#)
- [Supported Windows virtual machine operating systems.](#)
- [Supported Linux and FreeBSD virtual machine operating systems.](#)

ⓘ **Note:** The author created this article with assistance from AI. [Learn more](#)

Hyper-V features and terminology

08/11/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V has many features and terms that are specific to virtualization. This article provides an overview of the most common Hyper-V features and terminology, which can help you understand and optimize your virtualization environment and improve performance. The features and terms are grouped into categories for easier reference.

Virtual machine generations

Hyper-V supports two generations of virtual machines that determine available features and virtual hardware:

- **Generation 1** virtual machines use legacy BIOS firmware and provide compatibility with legacy applications that require older hardware support, including 32-bit systems and legacy hardware emulation such as IDE controllers and virtual floppy disk files.
- **Generation 2** virtual machines use modern UEFI firmware and offer enhanced security features like Secure Boot and vTPM, improved performance, SCSI boot support, and hot add/remove capabilities for network adapters and memory. Generation 2 virtual machines are recommended for most workloads.

You can't change a virtual machine's generation after creation, so choose carefully based on your requirements. For more information, see [Virtual machine generations](#).

Processor/CPU

Hyper-V supports various processor features and capabilities to optimize virtual machine performance and compatibility, which are covered in the following sections.

Processor compatibility

Processor instruction sets pass through to virtual machines from the Hyper-V host by default. Processor compatibility mode enables virtual machines to run on Hyper-V hosts with different processor generations by masking newer CPU features and instructions, providing a fixed commonality. With Hyper-V hosts running Windows Server, you can use live migration to move virtual machines between hosts with different processor models within the same vendor family.

Virtual machines remain portable across your Hyper-V infrastructure even when hardware varies, such as migrating virtual machines to a new cluster or replacing hardware.

Dynamic processor compatibility, introduced in Windows Server 2025, dynamically calculates the common set of processor features across all nodes, enabling virtual machines to take advantage of the maximum capabilities available across the cluster.

For more information, see [Processor compatibility mode](#).

Resource control

Resource control in Hyper-V enables you to manage and allocate CPU resources for virtual machines. You can set limits, reserves, and weights for CPU usage, ensuring that critical workloads receive the necessary resources while preventing resource contention among virtual machines.

NUMA

Non-Uniform Memory Access (NUMA) is a memory architecture that allows multiple processors to access memory in a way that optimizes performance. Hyper-V supports NUMA configurations, enabling virtual machines to take advantage of the underlying hardware's NUMA topology for improved memory access and performance.

NUMA spanning is a feature that allows virtual machines to use memory from multiple NUMA nodes, which can be beneficial for workloads that require large amounts of memory or have high memory bandwidth requirements. This feature is useful in scenarios where virtual machines need to access more memory or virtual processors than available in a single NUMA node. Virtual NUMA presents the host's NUMA topology to the virtual machine operating system.

Memory

Hyper-V provides several memory management features to optimize resource utilization and performance for virtual machines, which are covered in the following sections.

Dynamic memory

Hyper-V hosts can dynamically adjust the amount of memory allocated to a virtual machine based on its current workload while it's in operation, based on maximum, minimum, and startup values. Dynamically increasing or decreasing memory allocation as needed helps

optimize resource utilization by ensuring that virtual machines have the right amount of memory for their workloads without wasting resources.

The operating system of the virtual machine must support dynamic memory, and it must be enabled in the virtual machine settings. All supported Windows Server and Windows operating systems support dynamic memory, as do many Linux distributions.

Smart paging file

Smart paging provides temporary memory relief when a virtual machine's memory demand exceeds available physical memory during startup. Hyper-V creates a smart paging file on disk that serves as a temporary memory buffer, enabling the virtual machine to start successfully even under memory-constrained conditions. This feature is useful in environments where virtual machines might temporarily require more memory than initially allocated, helping to prevent startup failures while maintaining system stability.

Memory weight

Memory weight can prioritize memory allocation for virtual machines based on their importance or workload requirements. By assigning different memory weights to virtual machines, you can control how memory resources are distributed among them, ensuring that critical workloads receive the necessary memory while preventing resource contention.

Physical Address Extension kernel support

Physical Address Extension (PAE) technology allows a 32-bit Linux kernel to access a physical address space that is larger than 4GB. Older Linux distributions such as RHEL 5.x used to ship a separate kernel that was PAE enabled. Newer distributions include prebuilt PAE support.

Security

Hyper-V provides several security features to protect virtual machines and their data, which are covered in the following sections. These features help ensure the integrity, confidentiality, and availability of virtualized workloads.

Secure Boot

Secure Boot helps protect virtual machines from unauthorized code execution during the boot process. It ensures that only trusted and signed boot components are loaded, preventing malware or unauthorized software from compromising the virtual machine's integrity. Secure

Boot is available for Generation 2 virtual machines with both Windows and Linux operating systems and is enabled by default.

For more information, see [Secure Boot](#).

Encryption support

Encryption for virtual machines helps protect data at rest and in transit. This support includes options for a virtual Trusted Platform Module (TPM), virtual machine saved state, and live migration network traffic.

Virtual TPM is available for Generation 2 virtual machines and enables the virtual machine operating system to use hardware-based security functions similar to those functions available on physical machines. A TPM is useful for scenarios requiring enhanced security, such as running Windows 11 or BitLocker drive encryption, that require a TPM.

Both Generation 1 and Generation 2 virtual machines can encrypt the virtual machine saved state and live migration network traffic with a key protector. For more information, see [Key storage drive](#)

Shielded virtual machines

Shielded virtual machines provides a secure environment for sensitive workloads with an extra layer of security by protecting virtual machines from unauthorized access and tampering, such as from a compromised host. These virtual machines work with guarded fabric so they can run only on healthy and approved hosts in the fabric, and use encryption, secure boot, and a virtual TPM to ensure that only trusted code runs within the virtual machine. Shielding is available for Generation 2 virtual machines.

An example use case for shielded virtual machines and guarded fabric enables cloud service providers to provide a more secure environment for tenant virtual machines.

For more information, see [Guarded fabric and shielded VMs overview](#).

Storage

Hyper-V provides various storage features to manage virtual machine disks and optimize storage performance, which are covered in the following sections.

Storage architectures

Hyper-V supports a wide range of storage architectures to accommodate various storage requirements and performance needs. You can use:

- Storage area networks (SANs) for high-performance storage solutions that provide block-level access to virtual machine files.
- Storage Spaces Direct for building highly available and scalable storage solutions using local disks across multiple servers. Storage Spaces Direct is part of Windows Server and enables you to create a software-defined storage solution that provides redundancy, performance, and scalability, with hyperconverged or disaggregated storage architectures.
- Network-attached storage (NAS) for shared storage across multiple Hyper-V hosts, enabling high availability and failover clustering.
- Local storage, using NVMe, SSD or HDD, for fast access to virtual machine files.

You can use a combination of these storage architectures to meet your specific requirements. For more information, see [Storage architectures for Hyper-V](#).

Virtual hard disks

Virtual hard disks (VHDs) are files that represent a virtual machine's hard drive. Hyper-V supports two types of virtual hard disk formats:

- **VHD**: the original format, which supports up to 2TB of storage.
- **VHDX**: the newer format, which supports up to 64TB of storage. VHDX provides several benefits including better performance, enhanced data corruption protection, online resizing, support for larger logical sector sizes (4KB), and automatic data alignment and trimming for better storage utilization.

Virtual hard disks provide a flexible and efficient way to manage virtual machine storage with features such as:

- **Differencing disks**: create a new disk based on an existing parent disk. Changes made to the differencing disk don't affect the parent disk, enabling you to maintain a clean base image while allowing for modifications in the differencing disk. Differencing disks are useful for scenarios such as testing, development, or creating snapshots of virtual machines.
- **Pass-through disks**: physical disks that are directly attached to a virtual machine, bypassing the virtual hard disk layer. The virtual machine can access the physical disk directly, providing better performance for certain workloads that require low-level disk

access. Pass-through disks are typically used for high-performance applications or scenarios where direct access to the physical disk is necessary.

- **Shared drives:** multiple virtual machines can access the same virtual hard disk simultaneously for Generation 2 virtual machines. Shared drives are useful for scenarios such as clustering or high availability, where multiple virtual machines need to share data or resources.
- **Storage Quality of Service:** manage and control the performance of virtual machine storage by setting minimum and maximum Input/Output Operations Per Second (IOPS) limits for virtual hard disks. Critical workloads receive the necessary storage performance while preventing resource contention among virtual machines.

Virtual Fibre Channel

Support for Virtual Fibre Channel adapters enables virtual machines to connect to Fibre Channel storage area networks (SANs) directly, enabling them to access SAN resources as if they were physical servers. This feature is useful for scenarios where high-performance storage access is required, such as in enterprise environments with large databases or mission-critical applications.

Virtual Fibre Channel SAN aggregates multiple physical storage devices into a single logical storage pool. You can add a Virtual Fibre Channel adapter to a virtual machine to connect it to the Virtual Fibre Channel SAN.

Network

Hyper-V provides various networking features to manage virtual machine connectivity and optimize network performance, which are covered in the following sections. To learn more about Hyper-V networking, see [Plan for Hyper-V networking](#).

Virtual switch

A virtual switch is a software-based network switch that allows virtual machines to communicate with each other and with the external network. Hyper-V supports three types of virtual switches:

- **External:** connects virtual machines to the physical network, enabling them to communicate with external devices and services.
- **Internal:** allows communication between virtual machines on the same host and with the host operating system, but not with the external network.

- **Private:** allows communication only between virtual machines on the same host, without any connectivity to the host operating system or external network.

Virtual network adapter

Virtual network adapters are virtualized network interface cards (NICs) that allow virtual machines to connect to virtual switches and communicate with other virtual machines or external networks. By default, virtual network adapters are high performing and use synthetic drivers, which provide better performance and lower CPU overhead compared to legacy network adapters.

Legacy network adapters are also available that provide compatibility with older operating systems or applications that don't support synthetic adapters.

Virtual network adapters can be configured with various settings, such as:

- **Static or dynamic MAC addresses:** assign a static MAC address to a virtual network adapter or allow Hyper-V to dynamically assign one.
- **MAC address spoofing:** enable virtual machines to change their MAC addresses, which can be useful for scenarios such as network testing or security analysis.
- **VLAN support:** virtual network adapters can be configured to use VLANs (Virtual Local Area Networks) for network segmentation and isolation.
- **Bandwidth management:** set bandwidth limits for virtual network adapters to control network traffic and ensure fair resource allocation among virtual machines.
- **Network virtualization:** create isolated virtual networks that can coexist on the same physical infrastructure. Network virtualization is useful for scenarios such as multitenant environments or testing different network configurations without affecting the physical network.
- **IPsec task offloading:** virtual machines can offload IPsec processing to the host's network adapter, which improves performance and reduces CPU overhead for secure network communication.
- **DHCP guard:** prevent virtual machines from acting as DHCP servers, ensuring that only authorized DHCP servers can provide IP addresses to virtual machines on the network.
- **Router advertisement guard:** prevent virtual machines from sending router advertisements, ensuring that only authorized routers can advertise their presence on the network.

- **Protected network:** create a secure network environment for virtual machines, ensuring that only authorized virtual machines can communicate with each other and with the external network.
- **Port mirroring:** monitor network traffic by mirroring the traffic of a virtual network adapter to another virtual network adapter or a physical network adapter. Port mirroring is useful for scenarios such as network troubleshooting, performance monitoring, or security analysis.
- **NIC teaming:** combine multiple physical network adapters into a single logical adapter for improved performance and redundancy. This feature is useful for scenarios where high network throughput or fault tolerance is required.
- **NIC device naming:** assign custom names to virtual network adapters, making it easier to identify and manage them in the Hyper-V Manager or PowerShell. Only available for Generation 2 virtual machines.
- **NIC teaming in the guest:** configure NIC teaming within the virtual machine itself, enabling the virtual machine to take advantage of multiple network adapters for improved performance and redundancy.
- **Static IP injection for Hyper-V Replica failover:** inject static IP addresses into virtual machines during Hyper-V Replica failover, ensuring that the virtual machines can maintain their network connectivity after failover. It supports both IPv4 and IPv6 addresses.
- **Virtual machine queue (VMQ):** virtual machines can offload network processing to the host's network adapter, which improves network performance and reduces CPU overhead for network-intensive workloads.
- **Single Root I/O Virtualization (SR-IOV):** virtual machines can directly access physical network hardware, providing improved performance and reduced latency for network-intensive workloads. SR-IOV is useful for scenarios where low-latency network access is critical, such as in high-frequency trading or real-time data processing applications.

Controllers and ports

Hyper-V supports various controllers and ports to manage virtual machine hardware and connectivity.

- **SCSI controller:** connect virtual hard disks and DVD drives to a virtual machine as SCSI devices. It provides better performance and flexibility compared to IDE controllers, especially for Generation 2 virtual machines.

- **IDE controller:** connect virtual hard disks and DVD drives to a virtual machine as IDE devices. IDE controllers are only available for Generation 1 virtual machines.
- **COM ports:** connect serial devices or applications through a named pipe on a host or remote computer. COM ports are only available for Generation 1 virtual machines.
- **Fibre Channel adapter:** connect virtual machines to Fibre Channel storage area networks (SANs) for high-performance storage access, such as in enterprise environments with large databases or mission-critical applications.
- **Diskette drives:** connect virtual floppy disk (.vfd) files to Generation 1 virtual machines. Virtual floppy disks are used for booting from floppy disk images or for legacy applications that require floppy disk support.

Integration Services

Integration Services in Hyper-V are a set of services and drivers that enhance the performance and functionality of virtual machines. To learn more about Integration Services, see [Hyper-V Integration Services](#).

- **Operating system shutdown:** the host can gracefully shut down a virtual machine when the host itself is shutting down or when the virtual machine is being stopped. The guest operating system can perform a clean shutdown, preventing data loss or corruption.
- **Time synchronization:** synchronize the guest operating system's clock with the host's clock, ensuring accurate timekeeping within the virtual machine. Time synchronization is important for applications that rely on accurate timestamps or time-sensitive operations. Hyper-V benefits from time accuracy improvements, first introduced in Windows Server 2016, which provides more accurate time synchronization between the host and guest operating systems. For more information, see [Time accuracy improvements for Windows Server 2016](#).
- **Data Exchange:** a mechanism for exchanging data between the host and the guest operating system, enabling the host to retrieve information about the virtual machine, such as its name, IP address, and other configuration details.
- **Heartbeat:** a heartbeat mechanism that allows the host to monitor the health and status of the virtual machine. The host can detect if the guest operating system is responsive or unresponsive, enabling proactive management and troubleshooting.
- **Backup (volume shadow copy):** uses the Volume Shadow Copy Service (VSS) for creating application-consistent backups of virtual machines, ensuring that data is in a consistent state during backup operations.

- **Guest services:** provides an interface for the Hyper-V host to bidirectionally copy files to or from the virtual machine. Guest services isn't enabled by default.

Checkpoints

Checkpoints enable you to capture the state of a virtual machine at a specific point in time. You can use the checkpoint to roll back to that point, which can be useful for testing, development, or recovery purposes. There are two types of checkpoints:

- **Production checkpoints:** These checkpoints use the Volume Shadow Copy Service (VSS) to create application-consistent snapshots of the virtual machine. They're suitable for production environments and ensure that the virtual machine's data is in a consistent state.
- **Standard checkpoints:** captures the virtual machine's state (running or powered off), including memory, disk, and device states. They're suitable for testing and development scenarios but aren't application consistent.

Optional automatic checkpoints create a checkpoint automatically when a virtual machine is started and merged when the virtual machine is stopped, enabling you to quickly revert to a known good state if you forget to create a checkpoint manually.

To learn more about checkpoints, see [Hyper-V checkpoints](#).

Migration and replication

Migration and replication are essential features to help maintain high availability and flexibility in Hyper-V environments, which are covered in the following sections.

Live migration

Live migration allows you to move a running virtual machine from one Hyper-V host to another without downtime. Migrating virtual machines is useful for load balancing, hardware maintenance, or disaster recovery scenarios. Live migration can be performed over a network using a dedicated network SMB or TCP/IP, and it can also use RDMA (Remote Direct Memory Access) for faster transfers. Live migration is only available for Hyper-V in Windows Server.

Virtual machines can be migrated between hosts in the same cluster or between standalone hosts. In a cluster, automatic balancing can be configured to automatically migrate virtual machines based on host resource utilization, ensuring optimal performance and resource allocation.

To learn more about live migration, see [Live Migration Overview](#).

Storage migration

Storage migration allows you to move a virtual machine's storage files (such as virtual hard disks) from one location to another without downtime. Migrating storage is useful for scenarios such as moving virtual machines to different storage devices, optimizing storage performance, or reorganizing storage resources. Storage migration is only available for Hyper-V in Windows Server.

Replication

Hyper-V Replica enables you to replicate virtual machines asynchronously from one Hyper-V host to another to maintain a secondary copy of a virtual machine on a different host for disaster recovery purposes. Hyper-V Replica is only available for Hyper-V in Windows Server.

Replication can be configured to replicate virtual machines between hosts in the same cluster or between standalone hosts. It supports both full and incremental replication, allowing you to choose the level of replication based on your requirements.

Replication can be configured with various options, such as:

- **Replication frequency:** configure the frequency of replication, such as every 30 seconds, 5 minutes, or 15 minutes, to balance performance and data consistency.
- **Authentication:** supports Kerberos and certificate-based authentication for secure replication.
- **Compression:** compress data during transfer to reduce bandwidth usage.
- **Recovery points:** specify how many recovery points to retain for each replicated virtual machine, allowing you to recover to a specific point in time.
- **VSS snapshots:** Hyper-V Replica uses Volume Shadow Copy Service (VSS) to create application-consistent snapshots of virtual machines during replication. VSS ensures that the replicated data is in a consistent state, even for applications running within the virtual machine.
- **Exclusion of disks from replication:** You can exclude specific virtual hard disks from replication, allowing you to control which disks are replicated and which aren't.

To learn more about Hyper-V Replica, see [Set up Hyper-V Replica](#).

Graphics

GPUs can be used to enhance the graphics capabilities of virtual machines in Hyper-V environments, enabling scenarios that require high-performance graphics rendering, such as gaming, 3D modeling, or video editing.

Hyper-V supports two options for GPU usage in virtual machines:

- **Discrete Device Assignment:** assign a physical GPU directly to a virtual machine, providing it with exclusive access to the GPU's resources. Discrete Device Assignment is suitable for scenarios where high-performance graphics processing is required, such as running graphics-intensive applications or games within the virtual machine.
- **GPU partitioning:** allocate a portion of a physical GPU's resources to multiple virtual machines, enabling them to share the GPU for graphics processing. GPU partitioning is useful for scenarios where multiple virtual machines need access to graphics capabilities without requiring dedicated GPUs for each one.

To learn more about GPU acceleration in Hyper-V, see [Plan for GPU acceleration](#).

Nested virtualization

Nested virtualization enables you to run Hyper-V within a virtual machine to run more virtual machines, enabling scenarios such as testing, development, or training without requiring physical hardware. This feature is useful for scenarios where you want to create and manage virtual machines within a virtualized environment, such as running Hyper-V on a laptop or desktop computer.

Nested virtualization is available for Generation 2 virtual machines. To learn more about nested virtualization, see [What is Nested Virtualization](#).

Management

To manage Hyper-V environments and virtual machines, you can use various tools and interfaces:

- **Hyper-V Manager:** a built-in graphical user interface (GUI) tool for managing Hyper-V hosts and virtual machines. It provides an easy way to create, configure, and maintain virtual machines. To learn more about Hyper-V Manager, see [Remotely manage Hyper-V hosts with Hyper-V Manager](#).

- **Windows Admin Center:** a web-based management tool that provides a centralized interface for managing Hyper-V hosts and virtual machines, along with other Windows Server features. To learn more about using Windows Admin Center for Hyper-V management, see [Manage virtual machines by using Windows Admin Center](#).
- **PowerShell:** a powerful command-line interface that allows you to automate and script Hyper-V management tasks. PowerShell provides a rich set of cmdlets for managing virtual machines, hosts, and other Hyper-V features. To learn more about using PowerShell for Hyper-V management, see [Working with Hyper-V and Windows PowerShell](#).
- **System Center Virtual Machine Manager:** a comprehensive management solution for large-scale Hyper-V environments. Virtual Machine Manager provides advanced features for managing virtual machines, hosts, storage, and networking, along with support for multi-hypervisor environments. To learn more about Virtual Machine Manager, see [What is Virtual Machine Manager](#).

Console access

Console access provides a way to interact with virtual machines through a graphical user interface (GUI) or command-line interface (CLI). Hyper-V supports several methods for console access:

- **Virtual Machine Connection:** a built-in tool in Hyper-V Manager that provides console access for connecting to and managing virtual machines. It enables you to interact with the virtual machine's desktop, perform administrative tasks, and troubleshoot issues. It's also referred to as **VMConnect**. To learn more about Virtual Machine Connection, see [Hyper-V Virtual Machine Connection](#).
- **Enhanced session mode:** part of Virtual Machine Connection, connect to a virtual machine using Remote Desktop Protocol (RDP) over a direct connection through the Hyper-V host. As it uses RDP, it also supports *RemoteFX* that provides extra capabilities, such as clipboard sharing, drive redirection, and printer redirection. Enhanced session mode is available for Generation 2 virtual machines and requires support from the virtual machine operating system. To learn more about enhanced session mode, see [Use local resources on Hyper-V virtual machine with VMConnect](#).
- **PowerShell Direct:** run PowerShell commands directly on a virtual machine from the Hyper-V host without requiring network connectivity. It provides a secure way to manage virtual machines without exposing them to the network. To learn more about PowerShell Direct, see [Manage Windows virtual machines with PowerShell Direct](#).

Power management

Hyper-V enables you to control how your virtual machines start and stop on a host:

- **Automatic start action:** specifies what action to take when the Hyper-V service starts and the virtual machine was running when the service stopped. Options include **Nothing**, **Automatically start if it was running when the service stopped**, **Always start this virtual machine automatically**, and **Automatic start delay (in seconds)**.
- **Automatic stop action:** specifies what action to take when the Hyper-V service stops. Options include **Save the virtual machine state**, **Turn off the virtual machine**, and **Shut down the guest operating system**.

Related content

Here are some other resources to help you learn more about Hyper-V and its capabilities:

- [Hardware requirements for Hyper-V.](#)
- [Plan for Hyper-V networking.](#)
- [Storage architectures for Hyper-V.](#)
- [Virtual machine generations.](#)
- [Supported Windows virtual machine operating systems.](#)
- [Supported Linux and FreeBSD virtual machine operating systems.](#)

 **Note:** The author created this article with assistance from AI. [Learn more](#)

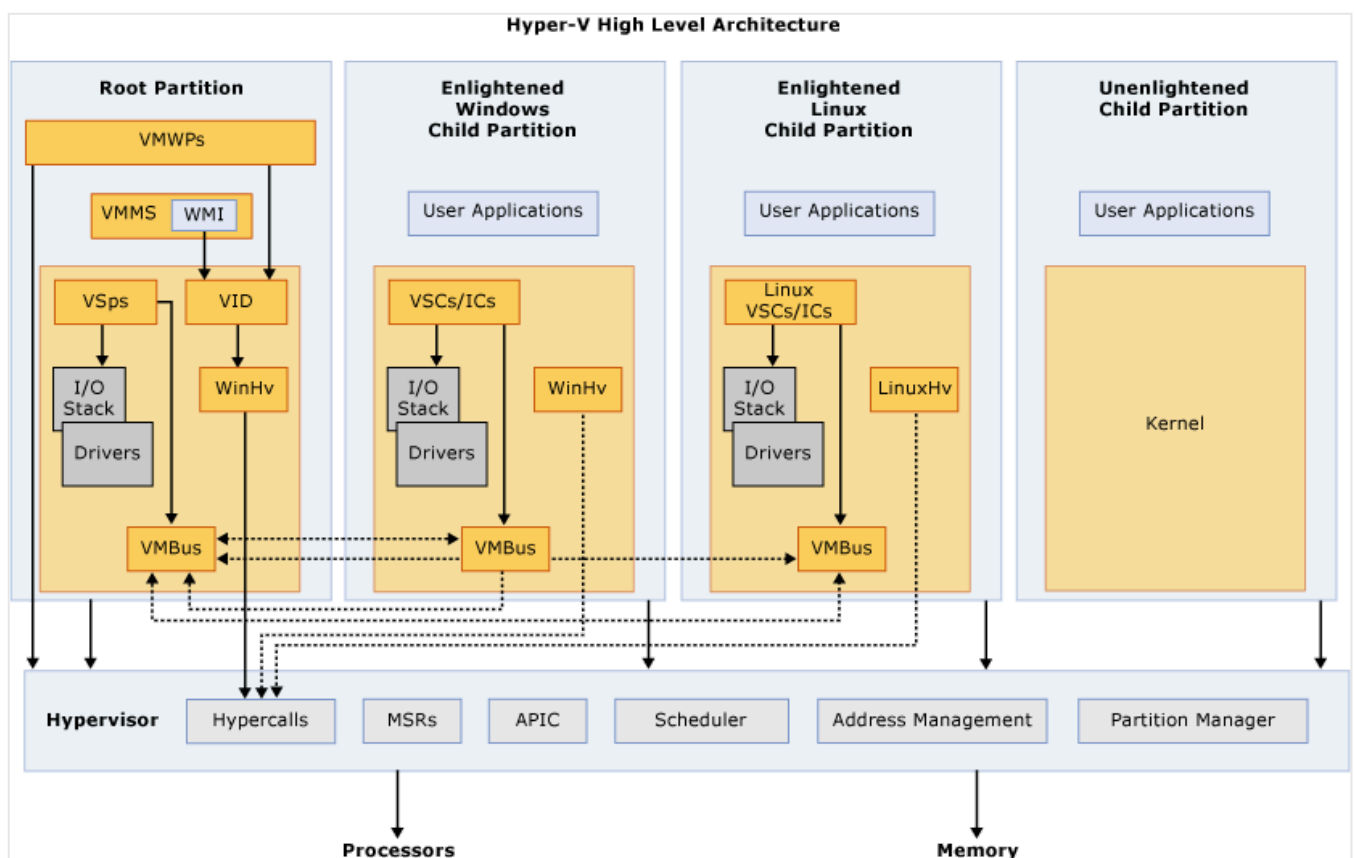
Hyper-V Architecture

Applies to:  Windows Server 2025,  Windows Server 2022,  Windows Server 2019,  Windows Server 2016,  Windows 11,  Windows 10,  Azure Local 2311.2 and later

Hyper-V is a hypervisor-based virtualization technology for certain x64 versions of Windows. The hypervisor is core to virtualization. It is the processor-specific virtualization platform that allows multiple isolated operating systems to share a single hardware platform.

Hyper-V requires a processor that includes hardware assisted virtualization, such as is provided with Intel VT or AMD Virtualization (AMD-V) technology.

The following diagram provides a high-level overview of the architecture of a Hyper-V environment.



Root and child partitions

Hyper-V supports isolation in terms of a partition. A partition is a logical unit of isolation, supported by the hypervisor, in which operating systems execute. The Microsoft hypervisor must have at least one parent, or root, partition, running Windows. The virtualization management stack runs in the parent partition and has direct access to hardware devices. The root partition then creates the child partitions which host the guest operating systems. A root partition creates child partitions using the hypercall application programming interface (API).

Child partitions don't have direct access to other hardware resources and are presented a virtual view of the resources, as virtual devices (VDevs). Requests to the virtual devices are redirected either via the VMBus or the hypervisor to the devices in the parent partition, which handles the requests.

Interrupt and memory handling

Partitions don't have access to the physical processor, nor do they handle the processor interrupts. Instead, they have a virtual view of the processor and run in a virtual memory address region that is private to each guest partition. The hypervisor handles the interrupts to the processor, and redirects them to the respective partition.

Hyper-V can also hardware accelerate the address translation between various guest virtual address spaces by using an Input Output Memory Management Unit (IOMMU) which operates independent of the memory management hardware used by the CPU. An IOMMU is used to remap physical memory addresses to the addresses that are used by the child partitions.

ⓘ Note

Second Level Address Translation (SLAT) is required for Hyper-V on Windows Server 2016 and later.

VMBus, VSP, and VSC

The VMBus is a logical inter-partition communication channel. The parent partition hosts Virtualization Service Providers (VSPs) which communicate over the VMBus to handle device access requests from child partitions. Child partitions host Virtualization Service Consumers (VSCs) which redirect device requests to VSPs in the parent partition via the VMBus. This entire process is transparent to the guest operating system.

Enlightened I/O and integration services

Virtual devices can also take advantage of a Windows Server Virtualization feature, named Enlightened I/O, for storage, networking, graphics, and input subsystems. Enlightened I/O is a specialized virtualization-aware implementation of high level communication protocols (such as SCSI) that utilize the VMBus directly, bypassing any device emulation layer. This makes the communication more efficient but requires an enlightened guest that is hypervisor and VMBus aware.

Hyper-V enlightened I/O and a hypervisor aware kernel is provided via Hyper-V integration services. Integration components, which include Virtual Server Client (VSC) drivers, are also available for other client operating systems.

Glossary

- **APIC** – Advanced Programmable Interrupt Controller – A device which allows priority levels to be assigned to its interrupt outputs.
- **Child Partition** – Partition that hosts a guest operating system - All access to physical memory and devices by a child partition is provided via the Virtual Machine Bus (VMBus) or the hypervisor.
- **Hypercall** – Interface for communication with the hypervisor - The hypercall interface accommodates access to the optimizations provided by the hypervisor.
- **Hypervisor** – A layer of software that sits between the hardware and one or more operating systems. Its primary job is to provide isolated execution environments called partitions. The hypervisor controls and arbitrates access to the underlying hardware.
- **IC** – Integration component – Component that allows child partitions to communication with other partitions and the hypervisor.
- **IOMMU** – Input/Output Memory Management Unit – A memory management unit that connects a direct-memory-access-capable I/O bus to main memory, remapping physical addresses to guest physical addresses for device isolation.
- **I/O stack** – Input/output stack
- **MSR** – Model-Specific Registers. Used for status and control values.
- **Root Partition** – Sometimes called parent partition. Manages machine-level functions such as device drivers, power management, and device hot addition/removal. The root (or parent) partition is the only partition that has direct access to physical memory and devices.
- **VDev** – Virtual Device – A virtualized representation of a hardware device presented to child partitions. VDevs abstract physical hardware, allowing guest operating systems to interact with devices through the VMBus or hypervisor.
- **VID** – Virtualization Infrastructure Driver – Provides partition management services, virtual processor management services, and memory management services for partitions.
- **VMBus** – Channel-based communication mechanism used for inter-partition communication and device enumeration on systems with multiple active virtualized partitions. The VMBus is installed with Hyper-V Integration Services.
- **VMMS** – Virtual Machine Management Service – Responsible for managing the state of all virtual machines in child partitions.
- **VMWP** – Virtual Machine Worker Process – A user mode component of the virtualization stack. The worker process provides virtual machine management services from the Windows Server 2008 instance in the parent partition to the guest operating systems in

the child partitions. The Virtual Machine Management Service spawns a separate worker process for each running virtual machine.

- **VSC** – Virtualization Service Client – A synthetic device instance that resides in a child partition. VSCs utilize hardware resources that are provided by Virtualization Service Providers (VSPs) in the parent partition. They communicate with the corresponding VSPs in the parent partition over the VMBus to satisfy a child partitions device I/O requests.
- **VSP** – Virtualization Service Provider – Resides in the root partition and provide synthetic device support to child partitions over the Virtual Machine Bus (VMBus).
- **WinHv** – Windows Hypervisor Interface Library - WinHv is essentially a bridge between a partitioned operating system's drivers and the hypervisor which allows drivers to call the hypervisor using standard Windows calling conventions
- **WMI** – The Virtual Machine Management Service exposes a set of Windows Management Instrumentation (WMI)-based APIs for managing and controlling virtual machines.

Hyper-V maximum scale limits in Windows Server

10/17/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Hyper-V can scale to provide your workloads with the resources they need. This article details Hyper-V scalability limits in supported versions of Windows Server for virtual machines (VMs) and hosts, including virtual processors, memory, storage, and checkpoints. Use these maximums to plan capacity and support demanding workloads.

As you plan your deployment, consider these maximums that apply to each virtual machine and host. Maximums continue to grow in Windows Server versions, in response to requests to support newer scenarios such as machine learning and data analytics.

Maximums for Hyper-V virtual machines

These maximums apply to each virtual machine for the version of Windows Server running on a Hyper-V host. Some components have different maximums depending on whether the VM is generation 1 or generation 2. For a comparison of the generations, see [Should I create a generation 1 or 2 virtual machine in Hyper-V?](#).

 **Important**

The guest operating system you run in the virtual machine might support less than Hyper-V's maximum. Check with your operating system vendor for its supported limits and choose the appropriate VM generation before deploying at scale. Vendor limits might change; verify against current updates.

Select the relevant tab to see the maximums for each supported version of Windows Server.

Windows Server 2025 host
Here are the limits for virtual machines on a Hyper-V host running Windows Server 2025:  Expand table

Component	Maximum Limit
Virtual processors	2,048 (generation 2 - for example: 64 sockets with 32 virtual processors per NUMA node, PowerShell only); 64 (generation 1).
Memory	240 TB (generation 2); 1 TB (generation 1).
Virtual hard disk capacity	64 TB (VHDX); 2,040 GB (VHD).
Virtual hard disks	256 (SCSI); 4 (IDE, generation 1 only).
Physical disk pass-through	The operating system determines the maximum size for the virtual machine.
Virtual network adapters	64 standard (generation 2); 8 standard + 4 legacy (generation 1).
Virtual SCSI controllers	4
Virtual SCSI devices	256. Each SCSI controller supports up to 64 virtual hard disks and/or DVD drives.
Virtual IDE controllers (generation 1 only)	2. Each IDE controller supports up to 2 IDE devices (hard disks and/or DVD drives) per controller.
Virtual IDE devices (generation 1 only)	4. For generation 1 VMs, the startup/boot disk must be attached to one of the IDE devices.
Virtual DVD drives	256 (SCSI); 4 (IDE, generation 1 only).
Virtual Fibre Channel adapters	4. We recommended that you connect each virtual Fibre Channel Adapter to a different virtual SAN.
Serial (COM) ports (generation 1 only)	2
Virtual floppy devices (generation 1 only)	1
Checkpoints	50, dependent on the amount of available storage.

Maximums for Hyper-V hosts

These maximums apply to each Hyper-V host running Windows Server. Select the relevant tab to see the maximums for each supported version of Windows Server.

Windows Server 2025

Here are the limits for a Hyper-V host running Windows Server 2025:

[Expand table](#)

Component	Maximum Limit
Running virtual machines per server	1,024
Logical processors	2,048
Virtual processors available to the host	2,048. This limit is applied to the host operating system using the Hyper-V root partition.
Virtual processors per logical processor	No ratio imposed by Hyper-V.
Virtual processors in-use per host	2,048
Memory	4 PB (5-level paging); 256 TB (4-level paging).
Physical network adapters	No limits imposed by Hyper-V.
Network adapter teams (NIC teaming)	No limits imposed by Hyper-V.
Virtual network switch ports per server	No limits imposed by Hyper-V. The practical limit depends on the available computing resources.
Virtual switches	No limits imposed by Hyper-V. The practical limit depends on the available computing resources.
Storage	No limits imposed by Hyper-V.
Virtual storage area networks (SANs)	No limits imposed by Hyper-V.

Maximums for Failover Cluster with Hyper-V hosts

These maximums apply in a Failover Cluster with Hyper-V hosts for all supported versions of Windows Server. It's important to do capacity planning to ensure that there's enough hardware resources to run all the virtual machines in a clustered environment.

[Expand table](#)

Component	Maximum Limit
Nodes per cluster	64

Component	Maximum Limit
Running virtual machines per cluster	8,000

 **Note:** The author created this article with assistance from AI. [Learn more](#)

System requirements for Hyper-V on Windows and Windows Server

07/25/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V has specific hardware requirements, and some Hyper-V features have other requirements. Use the details in this article to decide what requirements your system must meet so you can use Hyper-V the way you plan to.

Operating system requirements

Hyper-V can be enabled on these versions of Windows:

- Windows 10 Professional or Enterprise
- Windows 11 Professional or Enterprise

General requirements

Regardless of the Hyper-V features you want to use, you need:

- A 64-bit processor with second-level address translation (SLAT). To install the Hyper-V virtualization components such as Windows hypervisor, the processor must have SLAT. However, it's not required to install Hyper-V management tools like Virtual Machine Connection (VMConnect), Hyper-V Manager, and the Hyper-V cmdlets for Windows PowerShell. See [How to check for Hyper-V requirements](#) to find out if your processor has SLAT.
- VM Monitor Mode extensions.
- Enough memory. Plan for at least 4 GB of RAM. More memory is better. You need enough memory for the host and all virtual machines that you want to run at the same time.
- Virtualization support turned on in the BIOS or UEFI:
 - Hardware-assisted virtualization. This is available in processors that include a virtualization option: specifically processors with Intel Virtualization Technology (Intel VT) or AMD Virtualization (AMD-V) technology.
 - Hardware-enforced Data Execution Prevention (DEP) must be available and enabled. For Intel systems, this is the XD bit (execute disable bit). For AMD systems, this is the

NX bit (no execute bit).

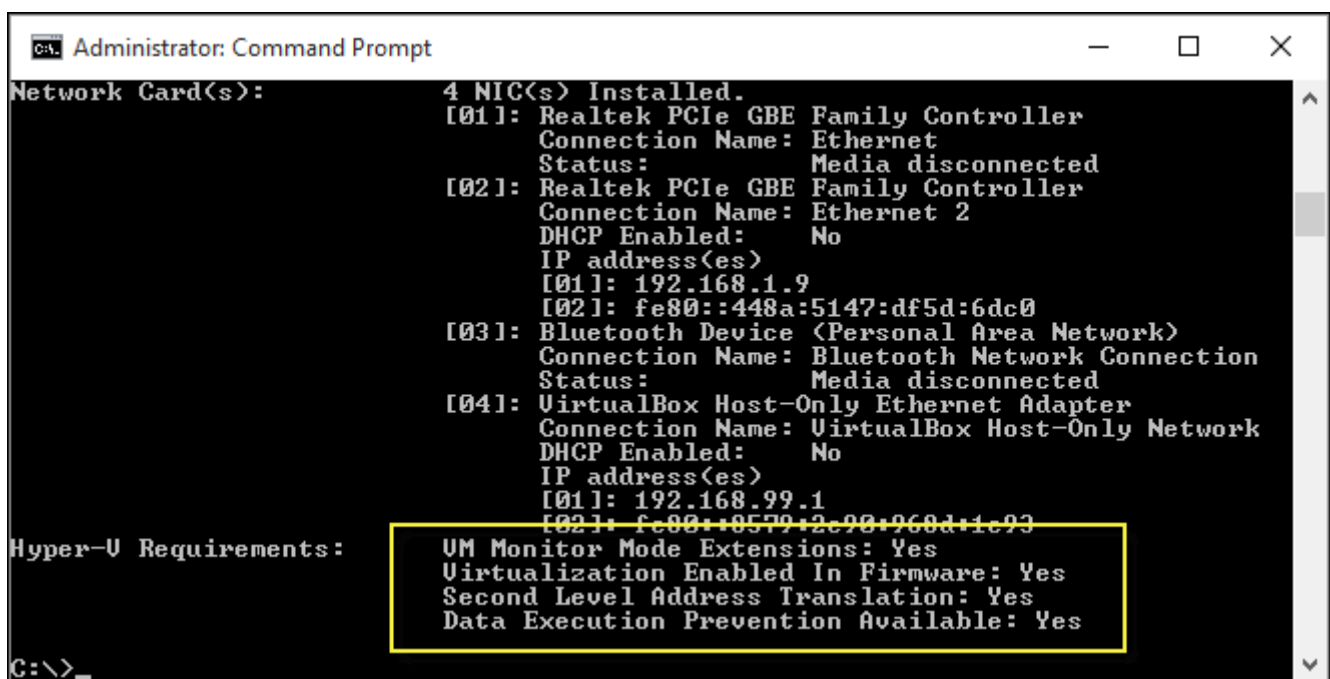
How to check for Hyper-V requirements

Open Windows PowerShell or a command prompt and type:

Windows Command Prompt

Systeminfo.exe

Scroll to the Hyper-V requirements section to review the report. If all listed Hyper-V requirements have a value of **Yes**, your system can run the Hyper-V role. If any item returns **No**, check the requirements listed in this document and make adjustments where possible.



```
Administrator: Command Prompt
Network Card(s):      4 NIC(s) Installed.
                      [01]: Realtek PCIe GBE Family Controller
                        Connection Name: Ethernet
                        Status:        Media disconnected
                      [02]: Realtek PCIe GBE Family Controller
                        Connection Name: Ethernet 2
                        DHCP Enabled:  No
                        IP address(es)
                        [01]: 192.168.1.9
                        [02]: fe80::448a:5147:df5d:6dc0
                      [03]: Bluetooth Device (Personal Area Network)
                        Connection Name: Bluetooth Network Connection
                        Status:        Media disconnected
                      [04]: VirtualBox Host-Only Ethernet Adapter
                        Connection Name: VirtualBox Host-Only Network
                        DHCP Enabled:  No
                        IP address(es)
                        [01]: 192.168.99.1
                        [02]: fe80::0579:2c90:968d:1c93
Hyper-V Requirements:  VM Monitor Mode Extensions: Yes
                      Virtualization Enabled In Firmware: Yes
                      Second Level Address Translation: Yes
                      Data Execution Prevention Available: Yes
C:\>
```

Requirements for specific features

This section lists the requirements for discrete device assignment and shielded virtual machines.

Discrete device assignment

Host requirements are similar to the existing requirements for the SR-IOV feature in Hyper-V.

- The processor must have either Intel's Extended Page Table (EPT) or AMD's Nested Page Table (NPT).
- The chipset must have:

- Interrupt remapping - Intel's VT-d with the Interrupt Remapping capability (VT-d2) or any version of AMD I/O Memory Management Unit (I/O MMU).
- DMA remapping - Intel's VT-d with Queued Invalidations or any AMD I/O MMU.
- Access control services (ACS) on PCI Express root ports.
- The firmware tables must expose the I/O MMU to the Windows hypervisor. This feature might be turned off in the UEFI or BIOS. For instructions, see the hardware documentation or contact your hardware manufacturer.

Devices need GPU or nonvolatile memory express (NVMe). For GPU, only certain devices support discrete device assignment. To verify, see the hardware documentation or contact your hardware manufacturer. For details about this feature, including how to use it and considerations, see [Discrete Device Assignment--Description and background](#) in the Virtualization blog.

Final check

If all OS, hardware, and compatibility requirements are met, you see **Hyper-V** in **Control Panel: Turn Windows features on or off**, and it has two options:

- Hyper-V platform
- Hyper-V Management Tools

ⓘ Note

If you see **Windows Hypervisor Platform** instead of **Hyper-V** in **Control Panel: Turn Windows features on or >off**, your system might not be compatible with Hyper-V. Check the preceding requirements. If you run **systeminfo** on an existing Hyper-V host, the Hyper-V Requirements section reads:

Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.

Hyper-V feature compatibility by generation and guest

06/19/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

The tables in this article show you the generations and operating systems that are compatible with some of the Hyper-V features, grouped by categories. In general, you'll get the best availability of features with a generation 2 virtual machine that runs the newest operating system.

Keep in mind that some features rely on hardware or other infrastructure. For hardware details, see [System requirements for Hyper-V on Windows Server](#). In some cases, a feature can be used with any supported guest operating system. For details on which operating systems are supported, see:

- [Supported Linux and FreeBSD virtual machines](#)
- [Supported Windows guest operating systems](#)

Availability and backup

 Expand table

Feature	Generation	Guest operating system
Checkpoints	1 and 2	Any supported guest
Guest clustering	1 and 2	Guests that run cluster-aware applications and have iSCSI target software installed
Replication	1 and 2	Any supported guest
Domain Controller	1 and 2	Any supported Windows Server guest using only production checkpoints. See Supported Windows Server guest operating systems

Compute

 Expand table

Feature	Generation	Guest operating system
Dynamic memory	1 and 2	Specific versions of supported guests. See Hyper-V Dynamic Memory Overview for versions older than Windows Server 2016 and Windows 10.
Hot add/removal of memory	1 and 2	Windows Server 2016, Windows 10
Virtual NUMA	1 and 2	Any supported guest

Development and test

 Expand table

Feature	Generation	Guest operating system
COM/Serial ports	1 and 2 Note: For generation 2, use Windows PowerShell to configure. For details, see Add a COM port for kernel debugging .	Any supported guest

Mobility

 Expand table

Feature	Generation	Guest operating system
Live migration	1 and 2	Any supported guest
Import/export	1 and 2	Any supported guest

Networking

 Expand table

Feature	Generation	Guest operating system
Hot add/removal of virtual network adapter	2	Any supported guest
Legacy virtual network adapter	1	Any supported guest

Feature	Generation	Guest operating system
Single root input/output virtualization (SR-IOV)	1 and 2	64-bit Windows guests, starting with Windows Server 2012 and Windows 8.
Virtual machine multi queue (VMMQ)	1 and 2	Any supported guest

Remote connection experience

 Expand table

Feature	Generation	Guest operating system
Discrete device assignment (DDA)	1 and 2	Windows Server 2012 and later, Windows 10 and Windows 11
Enhanced session mode	1 and 2	Windows Server 2012 R2 and later, and Windows 8.1 and later, with Remote Desktop Services enabled Note: You might need to also configure the host. For details, see Use local resources on Hyper-V virtual machine with VMConnect .
RemoteFx	1 and 2	Generation 1 on 32-bit and 64-bit Windows versions starting with Windows 8. Generation 2 on 64-bit Windows 10 and Windows 11 versions

Security

 Expand table

Feature	Generation	Guest operating system
Secure boot	2	Linux: Ubuntu 14.04 and later, SUSE Linux Enterprise Server 12 and later, Red Hat Enterprise Linux 7.0 and later, and CentOS 7.0 and later Windows: All supported versions that can run on a generation 2 virtual machine
Shielded virtual machines	2	Windows: All supported versions that can run on a generation 2 virtual machine

Storage

 Expand table

Feature	Generation	Guest operating system
Shared virtual hard disks (VHDX only)	1 and 2	Windows Server 2012 and later
SMB3	1 and 2	All that support SMB3
Storage spaces direct	2	Windows Server 2016 and later
Virtual Fibre Channel	1 and 2	Windows Server 2012 and later
VHDX format	1 and 2	Any supported guest

Guest operating system and application supportability on Hyper-V

Article • 12/05/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local, versions 23H2 and 22H2](#)

Hyper-V is a hypervisor that is broadly used in many Microsoft server products, including the Windows Server family (Datacenter, Standard, and Essentials editions) and Azure Local. Hyper-V provides a platform with broad ecosystem support and compatibility. This article clarifies which versions of Windows Server or Azure Local map to which Hyper-V build numbers. This helps understand supported scenarios where a guest operating system or application has been validated for Hyper-V.

Although the different products that include Hyper-V could contain variations in features, the common codebase provides a consistent platform for guest operating systems and applications running inside of a virtual machine to run on compatible products that share the same Hyper-V build number. This means that any support or compatibility statements for a guest operating system or application that is certified for specific builds of Hyper-V is compatible with all products that share the same build number for Hyper-V.

The table below shows which Hyper-V build numbers are available in which compatible products:

 Expand table

Hyper-V build	Compatible products
20348	Windows Server 2022 Datacenter Windows Server 2022 Standard Windows Server 2022 Essentials Azure Stack HCI version 21H2 Azure Stack HCI version 22H2
17763 and 17784	Windows Server 2019 Datacenter Windows Server 2019 Standard Windows Server 2019 Essentials Hyper-V Server 2019 Azure Stack HCI version 20H2
14393	Windows Server 2016 Datacenter Windows Server 2016 Standard

Hyper-V build	Compatible products
	Windows Server 2016 Essentials Hyper-V Server 2016

For more information, see:

- [Windows Server release information](#)
- [Supported Windows guest operating systems on Hyper-V](#)
- [Supported Linux and FreeBSD guest operating systems on Hyper-V](#)

Feedback

Was this page helpful?



Yes



No

Supported Windows guest operating systems for Hyper-V on Windows, Windows Server, and Azure Local

10/28/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V supports several versions of Windows Server, Windows, and Linux distributions to run in virtual machines, as guest operating systems. This article covers supported Windows Server and Windows guest operating systems. For Linux and FreeBSD distributions, see [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows](#).

Some operating systems have the integration services built-in. Others require that you install or upgrade integration services as a separate step after you set up the operating system in the virtual machine. For more information, see the following sections and [Integration Services](#).

Guest operating systems configurable components are confined based on the hosting operating system. To learn more about the maximum configurable components in Hyper-V, see [Plan for Hyper-V scalability in Windows Server](#).

Supported Windows Server guest operating systems

Following are the versions of Windows Server that are supported as guest operating systems for Hyper-V on Windows Server.

Windows Server 2025

- **Maximum number of virtual processors:**
 - 2,048 for generation 2
 - 64 for generation 1
 - 2,048 available to the host OS (root partition)
- **Integration Services:** Built-in
- **Host operating systems:**
 - Windows Server 2022 and later
 - Azure Local OS, version 23H2 and later with Windows Server subscription.
- **Notes:**

- Maximum number of virtual processors depends on the host OS. For example, Windows Server 2025 can provide up to 2,048 virtual processors.

Windows Server 2022

- **Maximum number of virtual processors:**
 - 1,024 for generation 2
 - 64 for generation 1
 - 1,024 available to the host OS (root partition)
- **Integration Services:** Built-in
- **Host operating systems:**
 - Windows Server 2019 and later
 - Azure Local OS, version 23H2 and later.

Windows Server 2019

- **Maximum number of virtual processors:**
 - 240 for generation 2
 - 64 for generation 1
 - 320 available to the host OS (root partition)
- **Integration Services:** Built-in

Windows Server 2016

- **Maximum number of virtual processors:**
 - 240 for generation 2
 - 64 for generation 1
 - 320 available to the host OS (root partition)
- **Integration Services:** Built-in

Windows Server 2012 R2

- **Maximum number of virtual processors:** 64
- **Integration Services:** Built-in

Windows Server 2012

- **Maximum number of virtual processors:** 64
- **Integration Services:** Built-in

Windows Server 2008 R2 with Service Pack 1 (SP1)

- **Maximum number of virtual processors:** 64
- **Integration Services:** Install all critical Windows updates after you set up the guest operating system.
- **Notes:** Datacenter, Enterprise, Standard, and Web editions.

Windows Server 2008 with Service Pack 2 (SP2)

- **Maximum number of virtual processors:** 8
- **Integration Services:** Install all critical Windows updates after you set up the guest operating system.
- **Notes:** Datacenter, Enterprise, Standard, and Web editions (32-bit and 64-bit).

Supported Windows client guest operating systems

Following are the versions of Windows client that are supported as guest operating systems for Hyper-V on Windows Server.

Windows 11

- **Maximum number of virtual processors:** 32
- **Integration Services:** Built-in
- **Host operating systems:**
 - Generation 2 virtual machine hosted on Windows Server 2019 and later
 - Generation 2 virtual machine hosted on Azure Local OS, version 23H2 and later
- **Notes:**
 - Windows 11 requires an increased CPU instruction for installation and boot (POPCNT and SSE4.2). To meet these requirements in Hyper-V:
 - On Windows 11, Windows Server 2022 (standalone or clustered hosts), or standalone Windows Server 2025 hosts, disable processor compatibility to enable these features. Disabling processor compatibility limits live migration to hosts with the same processor features.
 - On Windows Server 2025 clusters with CPUs supporting POPCNT and SSE4.2, enable dynamic processor compatibility to ensure these features are available while retaining live migration capabilities within the cluster. To learn more about dynamic processor compatibility, see [Processor compatibility for Hyper-V virtual machines](#).

Windows 10

- Maximum number of virtual processors: 32
- Integration Services: Built-in

Windows 8.1

- Maximum number of virtual processors: 32
- Integration Services: Built-in

Windows 7 with Service Pack 1 (SP1)

- Maximum number of virtual processors: 4
- Integration Services: Upgrade the integration services after you set up the guest operating system.
- Notes: Ultimate, Enterprise, and Professional editions (32-bit and 64-bit).

Supported Linux and FreeBSD

[Expand table](#)

Guest operating system
CentOS and Red Hat Enterprise Linux
Debian virtual machines on Hyper-V
SUSE
Oracle Linux
Ubuntu
FreeBSD

How Microsoft provides support for guest operating systems

Microsoft provides support for guest operating systems in the following manner:

- Issues found in Microsoft operating systems and in integration services are supported by Microsoft support.

- For issues found in other operating systems that have been certified by the operating system vendor to run on Hyper-V, support is provided by the vendor.
- For issues found in other operating systems, Microsoft submits the issue to the multi-vendor support community, [TSANet](#) .

In order to be supported, all operating systems (guest and host) must be up-to-date. Check Windows Update for critical updates.

Supported Linux and FreeBSD virtual machines for Hyper-V on Windows Server and Windows

06/19/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V supports both emulated and Hyper-V-specific devices for Linux and FreeBSD virtual machines. When running with emulated devices, no additional software is required to be installed. However emulated devices do not provide high performance and cannot leverage the rich virtual machine management infrastructure that the Hyper-V technology offers. In order to make full use of all benefits that Hyper-V provides, it is best to use Hyper-V-specific devices for Linux and FreeBSD. The collection of drivers that are required to run Hyper-V-specific devices are known as Linux Integration Services (LIS) or FreeBSD Integration Services (BIS).

LIS has been added to the Linux kernel and is updated for new releases. But Linux distributions based on older kernels may not have the latest enhancements or fixes. Microsoft provides a download containing installable LIS drivers for some Linux installations based on these older kernels. Because distribution vendors include versions of Linux Integration Services, it is best to install the latest downloadable version of LIS, if applicable, for your installation.

For other Linux distributions LIS changes are regularly integrated into the operating system kernel and applications so no separate download or installation is required.

For older FreeBSD releases (before 10.0), Microsoft provides ports that contain the installable BIS drivers and corresponding daemons for FreeBSD virtual machines. For newer FreeBSD releases, BIS is built in to the FreeBSD operating system, and no separate download or installation is required except for a KVP ports download that is needed for FreeBSD 10.0.

Tip

- Download [Windows Server](#)  from the Evaluation Center.

The goal of this content is to provide information that helps facilitate your Linux or FreeBSD deployment on Hyper-V. Specific details include:

- Linux distributions or FreeBSD releases that require the download and installation of LIS or BIS drivers.
- Linux distributions or FreeBSD releases that contain built-in LIS or BIS drivers.

- Feature distribution maps that indicate the features in major Linux distributions or FreeBSD releases.
- Known issues and workarounds for each distribution or release.
- Feature description for each LIS or BIS feature.

In this section

- [Supported CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V](#)
- [Supported Debian virtual machines on Hyper-V](#)
- [Supported Oracle Linux virtual machines on Hyper-V](#)
- [Supported SUSE virtual machines on Hyper-V](#)
- [Supported Ubuntu virtual machines on Hyper-V](#)
- [Supported FreeBSD virtual machines on Hyper-V](#)
- [Feature Descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best Practices for running Linux on Hyper-V](#)
- [Best practices for running FreeBSD on Hyper-V](#)

Supported CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

The following feature distribution maps indicate the features that are present in built-in and downloadable versions of Linux Integration Services. The known issues and workarounds for each distribution are listed after the tables.

The built-in Red Hat Enterprise Linux Integration Services drivers for Hyper-V are sufficient for Red Hat Enterprise Linux guests to run using the high performance synthetic devices on Hyper-V hosts. These built-in drivers are certified by Red Hat for this use. Certified configurations can be viewed in the [Red Hat Certification Catalog](#) .

It isn't necessary to download and install Linux Integration Services packages from the Microsoft Download Center, and doing so may limit your Red Hat support as described in [Red Hat Knowledgebase 1067](#) .

Because of potential conflicts between the built-in LIS support and the downloadable LIS support when you upgrade the kernel, disable automatic updates, uninstall the LIS downloadable packages, update the kernel, reboot, and then install the latest LIS release, and reboot again.

In this section:

- [RHEL/CentOS 10.x Series](#)
- [RHEL/CentOS 9.x Series](#)
- [RHEL/CentOS 8.x Series](#)
- [Notes](#)

Table legend

- **Built in** - LIS are included as part of this Linux distribution. The kernel module version numbers for the built-in LIS (as shown by **lsmod**, for example) are different from the

version number on the Microsoft-provided LIS download package. A mismatch doesn't indicate that the built-in LIS is out of date.

- ✓ - Feature available
- (blank) - Feature not available

RHEL/CentOS 10.x Series

 Expand table

Feature	Host OS	10.x
LIS Availability		Built in
Core	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Windows Server 2016 Accurate Time	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
>256 vCPUs		✓
Networking		
Jumbo frames	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
VLAN tagging and trunking	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Live Migration	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Static IP Injection	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 2
vRSS	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
TCP Segmentation and Checksum Offloads	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
SR-IOV	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Storage		
VHDX resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓

Feature	Host OS	10.x
Virtual Fibre Channel	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 3
Live virtual machine backup	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 5
TRIM support	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
SCSI WWN	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Memory		
PAE Kernel Support	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	
Configuration of MMIO gap	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Dynamic Memory - Hot-Add	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 9, 10
Dynamic Memory - Ballooning	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 9, 10
Runtime Memory Resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Video		
Hyper-V-specific video device	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Miscellaneous		
Key-Value Pair	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Non-Maskable Interrupt	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
File copy from host to guest	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Isvmbus command	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Hyper-V Sockets	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓

Feature	Host OS	10.x
PCI Passthrough/DDA	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Generation 2 virtual machines		
Boot using UEFI	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 14, 17
Secure boot	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓

RHEL/CentOS 9.x Series

 Expand table

Feature	Host OS	9.x
LIS Availability		Built in
Core	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Windows Server 2016 Accurate Time	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
>256 vCPUs		✓
Networking		
Jumbo frames	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
VLAN tagging and trunking	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Live Migration	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Static IP Injection	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 2
vRSS	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
TCP Segmentation and Checksum Offloads	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
SR-IOV	Windows Server 2025, 2022, 2019, 2016	✓

Feature	Host OS	9.x
	Azure Stack HCI	
Storage		
VHDX resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Virtual Fibre Channel	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 3
Live virtual machine backup	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 5
TRIM support	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
SCSI WWN	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Memory		
PAE Kernel Support	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	
Configuration of MMIO gap	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Dynamic Memory - Hot-Add	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 9, 10
Dynamic Memory - Ballooning	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 9,10
Runtime Memory Resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Video		
Hyper-V-specific video device	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Miscellaneous		
Key-Value Pair	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Non-Maskable Interrupt	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
File copy from host to guest	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓

Feature	Host OS	9.x
Isvmbus command	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Hyper-V Sockets	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
PCI Passthrough/DDA	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓
Generation 2 virtual machines		
Boot using UEFI	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 14, 17
Secure boot	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓

RHEL/CentOS 8.x Series

[Expand table](#)

Feature	Host OS	8.1-8.6+	8.0
LIS Availability		Built in	Built in
Core	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Windows Server 2016 Accurate Time	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
>256 vCPUs		✓	
Networking			
Jumbo frames	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
VLAN tagging and trunking	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Live Migration	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓

Feature	Host OS	8.1-8.6+	8.0
Static IP Injection	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 2	✓ Note 2
vRSS	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
TCP Segmentation and Checksum Offloads	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
SR-IOV	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Storage			
VHDX resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Virtual Fibre Channel	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 3	✓ Note 3
Live virtual machine backup	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 5	✓ Note 5
TRIM support	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
SCSI WWN	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Memory			
PAE Kernel Support	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	N/A	N/A
Configuration of MMIO gap	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓

Feature	Host OS	8.1-8.6+	8.0
Dynamic Memory - Hot-Add	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 9, 10	✓ Note 9, 10
Dynamic Memory - Ballooning	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 9, 10	✓ Note 9, 10
Runtime Memory Resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Video			
Hyper-V-specific video device	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Miscellaneous			
Key-Value Pair	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Non-Maskable Interrupt	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
File copy from host to guest	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Isvmbus command	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Hyper-V Sockets	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
PCI Passthrough/DDA	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓
Generation 2 virtual machines			
Boot using UEFI	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 14, 17	✓ Note 14

[illegible]

Feature	Host OS	7.5- 7.7	7.3- 7.4	7.0- 7.2	7.6- 7.9	7.5	7.4	7.3	7.2	7.1	7.0
and Checksum Offloads	2022, 2019, 2016 Azure Stack HCI										
SR-IOV	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓		✓	✓	✓				
Storage											
VHDX resize	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓	✓		
Virtual Fibre Channel	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3	✓ Note 3
Live virtual machine backup	Windows Server 2025, 2022, 2019, 2016 Azure	✓ Note 5	✓ Note 5	✓ Note 5	✓ Note 4,5	✓ Note 4,5	✓ Note 4, 5	✓ Note 4, 5	✓ Note 4, 5	✓ Note 4, 5	✓ Note 4, 5

[illegible]

[illegible]

Feature	Host OS	7.5- 7.7	7.3- 7.4	7.0- 7.2	7.6- 7.9	7.5	7.4	7.3	7.2	7.1	7.0
	2022, 2019, 2016 Azure Stack HCI										
Non-Maskable Interrupt	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
File copy from host to guest	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓	✓	✓ Note 4	✓ Note 4	✓ Note 4	✓ Note 4	✓ Note 4	✓ Note 4	
Isvmbus command	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓	✓							
Hyper-V Sockets	Windows Server 2025, 2022, 2019, 2016	✓	✓	✓							
PCI Passthrough/DDA	Windows Server 2025, 2022,	✓	✓		✓	✓	✓	✓			

Feature	Host OS	7.5- 7.7	7.3- 7.4	7.0- 7.2	7.6- 7.9	7.5	7.4	7.3	7.2	7.1	7.0
	2019, 2016 Azure Stack HCI										
Generation 2 virtual machines											
Boot using UEFI	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14	✓ Note 14
Secure boot	Windows Server 2025, 2022, 2019, 2016 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes

1. For this RHEL/CentOS release, VLAN tagging works but VLAN trunking doesn't.
2. Static IP injection may not work if Network Manager has been configured for a given synthetic network adapter on the virtual machine. For smooth functioning of static IP injection, make sure that either Network Manager is either turned off completely or has been turned off for a specific network adapter through its ifcfg-ethX file.
3. While using virtual fibre channel devices, make sure that logical unit number 0 (LUN 0) has been populated. If LUN 0 hasn't been populated, a Linux virtual machine might not be able to mount fibre channel devices natively.
4. For built-in LIS, the "hyperv-daemons" package must be installed for this functionality.

5. If there are open file handles during a live virtual machine backup operation, then in some corner cases, the backed-up VHDs might have to undergo a file system consistency check (fsck) on restore. Live backup operations can fail silently if the virtual machine has an attached iSCSI device or direct-attached storage (also known as a pass-through disk).
6. (Intentionally omitted for numbering consistency)
7. Dynamic memory support is only available on 64-bit virtual machines.
8. Hot-Add support isn't enabled by default in this distribution. To enable Hot-Add support you need to add a udev rule under `/etc/udev/rules.d/` as follows:
 - a. Create a file `/etc/udev/rules.d/100-balloon.rules`. You may use any other desired name for the file.
 - b. Add the following content to the file: `SUBSYSTEM=="memory", ACTION=="add", ATTR{state}="online"`
 - c. Reboot the system to enable Hot-Add support.

While the Linux Integration Services download creates this rule on installation, the rule is also removed when LIS is uninstalled, so the rule must be recreated if dynamic memory is needed after uninstallation.

9. Dynamic memory operations can fail if the guest operating system is running too low on memory. The following are some best practices:
 - Startup memory and minimal memory should be equal to or greater than the amount of memory that the distribution vendor recommends.
 - Applications that tend to consume the entire available memory on a system are limited to consuming up to 80 percent of available RAM.
10. If you're using Dynamic Memory on Windows Server, specify **Startup memory**, **Minimum memory**, and **Maximum memory** parameters in multiples of 128 megabytes (MB). Failure to do so can lead to hot-add failures, and you may not see any memory increase in a guest operating system.
11. (Intentionally omitted for numbering consistency)
12. To enable key/value pair (KVP) infrastructure, install the `hypervkvpd` or `hyperv-daemons` rpm package from your RHEL ISO. Alternatively the package can be installed directly from RHEL repositories.

13. The key/value pair (KVP) infrastructure might not function correctly without a Linux software update. Contact your distribution vendor to obtain the software update in case you see problems with this feature.
14. Windows Server Generation 2 virtual machines have secure boot enabled by default and some Linux virtual machines will not boot unless the secure boot option is disabled. You can disable secure boot in the **Firmware** section of the settings for the virtual machine in **Hyper-V Manager** or you can disable it using PowerShell:

PowerShell

```
Set-VMFirmware -VMName "VMname" -EnableSecureBoot Off
```

The Linux Integration Services download can be applied to existing Generation 2 VMs but doesn't impart Generation 2 capability.

15. RHEL 8.5 requires Windows Server 2019 or newer, or Azure Local 20H2 or newer.

Related content

- [Set-VMFirmware](#)
- [Supported Debian virtual machines on Hyper-V](#)
- [Supported Oracle Linux virtual machines on Hyper-V](#)
- [Supported SUSE virtual machines on Hyper-V](#)
- [Supported Ubuntu virtual machines on Hyper-V](#)
- [Supported FreeBSD virtual machines on Hyper-V](#)
- [Feature Descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best Practices for running Linux on Hyper-V](#)
- [Red Hat Hardware Certification](#) 

Supported Debian virtual machines on Hyper-V

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This article outlines the support offered for Debian virtual machines (VMs) on Hyper-V.

Table legend

The following feature distribution map indicates the features that are present in each version of Windows Server. The known issues and workarounds for each distribution are listed after the table.

- **Built-in LIS** - Linux integration services (LIS) are included as part of this Linux distribution. The Microsoft-provided LIS download package doesn't work for this distribution. Don't install the Microsoft package. The kernel module version numbers for the built-in LIS (as shown by `lsmod`, for example) are different from the version number on the Microsoft-provided LIS download package. A mismatch doesn't indicate that the built-in LIS is out of date.
- ✓ - Feature available
- (blank) - Feature not available

 Expand table

Feature	Windows Server Hyper-V version	12 (Bookworm)	11 (Bullseye)	10 (Buster)
Availability		Built-in LIS	Built-in LIS	Built-in LIS
Core	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Windows Server 2016 Accurate Time	2025, 2022, 2019, 2016	✓ Note 4	✓ Note 4	✓ Note 4
Networking				
Jumbo frames	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓

Feature	Windows Server Hyper-V version	12 (Bookworm)	11 (Bullseye)	10 (Buster)
VLAN tagging and trunking	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Live Migration	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Static IP Injection	2025, 2022, 2019, 2016, 2012 R2			
vRSS	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4	✓ Note 4	✓ Note 4
TCP Segmentation and Checksum Offloads	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4	✓ Note 4	✓ Note 4
SR-IOV	2025, 2022, 2019, 2016	✓ Note 4	✓ Note 4	✓ Note 4
Storage				
VHDX resize	2025, 2022, 2019, 2016, 2012 R2	✓ Note 1	✓ Note 1	✓ Note 1
Virtual Fibre Channel	2025, 2022, 2019, 2016, 2012 R2			
Live virtual machine backup	2025, 2022, 2019, 2016, 2012 R2	✓ Note 2	✓ Note 2	✓ Note 2
TRIM support	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4	✓ Note 4	✓ Note 4
SCSI WWN	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4	✓ Note 4	✓ Note 4
Memory				
PAE Kernel Support	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Configuration of MMIO gap	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Dynamic Memory - Hot-Add	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4	✓ Note 4	✓ Note 4
Dynamic Memory - Ballooning	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4	✓ Note 4	✓ Note 4
Runtime Memory Resize	2025, 2022, 2019, 2016	✓ Note 4	✓ Note 4	✓ Note 4

Feature	Windows Server Hyper-V version	12 (Bookworm)	11 (Bullseye)	10 (Buster)
Video				
Hyper-V-specific video device	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Miscellaneous				
Key-Value Pair	2025, 2022, 2019, 2016, 2012 R2	✓ Note 2	✓ Note 2	✓ Note 2
Non-Maskable Interrupt	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
File copy from host to guest	2025, 2022, 2019, 2016, 2012 R2	✓ Note 2	✓ Note 2	✓ Note 2
lsmbus command	2025, 2022, 2019, 2016, 2012 R2			
Hyper-V Sockets	2025, 2022, 2019, 2016	✓ Note 4	✓ Note 4	✓ Note 4
PCI Passthrough/DDA	2025, 2022, 2019, 2016	✓ Note 4	✓ Note 4	✓ Note 4
Generation 2 virtual machines				
Boot using UEFI	2025, 2022, 2019, 2016, 2012 R2	✓ Note 3	✓ Note 3	✓ Note 3
Secure boot	2025, 2022, 2019, 2016	✓	✓	✓

Notes

1. Creating file systems on VHDs larger than 2 TB isn't supported.
2. Starting with Debian 8.3, the manually installed Debian package "hyperv-daemons" contains the key-value pair, fcopy, and VSS daemons. On Debian 7.x and 8.0-8.2, the hyperv-daemons package must come from [Debian backports](#).
3. On Windows Server 2012 R2, Generation 2 virtual machines have secure boot enabled by default, and some Linux virtual machines don't boot unless the secure boot option is disabled. You can disable secure boot in the **Firmware** section of the settings for the virtual machine in **Hyper-V Manager**, or you can disable it by using PowerShell:

```
PowerShell
```

```
Set-VMFirmware -VMName "VMname" -EnableSecureBoot Off
```

4. The latest upstream kernel capabilities are only available by using the kernels available in the [Debian backports repository](#) [↗].

See also

- [Supported CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V](#)
- [Supported Oracle Linux virtual machines on Hyper-V](#)
- [Supported SUSE Linux Enterprise Server \(SLES\) virtual machines on Hyper-V](#)
- [Supported Ubuntu virtual machines on Hyper-V](#)
- [Supported FreeBSD virtual machines on Hyper-V](#)
- [Feature descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best practices for running Linux on Hyper-V](#)

Last updated on 01/21/2026

Supported Oracle Linux virtual machines on Hyper-V

06/20/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

The following feature distribution map indicates the features that are present in each version. The known issues and workarounds for each distribution are listed after the table.

In this section:

- [Oracle Linux 9.x Series](#)
- [Oracle Linux 8.x Series](#)

Table legend

- Built in** - LIS are included as part of this Linux distribution. The kernel module version numbers for the built in LIS (as shown by **lsmod**, for example) are different from the version number on the Microsoft-provided LIS download package. A mismatch doesn't indicate that the built in LIS is out of date.
- ✓ - Feature available
- (blank) - Feature not available
- RHCK** - Red Hat Compatible Kernel

Oracle Linux 9.x Series

 Expand table

Feature	Windows Server version	9.0 (RHCK)
Availability		
Core	2025, 2022, 2019, 2016, 2012 R2	✓
Windows Server 2016 Accurate Time	2025, 2022, 2019, 2016	✓
Networking		
Jumbo frames	2025, 2022, 2019, 2016, 2012 R2	✓

Feature	Windows Server version	9.0 (RHCK)
VLAN tagging and trunking	2025, 2022, 2019, 2016, 2012 R2	✓
Live Migration	2025, 2022, 2019, 2016, 2012 R2	✓
Static IP Injection	2025, 2022, 2019, 2016, 2012 R2	✓ Note 1
vRSS	2025, 2022, 2019, 2016, 2012 R2	✓
TCP Segmentation and Checksum Offloads	2025, 2022, 2019, 2016, 2012 R2	✓
SR-IOV	2025, 2022, 2019, 2016	✓
Storage		
VHDX resize	2025, 2022, 2019, 2016, 2012 R2	✓
Virtual Fibre Channel	2025, 2022, 2019, 2016, 2012 R2	✓ Note 2
Live virtual machine backup	2025, 2022, 2019, 2016, 2012 R2	✓ Note 3
TRIM support	2025, 2022, 2019, 2016, 2012 R2	✓
SCSI WWN	2025, 2022, 2019, 2016, 2012 R2	✓
Memory		
PAE Kernel Support	2025, 2022, 2019, 2016, 2012 R2	N/A
Configuration of MMIO gap	2025, 2022, 2019, 2016, 2012 R2	✓
Dynamic Memory - Hot-Add	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4, 5, 6
Dynamic Memory - Ballooning	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4, 5, 6
Runtime Memory Resize	2025, 2022, 2019, 2016	✓
Video		
Hyper-V-specific video device	2025, 2022, 2019, 2016, 2012 R2	✓
Miscellaneous		
Key-Value Pair	2025, 2022, 2019, 2016, 2012 R2	✓
Non-Maskable Interrupt	2025, 2022, 2019, 2016, 2012 R2	✓
File copy from host to guest	2025, 2022, 2019, 2016, 2012 R2	✓
Isvmbus command	2025, 2022, 2019, 2016, 2012 R2	✓
Hyper-V Sockets	2025, 2022, 2019, 2016	✓

Feature	Windows Server version	9.0 (RHCK)
PCI Passthrough/DDA	2025, 2022, 2019, 2016	✓
Generation 2 virtual machines		
Boot using UEFI	2025, 2022, 2019, 2016, 2012 R2	✓ Note 7
Secure boot	2025, 2022, 2019, 2016	✓

Oracle Linux 8.x Series

[Expand table](#)

Feature	Windows Server version	8.0-8.5 (RHCK)
Availability		
Core	2025, 2022, 2019, 2016, 2012 R2	✓
Windows Server 2016 Accurate Time	2025, 2022, 2019, 2016	✓
Networking		
Jumbo frames	2025, 2022, 2019, 2016, 2012 R2	✓
VLAN tagging and trunking	2025, 2022, 2019, 2016, 2012 R2	✓
Live Migration	2025, 2022, 2019, 2016, 2012 R2	✓
Static IP Injection	2025, 2022, 2019, 2016, 2012 R2	✓ Note 1
vRSS	2025, 2022, 2019, 2016, 2012 R2	✓
TCP Segmentation and Checksum Offloads	2025, 2022, 2019, 2016, 2012 R2	✓
SR-IOV	2025, 2022, 2019, 2016	✓
Storage		
VHDX resize	2025, 2022, 2019, 2016, 2012 R2	✓
Virtual Fibre Channel	2025, 2022, 2019, 2016, 2012 R2	✓ Note 2
Live virtual machine backup	2025, 2022, 2019, 2016, 2012 R2	✓ Note 3
TRIM support	2025, 2022, 2019, 2016, 2012 R2	✓
SCSI WWN	2025, 2022, 2019, 2016, 2012 R2	✓
Memory		

Feature	Windows Server version	8.0-8.5 (RHCK)
PAE Kernel Support	2025, 2022, 2019, 2016, 2012 R2	N/A
Configuration of MMIO gap	2025, 2022, 2019, 2016, 2012 R2	✓
Dynamic Memory - Hot-Add	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4, 5, 6
Dynamic Memory - Ballooning	2025, 2022, 2019, 2016, 2012 R2	✓ Note 4, 5, 6
Runtime Memory Resize	2025, 2022, 2019, 2016	✓
Video		
Hyper-V-specific video device	2025, 2022, 2019, 2016, 2012 R2	✓
Miscellaneous		
Key-Value Pair	2025, 2022, 2019, 2016, 2012 R2	✓
Non-Maskable Interrupt	2025, 2022, 2019, 2016, 2012 R2	✓
File copy from host to guest	2025, 2022, 2019, 2016, 2012 R2	✓
Isvmbus command	2025, 2022, 2019, 2016, 2012 R2	✓
Hyper-V Sockets	2025, 2022, 2019, 2016	✓
PCI Passthrough/DDA	2025, 2022, 2019, 2016	✓
Generation 2 virtual machines		
Boot using UEFI	2025, 2022, 2019, 2016, 2012 R2	✓ Note 7
Secure boot	2025, 2022, 2019, 2016	✓

Notes

1. Static IP injection may not work if Network Manager has been configured for a given synthetic network adapter on the virtual machine. For smooth functioning of static IP injection please make sure that either Network Manager is either turned off completely or has been turned off for a specific network adapter through its ifcfg-ethX file.
2. On Windows Server 2012 R2 while using virtual fibre channel devices, make sure that logical unit number 0 (LUN 0) has been populated. If LUN 0 has not been populated, a Linux virtual machine might not be able to mount fibre channel devices natively.
3. If there are open file handles during a live virtual machine backup operation, then in some corner cases, the backed-up VHDs might have to undergo a file system consistency check

(fsck) on restore. Live backup operations can fail silently if the virtual machine has an attached iSCSI device or direct-attached storage (also known as a pass-through disk).

4. Hot-Add support is not enabled by default in this distribution. To enable Hot-Add support you need to add a udev rule under `/etc/udev/rules.d/` as follows:
 - a. Create a file `/etc/udev/rules.d/100-balloon.rules`. You may use any other desired name for the file.
 - b. Add the following content to the file: `SUBSYSTEM=="memory", ACTION=="add", ATTR{state}="online"`
 - c. Reboot the system to enable Hot-Add support.

While the Linux Integration Services download creates this rule on installation, the rule is also removed when LIS is uninstalled, so the rule must be recreated if dynamic memory is needed after uninstallation.

5. Dynamic memory operations can fail if the guest operating system is running too low on memory. The following are some best practices:
 - Startup memory and minimal memory should be equal to or greater than the amount of memory that the distribution vendor recommends.
 - Applications that tend to consume the entire available memory on a system are limited to consuming up to 80 percent of available RAM.
6. If you are using Dynamic Memory on a Windows Server 2016 or Windows Server 2012 R2 operating system, specify **Startup memory**, **Minimum memory**, and **Maximum memory** parameters in multiples of 128 megabytes (MB). Failure to do so can lead to hot-add failures, and you may not see any memory increase in a guest operating system.
7. On Windows Server 2012 R2 Generation 2 virtual machines have secure boot enabled by default and some Linux virtual machines will not boot unless the secure boot option is disabled. You can disable secure boot in the **Firmware** section of the settings for the virtual machine in **Hyper-V Manager** or you can disable it using PowerShell:

Powershell

```
Set-VMFirmware -VMName "VMname" -EnableSecureBoot Off
```

The Linux Integration Services download can be applied to existing Generation 2 VMs but does not impart Generation 2 capability.

Related content

- [Set-VMFirmware](#)
- [Supported CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V](#)
- [Supported Debian virtual machines on Hyper-V](#)
- [Supported SUSE virtual machines on Hyper-V](#)
- [Supported Ubuntu virtual machines on Hyper-V](#)
- [Supported FreeBSD virtual machines on Hyper-V](#)
- [Feature Descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best Practices for running Linux on Hyper-V](#)

Supported SUSE Linux Enterprise Server (SLES) virtual machines on Hyper-V

06/20/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

SUSE Linux Enterprise Server (SLES) is supported as a guest operating system on Hyper-V. This article provides a detailed overview of the integration services and features available for each supported SLES version. Use the included feature matrix to compare support across versions, and review the notes for important details, known issues, and recommended workarounds. This information helps you plan, deploy, and manage SLES virtual machines on Hyper-V with confidence.

The built-in SUSE Linux Enterprise Service drivers for Hyper-V are certified by SUSE. An example configuration can be viewed in this bulletin: [SUSE YES Certification Bulletin](#).

Table legend for SLES virtual machine features

- **Built-in** - LIS are included as part of this Linux distribution. The Microsoft-provided LIS download package doesn't work for this distribution, so don't install it. The kernel module version numbers for the built-in LIS (as shown by **lsmod**, for example) are different from the version number on the Microsoft-provided LIS download package. A mismatch doesn't indicate that the built-in LIS is out of date.
- **✓** - Feature available
- *(blank)* - Feature not available

SLES12+ is 64-bit only.

 Expand table

Feature	Operating system version	SLES 15 SP1-SP4	SLES 15	SLES 12 SP3-SP5	SLES 12 SP2	SLES 12 SP1	SLES 11 SP4	SLES 11 SP3
Availability		Built-in	Built-in	Built-in	Built-in	Built-in	Built-in	Built-in
Core	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓

Feature	Operating system version	SLES 15 SP1- SP4	SLES 15	SLES 12 SP3- SP5	SLES 12 SP2	SLES 12 SP1	SLES 11 SP4	SLES 11 SP3
Windows Server 2016 Accurate Time	WS/Hyper-V 2025,2022,2019,2016	✓	✓	✓	✓			
Networking								
Jumbo frames	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
VLAN tagging and trunking	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
Live migration	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
Static IP Injection	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓Note 1	✓Note 1	✓Note 1	✓Note 1	✓Note 1	✓Note 1	✓Note 1
vRSS	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓		
TCP Segmentation and Checksum Offloads	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	
SR-IOV	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓			
Storage								
VHDX resize	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
Virtual Fibre Channel	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
Live virtual machine backup	WS/Hyper-V 2025,2022,2019,2016,2012	✓ Note	✓Note 2,3,8	✓ Note	✓ Note	✓ Note	✓ Note	✓ Note

Feature	Operating system version	SLES 15 SP1- SP4	SLES 15	SLES 12 SP3- SP5	SLES 12 SP2	SLES 12 SP1	SLES 11 SP4	SLES 11 SP3
	Azure Stack HCI	2,3,8		2,3,8	2,3,8	2,3,8	2,3,8	2,3,8
TRIM support	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	
SCSI WWN	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓			
Memory								
PAE Kernel Support	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	N/A	N/A	N/A	N/A	N/A	✓	✓
Configuration of MMIO gap	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
Dynamic Memory - Hot-Add	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓ Note 6	✓Note 6	✓ Note 6	✓ Note 6	✓ Note 6	✓ Note 4,5,6	✓ Note 4,5,6
Dynamic Memory - Ballooning	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓ Note 6	✓ Note 6	✓ Note 6	✓ Note 6	✓ Note 6	✓ Note 4,5,6	✓ Note 4,5,6
Runtime Memory Resize	WS/Hyper-V 2025,2022,2019,2016	✓ Note 6	✓ Note 6	✓ Note 6	✓ Note 6			
Video								
Hyper-V-specific video device	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓
Miscellaneous								
Key/value pair	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓ Note 7	✓ Note 7
Non-Maskable Interrupt	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	✓

Feature	Operating system version	SLES 15 SP1- SP4	SLES 15	SLES 12 SP3- SP5	SLES 12 SP2	SLES 12 SP1	SLES 11 SP4	SLES 11 SP3
File copy from host to guest	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓	✓	✓	
Isvmbus command	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓	✓	✓	✓			
Hyper-V Sockets	WS/Hyper-V 2025,2022,2019,2016	✓	✓	✓				
PCI Passthrough/DDA	WS/Hyper-V 2025,2022,2019,2016	✓	✓	✓	✓	✓		
Generation 2 virtual machines								
Boot using UEFI	WS/Hyper-V 2025,2022,2019,2016,2012 Azure Stack HCI	✓ Note 9	✓ Note 9	✓ Note 9	✓ Note 9	✓ Note 9	✓ Note 9	
Secure boot	WS/Hyper-V 2025,2022,2019,2016	✓	✓	✓	✓	✓		

Notes

1. Static IP injection may not work if **NetworkManager** is configured for a given Hyper-V-specific network adapter on the virtual machine as it can override static IP settings that are manually configured. To ensure smooth functioning of static IP injection ensure that Network Manager is turned off completely or is turned off for a specific network adapter through its **ifcfg-ethX** file.
2. If there are open file handles during a live virtual machine backup operation, then in some corner cases, the backed-up VHDs might have to undergo a file system consistency check (fsck) on restore.
3. Live backup operations can fail silently if the virtual machine has an attached iSCSI device or direct-attached storage (also known as a pass-through disk).
4. Dynamic memory operations can fail if the guest operating system is running too low on memory. The following are some best practices:

- Startup memory and minimal memory should be equal to or greater than the amount of memory that the distribution vendor recommends.
- Applications that tend to consume the entire available memory on a system are limited to consuming up to 80 percent of available RAM.

5. Dynamic memory support is only available on 64-bit virtual machines.

6. If you're using Dynamic Memory on Windows Server 2016 or Windows Server 2012 operating systems, specify **Startup memory**, **Minimum memory**, and **Maximum memory** parameters in multiples of 128 megabytes (MB). Failure to do so can lead to Hot-Add failures, and you may not see any memory increase in a guest operating system.

7. In Windows Server 2016 or Windows Server 2012 R2, the key/value pair infrastructure might not function correctly without a Linux software update. Contact your distribution vendor to obtain the software update in case you see problems with this feature.

8. VSS backup fails if a single partition is mounted multiple times.

9. On Windows Server 2012 R2, Generation 2 virtual machines have secure boot enabled by default and Generation 2 Linux virtual machines don't boot unless the secure boot option is disabled. You can disable secure boot in the **Firmware** section of the settings for the virtual machine in Hyper-V Manager or you can disable it using PowerShell:

Powershell

```
Set-VMFirmware -VMName "VMname" -EnableSecureBoot Off
```

See Also

- [Set-VMFirmware](#)
- [Supported CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V](#)
- [Supported Debian virtual machines on Hyper-V](#)
- [Supported Oracle Linux virtual machines on Hyper-V](#)
- [Supported Ubuntu virtual machines on Hyper-V](#)
- [Supported FreeBSD virtual machines on Hyper-V](#)
- [Feature Descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best Practices for running Linux on Hyper-V](#)

Supported Ubuntu virtual machines on Hyper-V

Article • 02/21/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2
and 22H2

The following feature distribution map indicates the features in each version. The known issues and workarounds for each distribution are listed after the table.

Table legend

- **Built in** - Linux Integration Services (LIS) is included as part of this Linux distribution. The Microsoft-provided LIS download package doesn't work for this distribution, so don't install it. The kernel module version numbers for the built in LIS (as shown by `lsmod`, for example) are different from the version number on the Microsoft-provided LIS download package. A mismatch doesn't indicate that the built in LIS is out of date.
-  - Feature available
- *(blank)* - Feature not available

 Expand table

Feature	Windows Server operating system version	24.04 LTS	22.04 LTS	20.04 LTS	18.04 LTS	16.04 LTS
Availability		Built-in	Built-in	Built-in	Built-in	Built-in
Core	2025, 2022, 2019, 2016					
Windows Server 2016 Accurate Time	2025, 2022, 2019, 2016					
Networking						
Jumbo frames	2025, 2022, 2019, 2016					
VLAN tagging and	2025, 2022, 2019,					

Feature	Windows Server operating system version	24.04 LTS	22.04 LTS	20.04 LTS	18.04 LTS	16.04 LTS
trunking	2016					
Live migration	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Static IP Injection	2025, 2022, 2019, 2016	✓ Note 1	✓ Note 1	✓ Note 1	✓ Note 1	✓ Note 1
vRSS	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
TCP Segmentation and Checksum Offloads	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
SR-IOV	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Storage						
VHDX resize	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Virtual Fibre Channel	2025, 2022, 2019, 2016	✓ Note 2	✓ Note 2	✓ Note 2	✓ Note 2	✓ Note 2
Live virtual machine backup	2025, 2022, 2019, 2016	✓ Note 3, 4, 5	✓ Note 3, 4, 5	✓ Note 3, 4, 5	✓ Note 3, 4, 5	✓ Note 3, 4, 5
TRIM support	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
SCSI WWN	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Memory						
PAE Kernel Support	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Configuration of MMIO gap	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Dynamic Memory - Hot-Add	2025, 2022, 2019, 2016	✓ Note 6, 7, 8	✓ Note 6, 7, 8	✓ Note 6, 7, 8	✓ Note 6, 7, 8	✓ Note 6, 7, 8
Dynamic Memory - Ballooning	2025, 2022, 2019, 2016	✓ Note 6, 7, 8	✓ Note 6, 7, 8	✓ Note 6, 7, 8	✓ Note 6, 7, 8	✓ Note 6, 7, 8

Feature	Windows Server operating system version	24.04 LTS	22.04 LTS	20.04 LTS	18.04 LTS	16.04 LTS
Runtime Memory Resize	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Video						
Hyper-V specific video device	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Miscellaneous						
Key/value pair	2025, 2022, 2019, 2016	✓ Note 5, 9	✓ Note 5, 9	✓ Note 5, 9	✓ Note 5, 9	✓ Note 5, 9
Non-Maskable Interrupt	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
File copy from host to guest	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
lsvmbus command	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Hyper-V Sockets	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
PCI Passthrough/DDA	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓
Generation 2 virtual machines						
Boot using UEFI	2025, 2022, 2019, 2016	✓	✓	✓ Note 10, 11	✓ Note 10, 11	✓ Note 10, 11
Secure boot	2025, 2022, 2019, 2016	✓	✓	✓	✓	✓

Notes

1. Static IP injection may not work if **NetworkManager** has been configured for a given Hyper-V-specific network adapter on the virtual machine as it can override static IP settings that have been manually configured. To ensure smooth functioning of static IP injection, ensure that Network Manager is turned off completely or has been turned off for a specific network adapter through its **ifcfg-ethX** file.

2. While using virtual fiber channel devices, ensure that logical unit number 0 (LUN 0) has been populated. If LUN 0 hasn't been populated, a Linux virtual machine might not be able to mount fiber channel devices natively.
3. If there are open file handles during a live virtual machine backup operation, then in some corner cases, the backed-up VHDs might have to undergo a file system consistency check (`fsck`) on restore.
4. Live backup operations can fail silently if the virtual machine has an attached iSCSI device or direct-attached storage (also known as a pass-through disk).
5. On long term support (LTS) releases use latest virtual Hardware Enablement (HWE) kernel for up to date Linux Integration Services.

To install the Azure-tuned kernel on Ubuntu, run the following commands as root (or sudo):

Bash

```
# apt-get update
# apt-get install linux-azure
```

6. Dynamic memory support is only available on 64-bit virtual machines.
7. Dynamic Memory operations can fail if the guest operating system is running too low on memory. The following are some best practices:
 - Startup memory and minimal memory should be equal to or greater than the amount of memory that the distribution vendor recommends.
 - Applications that tend to consume the entire available memory on a system are limited to consuming up to 80 percent of available RAM.
8. If you're using Dynamic Memory on Windows Server, specify **Startup memory**, **Minimum memory**, and **Maximum memory** parameters in multiples of 128 megabytes (MB). Failure to do so can lead to Hot-Add failures, and you might not see any memory increase on a guest operating system.
9. In Windows Server 2019, and Windows Server 2016, the key/value pair infrastructure might not function correctly without a Linux software update. Contact your distribution vendor to obtain the software update in case you see problems with this feature.
10. Generation 2 virtual machines have secure boot enabled by default and some Linux virtual machines won't boot unless the secure boot option is disabled. You

can disable secure boot in the **Firmware** section of the settings for the virtual machine in **Hyper-V Manager** or you can disable it using PowerShell:

Powershell

```
Set-VMFirmware -VMName "VMname" -EnableSecureBoot Off
```

11. Before attempting to copy the VHD of an existing Generation 2 VHD virtual machine to create new Generation 2 virtual machines, follow these steps:

a. Log in to the existing Generation 2 virtual machine.

b. Change directory to the boot EFI directory:

Bash

```
# cd /boot/efi/EFI
```

c. Copy the ubuntu directory in to a new directory named boot:

Bash

```
# sudo cp -r ubuntu/ boot
```

d. Change directory to the newly created boot directory:

Bash

```
# cd boot
```

e. Rename the shimx64.efi file:

Bash

```
# sudo mv shimx64.efi bootx64.efi
```

12. In order to perform live migrations for VMs that are Generation 2 configured, the **Migrate to a physical computer with a different processor version** option must be enabled under *Processor > Compatibility* in your VM settings. To learn more, see [Processor compatibility mode in Hyper-V](#).

See Also

- [Supported CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V](#)
- [Supported Debian virtual machines on Hyper-V](#)
- [Supported Oracle Linux virtual machines on Hyper-V](#)
- [Supported SUSE virtual machines on Hyper-V](#)
- [Feature Descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best Practices for running Linux on Hyper-V](#)
- [Set-VMFirmware](#)
- [Ubuntu 14.04 in a Generation 2 VM - Ben Armstrong's Virtualization Blog](#)

Feedback

Was this page helpful?

 Yes

 No

Supported FreeBSD virtual machines on Hyper-V

06/20/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

The following feature distribution map indicates the features in each version. The known issues and workarounds for each distribution are listed after the table.

Table legend

- **Built in** - BIS (FreeBSD Integration Service) are included as part of this FreeBSD release.
- **✓** - Feature available
- *(blank)* - Feature not available

 [Expand table](#)

Feature	Windows Server operating system version	13.0-13.5	12.0-12.4	11.0-11.4
Availability		Built in	Built in	Built in
Core	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Windows Server 2016 Accurate Time	2025, 2022, 2019, 2016	✓	✓	✓
Networking				
Jumbo frames	2025, 2022, 2019, 2016, 2012 R2	✓ Note 2	✓ Note 2	✓ Note 2
VLAN tagging and trunking	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Live migration	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Static IP Injection	2025, 2022, 2019, 2016, 2012 R2	✓ Note 3	✓ Note 3	✓ Note 3
vRSS	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
TCP Segmentation and Checksum Offloads	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓

Feature	Windows Server operating system version	13.0-13.5	12.0-12.4	11.0-11.4
Large Receive Offload (LRO)	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
SR-IOV	2025, 2022, 2019, 2016	✓	✓	✓
Storage		Note 1	Note1	Note 1
VHDX resize	2025, 2022, 2019, 2016, 2012 R2	✓ Note 6	✓ Note 6	✓ Note 6
Virtual Fibre Channel	2025, 2022, 2019, 2016, 2012 R2			
Live virtual machine backup	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
TRIM support	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
SCSI WWN	2025, 2022, 2019, 2016, 2012 R2			
Memory				
PAE Kernel Support	2025, 2022, 2019, 2016, 2012 R2			
Configuration of MMIO gap	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Dynamic Memory - Hot-Add	2025, 2022, 2019, 2016, 2012 R2			
Dynamic Memory - Ballooning	2025, 2022, 2019, 2016, 2012 R2			
Runtime Memory Resize	2025, 2022, 2019, 2016			
Video				
Hyper-V specific video device	2025, 2022, 2019, 2016, 2012 R2			
Miscellaneous				
Key/value pair	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
Non-Maskable Interrupt	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓
File copy from host to guest	2025, 2022, 2019, 2016, 2012 R2			
Isvmbus command	2025, 2022, 2019, 2016, 2012 R2			
Hyper-V Sockets	2025, 2022, 2019, 2016			
PCI Passthrough/DDA	2025, 2022, 2019, 2016	✓	✓	✓
Generation 2 virtual machines				
Boot using UEFI	2025, 2022, 2019, 2016, 2012 R2	✓	✓	✓

Feature	Windows Server operating system version	13.0-13.5	12.0-12.4	11.0-11.4
Secure boot	2025, 2022, 2019, 2016			

Notes

1. Suggest to [Label Disk Devices](#) to avoid ROOT MOUNT ERROR during startup.
2. 9126 is the maximum supported MTU size.
3. In a failover scenario, you cannot set a static IPv6 address in the replica server. Use an IPv4 address instead.

Related content

- [Feature Descriptions for Linux and FreeBSD virtual machines on Hyper-V](#)
- [Best practices for running FreeBSD on Hyper-V](#)

Best Practices for running Linux on Hyper-V

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This article has a list of recommendations for running Linux virtual machine on Hyper-V.

Tuning Linux File Systems on Dynamic VHDX Files

Some Linux file systems may consume significant amounts of real disk space even when the file system is mostly empty. To reduce the amount of real disk space usage of dynamic VHDX files, consider the following recommendations:

- When creating the VHDX, use 1 MB BlockSizeBytes (from the default 32 MB) in PowerShell, for example:

Powershell

```
PS > New-VHD -Path C:\MyVHDs\test.vhdx -SizeBytes 127GB -Dynamic -BlockSizeBytes 1MB
```

- The ext4 format is preferred to ext3 because ext4 is more space efficient than ext3 when used with dynamic VHDX files.
- When creating the filesystem specify the number of groups to be 4096, for example:

Bash

```
# mkfs.ext4 -G 4096 /dev/sdX1
```

Grub Menu Timeout on Generation 2 Virtual Machines

Because legacy hardware was removed from emulation in Generation 2 virtual machines, the grub menu timer counts down too quickly for the grub menu to be displayed, immediately loading the default entry. Until grub is fixed to use the EFI-supported timer, modify `/boot/grub/grub.conf`, `/etc/default/grub`, or equivalent to have `"timeout=100000"` instead of the default `"timeout=5"`.

PxE Boot on Generation 2 Virtual Machines

Because the PIT timer isn't present in Generation 2 Virtual Machines, network connections to the PXE TFTP server are terminated and prevent the bootloader from reading Grub configuration and loading a kernel from the server.

On RHEL 6.x, the legacy grub v0.97 EFI bootloader can be used instead of grub2 as described here:

https://access.redhat.com/documentation/Red_Hat_Enterprise_Linux/6/html/Installation_Guide/s1-netboot-pxe-config-efi.html ↗

On Linux distributions other than RHEL 6.x, similar steps can be followed to configure grub v0.97 to load Linux kernels from a PXE server.

Additionally, on RHEL/CentOS 6.6 keyboard and mouse input don't work with the preinstalled kernel which prevents specifying installation options in the menu. A serial console must be configured to allow choosing installation options.

- In the **efidefault** file on the PXE server, add the following kernel parameter
"console=ttyS1"
- On the VM in Hyper-V, set up a COM port using this PowerShell cmdlet:

Powershell

```
Set-VMComPort -VMName <Name> -Number 2 -Path \\.\pipe\dbg1
```

Specifying a kickstart file to the kernel would also avoid the need for keyboard and mouse input during installation.

Use static MAC addresses with failover clustering

Linux virtual machines deployed using failover clustering should be configured with a static media access control (MAC) address for each virtual network adapter. In some versions of Linux, the networking configuration may be lost after failover because a new MAC address is assigned to the virtual network adapter. To avoid losing the network configuration, ensure that each virtual network adapter has a static MAC address. You can configure the MAC address by editing the settings of the virtual machine in Hyper-V Manager or Failover Cluster Manager.

Use Hyper-V-specific network adapters, not the legacy network adapter

Configure and use the virtual Ethernet adapter, which is a Hyper-V-specific network card with enhanced performance. If both legacy and Hyper-V-specific network adapters are attached to a

virtual machine, the network names in the output of **ifconfig -a** might show random values such as `_tmp12000801310`. To avoid this issue, remove all legacy network adapters when using Hyper-V-specific network adapters in a Linux virtual machine.

Important

Hyper-V supports up to 64 virtual Ethernet adapters per guest. However, when more than 2 vCPUs are assigned to a Linux guest, the VM might not see the maximum of 64 virtual Ethernet adapters. For more information about Hyper-V scale limits, see [Hyper-V maximum scale limits in Windows Server](#).

Use I/O scheduler noop/none for better disk I/O performance

The Linux kernel offers two sets of disk I/O schedulers to reorder requests. One set is for the older 'blk' subsystem and one set is for the newer 'blk-mq' subsystem. In either case, with today's solid state disks it's recommended to use a scheduler that passes the scheduling decisions to the underlying Hyper-V hypervisor. For Linux kernels using the 'blk' subsystem, this is the "noop" scheduler. For Linux kernels using the 'blk-mq' subsystem, this is the "none" scheduler.

For a particular disk, the available schedulers can be seen at this file system location: `/sys/class/block/<diskname>/queue/scheduler`, with the currently selected scheduler in square brackets. You can change the scheduler by writing to this file system location. The change must be added to an initialization script in order to persist across reboots. Consult your Linux distro documentation for details.

NUMA

Linux kernel versions earlier than 2.6.37 don't support NUMA on Hyper-V with larger VM sizes. This issue primarily impacts older distributions using the upstream Red Hat 2.6.32 kernel, and was fixed in Red Hat Enterprise Linux (RHEL) 6.6 (kernel-2.6.32-504). Systems running custom kernels older than 2.6.37, or RHEL-based kernels older than 2.6.32-504 must set the boot parameter `numa=off` on the kernel command line in `grub.conf`. For more information, see [Red Hat KB 436883](#) .

Reserve more memory for kdump

In case the dump capture kernel ends up with a panic on boot, reserve more memory for the kernel. For example, change the parameter `crashkernel=384M-:128M` to `crashkernel=384M-:256M` in the Ubuntu grub configuration file.

Shrinking VHDX or expanding VHD and VHDX files can result in erroneous GPT partition tables

Hyper-V allows shrinking virtual disk (VHDX) files without regard for any partition, volume, or file system data structures that may exist on the disk. If the VHDX is shrunk to where the end of the VHDX comes before the end of a partition, data is lost, that partition can become corrupt, or invalid data is returned when the partition is read.

After resizing a VHD or VHDX, administrators should use a utility like fdisk or parted to update the partition, volume, and file system structures to reflect the change in the size of the disk. Shrinking or expanding the size of a VHD or VHDX that has a GUID Partition Table (GPT) causes a warning when a partition management tool is used to check the partition layout, and the administrator is warned to fix the first and secondary GPT headers. This manual step is safe to perform without data loss.

Related content

- [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows](#)
- [Best practices for running FreeBSD on Hyper-V](#)
- [Create Linux Images for Azure](#)
- [Optimize your Linux VM on Azure](#)

Last updated on 12/03/2025

Best practices for running FreeBSD on Hyper-V

Article • 11/01/2024 •

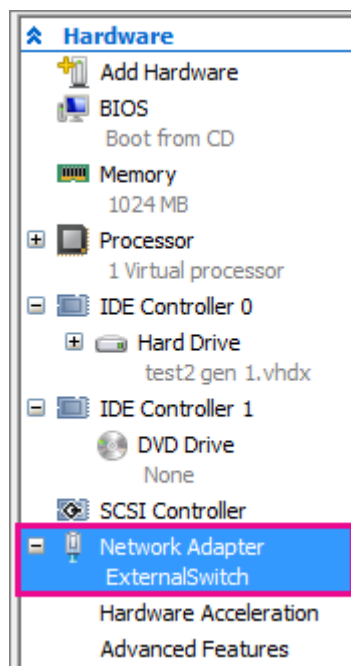
Applies ☒ Windows Server 2025, ☒ Windows Server 2022, ☒ Windows Server 2019, ☒ Windows Server 2016, ☒ Windows 11, ☒ Windows 10, ☒ Azure Local, versions 23H2 and 22H2

This topic contains a list of recommendations for running FreeBSD as a guest operating system on a Hyper-V virtual machine.

Enable CARP in FreeBSD 10.2 on Hyper-V

The Common Address Redundancy Protocol (CARP) allows multiple hosts to share the same IP address and Virtual Host ID (VHID) to help provide high availability for one or more services. If one or more hosts fail, the other hosts transparently take over so users won't notice a service failure. To use CARP in FreeBSD 10.2, follow the instructions in the [FreeBSD handbook](#) and do the following in Hyper-V Manager.

- Verify the virtual machine has a Network Adapter and it's assigned a virtual switch. Select the virtual machine and select **Actions** > **Settings**.



- Enable MAC address spoofing. To do this,
 1. Select the virtual machine and select **Actions** > **Settings**.
 2. Expand **Network Adapter** and select **Advanced Features**.

3. Select **Enable MAC Address spoofing**.

Create labels for disk devices

During startup, device nodes are created as new devices are discovered. This can mean that device names can change when new devices are added. If you get a ROOT MOUNT ERROR during startup, you should create labels for each IDE partition to avoid conflicts and changes. To learn how, see [Labeling Disk Devices](#). Below are examples.

Important

Make a backup copy of your fstab before making any changes.

1. Reboot the system into single user mode. This can be accomplished by selecting boot menu option 2 for FreeBSD 10.3+ (option 4 for FreeBSD 8.x), or performing a 'boot -s' from the boot prompt.
2. In Single user mode, create GEOM labels for each of the IDE disk partitions listed in your fstab (both root and swap). Below is an example of FreeBSD 10.3.

Bash

```
# cat /etc/fstab
# Device          Mountpoint      FStype  Options  Dump  Pass#
/dev/da0p2        /                ufs     rw       1     1
/dev/da0p3        none            swap    sw       0     0

# glabel label rootfs /dev/da0p2
# glabel label swap /dev/da0p3
# exit
```

Additional information on GEOM labels can be found at: [Labeling Disk Devices](#).

3. The system will continue with multi-user boot. After the boot completes, edit /etc/fstab and replace the conventional device names, with their respective labels. The final /etc/fstab will look like this:

```
# Device          Mountpoint      FStype  Options  Dump
Pass#
/dev/label/rootfs  /                ufs     rw       1
1
```

```
/dev/label/swap      none      swap      sw      0
0
```

4. The system can now be rebooted. If everything went well, it will come up normally and mount will show:

```
# mount
/dev/label/rootfs on / (ufs, local, journaled soft-updates)
devfs on /dev (devfs, local, mutilabel)
```

Use a wireless network adapter as the virtual switch

If the virtual switch on the host is based on wireless network adapter, reduce the ARP expiration time to 60 seconds by the following command. Otherwise the networking of the VM may stop working after a while.

```
# sysctl net.link.ether.inet.max_age=60
```

See also

- [Supported FreeBSD virtual machines on Hyper-V](#)

Feedback

Was this page helpful?

 Yes

 No

Should I create a generation 1 or 2 virtual machine in Hyper-V?

06/18/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Creating a generation 1 or generation 2 virtual machine depends on which guest operating system you want to install and the boot method you want to use to deploy the virtual machine. We recommend you create a generation 2 virtual machines to take advantage of features like Secure Boot unless one of the following statements is true:

- You're using an existing, prebuilt virtual hard drive (VHD or VHDX files), which isn't [compatible with UEFI](#).
- Generation 2 doesn't support the operating system you want to run on the virtual machine.
- Generation 2 doesn't support the boot method you want to use.

For more information about what features are available with generation 2 virtual machines, see [Hyper-V feature compatibility by generation and guest](#).

You can't change a virtual machine's generation once it's created. We recommend that you review the considerations here, and choose the operating system, boot method, and features you want to use before you choose a generation.

What are the advantages of using generation 2 virtual machines?

Here are some of the advantages you get when you use a generation 2 virtual machine:

- **Secure Boot**

Use Secure Boot to help prevent unauthorized firmware, operating systems, or UEFI drivers from running at boot time. Secure Boot verifies the boot loader is signed by a trusted authority in the UEFI database. Secure Boot is enabled by default for generation 2 virtual machines. If you need to run a guest operating system that Secure Boot doesn't support, you can disable it after you create the virtual machine. For more information, see [Secure Boot](#).

To Secure Boot generation 2 Linux virtual machines, you need to choose the UEFI CA Secure Boot template when you create the virtual machine.

- **Larger boot volume** The maximum boot volume for generation 2 virtual machines is 64 TB. This maximum boot volume is the maximum disk size supported by a `.VHDX`. For generation 1 virtual machines, the maximum boot volume is 2 TB for a `.VHDX` and 2040 GB for a `.VHD`. For more information, see [Hyper-V Virtual Hard Disk Format Overview](#).

You might also see a slight improvement in virtual machine boot and installation times with generation 2 virtual machines.

Which guest operating systems are supported?

Generation 1 virtual machines support most guest operating systems. Generation 2 virtual machines support most 64-bit versions of Windows and more current versions of Linux and FreeBSD operating systems. Use the following sections to see which generation of virtual machine supports the guest operating system you want to install.

- [Windows guest operating system support](#)
- [CentOS and Red Hat Enterprise Linux guest operating system support](#)
- [Debian guest operating system support](#)
- [FreeBSD guest operating system support](#)
- [Oracle Linux guest operating system support](#)
- [SUSE guest operating system support](#)
- [Ubuntu guest operating system support](#)

Windows guest operating system support

The following table shows which 64-bit versions of Windows you can use as a guest operating system for generation 1 and generation 2 virtual machines.

 Expand table

64-bit versions of Windows	Generation 1	Generation 2
Windows Server 2025	✓	✓
Windows Server 2022	✓	✓
Windows Server 2019	✓	✓
Windows Server 2016	✓	✓

64-bit versions of Windows	Generation 1	Generation 2
Windows Server 2012 R2	✓	✓
Windows Server 2012	✓	✓
Windows 11	✗	✓
Windows 10	✓	✓

The following table shows which 32-bit versions of Windows you can use as a guest operating system for generation 1 and generation 2 virtual machines.

[Expand table](#)

32-bit versions of Windows	Generation 1	Generation 2
Windows 10	✓	✗

CentOS and Red Hat Enterprise Linux guest operating system support

The following table shows which versions of Red Hat Enterprise Linux (RHEL) and CentOS you can use as a guest operating system for generation 1 and generation 2 virtual machines.

[Expand table](#)

Operating system versions	Generation 1	Generation 2
RHEL/CentOS 10.x series	✓	✓
RHEL/CentOS 9.x series	✓	✓
RHEL/CentOS 8.x series	✓	✓

For more information, see [CentOS and Red Hat Enterprise Linux virtual machines on Hyper-V](#).

Debian guest operating system support

The following table shows which versions of Debian you can use as a guest operating system for generation 1 and generation 2 virtual machines.

[Expand table](#)

Operating system versions	Generation 1	Generation 2
Debian 12.x (Bookworm) series	✓	✓
Debian 11.x (Bullseye) series	✓	✓
Debian 10.x (Buster) series	✓	✓

For more information, see [Debian virtual machines on Hyper-V](#).

FreeBSD guest operating system support

The following table shows which versions of FreeBSD you can use as a guest operating system for generation 1 and generation 2 virtual machines.

 Expand table

Operating system versions	Generation 1	Generation 2
FreeBSD 13.0 to 13.5	✓	✓
FreeBSD 12.0 to 12.4	✓	✓
FreeBSD 11.0 to 11.4	✓	✓

For more information, see [FreeBSD virtual machines on Hyper-V](#).

Oracle Linux guest operating system support

The following table shows which versions of Red Hat Compatible Kernel Series you can use as a guest operating system for generation 1 and generation 2 virtual machines.

 Expand table

Red Hat Compatible Kernel Series versions	Generation 1	Generation 2
Oracle Linux 9.x series	✓	✓
Oracle Linux 8.x series	✓	✓

The following table shows which versions of Unbreakable Enterprise Kernel you can use as a guest operating system for generation 1 and generation 2 virtual machines.

 Expand table

Unbreakable Enterprise Kernel (UEK) versions	Generation 1	Generation 2
Oracle Linux UEK R3 QU3	✓	✗
Oracle Linux UEK R3 QU2	✓	✗
Oracle Linux UEK R3 QU1	✓	✗

For more information, see [Oracle Linux virtual machines on Hyper-V](#).

SUSE guest operating system support

The following table shows which versions of SUSE you can use as a guest operating system for generation 1 and generation 2 virtual machines.

 Expand table

Operating system versions	Generation 1	Generation 2
SUSE Linux Enterprise Server 15 series	✓	✓
SUSE Linux Enterprise Server 12 series	✓	✓
SUSE Linux Enterprise Server 11 series	✓	✗
Open SUSE 12.3	✓	✗

For more information, see [SUSE virtual machines on Hyper-V](#).

Ubuntu guest operating system support

The following table shows which versions of Ubuntu you can use as a guest operating system for generation 1 and generation 2 virtual machines.

 Expand table

Operating system versions	Generation 1	Generation 2
Ubuntu 24.04	✓	✓
Ubuntu 22.04	✓	✓
Ubuntu 20.04	✓	✓
Ubuntu 18.04	✓	✓
Ubuntu 16.04	✓	✓

Operating system versions	Generation 1	Generation 2
Ubuntu 14.04	✓	✓

For more information, see [Ubuntu virtual machines on Hyper-V](#).

How can I boot the virtual machine?

Generation 1 and generation 2 VMs support different boot methods, these methods are shown in the following table.

 Expand table

Boot method	Generation 1	Generation 2
PXE boot by using a standard network adapter	✗	✓
PXE boot by using a legacy network adapter	✓	✗
Boot from a SCSI virtual hard disk (<code>.VHDX</code>) or virtual DVD (<code>.ISO</code>)	✗	✓
Boot from IDE Controller virtual hard disk (<code>.VHD</code>), virtual DVD (<code>.ISO</code>) or a physical CD/DVD drive	✓	✗
Boot from virtual floppy (<code>.VFD</code>)	✓	✗

What's the difference in device support?

The following table compares the devices available between generation 1 and generation 2 virtual machines.

 Expand table

Generation 1 Device	Generation 2 Replacement	Generation 2 Enhancements
IDE controller	Virtual SCSI controller	Boot from <code>.VHDX</code> (64 TB maximum size, and online resize capability)
IDE CD-ROM	Virtual SCSI CD-ROM	Support for up to 64 SCSI DVD devices per SCSI controller.
Legacy BIOS	UEFI firmware	Secure Boot

Generation 1 Device	Generation 2 Replacement	Generation 2 Enhancements
Legacy network adapter	Synthetic network adapter	Network boot with IPv4 and IPv6
Floppy controller and DMA controller	No floppy controller support	N/A
Universal asynchronous receiver/transmitter (UART) for COM ports	Optional UART for debugging	Faster and more reliable
i8042 keyboard controller	Software-based input	Uses fewer resources because there's no emulation. Also reduces the attack surface from the guest operating system.
PS/2 keyboard	Software-based keyboard	Uses fewer resources because there's no emulation. Also reduces the attack surface from the guest operating system.
PS/2 mouse	Software-based mouse	Uses fewer resources because there's no emulation. Also reduces the attack surface from the guest operating system.
S3 video	Software-based video	Uses fewer resources because there's no emulation. Also reduces the attack surface from the guest operating system.
PCI bus	No longer required	N/A
Programmable interrupt controller (PIC)	No longer required	N/A
Programmable interval timer (PIT)	No longer required	N/A
Super I/O device	No longer required	N/A

Considerations for using generation 1 and generation 2 virtual machines

Here are some more tips about using the different generations of virtual machines.

Create VMs with more than 64 logical CPUs

Hyper-V manager might fail to create a new generation 1 VM on a system with more than 64 logical CPUs. Hyper-V manager doesn't let you specify the number of virtual processors at VM creation time. For hosts with more than 64 logical processors, specify the number of virtual processors at VM creation using Windows Admin Center, PowerShell, or another tool.

Upload a virtual hard drive to Azure

Virtual hard drives created on generation 1 and generation 2 VMs can be uploaded to Azure as long as they use the VHD file format. The virtual hard drive must have a fixed (not dynamically expanding) sized disk. See [Generation 2 VMs on Azure](#) to learn more about generation 2 capabilities supported on Azure. For more information on uploading a Windows VHD or VHDX, see [Prepare a Windows VHD or VHDX to upload to Azure](#).

Attach or add a DVD drive

- You can't attach a physical CD or DVD drive to a generation 2 virtual machine. The virtual DVD drive in generation 2 virtual machines only supports ISO image files. To create an ISO image file of a Windows environment, you can use the *OScdimg* command line tool. For more information, see [Oscdimg Command-Line Options](#).
- When you create a new virtual machine with the `New-VM` Windows PowerShell cmdlet, the generation 2 virtual machine doesn't have a DVD drive. You can add a DVD drive while the virtual machine is running.

Use UEFI firmware

- Secure Boot or UEFI firmware isn't required on the physical Hyper-V host. For generation 2 VMs, Hyper-V provides virtual firmware to virtual machines that is independent of what's on the Hyper-V host.
- UEFI firmware in a generation 2 virtual machine doesn't support setup mode for Secure Boot.
- We don't support running a UEFI shell or other UEFI applications in a generation 2 virtual machine. Using a non-Microsoft UEFI shell or UEFI applications is technically possible if they're compiled directly from the sources. If these applications aren't digitally signed correctly, you must disable Secure Boot for the virtual machine.

Work with VHDX files

- You can resize a VHDX file that contains the boot volume for a generation 2 virtual machine while the virtual machine is running.

- We don't support or recommend that you create a single virtual disk (VHD or VHDX file) that is bootable to **both** generation 1 and generation 2 virtual machines. Instead, create bootable VHDX files that target only generation 1 **or** generation 2 virtual machines.
- The virtual machine generation is a property of the virtual machine, not a property of the virtual hard disk. You can't tell if a VHDX file was created as a generation 1 or a generation 2 virtual machine.
- A VHDX file created with a generation 2 virtual machine can be attached to the IDE controller or the SCSI controller of a generation 1 virtual machine. However, if the virtual hard drive is a bootable VHDX file, the generation 1 virtual machine fails to boot.

Use IPv6 instead of IPv4

When you boot from network with PXE, generation 2 virtual machines use IPv4 by default. To use IPv6 instead, run the [Set-VMFirmware](#) Windows PowerShell cmdlet. For example, the following command sets the preferred protocol to IPv6 for a virtual machine named TestVM:

PowerShell

```
Set-VMFirmware -VMName 'TestVM' -IPProtocolPreference IPv6
```

Add a COM port for kernel debugging

COM ports aren't available in generation 2 virtual machines until you add them. You can add COM ports with Windows PowerShell or Windows Management Instrumentation (WMI). These steps show you how to do it with Windows PowerShell.

To add a COM port:

1. Disable Secure Boot. Kernel debugging isn't compatible with Secure Boot. Make sure the virtual machine is in an Off state, then use the [Set-VMFirmware](#) cmdlet. For example, the following command disables Secure Boot on virtual machine TestVM:

PowerShell

```
Set-VMFirmware -VMName 'TestVM' -EnableSecureBoot Off
```

2. Add a COM port. Use the [Set-VMComPort](#) cmdlet to add a COM port. For example, the following command configures the first COM port on virtual machine, TestVM, to connect to the named pipe, TestPipe, on the local computer:

PowerShell

```
Set-VMComPort -VMName 'TestVM' -Number 1 -Path '\\.\pipe\TestPipe'
```

ⓘ Note

Configured COM ports aren't listed in the settings of a virtual machine in Hyper-V Manager.

Related content

- [Linux and FreeBSD Virtual Machines on Hyper-V](#)
- [Use local resources on Hyper-V virtual machine with VMConnect](#)
- [Plan for Hyper-V scalability in Windows Server 2016](#)

Generation 1 virtual machine security settings

07/01/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Use the generation 1 virtual machine security settings in Hyper-V Manager to help protect the data and state of a virtual machine.

Encryption support settings in Hyper-V Manager

You can help protect the data and state of the virtual machine by selecting the following encryption support option.

- **Encrypt state and virtual machine migration traffic** - Encrypts the virtual machine saved state when it's written to disk and the live migration traffic.

To enable this option, you must add a key storage drive for the virtual machine.

Key storage drive in Hyper-V Manager

A key storage drive provides a small drive to the virtual machine for a BitLocker key to be stored. This allows the virtual machine to encrypt its operating system disk without requiring a virtualized Trusted Platform Module (TPM) chip. The contents of the key storage drive are encrypted by using a Key Protector. The Key Protector authorizes the Hyper-V host to run the virtual machine. Both the contents of the key storage drive and the Key Protector are stored as part of the virtual machine's runtime state.

To decrypt the contents of the key storage drive and start the virtual machine, the Hyper-V host needs to be either:

- Part of an authorized guarded fabric for this virtual machine, or
- Have the private key from one of the virtual machine's guardians.

To learn more about guarded fabrics, please see the [Introducing Shielded VMs](#) section in [Security and Assurance](#).

You can add a key storage drive to an empty slot on one of the virtual machine's IDE controllers. To do this, click **Add Key Storage Drive** to add a key storage drive to the first free IDE controller slot of this virtual machine.

Additional References

- [Generation 2 virtual machine security settings in Hyper-V manager](#)
- [Security and Assurance](#)

Hyper-V generation 2 virtual machine security features

07/01/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V generation 2 virtual machines (VMs) provide robust security features designed to safeguard sensitive data and prevent unauthorized access or tampering. This article explores the security settings available in Hyper-V Manager for generation 2 VMs and shows you how to configure them. Learn how these features can help protect your VMs from threats and ensure compliance with security best practices.

The available security features for generation 2 VMs in Hyper-V include:

- Secure Boot.
- Encryption support for Trusted Platform Module (TPM), live migration, and saved state.
- Shielded VMs.
- Host Guardian Service (HGS).

These security features are designed to help protect the data and state of a virtual machine. You can protect VMs from inspection, theft, and tampering from both malware that might run on the host, and datacenter administrators. The level of security you get depends on the host hardware you run, the virtual machine generation, and whether you set up the companion Host Guardian Service (HGS), that authorizes hosts to start shielded VMs.

The Host Guardian Service was first introduced in Windows Server 2016. It identifies legitimate Hyper-V hosts and allows them to run a set of shielded VMs. You'd most commonly enable the Host Guardian Service for a datacenter, but you can also create a shielded VM to run it locally without setting up Host Guardian Service. You can later distribute the shielded virtual machine to the Host Guardian Service.

To learn how you can make your VMs more secure with the Host Guardian Service, see [Guarded Fabric and Shielded VMs](#) and [Harden the Fabric: Protecting Tenant Secrets in Hyper-V \(Ignite video\)](#) .

Secure Boot

Secure Boot is a feature available with generation 2 VMs that helps prevent unauthorized firmware, operating systems, or Unified Extensible Firmware Interface (UEFI) drivers (also known as option ROMs) from running at boot time. Secure Boot is enabled by default. You can use secure boot with generation 2 VMs that run Windows or Linux distribution operating systems.

There are three different templates available, depending on the VM's operating system and configuration. The following table lists each of these templates and refers to the certificates that you need to verify the integrity of the boot process:

 Expand table

Template name	Compatibility
Microsoft Windows	Windows operating systems.
Microsoft UEFI Certificate Authority	Linux distribution operating systems.
Open Source Shielded VM	Linux-based shielded VMs.

Encryption support

Hyper-V Generation 2 virtual machines offer robust encryption capabilities that provide multiple layers of protection for your virtualized infrastructure. The encryption support encompasses three critical areas: TPM (Trusted Platform Module) functionality, live migration network traffic, and saved state data. These security features work together to create a comprehensive defense against unauthorized access and data breaches, ensuring that sensitive information remains protected both at rest and in transit.

The virtualized TPM (vTPM) feature represents a significant advancement in VM security architecture. By adding a vTPM to your Generation 2 virtual machine, you enable the guest operating system to use hardware-based security functions similar to those functions available on physical machines. This virtualized security chip allows the guest OS to encrypt the entire virtual machine disk using BitLocker Drive Encryption, creating an extra layer of protection against unauthorized access. The vTPM can also support other security technologies that need a TPM, making it an essential component for enterprise environments that require strict compliance with security standards and regulations.

You can migrate a virtual machine with virtual TPM enabled to any host that runs a supported version of Windows Server or Windows. If you migrate it to another host, you might not be able to start it. You must update the Key Protector for that virtual machine to authorize the new host to run the virtual machine. For more information, see [Guarded Fabric and Shielded VMs](#) and [System requirements for Hyper-V on Windows Server](#).

Security Policy in Hyper-V Manager

Shielded virtual machines represent the highest level of security available for Hyper-V Generation 2 VMs, providing comprehensive protection against both external threats and

privileged access attacks. When you enable shielding on a virtual machine, you create a hardened environment that encrypts the VM's state and migration traffic while restricting administrative access to critical VM functions. This protection extends beyond traditional security measures by preventing even datacenter administrators and host-level malware from accessing the VM's memory, saved state, or network traffic during live migration operations.

The shielding feature automatically enforces several security requirements, including Secure Boot, TPM enablement, and encryption of saved state and migration traffic. Additionally, shielded VMs disable certain management capabilities such as console connections, PowerShell Direct, and specific integration components that attackers could potentially exploit. This approach creates a defense-in-depth security model where the virtual machine becomes effectively opaque to the host system, ensuring that sensitive workloads remain protected even in compromised hosting environments. Organizations can deploy shielded VMs with Host Guardian Service for enterprise-scale implementations or run them locally for enhanced security in smaller deployments.

You can run the shielded virtual machine locally without setting up a Host Guardian Service. If you migrate it to another host, you might not be able to start it. You must update the Key Protector for that virtual machine to authorize the new host to run the virtual machine. For more information, see [Guarded Fabric and Shielded VMs](#).

Related content

For more information, see the following articles:

- [Security and Assurance in Windows Server](#).
- [Should I create a generation 1 or 2 virtual machine in Hyper-V?](#)
- [Linux and FreeBSD Virtual Machines on Hyper-V](#)

 **Note:** The author created this article with assistance from AI. [Learn more](#)

Plan for Hyper-V networking in Windows Server

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This article covers what you need for basic networking in Hyper-V, including considerations for live migration and other server features.

Hyper-V networking basics

Hyper-V networking uses a virtual switch and a virtual network adapter. The virtual switch connects to an Ethernet-based network, and the virtual network adapter connects to a port on the virtual switch, enabling a virtual machine to use the network. When you create a virtual machine and connect the virtual machine to a virtual switch, a virtual network adapter is automatically added. All virtual switches are part of the Hyper-V host, but each virtual network adapter belongs to only one virtual machine. For instructions, see [Create a virtual switch for Hyper-V virtual machines](#).

The virtual switch is a software-based layer-2 Ethernet network switch. The virtual switch provides built-in features for monitoring, controlling, and segmenting traffic, as well as security, and diagnostics. You can add to the set of built-in features by installing plug-ins, also called *extensions*. Extensions are available from independent software vendors. For more information about the switch and extensions, see [Hyper-V Virtual Switch](#).

Switch and network adapter choices

Hyper-V offers three types of virtual switches and two types of virtual network adapters. You choose the type when you create the switch or adapter. You can use Hyper-V Manager or the Hyper-V module for Windows PowerShell to create and manage virtual switches and virtual network adapters. Some advanced networking capabilities, such as extended port access control lists (ACLs), can only be managed by using cmdlets in the Hyper-V module.

You can change an existing virtual switch to a different type after creation, but changing the virtual switch type affects the networking capabilities of all connected virtual machines. You can connect a virtual network adapter to a different virtual switch, but you can't change a virtual network adapter from one type to another; instead, add another virtual network adapter and choose the appropriate type.

Virtual switch types are:

- **External virtual switch** - Connects to a wired, physical network by binding to a physical network adapter.
- **Internal virtual switch** - Connects to a network that can be used only by the virtual machines running on the host that has the virtual switch, and between the host and the virtual machines.
- **Private virtual switch** - Connects to a network that can be used only by the virtual machines running on the host that has the virtual switch, but doesn't provide networking between the host and the virtual machines.

Virtual switch options:

 Expand table

Setting name	Description
Allow management operating system to share this network adapter	Allow the Hyper-V host to share the use of the virtual switch and NIC or NIC team with the virtual machine. With this enabled, the host can use any of the settings that you configure for the virtual switch, such as Quality of Service (QoS) settings, security settings, or other features of the Hyper-V virtual switch.
Enable single-root I/O virtualization (SR-IOV)	Allow virtual machine traffic to bypass the virtual machine switch and go directly to the physical NIC. SR-IOV is only available for virtual machines running a supported Windows Server guest . For more information, see Single-Root I/O Virtualization .

Virtual network adapter types:

 Expand table

Adapter type	Key constraints and when to use
Network adapter (Hyper-V specific network adapter)	Available for generation 1 and generation 2 VMs. Requires a driver included in Hyper-V integration services, provided only for supported guest operating systems . Faster performance; recommended unless you need network boot or are running an unsupported guest OS.
Legacy network adapter	Generation 1 VMs only; not available for generation 2 VMs. Emulates an Intel 21140-based PCI Fast Ethernet Adapter. Use when you need to boot to a network to install an operating system (for example, via Windows Deployment Services).

Hyper-V networking and related technologies

Some features rely on specific networking configurations or perform better under certain configurations.

Failover clustering - It's a best practice to isolate cluster traffic and use Hyper-V Quality of Service (QoS) on the virtual switch.

Live migration - Use performance options to reduce network and CPU usage and the time it takes to complete a live migration. For instructions, see [Set up hosts for live migration without Failover Clustering](#).

Storage Spaces Direct - Storage Spaces Direct relies on the SMB3.0 network protocol and RDMA. For details, see [Storage Spaces Direct overview](#).

Last updated on 01/21/2026

Network recommendations for Hyper-V in a failover cluster

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

When you deploy a highly available Hyper-V solution, you must consider and plan for several different types of network traffic. Design your network configuration with the following goals in mind:

- Ensure network quality of service
- Provide network redundancy
- Isolate traffic to defined networks
- Where applicable, take advantage of Server Message Block (SMB) Multichannel

This topic provides network configuration recommendations that are specific to a Hyper-V cluster running Windows Server. It includes an overview of the different network traffic types, recommendations for how to isolate traffic, recommendations for features such as NIC Teaming, Quality of Service (QoS), and Virtual Machine Queue (VMQ), and a Windows PowerShell script that shows an example of converged networking, where the network traffic on a Hyper-V cluster routes through one external virtual switch.

Windows Server supports the concept of converged networking, where different types of network traffic share the same Ethernet network infrastructure. Features such as Hyper-V QoS and the ability to add virtual network adapters to the management operating system enable you to consolidate the network traffic on fewer physical adapters. Combined with traffic isolation methods such as VLANs, you can isolate and control the network traffic.

 Note

If you use System Center Virtual Machine Manager (VMM) to create or manage Hyper-V clusters, you must use VMM to configure the network settings that are described in this topic.

Overview of different network traffic types

When you deploy a Hyper-V cluster, you must plan for several types of network traffic. The following table summarizes the different traffic types.

Network Traffic Type	Description
Management	Provides connectivity between the server that is running Hyper-V and basic infrastructure functionality. Use it to manage the Hyper-V management operating system and virtual machines.
Cluster	Use it for inter-node cluster communication such as the cluster heartbeat and Cluster Shared Volumes (CSV) redirection.
Live migration	Use it for virtual machine live migration.
Storage	Use it for SMB traffic or for iSCSI traffic.
Replica traffic	Use it for virtual machine replication through the Hyper-V Replica feature.
Virtual machine access	Use it for virtual machine connectivity. Typically requires external network connectivity to service client requests.

The following sections provide more detailed information about each network traffic type.

Management traffic

A management network provides connectivity between the operating system of the physical Hyper-V host (also known as the *management operating system*) and basic infrastructure functionality such as Active Directory Domain Services (AD DS), Domain Name System (DNS), and Windows Server Update Services (WSUS). It also manages the server that is running Hyper-V and the virtual machines.

The management network must have connectivity between all required infrastructure, and to any location from which you want to manage the server.

Cluster traffic

A failover cluster monitors and communicates the cluster state between all members of the cluster. This communication is very important to maintain cluster health. If a cluster node doesn't communicate a regular health check (known as the *cluster heartbeat*), the cluster considers the node down and removes the node from cluster membership. The cluster then transfers the workload to another cluster node.

Inter-node cluster communication also includes traffic that's associated with CSV. For CSV, where all nodes of a cluster can access shared block-level storage simultaneously, the nodes in the cluster must communicate to orchestrate storage-related activities. Also, if a cluster node loses its direct connection to the underlying CSV storage, CSV has resiliency features which redirect the storage I/O over the network to another cluster node that can access the storage.

Live migration traffic

Live migration enables the transparent movement of running virtual machines from one Hyper-V host to another without a dropped network connection or perceived downtime.

Use a dedicated network or VLAN for live migration traffic to ensure quality of service and for traffic isolation and security. Live migration traffic can saturate network links. This saturation can cause other traffic to experience increased latency. The time it takes to fully migrate one or more virtual machines depends on the throughput of the live migration network. Therefore, you must ensure that you configure the appropriate quality of service for this traffic. To provide the best performance, live migration traffic isn't encrypted.

You can designate multiple networks as live migration networks in a prioritized list. For example, you might have one migration network for cluster nodes in the same cluster that is fast (10 GB), and a second migration network for cross-cluster migrations that is slower (1 GB).

All Hyper-V hosts that can initiate or receive a live migration must have connectivity to a network that is configured to allow live migrations. Because live migration can occur between nodes in the same cluster, between nodes in different clusters, and between a cluster and a stand-alone Hyper-V host, make sure that all these servers can access a live migration-enabled network.

Storage traffic

For a virtual machine to be highly available, all members of the Hyper-V cluster must access the virtual machine state. This state includes the configuration state and the virtual hard disks. To meet this requirement, you must have shared storage.

In Windows Server, you can provide shared storage in two ways:

- **Shared block storage.** Shared block storage options include Fibre Channel, Fibre Channel over Ethernet (FCoE), iSCSI, and shared Serial Attached SCSI (SAS).
- **File-based storage.** Remote file-based storage is provided through SMB 3.0.

SMB 3.0 includes new functionality known as SMB Multichannel. SMB Multichannel automatically detects and uses multiple network interfaces to deliver high performance and highly reliable storage connectivity.

By default, SMB Multichannel is enabled and requires no additional configuration. Use at least two network adapters of the same type and speed so that SMB Multichannel is in effect. Network adapters that support RDMA (Remote Direct Memory Access) are recommended but not required.

SMB 3.0 also automatically discovers and takes advantage of available hardware offloads, such as RDMA. A feature known as SMB Direct supports the use of network adapters that have RDMA capability. SMB Direct provides the best performance possible while also reducing file server and client overhead.

Note

The NIC Teaming feature is incompatible with RDMA-capable network adapters. Therefore, if you intend to use the RDMA capabilities of the network adapter, don't team those adapters.

Both iSCSI and SMB use the network to connect the storage to cluster members. Because reliable storage connectivity and performance are very important for Hyper-V virtual machines, use multiple networks (physical or logical) to ensure that these requirements are achieved.

For more information about SMB Direct and SMB Multichannel, see [Improve Performance of a File Server with SMB Direct](#).

Replica traffic

Hyper-V Replica provides asynchronous replication of Hyper-V virtual machines between two hosting servers or Hyper-V clusters. Replica traffic occurs between the primary and Replica sites.

Hyper-V Replica automatically discovers and uses available network interfaces to transmit replication traffic. To throttle and control the replica traffic bandwidth, define QoS policies with minimum bandwidth weight.

If you use certificate-based authentication, Hyper-V Replica encrypts the traffic. If you use Kerberos-based authentication, traffic isn't encrypted.

Virtual machine access traffic

Most virtual machines require some form of network or Internet connectivity. For example, workloads that run on virtual machines typically need external network connectivity to service client requests. This connectivity can include tenant access in a hosted cloud implementation. Because multiple subclasses of traffic exist, such as traffic that's internal to the datacenter and traffic that's external (for example, to a computer outside the datacenter or to the Internet), these virtual machines need one or more networks to communicate.

To separate virtual machine traffic from the management operating system, use VLANs that aren't exposed to the management operating system.

How to isolate the network traffic on a Hyper-V cluster

To provide the most consistent performance and functionality and to improve network security, isolate the different types of network traffic.

Note

If you want to have a physical or logical network that's dedicated to a specific traffic type, you must assign each physical or virtual network adapter to a unique subnet. For each cluster node, Failover Clustering recognizes only one IP address per subnet.

Isolate traffic on the management network

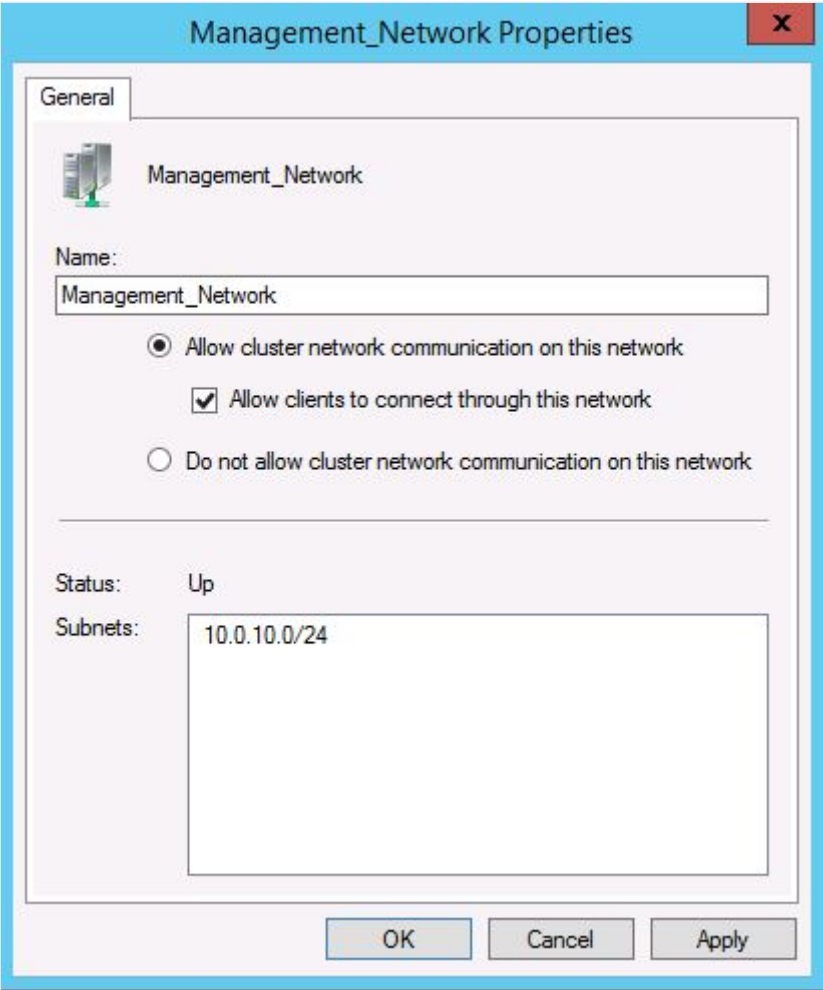
Use a firewall or IPsec encryption, or both, to isolate management traffic. In addition, use auditing to ensure that only defined and allowed communication is transmitted through the management network.

Isolate traffic on the cluster network

To isolate inter-node cluster traffic, configure a network to either allow cluster network communication or not to allow cluster network communication. For a network that allows cluster network communication, you can also configure whether to allow clients to connect through the network. This configuration includes client and management operating system access.

A failover cluster can use any network that allows cluster network communication for cluster monitoring, state communication, and for CSV-related communication.

To configure a network to allow or not to allow cluster network communication, use Failover Cluster Manager or Windows PowerShell. To use Failover Cluster Manager, select **Networks** in the navigation tree. In the **Networks** pane, right-click a network, then select **Properties**.



The following Windows PowerShell example configures a network named *Management Network* to allow cluster and client connectivity.

PowerShell

```
(Get-ClusterNetwork -Name "Management Network").Role = 3
```

The **Role** property has the following possible values.

 Expand table

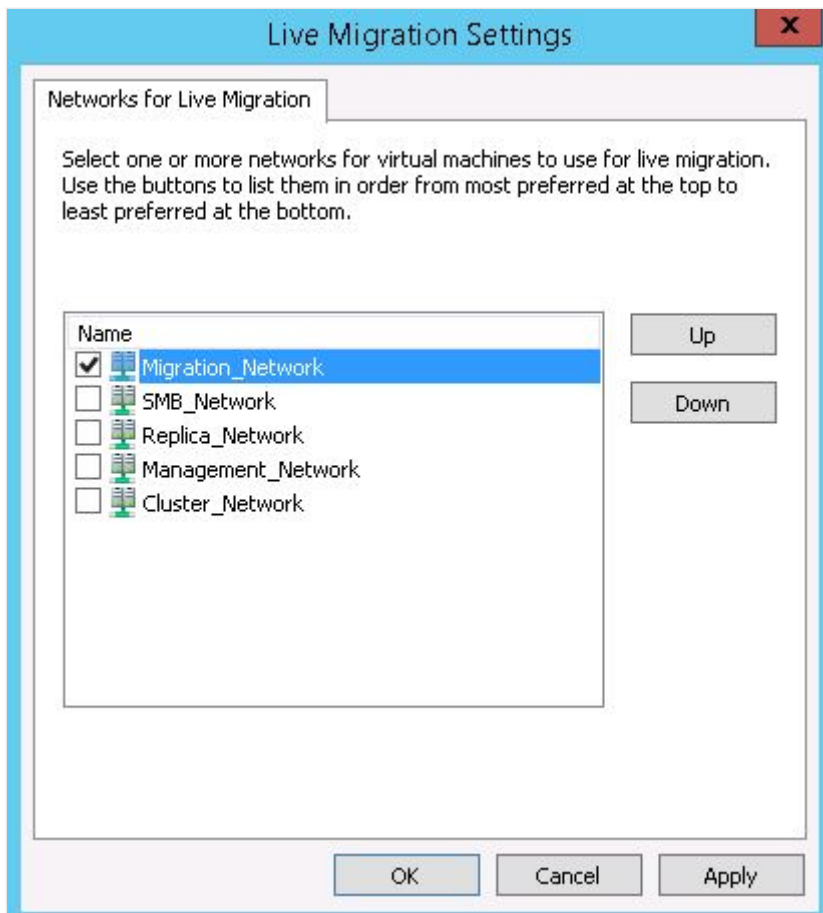
Value	Network Setting
0	Do not allow cluster network communication
1	Allow cluster network communication only
3	Allow cluster network communication and client connectivity

The following table shows the recommended settings for each type of network traffic. Virtual machine access traffic isn't listed because you should isolate these networks from the management operating system by using VLANs that aren't exposed to the host. Therefore, virtual machine networks shouldn't appear in Failover Cluster Manager as cluster networks.

Network Type	Recommended Setting
Management	Both of the following: - Allow cluster network communication on this network - Allow clients to connect through this network
Cluster	Allow cluster network communication on this network Note: Clear the Allow clients to connect through this network check box.
Live migration	Allow cluster network communication on this network Note: Clear the Allow clients to connect through this network check box.
Storage	Do not allow cluster network communication on this network
Replica traffic	Both of the following: - Allow cluster network communication on this network - Allow clients to connect through this network

Isolate traffic on the live migration network

By default, live migration traffic uses the cluster network topology to discover available networks and to establish priority. However, you can manually configure live migration preferences to isolate live migration traffic to only the networks that you define. To do this, you can use Failover Cluster Manager or Windows PowerShell. To use Failover Cluster Manager, in the navigation tree, right-click **Networks**, then select **Live Migration Settings**.



The following Windows PowerShell example enables live migration traffic only on a network named *Migration_Network*.

PowerShell

```
Get-ClusterResourceType -Name "Virtual Machine" | Set-ClusterParameter -Name  
MigrationExcludeNetworks -Value ([String]::Join(";",(Get-ClusterNetwork | Where-  
Object {$_.Name -ne "Migration_Network"}).ID))
```

Isolate traffic on the storage network

To isolate SMB storage traffic, use Windows PowerShell to set SMB Multichannel constraints. SMB Multichannel constraints restrict SMB communication between a given file server and the Hyper-V host to one or more defined network interfaces.

For example, the following Windows PowerShell command sets a constraint for SMB traffic from the file server *FileServer1* to the network interfaces *SMB1*, *SMB2*, *SMB3*, and *SMB4* on the Hyper-V host from which you run this command.

PowerShell

```
New-SmbMultichannelConstraint -ServerName "FileServer1" -InterfaceAlias "SMB1",  
"SMB2", "SMB3", "SMB4"
```

ⓘ Note

- You must run this command on each node of the Hyper-V cluster.
- To find the interface name, use the **Get-NetAdapter** cmdlet.

For more information, see [New-SmbMultichannelConstraint](#).

To isolate iSCSI traffic, configure the iSCSI target with interfaces on a dedicated network (logical or physical). Use the corresponding interfaces on the cluster nodes when you configure the iSCSI initiator.

Isolate traffic for replication

To isolate Hyper-V Replica traffic, use a different subnet for the primary and Replica sites.

If you want to isolate the replica traffic to a particular network adapter, define a persistent static route that redirects the network traffic to the defined network adapter. To specify a static route, use the following command:

```
route add <destination> mask <subnet mask and gateway> if <interface> -p
```

For example, to add a static route to the 10.1.17.0 network (example network of the Replica site) that uses a subnet mask of 255.255.255.0, a gateway of 10.0.17.1 (example IP address of the primary site), where the interface number for the adapter that you want to dedicate to replica traffic is 8, run the following command:

```
route add 10.1.17.1 mask 255.255.255.0 10.0.17.1 if 8 -p
```

NIC Teaming (LBFO) recommendations

Team physical network adapters in the management operating system. This configuration provides bandwidth aggregation and network traffic failover if a network hardware failure or outage occurs.

The NIC Teaming feature, also known as load balancing and failover (LBFO), provides two basic sets of algorithms for teaming.

- **Switch-dependent modes.** Requires the switch to participate in the teaming process. Typically requires all the network adapters in the team to be connected to the same switch.
- **Switch-independent modes.** Doesn't require the switch to participate in the teaming process. Although not required, team network adapters can be connected to different

switches.

Both modes provide bandwidth aggregation and traffic failover if a network adapter failure or network disconnection occurs. However, in most cases only switch-independent teaming provides traffic failover for a switch failure.

NIC Teaming also provides a traffic distribution algorithm that's optimized for Hyper-V workloads. This algorithm is referred to as the *Hyper-V port* load balancing mode. This mode distributes the traffic based on the MAC address of the virtual network adapters. The algorithm uses round robin as the load-balancing mechanism. For example, on a server that has two teamed physical network adapters and four virtual network adapters, the first and third virtual network adapter use the first physical adapter, and the second and fourth virtual network adapter use the second physical adapter. Hyper-V port mode also enables the use of hardware offloads such as virtual machine queue (VMQ) which reduces CPU overhead for networking operations.

NIC Teaming recommendations

For a clustered Hyper-V deployment, use the following settings when you configure the additional properties of a team.

 Expand table

Property Name	Recommended Setting
Teaming mode	Switch Independent (the default setting)
Load balancing mode	Hyper-V Port

 Note

NIC teaming disables the RDMA capability of the network adapters. To use SMB Direct and the RDMA capability of the network adapters, don't use NIC Teaming.

For more information about the NIC Teaming modes and how to configure NIC Teaming settings, see [Hyper-V Virtual Switch](#).

Quality of Service (QoS) recommendations

Use QoS technologies that are available in Windows Server to meet the service requirements of a workload or an application. QoS provides the following features:

- Measures network bandwidth, detects changing network conditions (such as congestion or availability of bandwidth), and prioritizes or throttles network traffic.
- Enables you to converge multiple types of network traffic on a single adapter.
- Includes a minimum bandwidth feature that guarantees a certain amount of bandwidth to a given type of traffic.

Configure appropriate Hyper-V QoS on the virtual switch to ensure that network requirements are met for all appropriate types of network traffic on the Hyper-V cluster.

ⓘ Note

You can use QoS to control outbound traffic, but not the inbound traffic. For example, with Hyper-V Replica, you can use QoS to control outbound traffic (from the primary server), but not the inbound traffic (from the Replica server).

QoS recommendations

For a Hyper-V cluster, configure Hyper-V QoS that applies to the virtual switch. When you configure QoS, complete the following steps:

- **Configure minimum bandwidth in weight mode instead of in bits per second.** Minimum bandwidth specified by weight is more flexible and it's compatible with other features, such as live migration and NIC Teaming. For more information, see the **MinimumBandwidthMode** parameter in [New-VMSwitch](#).
- **Enable and configure QoS for all virtual network adapters.** Assign a weight to all virtual adapters. For more information, see [Set-VMNetworkAdapter](#). To make sure that all virtual adapters have a weight, configure the **DefaultFlowMinimumBandwidthWeight** parameter on the virtual switch to a reasonable value. For more information, see [Set-VMSwitch](#).

The following table recommends some generic weight values. You can assign a value from 1 to 100. For guidelines to consider when you assign weight values, see [QoS minimum bandwidth best practices](#).

 Expand table

Network Classification	Weight
Default weight	0
Virtual machine access	1, 3, or 5 (low, medium, and high-throughput virtual machines)
Cluster	10

Network Classification	Weight
Management	10
Replica traffic	10
Live migration	40
Storage	40

Virtual machine queue (VMQ) recommendations

Virtual machine queue (VMQ) is a feature that's available to computers that have VMQ-capable network hardware. VMQ uses hardware packet filtering to deliver packet data from an external virtual network directly to virtual network adapters. This feature reduces the overhead of routing packets. When you enable VMQ, a dedicated queue is established on the physical network adapter for each virtual network adapter that requests a queue.

Not all physical network adapters support VMQ. Those that support VMQ have a fixed number of queues available, and the number varies. To determine whether a network adapter supports VMQ, and how many queues it supports, use the [Get-NetAdapterVmq](#) cmdlet.

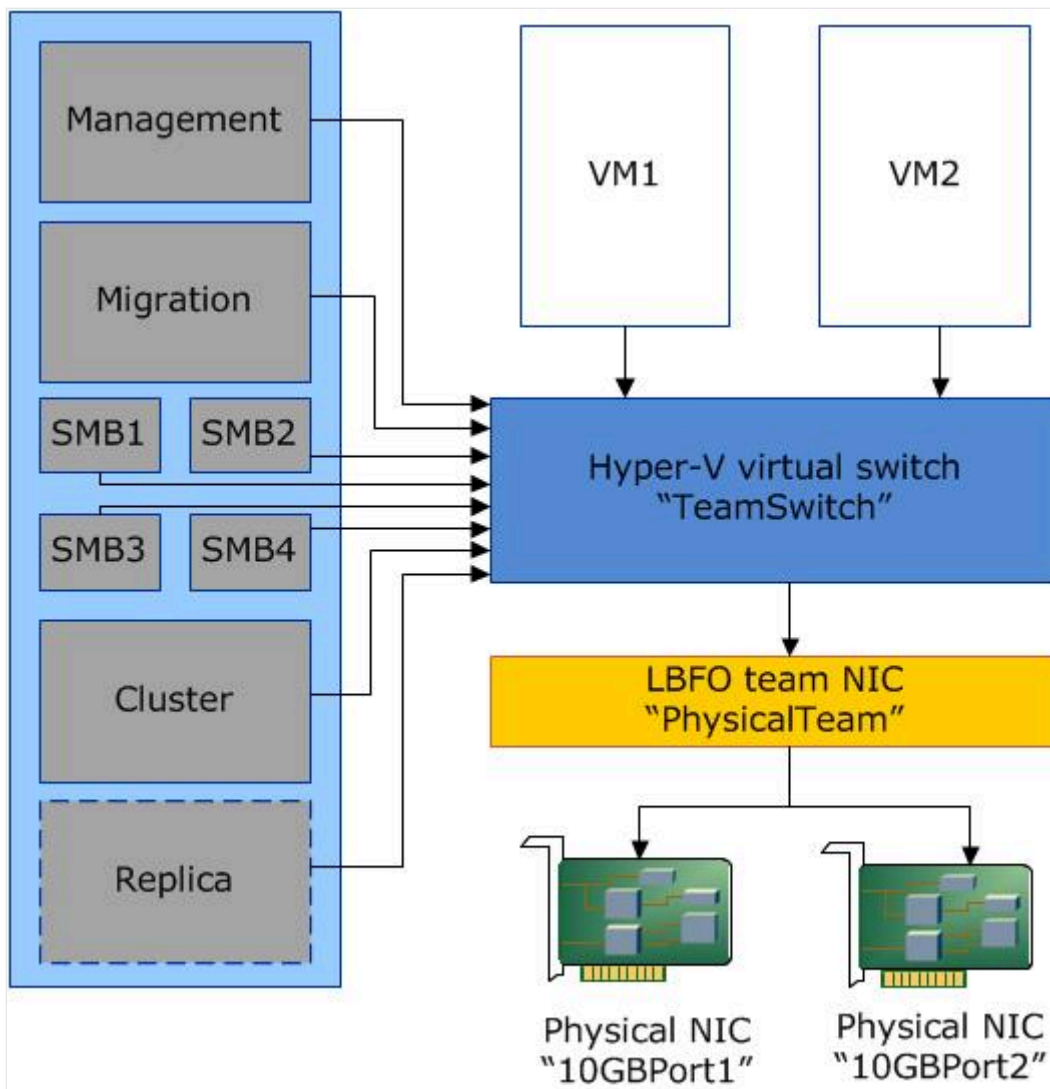
You can assign virtual machine queues to any virtual network adapter. This capability includes virtual network adapters that you expose to the management operating system. Assign queues according to a weight value, in a first-come first-serve manner. By default, all virtual adapters have a weight of 100.

VMQ recommendations

Increase the VMQ weight for interfaces with heavy inbound traffic, such as storage and live migration networks. To do this, use the [Set-VMNetworkAdapter](#) Windows PowerShell cmdlet.

Example of converged networking: routing traffic through one Hyper-V virtual switch

The following Windows PowerShell script shows an example of how you can route traffic on a Hyper-V cluster through one Hyper-V external virtual switch. The example uses two physical 10 GB network adapters that are teamed by using the NIC Teaming feature. The script configures a Hyper-V cluster node with a management interface, a live migration interface, a cluster interface, and four SMB interfaces. After the script, there is more information about how to add an interface for Hyper-V Replica traffic. The following diagram shows the example network configuration.



The example also configures network isolation which restricts cluster traffic from the management interface, restricts SMB traffic to the SMB interfaces, and restricts live migration traffic to the live migration interface.

PowerShell

```
# Create a network team using switch independent teaming and Hyper-V port mode
New-NetLbfoTeam "PhysicalTeam" -TeamMembers "10GBPort1", "10GBPort2" -TeamNicName
"PhysicalTeam" -TeamingMode SwitchIndependent -LoadBalancingAlgorithm HyperVPort

# Create a Hyper-V virtual switch connected to the network team
# Enable QoS in Weight mode
New-VMSwitch "TeamSwitch" -NetAdapterName "PhysicalTeam" -MinimumBandwidthMode
Weight -AllowManagementOS $false

# Configure the default bandwidth weight for the switch
# Ensures all virtual NICs have a weight
Set-VMSwitch -Name "TeamSwitch" -DefaultFlowMinimumBandwidthWeight 0

# Create virtual network adapters on the management operating system
# Connect the adapters to the virtual switch
# Set the VLAN associated with the adapter
# Configure the VMQ weight and minimum bandwidth weight
```

```
Add-VMNetworkAdapter -ManagementOS -Name "Management" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "Management" -Access -
VlanId 10
Set-VMNetworkAdapter -ManagementOS -Name "Management" -VmqWeight 80 -
MinimumBandwidthWeight 10

Add-VMNetworkAdapter -ManagementOS -Name "Cluster" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "Cluster" -Access -
VlanId 11
Set-VMNetworkAdapter -ManagementOS -Name "Cluster" -VmqWeight 80 -
MinimumBandwidthWeight 10

Add-VMNetworkAdapter -ManagementOS -Name "Migration" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "Migration" -Access -
VlanId 12
Set-VMNetworkAdapter -ManagementOS -Name "Migration" -VmqWeight 90 -
MinimumBandwidthWeight 40

Add-VMNetworkAdapter -ManagementOS -Name "SMB1" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "SMB1" -Access -VlanId
13
Set-VMNetworkAdapter -ManagementOS -Name "SMB1" -VmqWeight 100 -
MinimumBandwidthWeight 40

Add-VMNetworkAdapter -ManagementOS -Name "SMB2" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "SMB2" -Access -VlanId
14
Set-VMNetworkAdapter -ManagementOS -Name "SMB2" -VmqWeight 100 -
MinimumBandwidthWeight 40

Add-VMNetworkAdapter -ManagementOS -Name "SMB3" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "SMB3" -Access -VlanId
15
Set-VMNetworkAdapter -ManagementOS -Name "SMB3" -VmqWeight 100 -
MinimumBandwidthWeight 40

Add-VMNetworkAdapter -ManagementOS -Name "SMB4" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "SMB4" -Access -VlanId
16
Set-VMNetworkAdapter -ManagementOS -Name "SMB4" -VmqWeight 100 -
MinimumBandwidthWeight 40

# Rename the cluster networks if desired
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.10.0"}).Name =
"Management_Network"
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.11.0"}).Name =
"Cluster_Network"
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.12.0"}).Name =
"Migration_Network"
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.13.0"}).Name =
"SMB_Network1"
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.14.0"}).Name =
"SMB_Network2"
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.15.0"}).Name =
"SMB_Network3"
```

```
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.16.0"}).Name =
"SMB_Network4"

# Configure the cluster network roles
(Get-ClusterNetwork -Name "Management_Network").Role = 3
(Get-ClusterNetwork -Name "Cluster_Network").Role = 1
(Get-ClusterNetwork -Name "Migration_Network").Role = 1
(Get-ClusterNetwork -Name "SMB_Network1").Role = 0
(Get-ClusterNetwork -Name "SMB_Network2").Role = 0
(Get-ClusterNetwork -Name "SMB_Network3").Role = 0
(Get-ClusterNetwork -Name "SMB_Network4").Role = 0

# Configure an SMB Multichannel constraint
# This ensures that SMB traffic from the named server only uses SMB interfaces
New-SmbMultichannelConstraint -ServerName "FileServer1" -InterfaceAlias "vEthernet
(SMB1)", "vEthernet (SMB2)", "vEthernet (SMB3)", "vEthernet (SMB4)"

# Configure the live migration network
Get-ClusterResourceType -Name "Virtual Machine" | Set-ClusterParameter -Name
MigrationExcludeNetworks -Value ([String]::Join(";",(Get-ClusterNetwork | Where-
Object {$_.Name -ne "Migration_Network"}).ID))
```

Hyper-V Replica considerations

If you also use Hyper-V Replica in your environment, you can add another virtual network adapter to the management operating system for replica traffic. For example:

PowerShell

```
Add-VMNetworkAdapter -ManagementOS -Name "Replica" -SwitchName "TeamSwitch"
Set-VMNetworkAdapterVlan -ManagementOS -VMNetworkAdapterName "Replica" -Access -
VlanId 17
Set-VMNetworkAdapter -ManagementOS -Name "Replica" -VmqWeight 80 -
MinimumBandwidthWeight 10

# If the host is clustered - configure the cluster name and role
(Get-ClusterNetwork | Where-Object {$_.Address -eq "10.0.17.0"}).Name = "Replica"
(Get-ClusterNetwork -Name "Replica").Role = 3
```

ⓘ Note

If you're instead using policy-based QoS, where you can throttle outgoing traffic regardless of the interface on which it is sent, you can use either of the following methods to throttle Hyper-V Replica traffic: Create a QoS policy that is based on the destination port. In the following example, the network listener on the Replica server or cluster is configured to use port 8080 to receive replication traffic.

Appendix: Encryption

Cluster traffic encryption

By default, cluster communication isn't encrypted. You can enable encryption if you want. However, encryption adds performance overhead. To enable encryption, use the following Windows PowerShell command to set the security level for the cluster.

PowerShell

```
(Get-Cluster). SecurityLevel = 2
```

The following table shows the different security level values.

 Expand table

Security Description	Value
Clear text	0
Signed (default)	1
Encrypted	2

Live migration traffic encryption

Live migration traffic isn't encrypted. You can enable IPsec or other network layer encryption technologies if you want. However, encryption technologies typically affect performance.

SMB traffic

By default, SMB traffic isn't encrypted. Therefore, use a dedicated network (physical or logical) or use encryption. For SMB traffic, you can use SMB encryption, layer-2, or layer-3 encryption. SMB encryption is the preferred method.

Replica traffic encryption

If you use Kerberos-based authentication, Hyper-V Replica traffic isn't encrypted. Encrypt replication traffic that transits public networks over the WAN or the Internet. Use Secure Sockets Layer (SSL) encryption as the encryption method. You can also use IPsec. However, using IPsec might significantly affect performance.

Hyper-V Virtual Switch

07/25/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This topic provides an overview of Hyper-V Virtual Switch, which provides you with the ability to connect virtual machines (VMs) to networks that are external to the Hyper-V host, including your organization's intranet and the Internet.

You can also connect to virtual networks on the server that is running Hyper-V when you deploy Software Defined Networking (SDN).

Note

In addition to this topic, the following Hyper-V Virtual Switch documentation is available.

- [Remote Direct Memory Access \(RDMA\) and Switch Embedded Teaming \(SET\)](#)
- [Network Switch Team Cmdlets in Windows PowerShell](#)
- [What's New in VMM 2016](#)
- [Set up the VMM networking fabric](#)
- [Hyper-V forum](#)
- [Hyper-V: The WFP virtual switch extension should be enabled if it is required by third party extensions](#)

For more information about other networking technologies, see [Networking in Windows Server 2016](#).

Hyper-V Virtual Switch is a software-based layer-2 Ethernet network switch that is available in Hyper-V Manager when you install the Hyper-V server role.

Hyper-V Virtual Switch includes programmatically managed and extensible capabilities to connect VMs to both virtual networks and the physical network. In addition, Hyper-V Virtual Switch provides policy enforcement for security, isolation, and service levels.

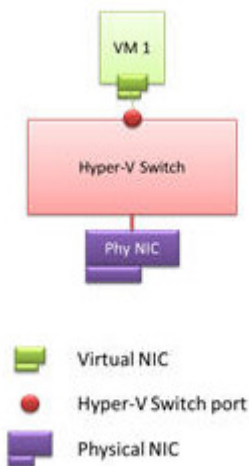
Note

Hyper-V Virtual Switch only supports Ethernet, and does not support any other wired local area network (LAN) technologies, such as Infiniband and Fibre Channel.

Hyper-V Virtual Switch includes tenant isolation capabilities, traffic shaping, protection against malicious virtual machines, and simplified troubleshooting.

With built-in support for Network Device Interface Specification (NDIS) filter drivers and Windows Filtering Platform (WFP) callout drivers, the Hyper-V Virtual Switch enables independent software vendors (ISVs) to create extensible plug-ins, called Virtual Switch Extensions, that can provide enhanced networking and security capabilities. Virtual Switch Extensions that you add to the Hyper-V Virtual Switch are listed in the Virtual Switch Manager feature of Hyper-V Manager.

In the following illustration, a VM has a virtual NIC that is connected to the Hyper-V Virtual Switch through a switch port.



Hyper-V Virtual Switch capabilities provide you with more options for enforcing tenant isolation, shaping and controlling network traffic, and employing protective measures against malicious VMs.

ⓘ Note

In Windows Server 2016, a VM with a virtual NIC accurately displays the maximum throughput for the virtual NIC. To view the virtual NIC speed in **Network Connections**, right-click the desired virtual NIC icon and then click **Status**. The virtual NIC **Status** dialog box opens. In **Connection**, the value of **Speed** matches the speed of the physical NIC installed in the server.

Uses for Hyper-V Virtual Switch

Following are some use case scenarios for Hyper-V Virtual Switch.

Displaying statistics: A developer at a hosted cloud vendor implements a management package that displays the current state of the Hyper-V virtual switch. The management

package can query switch-wide current capabilities, configuration settings, and individual port network statistics using WMI. The status of the switch is then displayed to give administrators a quick view of the state of the switch.

Resource tracking: A hosting company is selling hosting services priced according to the level of membership. Various membership levels include different network performance levels. The administrator allocates resources to meet the SLAs in a manner that balances network availability. The administrator programmatically tracks information such as the current usage of bandwidth assigned, and the number of virtual machine (VM) assigned virtual machine queue (VMQ) or IOV channels. The same program also periodically logs the resources in use in addition to the per-VM resources assigned for double entry tracking or resources.

Managing the order of switch extensions: An enterprise has installed extensions on their Hyper-V host to both monitor traffic and report intrusion detection. During maintenance, some extensions may be updated causing the order of extensions to change. A simple script program is run to reorder the extensions after updates.

Forwarding extension manages VLAN ID: A major switch company is building a forwarding extension that applies all policies for networking. One element that is managed is virtual local area network (VLAN) IDs. The virtual switch cedes control of the VLAN to a forwarding extension. The switch company's installation programmatically call a Windows Management Instrumentation (WMI) application programming interface (API) that turns on the transparency, telling the Hyper-V Virtual Switch to pass and take no action on VLAN tags.

Hyper-V Virtual Switch Functionality

Some of the principal features that are included in the Hyper-V Virtual Switch are:

- **ARP/ND Poisoning (spoofing) protection:** Provides protection against a malicious VM using Address Resolution Protocol (ARP) spoofing to steal IP addresses from other VMs. Provides protection against attacks that can be launched for IPv6 using Neighbor Discovery (ND) spoofing.
- **DHCP Guard protection:** Protects against a malicious VM representing itself as a Dynamic Host Configuration Protocol (DHCP) server for man-in-the-middle attacks.
- **Port ACLs:** Provides traffic filtering based on Media Access Control (MAC) or Internet Protocol (IP) addresses/ranges, which enables you to set up virtual network isolation.
- **Trunk mode to a VM:** Enables administrators to set up a specific VM as a virtual appliance, and then direct traffic from various VLANs to that VM.

- **Network traffic monitoring:** Enables administrators to review traffic that is traversing the network switch.
- **Isolated (private) VLAN:** Enables administrators to segregate traffic on multiple vlans, to more easily establish isolated tenant communities.

Following is a list of capabilities that enhance Hyper-V Virtual Switch usability:

- **Bandwidth limit and burst support:** Bandwidth minimum guarantees amount of bandwidth reserved. Bandwidth maximum caps the amount of bandwidth a VM can consume.
- **Explicit Congestion Notification (ECN) marking support:** ECN marking, also known as Data CenterTCP (DCTCP), enables the physical switch and operating system to regulate traffic flow such that the buffer resources of the switch are not flooded, which results in increased traffic throughput.
- **Diagnostics:** Diagnostics allow easy tracing and monitoring of events and packets through the virtual switch.

Host network requirements for Azure Local

Applies to: Azure Local 2311.2 and later

This topic discusses host networking considerations and requirements for Azure Local. For information on datacenter architectures and the physical connections between machines, see [Physical network requirements](#).

For information on how to simplify host networking using Network ATC, see [Simplify host networking with Network ATC](#).

Network traffic types

Azure Local network traffic can be classified by its intended purpose:

- **Management traffic:** Traffic to or from outside the local system. For example, traffic used by the administrator for management of the system like Remote Desktop, Windows Admin Center, Active Directory, etc.
- **Compute traffic:** Traffic originating from or destined to a virtual machine (VM).
- **Storage traffic:** Traffic using Server Message Block (SMB), for example Storage Spaces Direct or SMB-based live migration. This traffic is layer-2 traffic and is not routable.

Select a network adapter

Network adapters are qualified by the **network traffic types** (see above) they are supported for use with. As you review the [Windows Server Catalog](#), the Windows Server 2022 certification now indicates one or more of the following roles. Before purchasing a machine for Azure Local, you must minimally have *at least* one adapter that is qualified for management, compute, and storage as all three traffic types are required on Azure Local. You can then use [Network ATC](#) to configure your adapters for the appropriate traffic types.

For more information about this role-based NIC qualification, see this [Windows Server blog post](#).

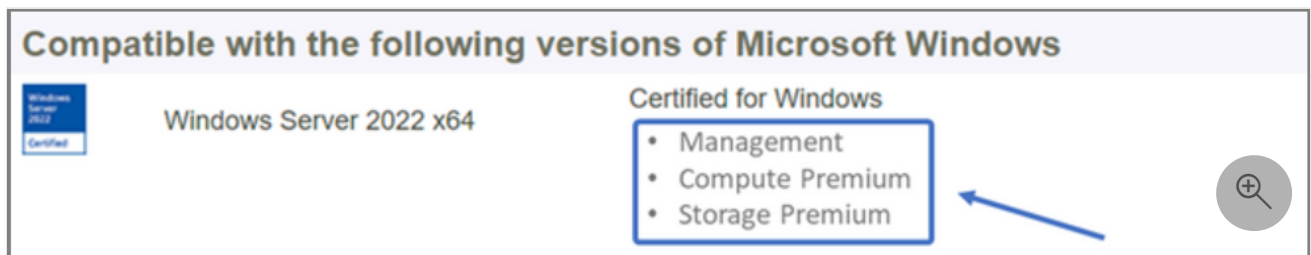
Important

Using an adapter outside of its qualified traffic type is not supported.

Level	Management Role	Compute Role	Storage Role
Role-based distinction	Management	Compute Standard	Storage Standard
Maximum Award	Not Applicable	Compute Premium	Storage Premium

ⓘ Note

The highest qualification for any adapter in our ecosystem will contain the **Management**, **Compute Premium**, and **Storage Premium** qualifications.



Driver Requirements

Inbox drivers are not supported for use with Azure Local. To identify if your adapter is using an inbox driver, run the following cmdlet. An adapter is using an inbox driver if the **DriverProvider** property is **Microsoft**.

Powershell

```
Get-NetAdapter -Name <AdapterName> | Select *Driver*
```

If inbox driver is being used, download and install the latest driver from your OEM partner's website.

Overview of key network adapter capabilities

Important network adapter capabilities used by Azure Local include:

- Dynamic Virtual Machine Multi-Queue (Dynamic VMMQ or d.VMMQ)
- Remote Direct Memory Access (RDMA)
- Guest RDMA
- Switch Embedded Teaming (SET)

Dynamic VMMQ

All network adapters with the Compute (Premium) qualification support Dynamic VMMQ. Dynamic VMMQ requires the use of Switch Embedded Teaming.

Applicable traffic types: compute

Certifications required: Compute (Premium)

Dynamic VMMQ is an intelligent, receive-side technology. It builds upon its predecessors of Virtual Machine Queue (VMQ), Virtual Receive Side Scaling (vRSS), and VMMQ, to provide three primary improvements:

- Optimizes host efficiency by using fewer CPU cores.
- Automatic tuning of network traffic processing to CPU cores, thus enabling VMs to meet and maintain expected throughput.
- Enables "bursty" workloads to receive the expected amount of traffic.

For more information on Dynamic VMMQ, see the blog post [Synthetic accelerations](#).

RDMA

RDMA is a network stack offload to the network adapter. It allows SMB storage traffic to bypass the operating system for processing.

RDMA enables high-throughput, low-latency networking, using minimal host CPU resources. These host CPU resources can then be used to run additional VMs or containers.

Applicable traffic types: host storage

Certifications required: Storage (Standard)

All adapters with Storage (Standard) or Storage (Premium) qualification support host-side RDMA. For more information on using RDMA with guest workloads, see the "Guest RDMA" section later in this article.

Azure Local supports RDMA with either the Internet Wide Area RDMA Protocol (iWARP) or RDMA over Converged Ethernet (RoCE) protocol implementations.

Important

RDMA adapters only work with other RDMA adapters that implement the same RDMA protocol (iWARP or RoCE).

Not all network adapters from vendors support RDMA. The following table lists those vendors (in alphabetical order) that offer certified RDMA adapters. However, there are hardware vendors not

included in this list that also support RDMA. See the [Windows Server Catalog](#) to find adapters with the Storage (Standard) or Storage (Premium) qualification which require RDMA support.

Note

InfiniBand (IB) is not supported with Azure Local.

 Expand table

NIC vendor	iWARP	RoCE
Broadcom	No	Yes
Intel	Yes	Yes (some models)
Marvell (Qlogic)	Yes	Yes
Nvidia	No	Yes

For more information on deploying RDMA for the host, we highly recommend you use Network ATC. For information on manual deployment see the [SDN GitHub repo](#).

iWARP

iWARP uses Transmission Control Protocol (TCP), and can be optionally enhanced with Priority-based Flow Control (PFC) and Enhanced Transmission Service (ETS).

Use iWARP if:

- You don't have experience managing RDMA networks.
- You don't manage or are uncomfortable managing your top-of-rack (ToR) switches.
- You won't be managing the solution after deployment.
- You already have deployments that use iWARP.
- You're unsure which option to choose.

RoCE

RoCE uses User Datagram Protocol (UDP), and requires PFC and ETS to provide reliability.

Use RoCE if:

- You already have deployments with RoCE in your datacenter.
- You're comfortable managing the DCB network requirements.

Guest RDMA

Guest RDMA is not supported on Azure Local.

Switch Embedded Teaming (SET)

SET is a software-based teaming technology that has been included in the Windows Server operating system since Windows Server 2016. SET is the only teaming technology supported by Azure Local and works well with compute, storage, and management traffic. SET supports up to eight adapters in a single team. Other NIC teaming methods, such as [Load Balancing/Failover \(LBFO\)](#) [↗](#), aren't supported.

In Azure Local, Network ATC automatically configures both the SET and the vSwitch. You shouldn't manually deploy SET using PowerShell, such as with the [New-VMSwitch](#) cmdlet. While this command enables Embedded Teaming by default when multiple adapters are listed, the recommended approach is to use Network ATC with intents.

Applicable traffic types: compute, storage, and management

Certifications required: Compute (Standard) or Compute (Premium)

SET is important for Azure Local because it's the only teaming technology that enables:

- Teaming of RDMA adapters (if needed).
- Guest RDMA.
- Dynamic VMMQ.
- Other key Azure Local features (see [Teaming in Azure Local](#) [↗](#)).

SET requires the use of symmetric (identical) adapters. Symmetric network adapters are those that have the same:

- make (vendor)
- model (version)
- speed (throughput)
- configuration

In 22H2, Network ATC will automatically detect and inform you if the adapters you've chosen are asymmetric. The easiest way to manually identify if adapters are symmetric is if the speeds and interface descriptions are **exact** matches. They can deviate only in the numeral listed in the description. Use the [Get-NetAdapterAdvancedProperty](#) cmdlet to ensure the configuration reported lists the same property values.

See the following table for an example of the interface descriptions deviating only by numeral (#):

Name	Interface description	Link speed
NIC1	Network Adapter #1	25 Gbps
NIC2	Network Adapter #2	25 Gbps
NIC3	Network Adapter #3	25 Gbps
NIC4	Network Adapter #4	25 Gbps

ⓘ Note

SET supports only switch-independent configuration by using either Dynamic or Hyper-V Port load-balancing algorithms. For best performance, Hyper-V Port is recommended for use on all NICs that operate at or above 10 Gbps. Network ATC makes all the required configurations for SET.

RDMA traffic considerations

If you implement DCB, you must ensure that the PFC and ETS configuration is implemented properly across every network port, including network switches. DCB is required for RoCE and optional for iWARP.

For detailed information on how to deploy RDMA, download the document from the [SDN GitHub repo](#).

RoCE-based Azure Local implementations require the configuration of three PFC traffic classes, including the default traffic class, across the fabric and all hosts.

System traffic class

This traffic class ensures that there's enough bandwidth reserved for system heartbeats:

- Required: Yes
- PFC-enabled: No
- Recommended traffic priority: Priority 7
- Recommended bandwidth reservation:
 - 10 GbE or lower RDMA networks = 2 percent
 - 25 GbE or higher RDMA networks = 1 percent

RDMA traffic class

This traffic class ensures that there's enough bandwidth reserved for lossless RDMA communications by using SMB Direct:

- Required: Yes
- PFC-enabled: Yes
- Recommended traffic priority: Priority 3 or 4
- Recommended bandwidth reservation: 50 percent

Default traffic class

This traffic class carries all other traffic not defined in the system or RDMA traffic classes, including VM traffic and management traffic:

- Required: By default (no configuration necessary on the host)
- Flow control (PFC)-enabled: No
- Recommended traffic class: By default (Priority 0)
- Recommended bandwidth reservation: By default (no host configuration required)

Storage traffic models

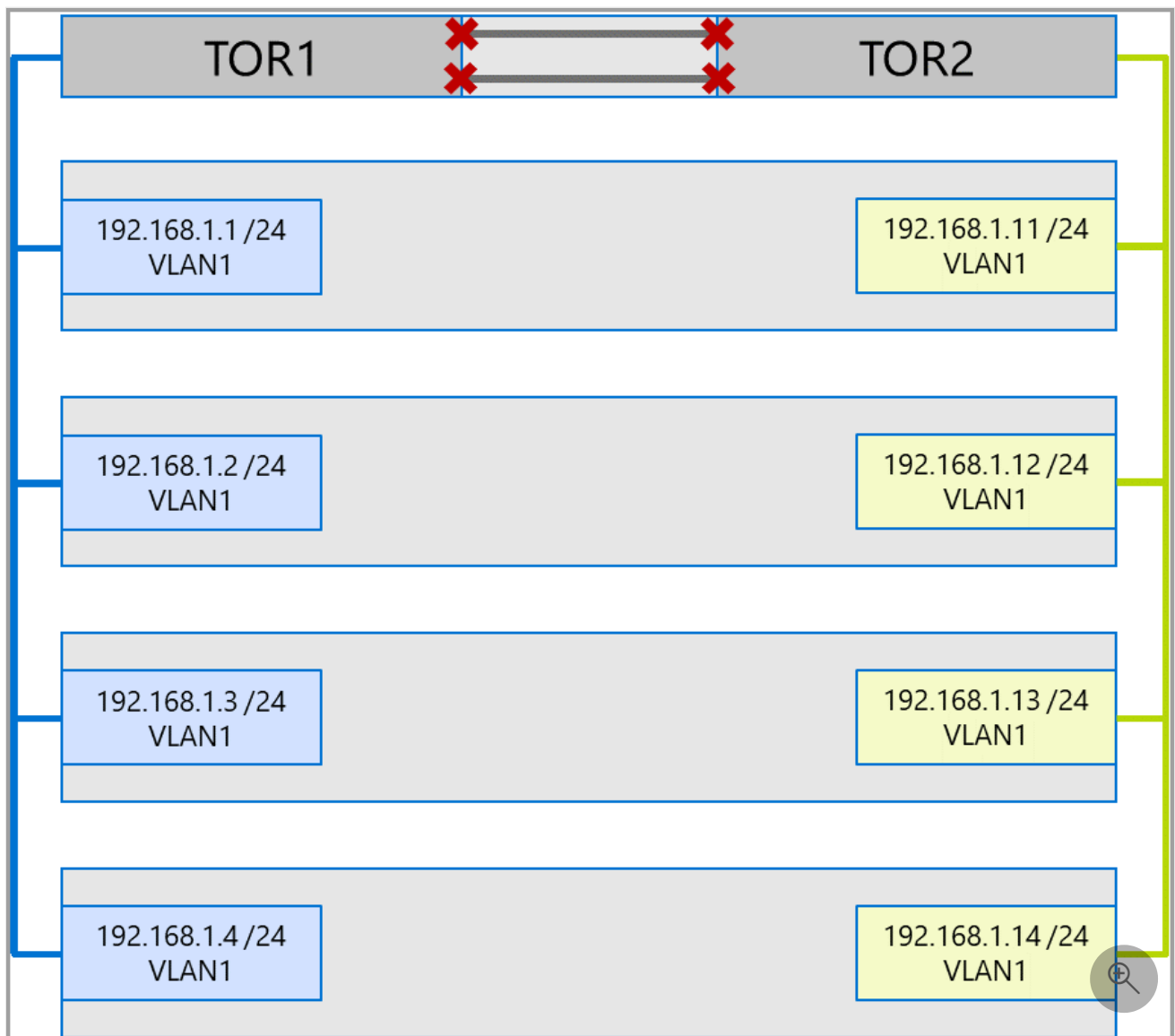
SMB provides many benefits as the storage protocol for Azure Local, including SMB Multichannel. SMB Multichannel isn't covered in this article, but it's important to understand that traffic is multiplexed across every possible link that SMB Multichannel can use.

ⓘ Note

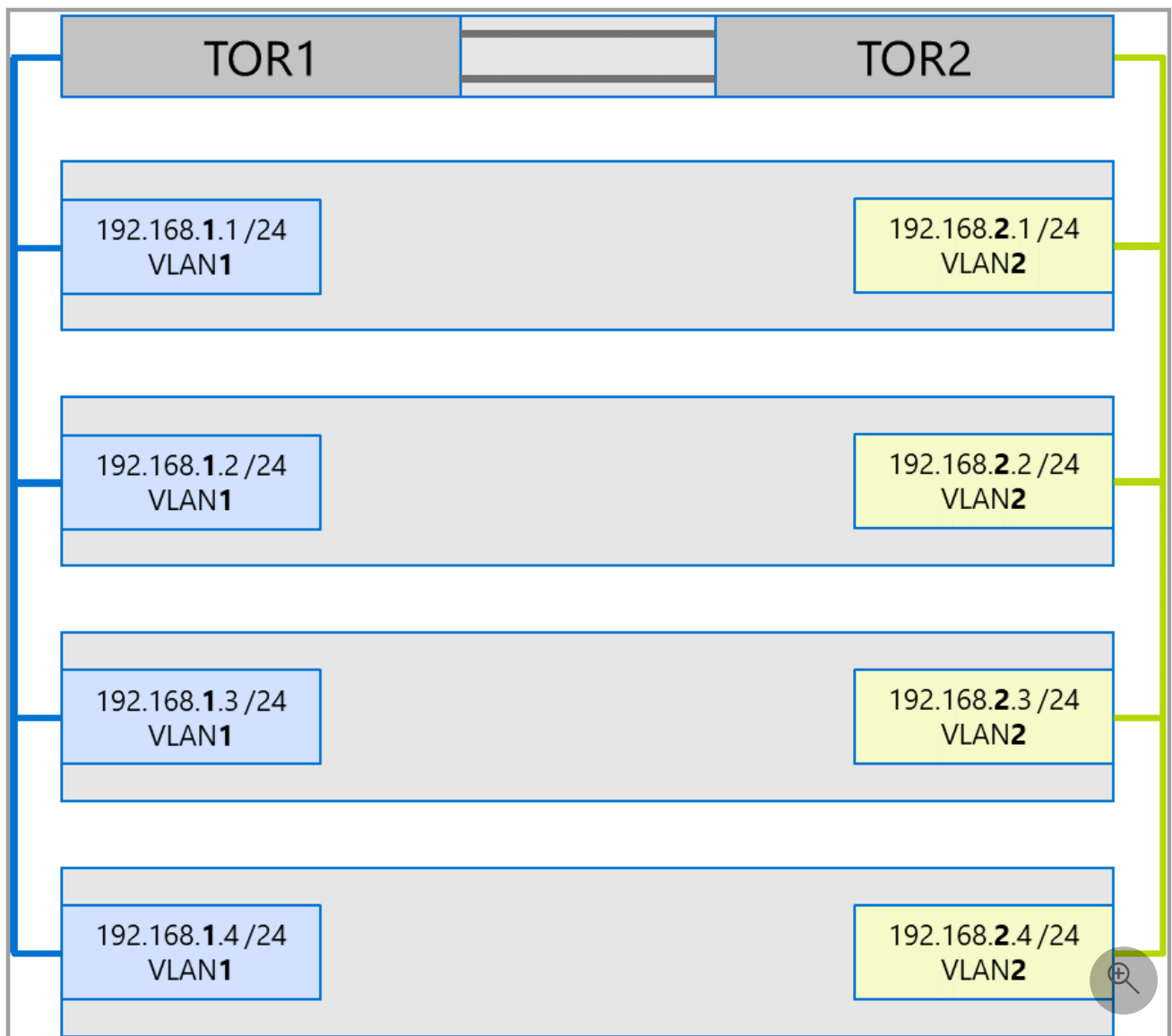
We recommend using multiple subnets and VLANs to separate storage traffic in Azure Local.

Consider the following example of a four node system. Each machine has two storage ports (left and right side). Because each adapter is on the same subnet and VLAN, SMB Multichannel will spread connections across all available links. Therefore, the left-side port on the first machine (192.168.1.1) will make a connection to the left-side port on the second machine (192.168.1.2). The right-side port on the first machine (192.168.1.12) will connect to the right-side port on the second machine. Similar connections are established for the third and fourth machines.

However, this creates unnecessary connections and causes congestion at the interlink (multi-chassis link aggregation group or MC-LAG) that connects the ToR switches (marked with Xs). See the following diagram:



The recommended approach is to use separate subnets and VLANs for each set of adapters. In the following diagram, the right-hand ports now use subnet 192.168.2.x /24 and VLAN2. This allows traffic on the left-side ports to remain on TOR1 and the traffic on the right-side ports to remain on TOR2.



Traffic bandwidth allocation

The following table shows example bandwidth allocations of various traffic types, using common adapter speeds, in Azure Local. Note that this is an example of a *converged solution*, where all traffic types (compute, storage, and management) run over the same physical adapters, and are teamed by using SET.

Because this use case poses the most constraints, it represents a good baseline. However, considering the permutations for the number of adapters and speeds, this should be considered an example and not a support requirement.

The following assumptions are made for this example:

- There are two adapters per team.
- Storage Bus Layer (SBL), Cluster Shared Volume (CSV), and Hyper-V (Live Migration) traffic:
 - Use the same physical adapters.
 - Use SMB.

- SMB is given a 50 percent bandwidth allocation by using DCB.
 - SBL/CSV is the highest priority traffic, and receives 70 percent of the SMB bandwidth reservation.
 - Live Migration (LM) is limited by using the `Set-SMBBandwidthLimit` cmdlet, and receives 29 percent of the remaining bandwidth.
 - If the available bandwidth for Live Migration is ≥ 5 Gbps, and the network adapters are capable, use RDMA. Use the following cmdlet to do so:

Powershell

```
Set-VMHost -VirtualMachineMigrationPerformanceOption SMB
```

- If the available bandwidth for Live Migration is < 5 Gbps, use compression to reduce blackout times. Use the following cmdlet to do so:

Powershell

```
Set-VMHost -VirtualMachineMigrationPerformanceOption Compression
```

- If you're using RDMA for Live Migration traffic, ensure that Live Migration traffic can't consume the entire bandwidth allocated to the RDMA traffic class by using an SMB bandwidth limit. Be careful, because this cmdlet takes entry in bytes per second (Bps), whereas network adapters are listed in bits per second (bps). Use the following cmdlet to set a bandwidth limit of 6 Gbps, for example:

Powershell

```
Set-SMBBandwidthLimit -Category LiveMigration -BytesPerSecond 750MB
```

ⓘ Note

750 MBps in this example equates to 6 Gbps.

Here is the example bandwidth allocation table:

[Expand table](#)

NIC speed	Teamed bandwidth	SMB bandwidth reservation**	SBL/CSV %	SBL/CSV bandwidth	Live Migration %	Max Live Migration bandwidth	Heartbeat %	Heartbeat bandwidth
10	20 Gbps	10 Gbps	70%	7 Gbps	*	200 Mbps		

NIC speed	Teamed bandwidth	SMB bandwidth reservation**	SBL/CSV %	SBL/CSV bandwidth	Live Migration %	Max Live Migration bandwidth	Heartbeat %	Heartbeat bandwidth
Gbps								
25 Gbps	50 Gbps	25 Gbps	70%	17.5 Gbps	29%	7.25 Gbps	1%	250 Mbps
40 Gbps	80 Gbps	40 Gbps	70%	28 Gbps	29%	11.6 Gbps	1%	400 Mbps
50 Gbps	100 Gbps	50 Gbps	70%	35 Gbps	29%	14.5 Gbps	1%	500 Mbps
100 Gbps	200 Gbps	100 Gbps	70%	70 Gbps	29%	29 Gbps	1%	1 Gbps
200 Gbps	400 Gbps	200 Gbps	70%	140 Gbps	29%	58 Gbps	1%	2 Gbps

* Use compression rather than RDMA, because the bandwidth allocation for Live Migration traffic is <5 Gbps.

** 50 percent is an example bandwidth reservation.

Rack aware clustering

Rack aware clustering improves fault tolerance and data distribution in an Azure Local instance. This architecture allows you to cluster nodes that are strategically placed across two physical racks in separate rooms or buildings, connected by high bandwidth and low latency networking. For an overview of this feature, see [Azure Local rack aware clustering overview](#).

For network requirements and architecture, see [Azure Local rack aware cluster reference architecture](#).

Next steps

- Learn about network switch and physical network requirements. See [Physical network requirements](#).
- Learn how to simplify host networking using Network ATC. See [Simplify host networking with Network ATC](#).
- Brush up on [failover clustering networking basics](#) [↗](#).
- See [Deploy using Azure portal](#).
- See [Deploy using Azure Resource Manager template](#).

Network QoS minimum bandwidth best practices for Hyper-V

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Use these best practices to configure Hyper-V network Quality of Service (QoS) Minimum Bandwidth on Windows Server. The guidance helps you choose the appropriate mode (weight or bits per second), assign effective weights, and avoid configuration pitfalls with NIC Teaming and Data Center Bridging (DCB).

The article covers: weight assignment strategy, when to prefer weight over absolute bandwidth, interaction with teaming algorithms, coexistence considerations for QoS and DCB, and mitigation techniques for uneven bandwidth distribution.

Guidelines for using Minimum Bandwidth

Following are guidelines for configuring Minimum Bandwidth.

1. **Keep the sum of the weights near or under 100.** The larger the value for weight, the higher rounding error or quantization error there is when QoS Packet Scheduler partitions the bandwidth among workloads. For example, if there are 20 VMs that you want to share the network bandwidth, rather than assigning a weight of 10 to each of them (sum would be 200), assign each VM a weight of 1.
2. **Assign a relatively large weight to critical workloads even if they don't require that percentage of bandwidth.** For example, management and cluster heartbeat traffic on a computer that is running Hyper-V rarely require more than 1–2% of bandwidth over a 10 GbE NIC—but they're critical. Instead of assigning them a weight of 1 or 2, assign each a weight of 5 or more.
3. **Gap the weight assignment to differentiate service levels.** Don't use consecutive numbers (1,2,3). For example, if you have VMs named Gold, Silver, and Bronze on the same Hyper-V virtual switch, assign weights 5 (Gold), 3 (Silver), 1 (Bronze) rather than 3,2,1.
4. **Account for unfiltered / catch-all traffic.** For example, if Storage, Live Migration, and Cluster are filtered out with workload-specific filters, group the rest of the network traffic with a wildcard filter as a single flow that has a weight assignment.

Minimum Bandwidth modes

When you create a Hyper-V virtual switch with PowerShell, choose one of two Minimum Bandwidth configuration modes. These modes determine how you quantify minimum bandwidth for a workload:

- Weight (integer 1–100)
- Bits per second (BPS)

Tip

Configure Minimum Bandwidth by weight rather than BPS. Weight is more flexible and compatible with features such as Live Migration and NIC Teaming.

Example scenario: A VM configured with a Minimum Bandwidth of 500 Mbps might fail Live Migration to a destination host that can't guarantee 500 Mbps spare bandwidth. If you specify Minimum Bandwidth by weight (for example, 3), the sum of weights simply increases by 3 and migration succeeds (assuming no other constraints).

If you must specify minimum bandwidth in an absolute number (BPS), use these guidelines:

- The minimum unit is 1% of link capacity. On a 10 GbE NIC, the minimum allocatable bandwidth per virtual network adapter is 100 Mbps.
- Values are rounded to the nearest percent point. For example, 234 Mbps rounds down to 200 Mbps (2% of a 10 GbE link).

Use the `-MinimumBandwidthMode` parameter of `New-VMSwitch` to specify the mode:

PowerShell

```
New-VMSwitch "QoS Switch by Weight" -NetAdapterName "External NIC" -  
MinimumBandwidthMode Weight
```

PowerShell

```
New-VMSwitch "QoS Switch by BPS" -NetAdapterName "External NIC" -  
MinimumBandwidthMode Absolute
```

For more information, see the PowerShell reference topic `New-VMSwitch`.

Minimum Bandwidth and Data Center Bridging (DCB)

QoS Minimum Bandwidth (enforced by QoS Packet Scheduler in Windows) and Data Center Bridging (DCB, enforced by a capable NIC) both ensure a workload receives its fair share of bandwidth under congestion. They're **not** designed to work together for the same networking stack or NIC.

Important

Don't enable both Minimum Bandwidth and DCB for workloads that share the same networking stack or NIC.

You can enable both Minimum Bandwidth and DCB on one server only when they apply to separate networking stacks or separate NICs.

Configuring QoS and NIC Teaming

NIC Teaming provides:

- Bandwidth aggregation (distribution across team members)
- Traffic failover (maintains connectivity on component failure)

NIC Teaming distribution algorithms:

- Hyper-V virtual switch port (hash based on switch port of a VM or management OS virtual NIC)
- Address hashing (hash based on MAC, IP, and transport port fields)

QoS interaction with NIC Teaming:

- **Classification and Tagging** – Works seamlessly; traffic is tagged before hashing.
- **Priority-based Flow Control (PFC)** – Works seamlessly; ensure consistent enablement across all team members.
- **Bandwidth Management** – Details in the following section.

Maximum Bandwidth

Works seamlessly: QoS Packet Scheduler throttles flows before NIC Teaming hashes them to team members.

Minimum Bandwidth (weight mode)

QoS Packet Scheduler partitions bandwidth among flows by assigned weights before hashing. A problem can occur when the NIC Teaming algorithm is Hyper-V switch port: each VM (single

vNIC) becomes a single flow. Teaming could hash high-weight VMs to one member and low-weight VMs to another, resulting in effective equal share rather than proportional share.

This issue is less pronounced when:

- Many VMs share the virtual switch
- The NIC Teaming algorithm is Address hashing (more granular flows – e.g., each TCP connection)

Mitigation: script VM startup order so groups of same-weight VMs are brought up together. NIC Teaming hashes virtual NICs round-robin; grouping preserves more even weight distribution.

Hardware enforced Minimum Bandwidth (DCB)

The same uneven distribution issue can arise with multiple DCB-capable NICs if hashing is skewed.

Absolute (BPS) Minimum Bandwidth with NIC Teaming

If a Hyper-V switch is bound to a two-member 1 Gb NIC team (aggregate 2 Gb), you can't safely allocate more than 1 Gb to a VM—upon single NIC failure only 1 Gb remains. Weight mode avoids this complexity, as remaining bandwidth is repartitioned proportionally.

 **Tip**

Prefer weight mode for most deployments; it gracefully handles migration, teaming, and failure scenarios.

Summary: QoS and NIC Teaming

 Expand table

QoS feature	Functionality with NIC Teaming
Maximum Bandwidth	Works seamlessly
Minimum Bandwidth (weight)	Potential uneven enforcement if switch-port algorithm skews distribution
Hardware enforced Minimum Bandwidth (DCB)	Potential uneven enforcement if traffic distribution is skewed

QoS feature	Functionality with NIC Teaming
Classification and Tagging	Works seamlessly
Priority-based Flow Control	Works seamlessly

See also

- [Hyper-V QoS overview \(newer versions\)](#) *(add link when available)*
 - `New-VMSwitch` PowerShell reference
 - [NIC Teaming overview](#)
-

Last updated on 10/28/2025

Plan for Hyper-V security in Windows Server

Article • 11/01/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 

to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2 and 22H2

Secure the Hyper-V host operating system, the virtual machines, configuration files, and virtual machine data. Use the following list of recommended practices as a checklist to help you secure your Hyper-V environment.

Secure the Hyper-V host

- **Keep the host OS secure.**
 - Minimize the attack surface by using the minimum Windows Server installation option that you need for the management operating system. For more information, see the [Installation Options section](#) of the Windows Server technical content library. We don't recommend that you run production workloads on Hyper-V on Windows 10.
 - Keep the Hyper-V host operating system, firmware, and device drivers up to date with the latest security updates. Check your vendor's recommendations to update firmware and drivers.
 - Don't use the Hyper-V host as a workstation or install any unnecessary software.
 - Remotely manage the Hyper-V host. If you must manage the Hyper-V host locally, use Credential Guard. For more information, see [Protect derived domain credentials with Credential Guard](#).
 - Enable code integrity policies. Use virtualization-based security protected Code Integrity services. For more information, see [Device Guard Deployment Guide](#).
- **Use a secure network.**
 - Use a separate network with a dedicated network adapter for the physical Hyper-V computer.
 - Use a private or secure network to access VM configurations and virtual hard disk files.
 - Use a private/dedicated network for your live migration traffic. Consider enabling IPsec on this network to use encryption and secure your VM's data going over the network during migration. For more information, see [Set up hosts for live migration without Failover Clustering](#).

- **Secure storage migration traffic.**

Use SMB 3.0 for end-to-end encryption of SMB data and data protection tampering or eavesdropping on untrusted networks. Use a private network to access the SMB share contents to prevent man-in-the-middle attacks. For more information, see [SMB Security Enhancements](#).

- **Configure hosts to be part of a guarded fabric.**

For more information, see [Guarded fabric](#).

- **Secure devices.**

Secure the storage devices where you keep virtual machine resource files.

- **Secure the hard drive.**

Use BitLocker Drive Encryption to protect resources.

- **Harden the Hyper-V host operating system.**

Use the baseline security setting recommendations described in the [Windows Server Security Baseline](#).

- **Grant appropriate permissions.**

- Add users that need to manage the Hyper-V host to the Hyper-V administrators group.
- Don't grant virtual machine administrators permissions on the Hyper-V host operating system.

- **Configure anti-virus exclusions and options for Hyper-V.**

Windows Defender already has [automatic exclusions](#) configured. For more information about exclusions, see [Recommended antivirus exclusions for Hyper-V hosts](#) [↗](#).

- **Don't mount unknown VHDs.** This can expose the host to file system level attacks.

- **Don't enable nesting in your production environment unless it's required.**

If you enable nesting, don't run unsupported hypervisors on a virtual machine.

For more secure environments:

- **Use hardware with a Trusted Platform Module (TPM) 2.0 chip to set up a guarded fabric.**

For more information, see [System requirements for Hyper-V on Windows Server 2016](#).

Secure virtual machines

- **Create generation 2 virtual machines for supported guest operating systems.**

For more information, see [Generation 2 security settings](#).

- **Enable Secure Boot.**

For more information, see [Generation 2 security settings](#).

- **Keep the guest OS secure.**
 - Install the latest security updates before you turn on a virtual machine in a production environment.
 - Install integration services for the supported guest operating systems that need it and keep it up to date. Integration service updates for guests that run supported Windows versions are available through Windows Update.
 - Harden the operating system that runs in each virtual machine based on the role it performs. Use the baseline security setting recommendations that are described in the [Windows Security Baseline](#).

- **Use a secure network.**

Make sure virtual network adapters connect to the correct virtual switch and have the appropriate security setting and limits applied.

- **Store virtual hard disks and snapshot files in a secure location.**

- **Secure devices.**

Configure only required devices for a virtual machine. Don't enable discrete device assignment in your production environment unless you need it for a specific scenario. If you do enable it, make sure to only expose devices from trusted vendors.

- **Configure antivirus, firewall, and intrusion detection software** within virtual machines as appropriate based on the virtual machine role.
- **Enable virtualization based security for guests that run Windows 10 or Windows Server 2016 or later.**

For more information, see the [Device Guard Deployment Guide](#).

- Only enable Discrete Device Assignment if needed for a specific workload.

Due to the nature of passing through a physical device, work with the device manufacturer to understand if it should be used in a secure environment.

For more secure environments:

- Deploy virtual machines with shielding enabled and deploy them to a guarded fabric.

For more information, see [Generation 2 security settings](#) and [Guarded fabric](#).

Feedback

Was this page helpful?



Plan for GPU acceleration in Windows Server

Article • 11/01/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local](#), versions 23H2 and 22H2

This article introduces the graphics virtualization capabilities available in Windows Server.

When to use GPU acceleration

Depending on your workload, you might want to consider GPU acceleration. Here's what you should consider before choosing GPU acceleration:

- **App and desktop remoting (VDI/DaaS) workloads:** If you're building an app or desktop remoting service with Windows Server, consider the catalog of apps you expect your users to run. Some types of apps, such as CAD/CAM apps, simulation apps, games, and rendering/visualization apps, rely heavily on 3D rendering to deliver smooth and responsive interactivity. Most customers consider GPUs a necessity for a reasonable user experience with these kinds of apps.
- **Remote rendering, encoding, and visualization workloads:** These graphics-oriented workloads tend to rely heavily on a GPU's specialized capabilities, such as efficient 3D rendering and frame encoding/decoding, in order to achieve cost-effectiveness and throughput goals. For this kind of workload, a single GPU-enabled virtual machine (VM) might be able to match the throughput of many CPU-only VMs.
- **HPC and ML workloads:** For highly data-parallel computational workloads, such as high-performance compute and machine learning model training or inference, GPUs can dramatically shorten time to result, time to inference, and training time. Alternatively, they might offer better cost-effectiveness than a CPU-only architecture at a comparable performance level. Many High Performance Compute (HPC) and machine learning frameworks can use GPU acceleration; consider whether GPU acceleration might benefit your specific workload.

GPU virtualization in Windows Server

GPU virtualization technologies enable GPU acceleration in a virtualized environment, typically within virtual machines. If your workload is virtualized with Hyper-V, then you need to employ graphics virtualization in order to provide GPU acceleration from the

physical GPU to your virtualized apps or services. However, if your workload runs directly on physical Windows Server hosts, then you have no need for graphics virtualization; your apps and services already have access to the GPU capabilities and APIs natively supported in Windows Server.

The following graphics virtualization technologies are available to Hyper-V VMs in Windows Server:

- [Discrete Device Assignment \(DDA\)](#)
- [GPU Partitioning \(GPU-P\)](#)

In addition to VM workloads, Windows Server also supports GPU acceleration of containerized workloads within Windows Containers. For more information, see [GPU Acceleration in Windows containers](#).

Discrete Device Assignment (DDA)

Discrete Device Assignment (DDA) allows you to dedicate one or more physical GPUs to a virtual machine. In a DDA deployment, virtualized workloads run on the native driver and typically have full access to the GPU's functionality. DDA offers the highest level of app compatibility and potential performance. DDA can also provide GPU acceleration to Linux VMs, subject to support.

A DDA deployment can accelerate only a limited number of virtual machines, since each physical GPU can provide acceleration to at most one VM. If you're developing a service whose architecture supports shared virtual machines, consider hosting multiple accelerated workloads per VM. For example, if you're building a Remote Desktop Services solution, you can improve user scale by using the multi-session capabilities of Windows Server to host multiple user desktops on each VM. These users share the benefits of GPU acceleration.

For more information, see these articles:

- [Plan for deploying Discrete Device Assignment](#)
- [Deploy graphics devices using Discrete Device Assignment](#)

GPU Partitioning (GPU-P)

Beginning with Windows Server 2025, GPU partitioning allows you to share a physical GPU device with multiple virtual machines (VMs). With GPU partitioning or GPU virtualization, each VM gets a dedicated fraction of the GPU instead of the entire GPU.

The GPU partitioning uses the [Single Root IO Virtualization \(SR-IOV\) interface](#), which provides a hardware-backed security boundary with predictable performance for each VM. Each VM can access only the GPU resources dedicated to them and the secure hardware partitioning prevents unauthorized access by other VMs.

To learn more about GPU partitioning, see these articles:

- [GPU partitioning](#)
- [Partition and assign GPUs to a virtual machine](#)

Comparing DDA and GPU partitioning

Consider the following functionality and support differences between graphics virtualization technologies when planning your deployment:

 [Expand table](#)

Description	Discrete Device Assignment	GPU Partitioning
GPU resource model	Dedicated only	Partitioned
VM density	Low (one or more GPUs to one VM)	High (one or more GPUs to many VMs)
App compatibility	All GPU capabilities provided by vendor (DX 12, OpenGL, CUDA)	All GPU capabilities provided by vendor (DX 12, OpenGL, CUDA)
AVC444	Available through Group Policy	Available through Group Policy
GPU VRAM	Up to VRAM supported by the GPU	Up to VRAM supported by the GPU per partition
GPU driver in guest	GPU vendor driver (NVIDIA, AMD, Intel)	GPU vendor driver (NVIDIA, AMD, Intel)

Feedback

Was this page helpful? [Yes](#) [No](#)

What is Nested Virtualization?

07/25/2025

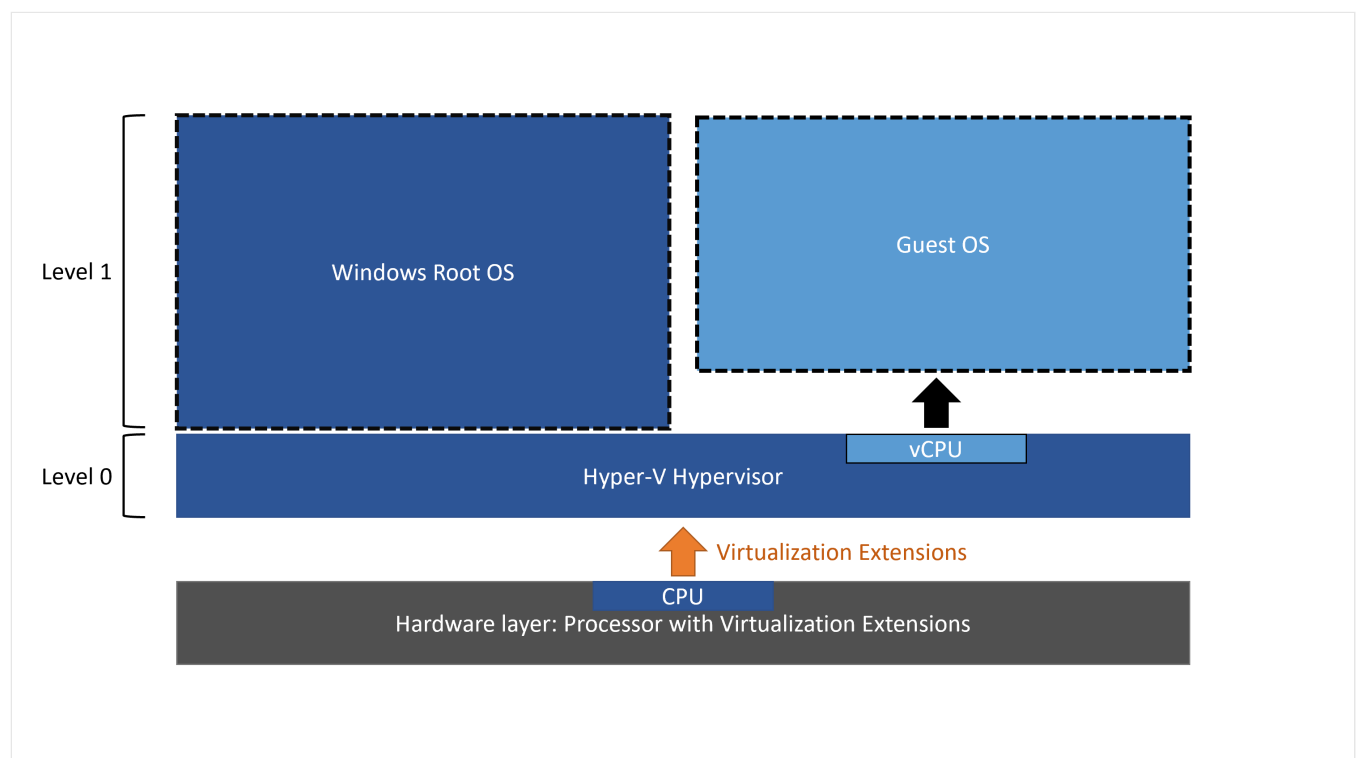
Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Nested virtualization is a feature that lets you run Hyper-V inside a Hyper-V virtual machine (VM). This article explains how nested virtualization works, its benefits, and when to use it in your environment. For example, nested virtualization can be useful for:

- Running applications or emulators in a nested VM
- Testing software releases on VMs
- Reducing deployment times for training environments
- Using Hyper-V isolation for containers

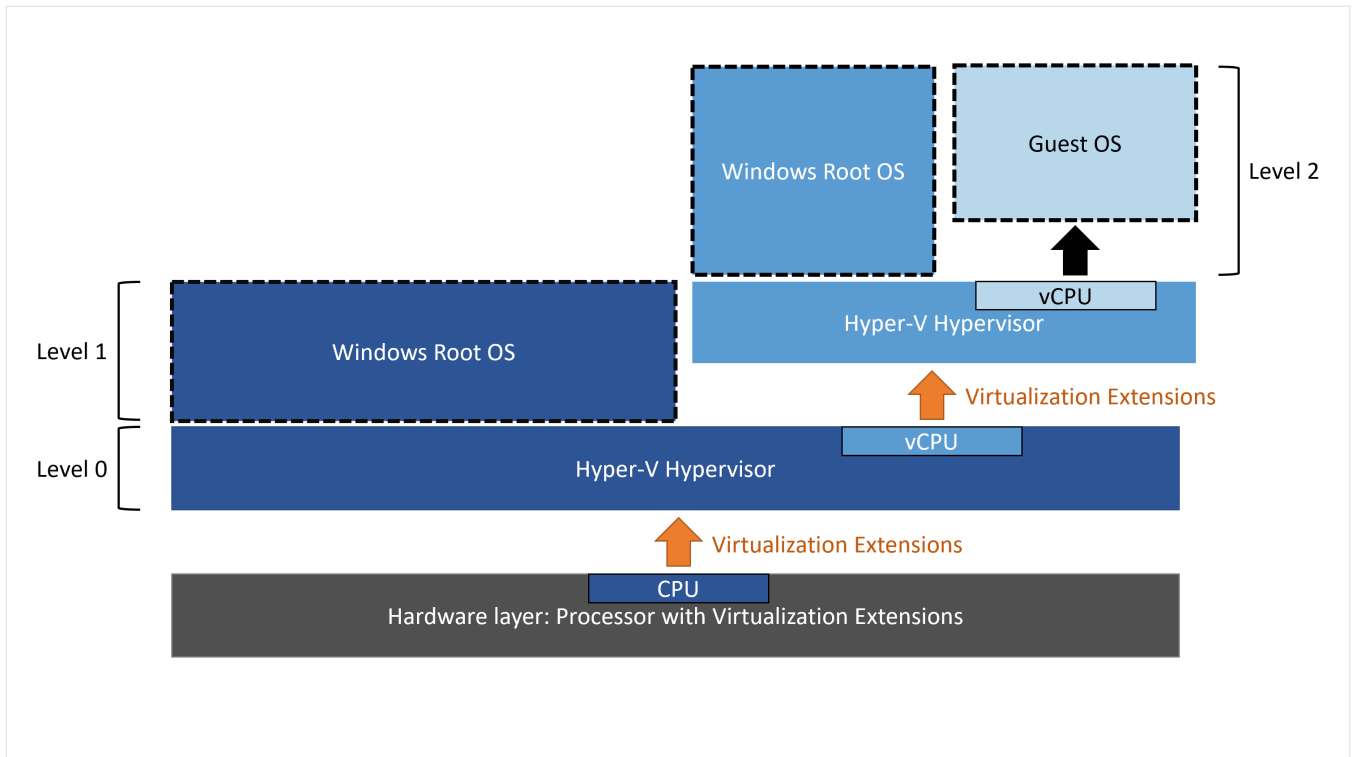
Modern processors include hardware features that make virtualization faster and more secure. Hyper-V relies on these processor extensions to run virtual machines, for example, Intel VT-x and AMD-V. Nested virtualization makes this hardware support available to guest virtual machines.

The following diagram shows Hyper-V without nesting. The Hyper-V hypervisor takes full control of the hardware virtualization capabilities (orange arrow), and doesn't expose them to the guest operating system.



In contrast, the following diagram shows Hyper-V with Nested Virtualization enabled. In this case, Hyper-V exposes the hardware virtualization extensions to its virtual machines. With

nesting enabled, a guest virtual machine can install its own hypervisor and run its own guest VMs.



Dynamic Memory and Runtime Memory Resize

When Hyper-V is running inside a virtual machine, the virtual machine must be turned off to adjust its memory. Meaning that even if dynamic memory is enabled, the amount of memory doesn't fluctuate. Simply enabling nested virtualization has no effect on dynamic memory or runtime memory resize.

For virtual machines without dynamic memory enabled, attempting to adjust the amount of memory while running fails. The incompatibility only occurs while Hyper-V is running in the VM.

Non-Microsoft virtualization apps

Virtualization applications other than Hyper-V aren't supported in Hyper-V virtual machines, and are likely to fail. Virtualization applications include any software that requires hardware virtualization extensions.

Supported scenarios

Using a nested Hyper-V VM in production is supported for both Azure and on-premises in the following scenarios. We also recommend you make sure that your services and applications are

also supported.

Nested Virtualization isn't suitable for Windows Server Failover Clustering, and performance sensitive applications. We recommended you fully evaluate the services and applications.

Hyper-V VMs on Hyper-V VMs

Running Hyper-V VMs nested on Hyper-V VMs is great for test labs and evaluation environments. Especially where configurations can be easily modified and saved states can be used to revert to specific configurations. Test labs don't typically require the same service level agreement (SLA) as production environments.

Production environments running Hyper-V VMs running on Hyper-V VMs are supported. Ensure that your services and applications are also supported. If you use a nested Hyper-V VM in production, make sure you fully evaluate whether your services or applications provide the expected behavior.

To learn more about setting up Nested Virtualization on Azure, see our Tech Community blog [How to Setup Nested Virtualization for Azure VM/VHD](#).

Non-Microsoft virtualization on Hyper-V virtualization

While it might be possible for non-Microsoft virtualization to run on Hyper-V, Microsoft doesn't test this scenario. Non-Microsoft virtualization on Hyper-V virtualization isn't supported. Ensure your hypervisor vendor supports this scenario.

Hyper-V virtualization on non-Microsoft virtualization

While it might be possible for Hyper-V virtualization to run on non-Microsoft virtualization, Microsoft doesn't test this scenario. Hyper-V virtualization on non-Microsoft virtualization isn't supported. Ensure your hypervisor vendor supports this scenario.

Azure Local nested on Hyper-V VMs

Azure Local is designed and tested to run on validated physical hardware. Azure Local can run nested in a virtual machine for evaluation, but production environments in a nested configuration aren't supported.

To learn more about Azure Local nested on Hyper-V VMs, see [Nested virtualization in Azure Local](#).

Hyper-V isolated containers running nested on Hyper-V

Microsoft offers Hyper-V isolation for containers. This isolation mode offers enhanced security and broader compatibility between host and container versions. With Hyper-V isolation, multiple container instances run concurrently on a host. Each container runs inside of a highly optimized virtual machine and effectively gets its own kernel. Since a Hyper-V isolated container offers isolation through a hypervisor layer between itself and the container host, when the container host is a Hyper-V based virtual machine, there's performance overhead. The associated performance overhead occurs in terms of container start-up time, storage, network, and CPU operations.

When a Hyper-V isolated container is run in a Hyper-V VM, it's running nested. Using a Hyper-V VM opens many useful scenarios but also increases latency, as there are two levels of hypervisors running above the physical host.

Running Hyper-V isolated containers nested on Hyper-V is supported. One level of nested virtualization is supported in production, which allows for isolated container deployments.

To learn more about Nested Hyper-V Containers, see [Performance tuning Windows Server Containers](#).

Run WSL2 in a Hyper-V VM running nested on Hyper-V

Windows Subsystem for Linux (WSL) is a feature of the Windows operating system that enables you to run a Linux file system, along with Linux command-line tools and GUI apps, directly on Windows.

Running WSL2 in a Hyper-V VM running nested on Hyper-V is supported.

To learn more about how to enable WSL 2 to run in a VM, see [Frequently Asked Questions about Windows Subsystem for Linux](#).

Related content

- [Run Hyper-V in a Virtual Machine with Nested Virtualization](#)

Hyper-V Virtual Fibre Channel

Article • 04/26/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local, versions 23H2 and 22H2](#)

Hyper-V provides Fibre Channel ports within guest operating systems (OSes) that let you connect to Fibre Channel directly from your virtual machines (VMs). This feature lets you virtualize workloads that use direct access to Fibre Channel storage, cluster guests over Fibre Channel, and gives you more storage options for servers hosted in your virtualization infrastructure. This article gives a brief overview of Fibre Channel on Hyper-V so you can understand how to implement the Virtual Fibre Channel into your Hyper-V planning.

Prerequisites

In order to use Virtual Fibre Channel on Hyper-V, your deployment must have the following:

- One or more installations of Windows Server 2012 or later with the Hyper-V role installed.
- A device with one or more Fibre Channel host bus adapters (HBAs) and an updated HBA driver that supports Virtual Fibre Channel.
 - HBA ports must also have a Fibre Channel topography that can support N_Port ID Virtualization (NPIV), a maximum transfer size of at least 0.5 MB, and data transfers of at least 128 physical pages.
 - The maximum transfer length of the adapter determines the LUN limit. For example, a maximum transfer length of 512k allows about 2,250 LUNs. You need to configure your LUNs and multi-storage paths to maintain these limits.
- An NPIV-enabled Storage Area Network (SAN).
- VMs that use Windows Server 2012 or later as their guest OS and can support virtual Fibre Channel adapters.
- Storage accessible through Virtual Fibre Channel support devices that present logical units. Virtual Fibre Channel logical units can't be used as boot media.

How Virtual Fibre Channel works

Virtual Fibre Channel for Hyper-V provides the guest OS with unmediated access to a SAN by using a standard World Wide Name (WWN) associated with a virtual machine. Hyper-V users can use Fibre Channel SANs to virtualize workloads that require direct access to SAN logical unit numbers (LUNs). Fibre Channel SANs also let you operate in new scenarios, such as running failover clustering inside the guest OS of a VM connected to shared Fibre Channel storage.

Physical deployments use the advanced storage functionality of mid-range and high-end storage arrays to offload certain management tasks from hosts to the Windows software virtual hard disk stack. In virtualized deployments, Virtual Fibre Channel plays a similar role, letting you use your SAN's functionality directly from Hyper-V VMs to offload functionality. For example, when you use Hyper-V to take a snapshot of a LUN, you can offload storage functionality on the SAN hardware by using a hardware Volume Shadow Copy Service (VSS) provider from within a Hyper-V VM.

The following sections explain which Virtual Fibre Channel features help it offload management and storage tasks.

NPIV support

Virtual Fibre Channel for Hyper-V guests uses the existing N_Port ID Virtualization (NPIV) T11 standard to map multiple virtual N_Port IDs to a single physical Fibre Channel N_port. A new NPIV port is created on the host every time you start a VM configured with a virtual HBA. When the VM stops running on the host, the system removes the NPIV port. You should configure any HBA ports that you use for Virtual Fibre Channel in a Fibre Channel topology that supports NPIV, and your SAN should also support NPIV ports.

Virtual SAN support

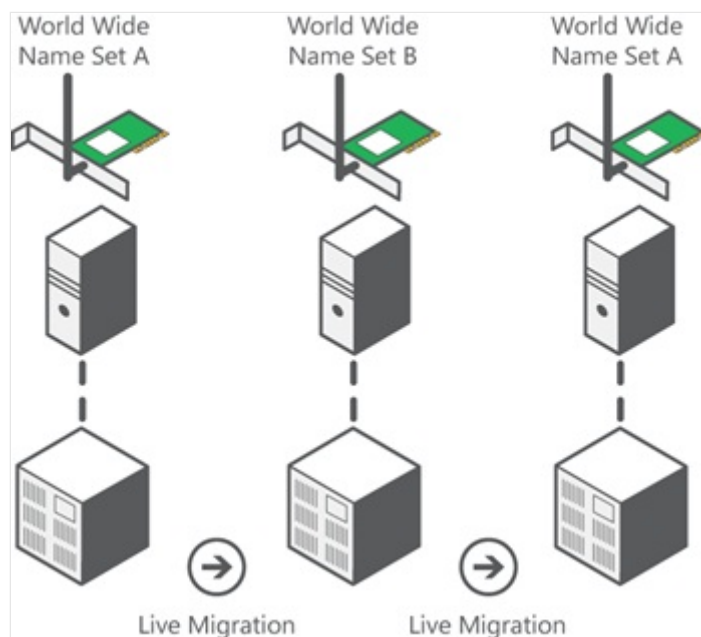
A virtual SAN defines a named group of physical Fibre Channel ports connected to the same physical SAN. Hyper-V lets you define virtual SANs on the host in scenarios where you connect a single Hyper-V host to different SANs through many Fibre Channel ports.

For example, if a Hyper-V host is connected to a production SAN and a test SAN, then the host is connected to each SAN through two physical Fibre Channel ports. In this scenario, you can configure two virtual SANs. The virtual production SAN has the two physical Fibre Channel ports connected to the physical production SAN. The virtual test SAN has two physical Fibre Channel ports connected to the test SAN. You can use the same technique to name two different paths to the same storage target.

You can configure up to four virtual Fibre Channel adapters on a VM and associate each one with a virtual SAN. Each virtual Fibre Channel adapter connects with one WWN address, and you can set each WWN address automatically or manually. However, in live migration scenarios, two WWN addresses are assigned to each adapter.

Live migration

To support live migration of virtual machines across Hyper-V hosts while maintaining Fibre Channel connectivity, two WWNs are configured for each virtual Fibre Channel adapter, as shown in the following diagram. Hyper-V automatically alternates between the Set A and Set B WWN addresses during a live migration. This ensures that all LUNs are available on the destination host before the migration and that no downtime occurs during the migration.



Tape library support

Windows Server only supports virtual tape libraries configured with virtual Fibre Channel adapters when using System Center Data Protection Manager 2012 R2 U3 or later with certified hardware. You can check if your virtual Fibre Channel adapter supports a tape library by contacting the tape library hardware vendor. You can also check compatibility by running the Data Protection Management (DPM) Tape Library Compatibility Test tool on the virtual tape library in a guest VM. For more information about the DPM Tape Library Compatibility Test, see [Verify tape library compatibility](#).

MPIO functionality

Hyper-V in Windows Server can use the multipath I/O (MPIO) functionality to ensure continuous connectivity to Fibre Channel storage from within a VM. You can use MPIO functionality with Fibre Channel in the following ways:

- Use MPIO for host access. To use this functionality, you install multiple Fibre Channel ports on the host, then use MPIO to provide highly available connectivity to the LUNs the host can access.
- Virtualize workloads that use MPIO. To use this functionality, configure multiple virtual Fibre Channel adapters inside a VM, then use a separate copy of MPIO within the VM operating system to connect to the LUNs that the VM can access. This configuration can coexist with a host MPIO setup.
- Use different device-specific modules (DSMs) for the host or each VM. This approach allows live migration of the virtual machine configuration, including DSM, connectivity between hosts, and compatibility with existing server configurations and DSMs.

Note

Live migration scenarios where the guest VM uses MPIO don't support Asymmetric Logical Unit Assignment (ALUA).

Related content

- [Plan for Hyper-V on Windows Server](#)
- [Plan for Hyper-V scalability on Windows Server](#)

Feedback

Was this page helpful?

 Yes

 No

Guarded fabric and shielded VMs

Article • 11/01/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#)

One of the most important goals of providing a hosted environment is to guarantee the security of the virtual machines running in the environment. As a cloud service provider or enterprise private cloud administrator, you can use a guarded fabric to provide a more secure environment for VMs. A guarded fabric consists of one Host Guardian Service (HGS) - typically, a cluster of three nodes - plus one or more guarded hosts, and a set of shielded virtual machines (VMs).

Important

Ensure that you have installed the latest cumulative update before you deploy shielded virtual machines in production.

Videos, blog, and overview topic about guarded fabrics and shielded VMs

- Video: [How to protect your virtualization fabric from insider threats with Windows Server 2019](#)
- Video: [Introduction to Shielded Virtual Machines in Windows Server 2016](#)
- Video: [Dive into Shielded VMs with Windows Server 2016 Hyper-V](#)
- Video: [Deploying Shielded VMs and a Guarded Fabric with Windows Server 2016](#) 
- Blog: [Datacenter and Private Cloud Security Blog](#)
- Overview: [Guarded fabric and shielded VMs overview](#)

Planning topics

- [Planning guide for hosters](#)
- [Planning guide for tenants](#)

Deployment topics

- [Deployment Guide](#)
 - [Quick start](#)

- [Deploy HGS](#)
- [Deploy guarded hosts](#)
 - [Configuring the fabric DNS for hosts that will become guarded hosts](#)
 - [Deploy a guarded host using AD mode](#)
 - [Deploy a guarded host using TPM mode](#)
 - [Confirm guarded hosts can attest](#)
 - [Shielded VMs - Hosting service provider deploys guarded hosts in VMM](#)
- [Deploy shielded VMs](#)
 - [Create a shielded VM template](#)
 - [Prepare a VM Shielding helper VHD](#)
 - [Set up Windows Azure Pack](#)
 - [Create a shielding data file](#)
 - [Deploy a shielded VM by using Windows Azure Pack](#)
 - [Deploy a shielded VM by using Virtual Machine Manager](#)

Operations and management topic

- [Managing the Host Guardian Service](#)
-

Feedback

Was this page helpful?

 Yes

 No

Hyper-V Backup Approaches

08/14/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V allows you to backup virtual machines (VMs), from the host operating system, without the need to run custom backup software inside the virtual machine. There are several approaches that are available for developers to utilize depending on their needs.

You can do a VM backup or a data-only backup:

- **VM backup:** This approach backs up the entire virtual machine, including the configuration and all data. It's useful when you want to restore the entire VM to a previous state. Each virtual machine backup (full or incremental) must copy over all associated virtual machine configuration and virtual machine state files including virtual machine configuration (`.VMCX`), virtual machine guest state (`.VMGS`), and virtual machine runtime state (`.VMRS`) files. If you don't copy over all these files, the virtual machine state isn't fully captured.
- **Data-only backup:** This approach backs up only the data stored on the virtual hard disks of the virtual machine. To restore a data-only backup, you create a new virtual machine and restore the data to it.

Hyper-V VSS Writer

Hyper-V implements a Volume Shadow Copy Service (VSS) writer on all versions of Windows Server where Hyper-V is supported. This VSS writer allows developers to utilize the existing VSS infrastructure to backup virtual machines. However, it's designed for small scale backup operations where all virtual machines on a server are backed up simultaneously.

Hyper-V WMI Based Backup

Starting in Windows Server 2016, Hyper-V supports backup through the Hyper-V Windows Management Instrumentation (WMI) API. This approach still utilizes VSS inside the virtual machine for backup purposes, but no longer uses VSS in the host operating system. Instead, a combination of reference points and resilient change tracking (RCT) is used to allow developers to access the information about backed up virtual machines in an efficient manner. This approach is more scalable than using VSS in the host, however it's only available on Windows Server 2016 and later.

Learn more in the [Hyper-V WMI provider reference](#).

Methods for reading backups from WMI Based Backup

When you create virtual machine backups using Hyper-V WMI, there are three methods for reading the actual data from the backup. Each has unique advantages and disadvantages.

WMI Export

Developers can export the backup data through the Hyper-V WMI interfaces (as used in the previous example). Hyper-V compiles the changes into a virtual hard drive and copy the file to the requested location. This method is easy to use, works for all scenarios, and works remotely. However, the virtual hard drive generated often creates a large amount of data to transfer over the network.

Win32 APIs

Developers can use the `SetVirtualDiskInformation`, `GetVirtualDiskInformation`, and `QueryChangesVirtualDisk` APIs on the Virtual Hard Disk Win32 API set as documented in the [Virtual Hard Disk Win32 reference](#). To use these APIs, Hyper-V WMI still needs to be used to create reference points on associated virtual machines. These Win32 APIs then allow for efficient access to the data of the backed up virtual machine. The Win32 APIs do have several limitations:

- They can only be accessed locally
- They don't support reading data from shared virtual hard disk files
- They return data addresses that are relative to the internal structure of the virtual hard disk

Remote Shared Virtual Disk Protocol

Finally, if a developer needs to efficiently access backup data information from a shared virtual hard disk file, they need to use the Remote Shared Virtual Disk Protocol. This protocol is documented in the [Remote Shared Virtual Disk Protocol reference](#).

Install Hyper-V

Article • 05/26/2025 •

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This article shows how to install Hyper-V in **Windows** and **Windows Server**. Follow the set of instructions that fit your needs.

Enable Hyper-V to create virtual machines on Windows. Hyper-V can be enabled in many ways including using the **Windows control panel**, **PowerShell**, or using the **Deployment Imaging Servicing and Management tool (DISM)**. This article walks through each option.

Note

Hyper-V is built into Windows as an optional feature--there's no Hyper-V download.

Check requirements for Windows

- Windows 10 (Pro or Enterprise), or Windows 11 (Pro or Enterprise)
- 64-bit Processor with Second Level Address Translation (SLAT).
- CPU support for VM Monitor Mode Extension (VT-c on Intel CPUs).
- Minimum of 4-GB memory.

Note

The Hyper-V role **can't** be installed on Windows 10 Home or Windows 11 Home.

For more information and troubleshooting, see [Windows Hyper-V System Requirements](#).

Enable Hyper-V

PowerShell

To enable Hyper-V on Windows using PowerShell:

1. On the Windows desktop, select the Start button and type any part of the name **Windows PowerShell**.
2. Right-click Windows PowerShell and select **Run as Administrator**.

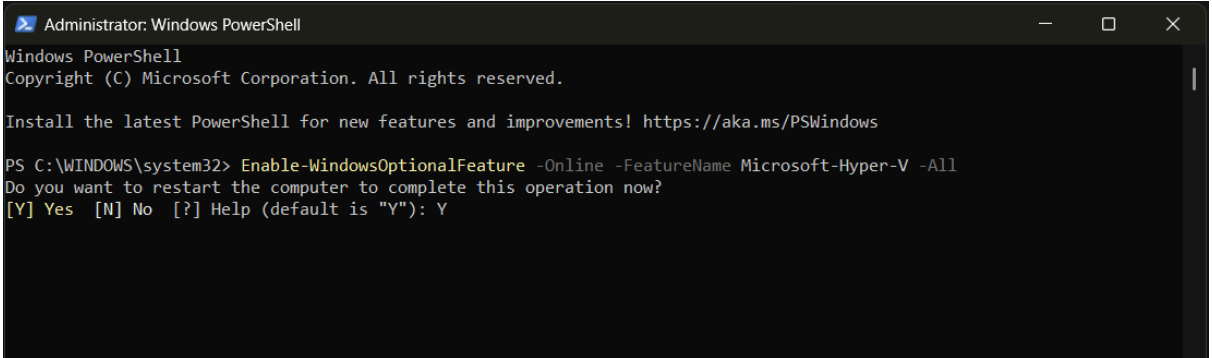
Important

You must run PowerShell as an Administrator, or the commands fail.

3. Run the following command:

PowerShell

```
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
Do you want to restart the computer to complete this operation now?
[Y] Yes  [N] No  [?] Help (default is "Y"): Y
```

4. Enter **Y** to have the computer restart to complete the installation.

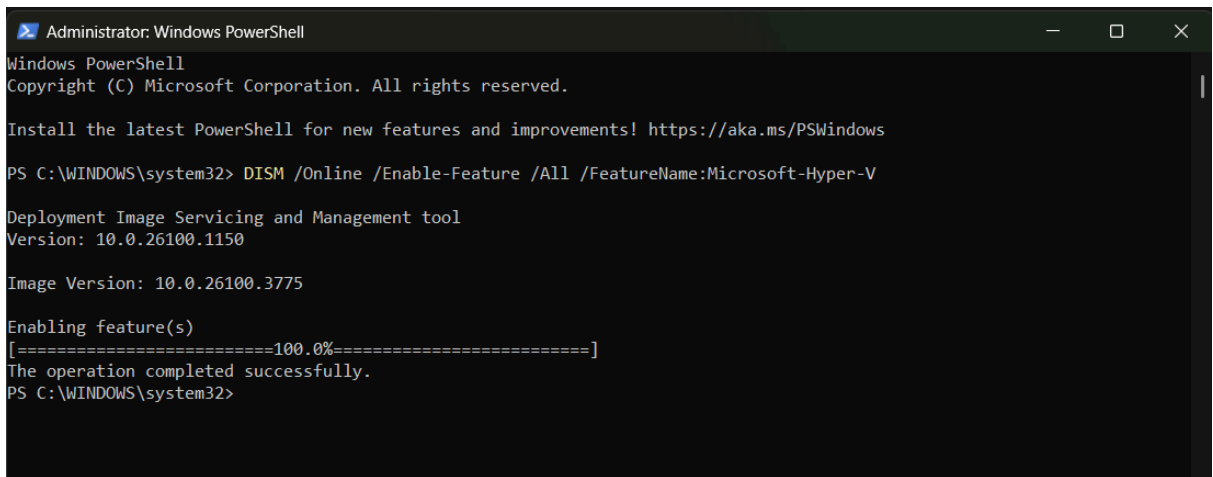
To enable the Hyper-V role on Windows using DISM:

The Deployment Image Servicing and Management tool (DISM) helps configure Windows and Windows images. Among its many applications, DISM can enable Windows features while the operating system is running.

1. On the Windows desktop, select the Start button and type any part of the name **Windows PowerShell**.
2. Right-click Windows PowerShell and select **Run as Administrator**.
3. Type the following command:

PowerShell

```
DISM /Online /Enable-Feature /All /FeatureName:Microsoft-Hyper-V
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> DISM /Online /Enable-Feature /All /FeatureName:Microsoft-Hyper-V

Deployment Image Servicing and Management tool
Version: 10.0.26100.1150

Image Version: 10.0.26100.3775

Enabling feature(s)
[=====100.0%=====]
The operation completed successfully.
PS C:\WINDOWS\system32>
```

4. You see that the feature is enabled and that "The operation completed successfully."

For more information about DISM, see the [DISM Technical Reference](#).

Next step

- [Create a virtual machine in Hyper-V](#)

Create and configure a virtual switch with Hyper-V

Article • 01/29/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2 and 22H2

This article shows you how to create and configure your virtual switch using Hyper-V Manager or PowerShell. A virtual switch allows virtual machines created on Hyper-V hosts to communicate with other computers.

Prerequisites

Before you can create and configure your virtual switch, your computer must meet the following prerequisites:

- Hyper-V must be enabled
- Virtual switch types are: external, internal, and private. Determine [what type of virtual switch you need to create](#).
- Identify which network you connect your computer to. Review the [Core network planning](#) article for more information.
- Have administrative rights.

Create a virtual switch

Once you complete the prerequisites, you're ready to create your virtual switch. In this section, we create a basic virtual switch by following these steps.

Hyper-V Manager

Here's how to create a virtual switch using Hyper-V Manager.

1. Open Hyper-V Manager.
2. From the **Actions** pane, select **Virtual Switch Manager**.
3. Choose the type of virtual switch, then select **Create Virtual Switch**.
4. Enter a name for the virtual switch, then perform one of the following steps.

a. If you selected *external*, choose the network adapter (NIC) that you want to use, then select **OK**.

You're prompted with a warning that the change may disrupt your network connectivity; select **Yes** if you're happy to continue.

b. If you selected *internal* or *private*, select **OK**.

Management Operating System sharing

An external virtual switch allows your virtual machines to connect to an external network. You can also allow the management operating system to share the same selected network adapter. To begin, follow these steps.

Hyper-V Manager

Here's how to allow the management operating system to share the selected network adapter switch using Hyper-V Manager.

1. Open Hyper-V Manager.
2. From the **Actions** pane, select **Virtual Switch Manager**.
3. Select the virtual switch you wish to configure, check the **Allow management operating system to share this network adapter** and select **OK**.

You'll be prompted with a warning that the change may disrupt your network connectivity; select **Yes** if you're happy to continue.

Virtual LAN (VLAN) identification

You can specify the VLAN identification (ID) used by virtual machines network adapters and virtual switches. For virtual switches connected to either an external or internal network, you can specify the (VLAN) ID. The VLAN ID number is used by the management operating system and virtual machines communicating through this virtual switch.

You can also configure your virtual switch with other VLAN options, such port mode and the native VLAN ID. For these options, you need to use PowerShell and ensure the configuration is compatible with your networks configuration.

To configure the VLAN identification for the switch, follow these steps.

Hyper-V Manager

Here's how to specify the VLAN ID using the Hyper-V Manager.

1. Open Hyper-V Manager.
2. From the **Actions** pane, select **Virtual Switch Manager**.
3. Select the virtual switch you wish to configure, check the **Enable virtual LAN identification for management operating system**.
 - a. You can enter any VLAN ID number or leave the default, then select **OK**.

You're prompted to warn you that the change may disrupt your network connectivity, select **Yes** if you're happy to continue.

VLAN identifiers should be consistent with your network to ensure compatibility between your computer, virtual machines, and other network devices.

NAT networking

Network Address Translation (NAT) gives a virtual machine access to your computer's network by combining the host computer's IP address with a port through an internal Hyper-V Virtual Switch.

This has a few useful properties:

1. NAT Conserves IP addresses by mapping an external IP address and port to a much larger set of internal IP addresses.
2. NAT allows multiple virtual machines to host applications that require identical (internal) communication ports by mapping these to unique external ports.
3. NAT uses an internal switch -- creating an internal switch doesn't cause you to use network connection and tends to interfere less with a computer's networking.

To set up a NAT network and connect it to a virtual machine, follow the [NAT networking user guide](#).

Related content

Now that you set up a virtual switch, here are other articles to help you continue with Hyper-V.

- Learn about [Switch Embedded Teaming \(SET\)](#).
 - Learn how to [create a virtual machine in Hyper-V](#).
 - Learn about other configuration options in the [Set-VMSwitch](#) and [Set-VMNetworkAdapterVlan](#) PowerShell reference articles.
-

Feedback

Was this page helpful?

 Yes

 No

Create a virtual machine in Hyper-V

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Learn how to create a virtual machine by using Hyper-V Manager and Windows PowerShell and what options you have when you create a virtual machine in Hyper-V Manager.

Prerequisites

Before you create a virtual machine, make sure you have the following prerequisites:

- A computer running Windows Server or Windows client with Hyper-V enabled. For more information, see [Install the Hyper-V role on Windows Server](#).
- Membership in the local **Administrators** group or the **Hyper-V Administrators** group.
- Sufficient physical memory on the host to allocate to the virtual machine.
- Sufficient disk space for virtual machine configuration files and virtual hard disks.
- (Optional) A virtual switch configured if you want network connectivity for the virtual machine. See [Create a virtual switch for Hyper-V virtual machines](#).
- (Optional) An operating system installation media (.iso file) or an existing virtual hard disk (.vhd or .vhdx) if you want to install an operating system.

Create a virtual machine

To create a virtual machine, use either Hyper-V Manager or Windows PowerShell. Select the method you prefer:

Hyper-V Manager

To create a virtual machine in Hyper-V Manager, follow these steps:

1. Select the **Start** button, type **Hyper-V Manager**, and then select it from the list.

Note

If you don't see Hyper-V Manager, make sure that the Hyper-V role is installed. For more information, see [Install the Hyper-V role on Windows Server](#).

2. In the left pane, under **Hyper-V Manager**, select your server.
3. From the **Actions** pane, select **New**, then select **Virtual Machine**.

4. From the **New Virtual Machine Wizard**, select **Next**.

5. For the **Specify Name and Location** page, enter a name for the virtual machine and choose a location to store the virtual machine configuration files. Select **Next**.

(Optionally) If you want to store the virtual machine configuration files in a different location, select the **Store the virtual machine in a different location** check box and then browse to the folder where you want to store the files.

6. In the **Specify Generation** page, select the generation of the virtual machine. For more information, see [Should I create a generation 1 or 2 virtual machine in Hyper-V?](#). Create a generation 2 virtual machine unless you have a specific reason to create a generation 1 virtual machine. Select **Next**.

7. For the **Assign Memory** page, enter the amount of memory to allocate to the virtual machine at startup. You can also choose to enable Dynamic Memory. For more information, see [Hyper-V Dynamic Memory Overview](#). Select **Next**.

 Note

The minimum amount of memory you can assign is 32 MB and the maximum is 5,902 MB.

8. For the **Configure Networking** page, select a virtual switch to connect the virtual machine to the network. If you don't have a virtual switch created, you can skip this step and configure networking later. Select **Next**.

9. In the **Connect Virtual Hard Disk** page, choose one of the following options, then select **Next**:

- **Create a virtual hard disk** - Enter a name for the virtual hard disk, choose a location to store it, and specify its size.
- **Use an existing virtual hard disk** - Browse to the location of an existing virtual hard disk (`.vhd` or `.vhdx`).
- **Attach a virtual hard disk later** - You can attach a virtual hard disk later.

10. For the **Installation Options** page, choose how you want to install the operating system on the virtual machine. You can choose one of the following options, then select **Next**:

- **Install an operating system later**

- **Install an operating system from a bootable image file** - Browse to the location of an `.iso` file that contains the operating system installation media.
- **Install an operating system from a bootable floppy disk** - Select a floppy disk drive that contains the operating system installation media.
- **Install an operating system from a network-based installation server** - This option is for using Windows Deployment Services (WDS) or another network-based installation service.

11. After verifying your choices in the **Summary** page, select **Finish**.

Start and connect to the virtual machine

After you create the virtual machine, you can start it and connect to it. You can use either Hyper-V Manager or Windows PowerShell. Select the method you prefer:

Hyper-V Manager

To start and connect to the virtual machine in Hyper-V Manager, follow these steps:

1. In Hyper-V Manager, right-click the virtual machine and select **Connect....**
2. In the Virtual Machine Connection window, select **Action > Start**.

Related content

- [New-VM](#)
- [Supported virtual machine configuration versions](#)
- [Should I create a generation 1 or 2 virtual machine in Hyper-V?](#)
- [Create a virtual switch for Hyper-V virtual machines](#)

Configure virtual local area networks for Hyper-V

08/20/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Virtual local area networks (VLANs) offer one way to isolate network traffic. VLANs are configured in switches and routers that support 802.1q. If you configure multiple VLANs and want communication to occur between them, you need to configure the network devices to allow that.

To set up VLANs, make sure you have:

- A physical network adapter and driver that supports 802.1q VLAN tagging.
- A physical network switch that supports 802.1q VLAN tagging.

On the host, set up the virtual switch to let network traffic on the physical switch port for the VLAN IDs you use with virtual machines. Then, set up the virtual machine to specify the VLAN it uses for all network communications.

Allow a virtual switch to use a VLAN

1. In Hyper-V Manager, select **Virtual Switch Manager** from the **Actions** pane on the right.
2. In **Virtual Switch Manager**, under **Virtual Switches** on the left, select a virtual switch connected to a physical network adapter that supports VLANs.
3. Under **VLAN ID** in the right pane, select **Enable virtual LAN identification for management operating system** and then type a number for the VLAN ID.
4. Select **OK**.

All traffic that goes through the physical network adapter connected to the virtual switch is tagged with the VLAN ID you set on the virtual switch and virtual machines.

Allow a virtual machine to use a VLAN

1. In Hyper-V Manager, under **Virtual Machines**, right-click the appropriate virtual machine and select **Settings**. Or, select the machine and then select **Settings** under the machine name in the right pane.

2. On the **Settings** screen, under **Hardware** in the left pane, select a **Network Adapter** that has a virtual switch that's set up with a VLAN.
3. Under **VLAN ID** in the right pane, select **Enable virtual LAN identification**, and then type the same VLAN ID as the one you specified for the virtual switch.
4. Select **OK**.

If the virtual machine needs to use more VLANs, do one of the following:

- Connect more virtual network adapters to appropriate virtual switches and assign the VLAN IDs. Make sure to configure the IP addresses correctly, and that the traffic you want to route through the VLAN also uses the correct IP address.
- Configure the virtual network adapter in trunk mode by using the [Set-VMNetworkAdapterVlan](#) cmdlet.

Related content

[Hyper-V virtual switch](#)

Set up a NAT network

08/14/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Windows 10 and Windows 11 Hyper-V allows native network address translation (NAT) for a virtual network.

This guide walks you through how-to:

- Create a NAT network
- Connect an existing virtual machine to your new network
- Confirm that the virtual machine is connected correctly

Requirements:

- Windows 10 Anniversary Update or later
- Hyper-V is enabled. Follow the instructions to [Enable Hyper-V](#)

Note

Currently, you're limited to one NAT network per host. For additional details on the Windows NAT (WinNAT) implementation, capabilities, and limitations, reference the [WinNAT capabilities and limitations blog](#) 

NAT Overview

NAT gives a virtual machine access to network resources using the host computer's IP address and a port through an internal Hyper-V Virtual Switch.

Network Address Translation (NAT) is a networking mode designed to conserve IP addresses by mapping an external IP address and port to a much larger set of internal IP addresses. Basically, a NAT uses a flow table to route traffic from an external (host) IP Address and port number to the correct internal IP address associated with an endpoint on the network (virtual machine, computer, container, etc.).

Also, NAT allows multiple virtual machines to host applications that require identical (internal) communication ports by mapping these to unique external ports.

For all of these reasons, NAT networking is very common for container technology (see [Container Networking](#)).

Create a NAT virtual network

Let's walk through setting up a new NAT network.

1. Open a PowerShell console as Administrator.
2. Create an internal switch.

PowerShell

```
New-VMSwitch -SwitchName "SwitchName" -SwitchType Internal
```

3. Find the interface index of the virtual switch you just created.

Console

```
PS C:\> Get-NetAdapter
```

Name	InterfaceDescription	ifIndex	Status
MacAddress	LinkSpeed		
----	-----	-----	-----
-----	-----		
vEthernet (intSwitch)	Hyper-V Virtual Ethernet Adapter	24	Up
00-15-5D-00-6A-01	10 Gbps		
Wi-Fi	Marvell AVASTAR Wireless-AC Net...	18	Up
98-5F-D3-34-0C-D3	300 Mbps		
Bluetooth Network ...	Bluetooth Device ...	21	Disconnected
98-5F-D3-34-0C-D4	3 Mbps		

The internal switch has a name like `vEthernet (SwitchName)` and an Interface Description of `Hyper-V Virtual Ethernet Adapter`. Take note of its `ifIndex` to use in the next step.

4. Configure the NAT gateway using [New-NetIPAddress](#).

Here is the generic command:

PowerShell

```
New-NetIPAddress -IPAddress <NAT Gateway IP> -PrefixLength <NAT Subnet Prefix Length> -InterfaceIndex <ifIndex>
```

In order to configure the gateway, you'll need a bit of information about your network:

- **IPAddress** -- NAT Gateway IP specifies the IPv4 or IPv6 address to use as the NAT gateway IP.

The generic form is a.b.c.1 (e.g. 172.16.0.1). While the final position doesn't have to be .1, it usually is (based on prefix length). This IP address is in the range of addresses used by the guest virtual machines. For example, if the guest VMs use IP range 172.16.0.0, then you can use an IP address 172.16.0.100 as the NAT Gateway. A common gateway IP is 192.168.0.1

- **PrefixLength** -- NAT Subnet Prefix Length defines the NAT local subnet size (subnet mask). The subnet prefix length is an integer value between 0 and 32.

0 would map the entire internet, 32 would only allow one mapped IP. Common values range from 24 to 12 depending on how many IPs need to be attached to the NAT.

A common PrefixLength is 24 -- this is a subnet mask of 255.255.255.0

- **InterfaceIndex** -- ifIndex is the interface index of the virtual switch, which you determined in the previous step.

Run the following to create the NAT Gateway:

PowerShell

```
New-NetIPAddress -IPAddress 192.168.0.1 -PrefixLength 24 -InterfaceIndex 24
```

5. Configure the NAT network using [New-NetNat](#).

Here is the generic command:

PowerShell

```
New-NetNat -Name <NATOutsideName> -InternalIPInterfaceAddressPrefix <NAT subnet prefix>
```

In order to configure the gateway, you'll need to provide information about the network and NAT Gateway:

- **Name** -- NATOutsideName describes the name of the NAT network. You'll use this to remove the NAT network.
- **InternalIPInterfaceAddressPrefix** -- NAT subnet prefix describes both the NAT Gateway IP prefix and the NAT Subnet Prefix Length.

The generic form is a.b.c.0/NAT Subnet Prefix Length

For this example, we'll use 192.168.0.0/24

Run the following to setup the NAT network:

PowerShell

```
New-NetNat -Name MyNATnetwork -InternalIPInterfaceAddressPrefix  
192.168.0.0/24
```

Congratulations! You now have a virtual NAT network!

Connect a virtual machine

To connect a virtual machine to your new NAT network, connect the internal switch you created in the first step of this article to your virtual machine using the VM Settings menu.

Since WinNAT by itself does not allocate and assign IP addresses to an endpoint (e.g. VM), you'll need to do this manually from within the VM itself - i.e. set IP address within range of NAT internal prefix, set default gateway IP address, set DNS server information. The only caveat to this is when the endpoint is attached to a container. In this case, the Host Network Service (HNS) allocates and uses the Host Compute Service (HCS) to assign the IP address, gateway IP, and DNS info to the container directly.

Configuration Example: Attaching VMs and Containers to a NAT network

If you need to attach multiple VMs and containers to a single NAT, you will need to ensure that the NAT internal subnet prefix is large enough to encompass the IP ranges being assigned by different applications or services (e.g. Docker for Windows and Windows Container – HNS). This requires either application-level assignment of IPs and network configuration or manual configuration which must be done by an admin and guaranteed not to re-use existing IP assignments on the same host.

Docker for Windows (Linux VM) and Windows Containers

Follow along in this section to allow both Docker for Windows, (Linux VM running Linux containers) and Windows Containers to share the same WinNAT instance using separate internal vSwitches. Connectivity between both Linux and Windows containers work.

User has connected VMs to a NAT network through an internal vSwitch named "VMNAT" and now wants to install Windows Container feature with docker engine:

Output

```

PS C:\> Get-NetNat "VMNAT" | Remove-NetNat (this removes the NAT but keep the
internal vSwitch).
Install Windows Container Feature
DO NOT START Docker Service (daemon)
Edit the arguments passed to the docker daemon (dockerd) by adding -fixed-cidr=
<container prefix> parameter. This tells docker to create a default nat network
with the IP subnet <container prefix> (e.g. 192.168.1.0/24) so that HNS can
allocate IPs from this prefix.
PS C:\> Start-Service Docker; Stop-Service Docker
PS C:\> Get-NetNat | Remove-NetNat (again, this removes the NAT but keep the
internal vSwitch)
PS C:\> New-NetNat -Name SharedNAT -InternalIPInterfaceAddressPrefix <shared
prefix>
PS C:\> Start-Service docker

```

Docker/HNS assigns IPs to Windows containers and Admin assigns IPs to VMs from the difference set of the two.

User has installed Windows Container feature with docker engine running and now wants to connect VMs to the NAT network:

Output

```

PS C:\> Stop-Service docker
PS C:\> Get-ContainerNetwork | Remove-ContainerNetwork -force
PS C:\> Get-NetNat | Remove-NetNat (this will remove the NAT but keep the internal
vSwitch)
Edit the arguments passed to the docker daemon (dockerd) by adding -b "none"
option to the end of docker daemon (dockerd) command to tell docker not to create
a default NAT network.
PS C:\> New-ContainerNetwork -name nat -Mode NAT -subnetprefix <container prefix>
(create a new NAT and internal vSwitch - HNS will allocate IPs to container
endpoints attached to this network from the <container prefix>)
PS C:\> Get-Netnat | Remove-NetNAT (again, this will remove the NAT but keep the
internal vSwitch)
PS C:\> New-NetNat -Name SharedNAT -InternalIPInterfaceAddressPrefix <shared
prefix>
PS C:\> New-VirtualSwitch -Type internal (attach VMs to this new vSwitch)
PS C:\> Start-Service docker

```

Docker/HNS assigns IPs to Windows containers and Admin assigns IPs to VMs from the difference set of the two.

In the end, you'll have two internal VM switches and one NetNat shared between them.

Multiple Applications using the same NAT

Some scenarios require multiple applications or services to use the same NAT. In this case, the following workflow must be followed so that multiple applications / services can use a larger NAT internal subnet prefix

We'll detail the Docker 4 Windows - Docker Beta - Linux VM co-existing with the Windows Container feature on the same host as an example. This workflow is subject to change

1. PowerShell

```
net stop docker
```

2. PowerShell

```
Stop Docker4Windows MobyLinux VM
```

3. PowerShell

```
Get-ContainerNetwork | Remove-ContainerNetwork -force
```

4. Remove any previously existing container networks (i.e. deletes vSwitch, deletes NetNat, cleans up).

PowerShell

```
Get-NetNat | Remove-NetNat
```

5. Create an internal vSwitch named nat, and a NAT network with IP prefix 10.0.76.0/24.

PowerShell

```
New-ContainerNetwork -Name nat -Mode NAT -subnetprefix 10.0.76.0/24
```

6. Remove both DockerNAT and nat NAT networks (keeps internal vSwitches).

PowerShell

```
Remove-NetNAT
```

7. Create a NAT network named DockerNAT with larger prefix 10.0.0.0/17 for both D4W and containers to share.

PowerShell


```
New-NetNat -Name DockerNAT -InternalIPInterfaceAddressPrefix 10.0.0.0/17
```

8. Run Docker4Windows (MobyLinux.ps1)

9. Use the user-defined NAT network as the default to connect Windows containers.

```
PowerShell
```

```
Net start docker
```

In the end, you have two internal vSwitches – one named **DockerNAT** and the other named **nat**. You only have one NAT network (10.0.0.0/17) confirmed by running `Get-NetNat`. IP addresses for Windows containers are assigned by the Windows Host Network Service (HNS) from the 10.0.76.0/24 subnet. Based on the existing MobyLinux.ps1 script, IP addresses for Docker 4 Windows are assigned from the 10.0.75.0/24 subnet.

Troubleshooting

Multiple NAT networks are not supported

This guide assumes that there are no other NATs on the host. However, applications or services require the use of a NAT and may create one as part of setup. Since Windows (WinNAT) only supports one internal NAT subnet prefix, trying to create multiple NATs places the system into an unknown state.

To see if this may be the problem, make sure you only have one NAT:

```
PowerShell
```

```
Get-NetNat
```

If a NAT already exists, delete it:

```
PowerShell
```

```
Get-NetNat | Remove-NetNat
```

Make sure you only have one “internal” vmSwitch for the application or feature (e.g. Windows containers). Record the name of the vSwitch:

```
PowerShell
```

Get-VMSwitch

Check to see if there are private IP addresses (e.g. NAT default Gateway IP Address – usually x.y.z.1) from the old NAT still assigned to an adapter:

PowerShell

```
Get-NetIPAddress -InterfaceAlias "vEthernet (<name of vSwitch>)"
```

If an old private IP address is in use, please delete it:

PowerShell

```
Remove-NetIPAddress -InterfaceAlias "vEthernet (<name of vSwitch>)" -IPAddress  
<IPAddress>
```

Removing Multiple NATs We've seen reports of multiple NAT networks created inadvertently. If you see multiple NAT networks, after running `docker network ls` or `Get-ContainerNetwork`, perform the following from an elevated PowerShell:

PowerShell

```
$keys = Get-ChildItem  
"HKLM:\SYSTEM\CurrentControlSet\Services\vmtoolsd\parameters\SwitchList"  
foreach($key in $keys)  
{  
    if ($key.GetValue("FriendlyName") -eq 'nat')  
    {  
        $newKeyPath = $key.Path+"\$key.PSChildName  
        Remove-Item -Path $newKeyPath -Recurse  
    }  
}  
Remove-NetNat -Confirm:$false  
Get-ContainerNetwork | Remove-ContainerNetwork  
Get-VMSwitch -Name nat | Remove-VMSwitch # failure is expected  
Stop-Service docker  
Set-Service docker -StartupType Disabled
```

Reboot the operating system prior executing the subsequent commands (`Restart-Computer`)

PowerShell

```
Get-NetNat | Remove-NetNat  
Set-Service docker -StartupType Automatic  
Start-Service docker
```

Export and import virtual machines

Article • 05/14/2025 •

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This article shows you how to export and import a virtual machine, which is a quick way to move or copy them. This article also discusses some of the choices to make when doing an export or import.

Export a virtual machine

An export gathers all required files into one unit--virtual hard disk files, virtual machine configuration files, and any checkpoint files. You can do this on a virtual machine that is in either a started or stopped state.

Hyper-V Manager

To create a virtual machine export:

1. In Hyper-V Manager, right-click the virtual machine and select **Export**.
2. Choose where to store the exported files, and click **Export**.

When the export is done, you can see all exported files under the export location.

Import a virtual machine

Importing a virtual machine registers the virtual machine with the Hyper-V host. You can import back into the host, or new host. If you're importing to the same host, you don't need to export the virtual machine first, because Hyper-V tries to recreate the virtual machine from available files. Importing a virtual machine registers it so it can be used on the Hyper-V host.

Important

Hyper-V virtual machine configurations have a specific version number. You can only import a virtual machine if the Hyper-V host supports that configuration version. Typically, this means that you can import a virtual machine to a Hyper-V host running a newer version of Hyper-V, but you can't import a virtual machine created on a newer version of Hyper-V to an older version of Hyper-V. For more information, see [Supported virtual machine configuration versions](#).

The Import Virtual Machine wizard also helps you fix incompatibilities that can exist when moving from one host to another. This is commonly differences in physical hardware, such as memory, virtual switches, and virtual processors.

Hyper-V Manager

To import a virtual machine:

1. From the **Actions** menu in Hyper-V Manager, select **Import Virtual Machine**.
2. Click **Next**.
3. Select the folder that contains the exported files, and select **Next**.
4. Select the virtual machine to import.
5. Choose the import type, and select **Next**. For descriptions, see [Import types](#).
6. Select **Finish**.

Import types

Hyper-V offers three import types:

- **Register in-place** – The register in-place type assumes export files are in the location where you store and run the virtual machine. The imported virtual machine has the same ID as it did at the time of export. Because of this, if the virtual machine is already registered with Hyper-V, it needs to be deleted before the import works. When the import has completed, the export files become the running state files and can't be removed.
- **Restore the virtual machine** – Restore the virtual machine to a location you choose, or use the default to Hyper-V. This import type creates a copy of the exported files and moves them to the selected location. When imported, the virtual machine has the same ID as it did at the time of export. Because of this, if the virtual machine is already running in Hyper-V, it needs to be deleted before the import can be completed. When the import has completed, the exported files remain intact and can be removed or imported again.
- **Copy the virtual machine** – The copy type is similar to the restore type in that you select a location for the files. The difference is that the imported virtual machine has a new unique ID, which means you can import the virtual machine to the same host multiple times.

Set up hosts for live migration without Failover Clustering

08/15/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Live migration lets you move running virtual machines between servers with minimal downtime. This article shows you how to set up nonclustered hosts for live migration in Hyper-V, including authentication options, network configuration, and security best practices. Use these steps if you didn't configure live migration during Hyper-V installation, or need to update your settings.

Requirements for setting up live migration

To set up nonclustered hosts for live migration, you need:

- Membership in the local Hyper-V Administrators group or the Administrators group on both the source and destination computers meets this requirement, unless you're setting up constrained delegation. Membership in the Domain Administrators group is required to set up constrained delegation.
- The Hyper-V role in Windows Server installed on the source and destination servers. You can run a live migration between hosts running Windows Server 2012 R2 and later. For installation instructions, see [Install the Hyper-V role on Windows Server](#).
- Source and destination computers that belong to the same Active Directory domain, or to domains that trust each other.
- The Hyper-V management tools installed on a computer running at least Windows Server 2016 or Windows 10, unless the tools are installed on the source or destination server and you run the tools from the server.

Consider options for authentication and networking

Decide how you want to set up authentication, performance, and networking options.

- **Authentication:** Which protocol do you use to authenticate live migration traffic between the source and destination server? The protocol you choose determines whether you need to sign in to the source server before starting a live migration:

- Kerberos lets you avoid signing in to the server, but requires you to set up constrained delegation. See the following details for instructions.
- CredSSP lets you avoid setting up constrained delegation, but requires you to sign in to the source server. You can do this through a local console session, a Remote Desktop session, or a remote Windows PowerShell session.

CredSSP requires you to sign in for situations that aren't always obvious. For example, if you sign in to TestServer01 to move a virtual machine to TestServer02, and then want to move the virtual machine back to TestServer01, you need to sign in to TestServer02 before you try to move the virtual machine back to TestServer01. If you don't, the authentication attempt fails, an error occurs, and you see the following message:

"Virtual machine migration operation failed at migration Source. Failed to establish a connection with host *computer name*: No credentials are available in the security package 0x8009030E."

- **Performance:** Should you set up performance options? These options can reduce network and CPU usage, and make live migrations go faster. Check your requirements and infrastructure, and test different configurations to decide. The options are described at the end of step 2.
- **Network preference:** Do you allow live migration traffic through any available network, or do you isolate the traffic to specific networks? As a security best practice, isolate the traffic onto trusted, private networks because live migration traffic isn't encrypted when sent over the network. You can isolate the network by using a physically isolated network or another trusted networking technology like VLANs.

Upgrading to Windows Server 2025

Starting with Windows Server 2025, [Credential Guard is enabled by default](#) on all domain-joined servers that aren't domain controllers. Because of this, you can't use CredSSP-based live migration with Hyper-V after upgrading to Windows Server 2025. CredSSP-based delegation is the default for Windows Server 2022 and earlier for live migration. Instead, use Kerberos constrained delegation, as described in the following section. For more information, see [Live migration with Hyper-V breaks when upgrading to Windows Server 2025](#).

Step 1: Configure constrained delegation (optional)

If you decide to use Kerberos to authenticate live migration traffic, configure constrained delegation using an account that is a member of the Domain Administrators group.

Use the Users and Computers snap-in to set up constrained delegation

1. Open the Active Directory Users and Computers snap-in. In Server Manager, select the server, then select **Tools > Active Directory Users and Computers**.
2. In the navigation pane in **Active Directory Users and Computers**, select the domain and then double-click the **Computers** folder.
3. In the **Computers** folder, right-click the computer account of the source server, and then select **Properties**.
4. In **Properties**, select the **Delegation** tab.
5. On the **Delegation** tab, select **Trust this computer for delegation to the specified services only**, and then select **Use any authentication protocol**.
6. Select **Add**.
7. In **Add Services**, select **Users or Computers**.
8. From **Select Users or Computers**, type the name of the destination server. Select **Check Names** to verify it, and then select **OK**.
9. From **Add Services**, in the list of available services, do the following and then select **OK**:
 - To move virtual machine storage, select **cifs**. This is required if you want to move the storage along with the virtual machine, and if you want to move only a virtual machine's storage. If the server is configured to use SMB storage for Hyper-V, this should already be selected.
 - To move virtual machines, select **Microsoft Virtual System Migration Service**.
10. On the **Delegation** tab of the Properties dialog box, check that the services you selected in the previous step are listed as the services to which the destination computer can present delegated credentials. Select **OK**.
11. In the **Computers** folder, select the computer account of the destination server and repeat the process. In the **Select Users or Computers** dialog box, make sure to specify the name of the source server.

The configuration changes take effect after:

- The changes replicate to the domain controllers that the servers running Hyper-V are logged into.
- The domain controller issues a new Kerberos ticket.

Step 2: Set up the source and destination computers for live migration

This step covers options for authentication and networking. For better security, select specific networks for live migration traffic, as discussed earlier. You also choose a performance option in this step.

Use Hyper-V Manager to set up the source and destination computers for live migration

1. Open Hyper-V Manager. In Server Manager, select **Tools > Hyper-V Manager**.
2. In the navigation pane, select a server. If it isn't listed, right-click **Hyper-V Manager**, select **Connect to Server**, enter the server name, and select **OK**. Repeat to add more servers.
3. In the **Action** pane, select **Hyper-V Settings > Live Migrations**.
4. In the **Live Migrations** pane, select **Enable incoming and outgoing live migrations**.
5. Under **Simultaneous live migrations**, enter a different number if you don't want to use the default of 2.
6. Under **Incoming live migrations**, if you want to use specific network connections to accept live migration traffic, select **Add** to type the IP address. To set Kerberos and performance options, expand **Live Migrations** and select **Advanced Features**.
 - If you set up constrained delegation, under **Authentication protocol**, select **Kerberos**.
 - Under **Performance options**, review the details and select a different option if it fits your environment. **ce options****, review the details and choose a different option if it's appropriate for your environment.
7. Select **OK**.
8. Select the other server in Hyper-V Manager and repeat these steps.

Use Windows PowerShell to set up the source and destination computers for live migration

You can use three cmdlets to set up live migration on nonclustered hosts: [Enable-VMMigration](#), [Set-VMMigrationNetwork](#), and [Set-VMHost](#). This example uses all three to:

- Set up live migration on the local host

- Let incoming migration traffic only on a specific network
- Choose Kerberos as the authentication protocol

Each line is a separate command.

PowerShell

```
PS C:\> Enable-VMMigration
```

```
PS C:\> Set-VMMigrationNetwork 192.168.10.1
```

```
PS C:\> Set-VMHost -VirtualMachineMigrationAuthenticationType Kerberos
```

`Set-VMHost` also lets you choose a performance option and other host settings. For example, to choose SMB but keep the authentication protocol set to the default of CredSSP, enter:

PowerShell

```
PS C:\> Set-VMHost -VirtualMachineMigrationPerformanceOption SMB
```

This table describes how the performance options work.

 Expand table

Option	Description
TCP/IP	Copies the memory of the virtual machine to the destination server over a TCP/IP connection.
Compression	Compresses the memory content of the virtual machine before copying it to the destination server over a TCP/IP connection. Compression is the default setting.
SMB	Copies the memory of the virtual machine to the destination server over an SMB 3.0 connection. - SMB Direct is used when the network adapters on the source and destination servers have Remote Direct Memory Access (RDMA) enabled. - SMB Multichannel automatically detects and uses multiple connections when a proper SMB Multichannel configuration is set up. For more information, see Improve Performance of a File Server with SMB Direct.

Next steps

After you set up the hosts, you're ready to do a live migration. For instructions, see [Use live migration without Failover Clustering to move a virtual machine](#).

Upgrade virtual machine version in Hyper-V on Windows or Windows Server

06/30/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Make the latest Hyper-V features available on your virtual machines by upgrading the configuration version. Don't do this until:

- You upgrade your Hyper-V hosts to the latest version of Windows or Windows Server.
- You upgrade the cluster functional level.
- You're sure that you won't need to move the virtual machine back to a Hyper-V host that runs a previous version of Windows or Windows Server.

For more information, see [Cluster Operating System Rolling Upgrade](#) and [Perform a rolling upgrade of a Hyper-V host cluster in VMM](#).

Step 1: Check the virtual machine configuration versions

1. On the Windows desktop, select the Start button and type any part of the name **Windows PowerShell**.
2. Right-click Windows PowerShell and select **Run as Administrator**.
3. Use the [Get-VM](#) cmdlet. Run the following command to get the versions of your virtual machines.

PowerShell

```
Get-VM * | Format-Table Name, Version
```

You can also see the configuration version in Hyper-V Manager by selecting the virtual machine and looking at the **Summary** tab.

Step 2: Upgrade the virtual machine configuration version

1. Shut down the virtual machine in Hyper-V Manager.
2. Select Action > Upgrade Configuration Version. If this option isn't available for the virtual machine, then it's already at the highest configuration version supported by the Hyper-V

host.

To upgrade the virtual machine configuration version by using Windows PowerShell, use the [Update-VMVersion](#) cmdlet. Run the following command where `vmname` is the name of the virtual machine.

PowerShell

```
Update-VMVersion <vmname>
```

Supported virtual machine configuration versions

Using the PowerShell cmdlet [Get-VMHostSupportedVersion](#) you can see what virtual machine configuration versions your Hyper-V Host supports. When you create a virtual machine, it's created with the default configuration version. To see which virtual machine configuration versions your Hyper-V Host supports and what the default is, run the following command.

PowerShell

```
Get-VMHostSupportedVersion
```

If you need to create a virtual machine that you can move to a Hyper-V Host that runs an older version of Windows, use the [New-VM](#) cmdlet with the `-Version` parameter. For example, to create a virtual machine named "WindowsCV5" with configuration version 5.0, run the following command:

PowerShell

```
New-VM -Name "WindowsCV5" -Version 5.0
```

ⓘ Note

You can only import a virtual machine if the Hyper-V host supports that configuration version. Typically, this means that you can import a virtual machine to a Hyper-V host running a newer version of Hyper-V, but you cannot import a virtual machine created on a newer version of Hyper-V to an older version of Hyper-V.

If the VM's configuration version is not listed as supported for your Hyper-V host OS in the table below, you either have to upgrade the VM configuration version to a newer version or create a new VM of the same generation using the existing virtual hard disks before you can start the VM.

Supported VM configuration versions for long-term servicing hosts

The following table lists the VM configuration versions for hosts running a long-term servicing version of Windows.

Expand table

[illegible]

Hyper-V host Windows version	10.0	9.3	9.2	9.1	9.0	8.3	8.2	8.1	8.0	7.1	7.0	6.2	5.0
Windows 10 November 2019 Update (version 1909)	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Windows Server, version 1903	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Windows 10 May 2019 Update (version 1903)	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Windows Server, version 1809	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓
Windows 10 October 2018 Update (version 1809)	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓
Windows Server, version 1803	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓
Windows 10 April 2018 Update (version 1803)	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓
Windows 10 Fall Creators Update (version 1709)	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Windows 10 Creators Update (version 1703)	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
Windows 10 Anniversary Update (version 1607)	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓

Why should I upgrade the virtual machine configuration version?

When you move or import a virtual machine to a computer that runs Hyper-V on Windows Server 2019, Windows Server 2016, or Windows 10, the virtual machine's configuration isn't automatically updated. This means that you can move the virtual machine back to a Hyper-V host that runs a previous version of Windows or Windows Server. But, this also means that you can't use some of the new virtual machine features until you manually update the configuration version.

Important

You can't downgrade a virtual machine configuration version after you've upgraded it.

The virtual machine configuration version represents the compatibility of the virtual machine's configuration, saved state, and snapshot files with the version of Hyper-V. When you update the configuration version, you change the file structure that is used to store the virtual machines configuration and the checkpoint files. You also update the configuration version to the latest version supported by that Hyper-V host. Upgraded virtual machines use a new configuration file format, which is designed to increase the efficiency of reading and writing virtual machine configuration data. The upgrade also reduces the potential for data corruption in the event of a storage failure.

The following table lists descriptions file name extensions, and default locations for each type of file that's used for new or upgraded virtual machines.

 Expand table

Virtual machine file types	Description
Configuration	Virtual machine configuration information that is stored in binary file format. File name extension: .vmcx Default location: C:\ProgramData\Microsoft\Windows\Hyper-V\Virtual Machines
Runtime state	Virtual machine runtime state information that is stored in binary file format. File name extension: .vmrs and .vmgs Default location: C:\ProgramData\Microsoft\Windows\Hyper-V\Virtual Machines
Virtual hard disk	Stores virtual hard disks for the virtual machine. File name extension: .vhd or .vhdx Default location: C:\ProgramData\Microsoft\Windows\Hyper-V\Virtual Hard Disks
Automatic virtual hard disk	Differencing disk files used for virtual machine checkpoints. File name extension: .avhdx Default location: C:\ProgramData\Microsoft\Windows\Hyper-V\Virtual Hard Disks
Checkpoint	Checkpoints are stored in multiple checkpoint files. Each checkpoint creates a configuration file and runtime state file. File name extensions: .vmrs and .vmcx Default location: C:\ProgramData\Microsoft\Windows\Snapshots

What happens if I don't upgrade the virtual machine configuration version?

If you have virtual machines that you created with an earlier version of Hyper-V, some features that are available on the newer host OS might not work with those virtual machines until you update the configuration version.

As a general guidance, we recommend updating the configuration version once you have successfully upgraded the virtualization hosts to a newer version of Windows and feel confident that you don't need to roll back. When you're using the [cluster OS rolling upgrade](#) feature, this would typically be after updating the cluster functional level. This way, you'll benefit from new features and internal changes and optimizations as well.

ⓘ **Note**

Once the VM configuration version is updated, the VM won't be able to start on hosts that do not support the updated configuration version.

The following table shows the minimum virtual machine configuration version required to use some Hyper-V features.

 [Expand table](#)

Feature	Minimum VM configuration version
GPU partitioning	12.0
Support for ARM64 in Windows 11	11.0
Dynamic processor compatibility mode	10.0
Nested virtualization for AMD processors	9.3
vNUMA topology improvements	9.2
AMD topology improvements	9.1
Allow additional processor features for Perfmon	9.0
Automatically expose simultaneous multithreading configuration for VMs running on hosts using the Core Scheduler	9.0
Hibernation support	9.0
Increase the default maximum number for virtual devices to 64 per device (for example, networking and assigned devices)	8.3
Guest Virtualization-Based Security support (VBS)	8.0
Key storage drive	8.0
Large memory VMs	8.0
Nested Virtualization	8.0

Feature	Minimum VM configuration version
Virtual processor count	8.0
XSAVE support	8.0
Virtual Machine Multi Queues (VMMQ)	7.1
Virtual Trusted Platform Module (vTPM)	7.0
Hot Add/Remove Memory	6.2
PowerShell Direct	6.2
Production Checkpoints	6.2
Secure Boot for Linux VMs	6.2
Virtual Machine Grouping	6.2

For more information about these features, see [What's new in Hyper-V on Windows Server](#).

GPU partitioning

07/24/2025 Applies to:  [Windows Server 2025](#)

GPU partitioning allows you to share a physical GPU device with multiple virtual machines (VMs). With GPU partitioning or GPU virtualization, each VM gets a dedicated fraction of the GPU instead of the entire GPU.

The GPU partitioning feature uses the [Single Root IO Virtualization \(SR-IOV\) interface](#), which provides a hardware-backed security boundary with predictable performance for each VM. Each VM can access only the GPU resources dedicated to them and the secure hardware partitioning prevents unauthorized access by other VMs.

Beginning with Windows Server 2025, live migration is supported with GPU partitioning, enabling greater flexibility for managing virtual machines. To utilize live migration with GPU partitioning, ensure your setup meets the requirements outlined in this article. Live migration allows you to move VMs between hosts without downtime, which is essential for maintenance and load balancing in a production environment.

This feature allows planned VM migrations while maintaining GPU resource allocation, ensuring minimal downtime and consistent performance.

GPU partitioning is designed for standalone servers. You can live migrate VMs between standalone nodes for planned downtime; however, for customers that require clustering for unplanned downtime, you must use Windows Server 2025 Datacenter.

When to use GPU partitioning

Some workloads, such as virtual desktop infrastructure (VDI), Artificial Intelligent (AI) and Machine Learning (ML) inferencing require GPU acceleration, GPU partitioning can help reduce your total cost of ownership for your overall infrastructure.

For example:

- VDI applications: Distributed edge customers run basic productivity apps, such as Microsoft Office and graphics-heavy visualization workloads in their VDI environments, which require GPU acceleration. For such workloads, you can achieve the required GPU acceleration via DDA or GPU partitioning. With GPU partitioning, you can create multiple partitions and assign each partition to VM hosting a VDI environment. GPU partitioning helps you achieve the desired density and scale the number of supported users by an order of magnitude.

- Inference with ML: Customers in retail stores and manufacturing plants can run inference at the edge, which requires GPU support for their servers. Using GPU on your servers, you can run ML models to get quick results that can be acted on before the data is sent to the cloud. The full data set can optionally be transferred to continue to retrain and improve your ML models. Along with DDA, where you dedicate an entire physical GPU to a VM, GPU partitioning enables you to run multiple inference applications simultaneously on the same GPU, but in separate hardware partitions, maximizing GPU utilization.

Requirements

To use GPU partitioning with live migration, you need to have a supported CPU, operating system, and GPU. The following sections describe the requirements.

CPU requirements

Your cluster hosts need to have Input/Output Memory Management Unit (IOMMU) DMA bit tracking capable processors. For example, processors supporting Intel VT-D or AMD-Vi. If you use Windows Server and live migration without IOMMU enabled processors, the VMs are automatically restarted where GPU resources are available.

Examples of processors that support IOMMU DMA bit tracking include:

- AMD EPYC 7002 and later (Milan)
- 4th Gen Intel Xeon SP (Sapphire Rapids)

Supported guest operating systems

GPU partitioning on Windows Server 2025 and later supports these guest operating systems:

- Windows 10 or later
- Windows 10 Enterprise multi-session or later
- Windows Server 2019 or later
- Linux Ubuntu 18.04 LTS, Linux Ubuntu 20.04 LTS, Linux Ubuntu 22.04 LTS

Supported GPUs

The following GPUs support GPU partitioning:

- NVIDIA A2
- NVIDIA A10
- NVIDIA A16

- NVIDIA A40
- NVIDIA L2
- NVIDIA L4
- NVIDIA L40
- NVIDIA L40S

To use live migration with GPU partitioning, you must use the driver included in the NVIDIA vGPU Software v18.x or later. The NVIDIA driver provides the necessary support for GPU partitioning and live migration capabilities.

We recommend working with your Original Equipment Manufacturer (OEM) partners to plan and order systems tailored to your workloads. Additionally, consult GPU Independent Hardware Vendors (IHVs) to ensure you have the appropriate configurations and necessary software for your setup. However, we support more GPUs if you want to use GPU acceleration via Discrete Device Assignment (DDA). Reach out to your OEM partners and IHVs to get a list of GPUs that support DDA. For more information about using GPU acceleration via DDA, see [Discrete Device Assignment \(DDA\)](#).

For best performance, we recommend that you create a homogeneous configuration for GPUs across all the servers in your cluster. A homogeneous configuration consists of installing the same make and model of the GPU, and configuring the same partition count in the GPUs across all the servers in the cluster. For example, in a cluster of two servers with one or more GPUs installed, all the GPUs must have the same make, model, and size. The partition count on each GPU must also match.

Limitations

Consider the following limitations when using the GPU partitioning feature:

- GPU partitioning is unsupported if your configuration isn't homogeneous. Here are some examples of unsupported configurations:
 - Mixing GPUs from different vendors in the same cluster.
 - Using different GPU models from different product families from the same vendor in the same cluster.
- You can't assign a physical GPU as both [Discrete Device Assignment \(DDA\)](#) or partitionable GPU. You can either assign it as DDA or as partitionable GPU, but not both.
- If you assign more than one GPU partition to a VM, each partition appears as another GPU.

- Partitions are autoassigned to the VMs. You can't choose a specific partition for a specific VM.
- You can partition your GPU using Windows Admin Center or using PowerShell. We recommend that you use Windows Admin Center to configure and assign GPU partitions. Windows Admin Center automatically validates for a homogeneous configuration of the GPUs across all the servers in your cluster. It provides appropriate warnings and errors to take any corrective action needed.
- If using PowerShell to enable GPU partitioning, you must perform the same configuration steps on each server in the cluster. You must manually ensure that the homogeneous configuration is maintained for GPUs across all the servers in your cluster.
- When live migrating a virtual machine with a GPU partition assigned, Hyper-V live migration automatically falls back to using TCP/IP with compression. Migrating a virtual machine has the potential effect of increasing the CPU utilization of a host. In addition, live migrations could take longer than with virtual machines without GPU partitions attached.

Related content

For more information on using GPUs with your VMs and GPU partitioning, see:

- [Partition and assign GPUs to a virtual machine](#)
- [Use GPUs with clustered VMs](#)

Partition and assign GPUs to a virtual machine

Article • 01/09/2025 • Applies to:  [Windows Server 2025](#)

This article describes how to configure graphics processing unit (GPU) partitions and assign a partition to a virtual machine (VM). It provides instructions on how to configure GPU partition count, assign GPU partitions, and unassign GPU partitions via Windows Admin Center and PowerShell.

To provision the GPU partitioning feature, you need to complete the following steps:

- [Complete all the prerequisites.](#)
- [Verify GPU driver installation.](#)
- [Configure partition count.](#)
- [Assign GPU partition to a VM.](#)
- [If necessary, unassign a GPU partition from a VM.](#)

Prerequisites

There are several requirements and things to consider before you begin to use the GPU partitioning feature:

Prerequisites for the host server

- You must have Windows Server 2025 installed on the host server. If clustering is required when live migrating a virtual machine host, Windows Server 2025 Datacenter must be installed.
- The Hyper-V role installed and configured on your server. See [Install the Hyper-V role on Windows Server](#) to find out how to get started.
- Install the physical GPU device of the same make, model, and size on every server of the cluster. Refer to your OEM-provided documentation when installing the GPU device on your physical servers in the cluster.
- Install the GPU drivers on every server of the cluster by following instructions from your GPU IHVs. For NVIDIA GPU drivers, see the [NVIDIA vGPU documentation](#).
- Ensure that the virtualization support and SR-IOV are enabled in the BIOS of each server in the cluster. Reach out to your system vendor if you're unable to identify the correct setting in your BIOS.

- Cluster hosts need to have Input/Output Memory Management Unit (IOMMU) DMA bit tracking capable processors. For example, processors supporting Intel VT-D or AMD-Vi.

ⓘ Note

When live migrating a virtual machine with a GPU partition assigned, Hyper-V live migration will automatically fall back to using TCP/IP with compression. This has the potential effect of increasing the CPU utilization of a host. In addition, live migrations could take longer than with virtual machines without GPU partitions attached.

Prerequisites for the VMs

- Deploy a VM using a guest operating system from the [Supported guest operating systems](#) list.
- Install the GPU drivers on the VM by following instructions from your GPU IHVs. For NVIDIA GPU drivers, see the [NVIDIA vGPU documentation](#).

Prerequisites for Windows Admin Center

If you're using Windows Admin Center to provision GPU partitioning, you must install the latest version of [Windows Admin Center](#) with the **GPUs** extension, version 2.8.0 or later. For instructions on how to install the **GPUs** extensions in Windows Admin Center, see [Installing an extension](#).

After you install the extension, it appears under the **Installed extensions** tab as shown in the following screenshot. Make sure the version of the **GPUs** extension is 2.8.0 or later.

Extensions
Windows Admin Center might restart after installing an extension, temporarily affecting anyone using this instance of Windows Admin Center.
Automatically update extensions ☒ On

[Available extensions](#) **Installed extensions** [Feeds](#)

Uninstall Update

41 items

Search

Name ↑	Version	Created by	Status
Cluster Creation	2.70.0	Not Available	Update available (2.88.0)
Cluster Manager	2.120.0	Microsoft	Installed
Developer Guide	2.297.0	Microsoft	Installed
Devices	2.15.0	Microsoft	Installed
Events	2.36.0	Microsoft	Installed
Failover cluster tools	2.81.0	Microsoft	Installed
Files & file sharing	2.116.0	Microsoft	Installed
Firewall	2.11.0	Microsoft	Installed
GPUs	2.8.0	Microsoft	Installed
Local users & groups	2.19.0	Microsoft	Installed
Microsoft Defender for Cloud	3.0.1	Microsoft	Installed
Network Controller tools and SDN Virtual networks	1.29.0	Microsoft	Installed

Prerequisites for PowerShell

If you're using PowerShell to provision GPU partitioning, you must run all PowerShell commands as the Administrator user.

For detailed information on how to use PowerShell commands for GPU partitioning, see the [Add-VMGpuPartitionAdapter](#), [Get-VMGpuPartitionAdapter](#), and [Remove-VMGpuPartitionAdapter](#) reference documentation.

Verify GPU driver installation

After you complete all the [prerequisites](#), you must verify if the GPU driver is installed and partitionable.

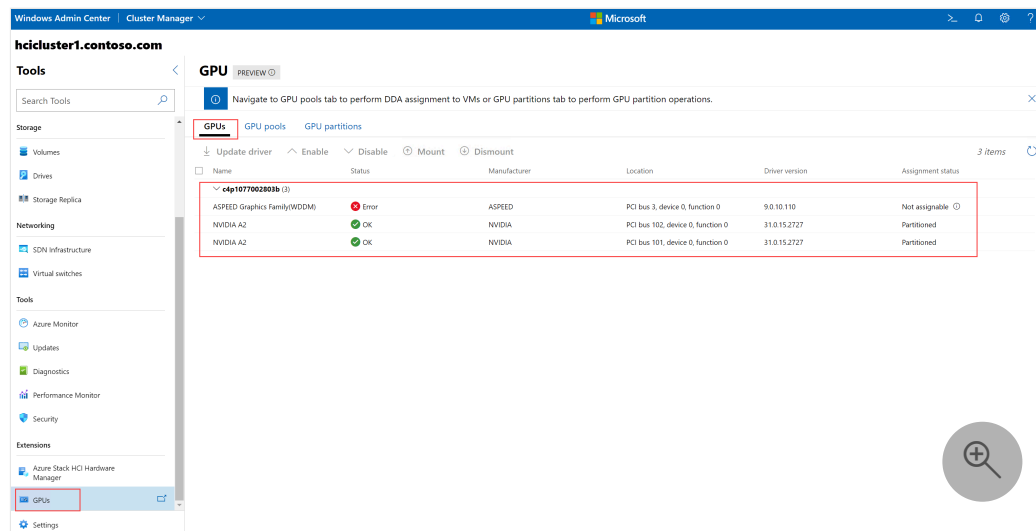
Windows Admin Center

Follow these steps to verify if the GPU driver is installed and partitionable using Windows Admin Center:

1. Launch Windows Admin Center and make sure the **GPUs** extension is already installed.
2. Select **Cluster Manager** from the top dropdown menu and connect to your cluster.
3. From the **Settings** menu, select **Extensions > GPUs**.

The **GPUs** tab on the **GPU** page displays inventory of all the servers and the physical GPUs that are installed on each server.

4. Check the **Assigned status** column for each GPU for all the servers. The **Assigned status** column can have one of these statuses:
 - **Ready for DDA assignment.** Indicates that the GPU is available for DDA assignment. You can't use it for GPU partitioning.
 - **Partitioned.** Indicates that the GPU is partitionable.
 - **Paravirtualization.** Indicates that the GPU has the partitioned driver capability installed but SR-IOV on the server isn't enabled.
 - **Not assignable.** Indicates that the GPU isn't assignable because it's an older PCI-style device or switch port.



- Proceed further in the GPU partitioning workflow only if the **Assigned status** column shows **Partitioned** for the GPUs in all the servers in your cluster.

Configure GPU partition count

Each partitionable GPU comes with a set of valid partition counts that's predefined by its OEM. You can't define the number of partitions a GPU can have. However, you can configure the partition count to any of the valid count from within the supported set.

Windows Admin Center

Follow these steps to configure partition count via Windows Admin Center:

- Select the **GPU partitions** tab to configure partition counts. You can also assign partition to VMs and unassign partitions from VMs using this tab.

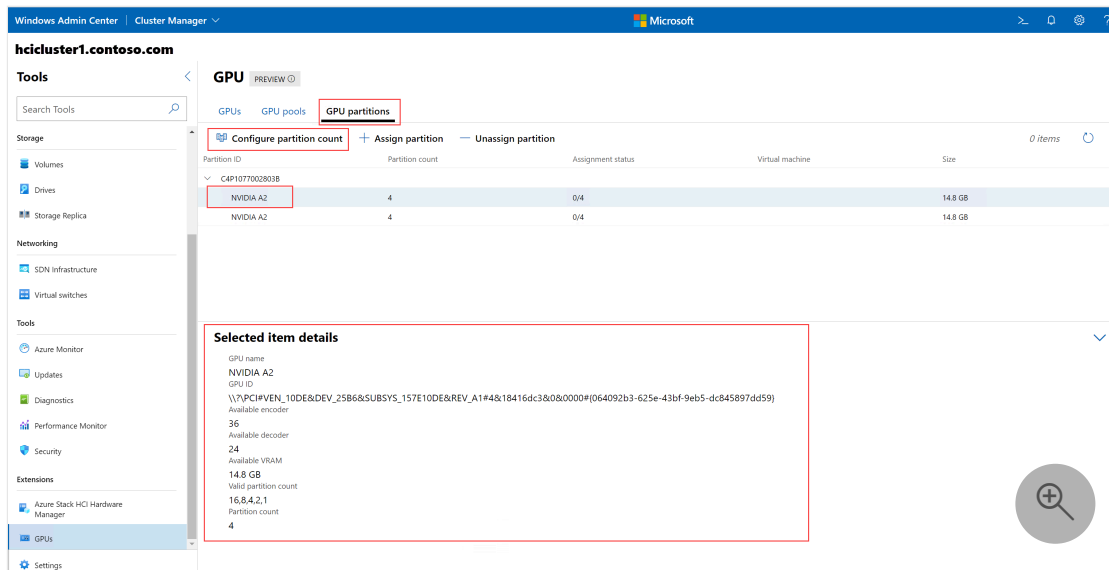
ⓘ Note

If there are no partitionable GPUs available in your cluster or the correct GPU partitioning driver isn't installed, the GPU partitions tab displays the following message:

No partitionable GPUs have been found. Please check that you have a GPU with the correct GPU-P driver to proceed.

- Select a GPU or a GPU partition to display its details in the bottom section of the page, under **Selected item details**. For example, if you select a GPU, it displays the GPU name, GPU ID, available encoder and decoder, available

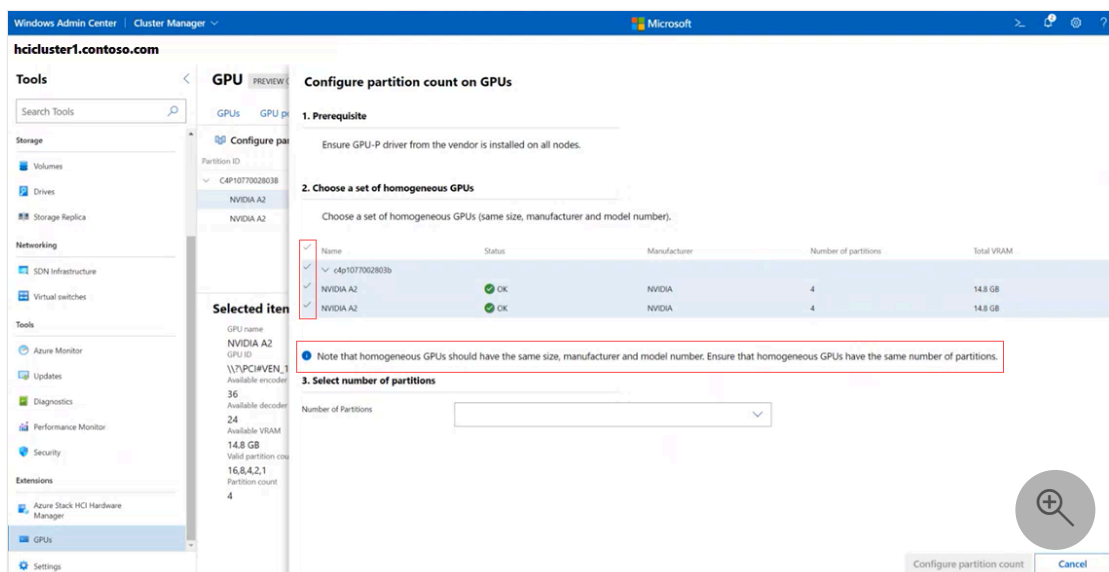
VRAM, valid partition count, and the current partition count. If you select a GPU partition, it displays the partition ID, VM ID, instance path, partition VRAM, partition encode, and partition decode.



3. Select Configure partition count.

The **Configure partition count on GPUs** page is displayed. For each server, it displays the GPU devices installed on them.

4. Select a set of homogeneous GPUs. A set of homogeneous GPUs is the one that has GPUs of the same size, manufacturer, model number, and number of partitions. By default, Windows Admin Center automatically selects a set of homogenous GPUs if it detects one, as shown in the following screenshot:



You might see a warning or an error depending on what selections you make:

- **Warning.** If you deselect one or more GPUs from the homogeneous set of GPUs, Windows Admin Center gives you a warning, but doesn't stop

you from proceeding further. Warning text indicates that you're not selecting all the GPUs and it might result in different partition count, which isn't recommended.

2. Choose a set of homogeneous GPUs

Choose a set of homogeneous GPUs (same size, manufacturer and model number).

☐	Name	Status	Manufacturer	Number of partitions	Total VRAM
☒	c4p1077002803b				
☒	NVIDIA A2	OK	NVIDIA	4	14.8 GB
☐	NVIDIA A2	OK	NVIDIA	4	14.8 GB

⚠️ Not all GPUs have been selected. This may result in different partition counts which is not recommended.

💡 Note that homogeneous GPUs should have the same size, manufacturer and model number. Ensure that homogeneous GPUs have the same number of partitions.

- **Warning.** If not all the GPUs across all the servers have the same configuration, Windows Admin Center gives a warning. You must manually select the GPUs with the same configuration to proceed further.

2. Choose a set of homogeneous GPUs

Choose a set of homogeneous GPUs (same size, manufacturer and model number).

☐	Name	Status	Manufacturer	Number of partitions	Total VRAM
☒	8S-3wp16r0125				
☒	NVIDIA A2	OK	Advanced Micro Devices, Inc.	8	16 GB
☒	8S-3wp16r0127				
☐	NVIDIA A40	OK	Advanced Micro Devices, Inc.	8	40 GB

⚠️ You have GPUs with different configuration. You must select similar GPUs to proceed.

💡 Note that homogeneous GPUs should have the same size, manufacturer and model number. Ensure that homogeneous GPUs have the same number of partitions.

- **Error.** If you select GPUs with different configurations, Windows Admin Center gives you an error, and doesn't let you proceed.

2. Choose a set of homogeneous GPUs

Choose a set of homogeneous GPUs (same size, manufacturer and model number).

☒	Name	Status	Manufacturer	Number of partitions	Total VRAM
☒	8S-3wp16r0125				
☒	NVIDIA A2	OK	Advanced Micro Devices, Inc.	8	16 GB
☒	8S-3wp16r0127				
☒	NVIDIA A40	OK	Advanced Micro Devices, Inc.	8	40 GB

❌ You have selected GPUs with different configuration. You must select similar GPUs to proceed.

⚠️ You have GPUs with different configuration. You must select similar GPUs to proceed.

💡 Note that homogeneous GPUs should have the same size, manufacturer and model number. Ensure that homogeneous GPUs have the same number of partitions.

- **Error.** If you select a GPU partition that is already assigned to a VM, Windows Admin Center gives you an error, and doesn't let you proceed. You must first unassign the partition from the VM before proceeding further. See [Unassign a partition from a VM](#).

2. Choose a set of homogeneous GPUs

Choose a set of homogeneous GPUs (same size, manufacturer and model number).

☒	Name	Status	Manufacturer	Number of partitions	Total VRAM
☒	c4p1077002803b				
☒	NVIDIA A2	OK	NVIDIA	4	14.8 GB
☒	NVIDIA A2	OK	NVIDIA	4	14.8 GB

❌ You have selected one or more GPUs with assigned partitions. You must remove the partitions from the GPUs partitions tab to proceed.

💡 Note that homogeneous GPUs should have the same size, manufacturer and model number. Ensure that homogeneous GPUs have the same number of partitions.

5. After you select a homogeneous set of GPUs, select the partition count from the **Number of Partitions** dropdown list. This list automatically populates the partition counts configured by your GPU manufacturer. The counts displayed in the list can vary depending on the type of GPU you selected.

As soon as you select a different partition count, a tooltip appears below the dropdown list, which dynamically displays the size of VRAM that each partition gets. For example, if the total VRAM is 16 GB for 16 partitions in the GPU, changing the partition count from 16 to 8 assigns each partition with 1.85 GB of VRAM.

3. Select number of partitions

Number of Partitions

 Each partition will have 1.85 GB of VRAM.

6. Select **Configure partition count**.

After the partition count is configured, Windows Admin Center notifies you that the partition count is successfully configured and displays the **GPU partitions** tab again. You can see the new partition count for the GPU partition under the **Partition count** column.

Assign GPU partition to a VM

Save your workloads before assigning partition to the VM.

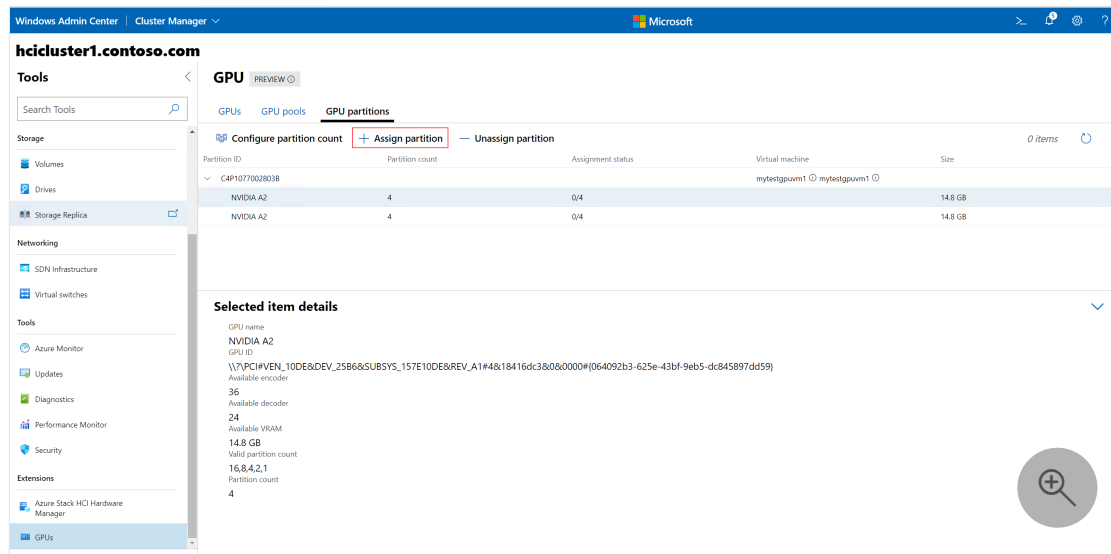
Note

Currently, you can assign only a single GPU partition to a VM. Both the VM and the GPU partition need to be on the same host machine. We recommend that you plan ahead and determine the GPU partition size based on your workload performance requirements.

Windows Admin Center

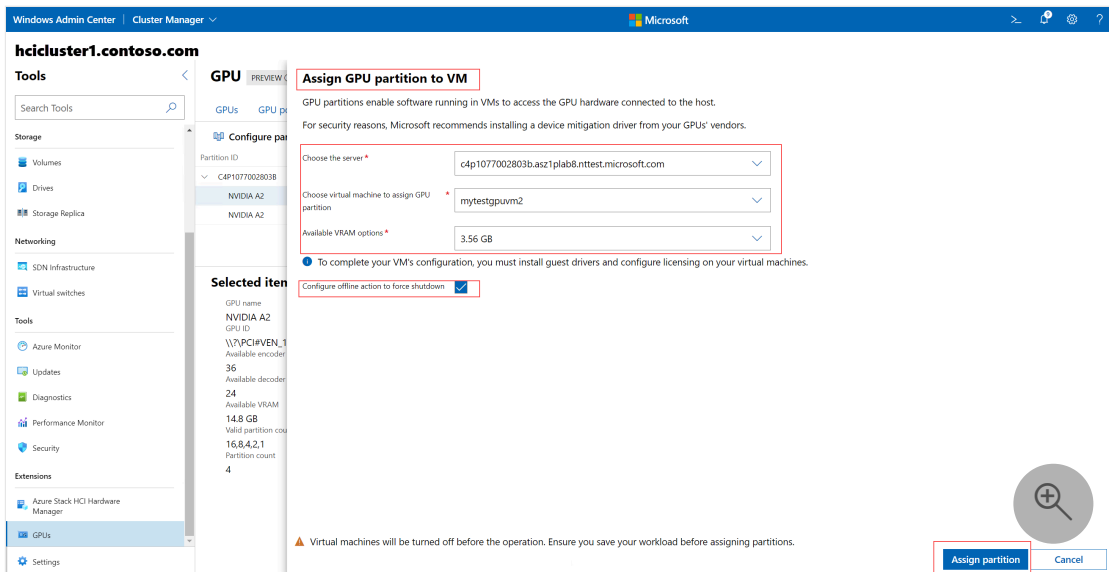
You must save your workloads before assigning partitions. If your VM is currently turned on or running, Windows Admin Center automatically turns it off, assigns the partition, and then automatically turns it on.

1. On the **GPU partitions** tab, select **+ Assign partition**.



The **Assign GPU partition to VM** page is displayed.

2. From **Choose the server** list, select the server where the VM resides. This list displays all the servers in your cluster.
3. Search for and select the VM to assign the GPU partition to. The list automatically populates the VMs that reside on the server that you selected in step 2.
 - If a GPU partition is already assigned to a VM, that VM appears as grayed out.
 - Select all the VMs at once by selecting the **Select All** checkbox.
4. Select the available VRAM options. The value in this field must match the size of the partition count that you configured.
5. (Optional, but recommended) Select the **Configure offline action for force shutdown** checkbox if you want your VM to be highly available and fail over if its host server goes down.
6. Select **Assign partition**. This assigns partition of the selected VRAM size to the selected VM on the selected host server.



After the partition is assigned, Windows Admin Center notifies you that the partition is successfully assigned and displays the **GPU partitions** tab again. On the **GPU partitions** tab, the VM appears on the GPU partition row under the server it's installed on.

Unassign a partition from a VM

You can unassign a GPU partition from the VM if you no longer need it to run your workloads. Unassigning the partition frees up the GPU partition resource, which you can reassign to another VM later.

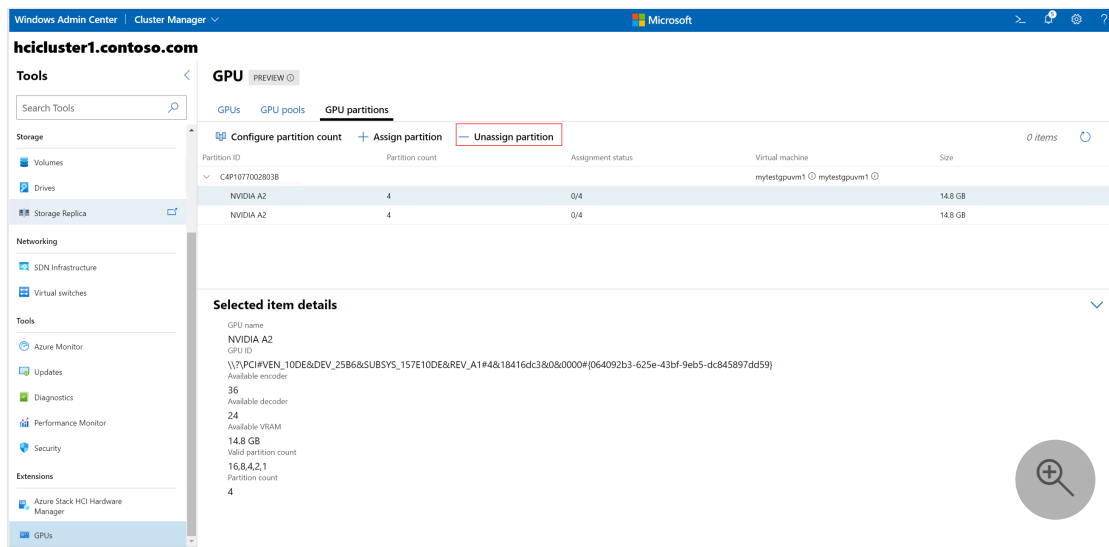
You must save your workloads before unassigning partitions.

Windows Admin Center

If your VM is currently turned on or running, Windows Admin Center automatically turns it off first, unassigns the partition, then automatically turns it on.

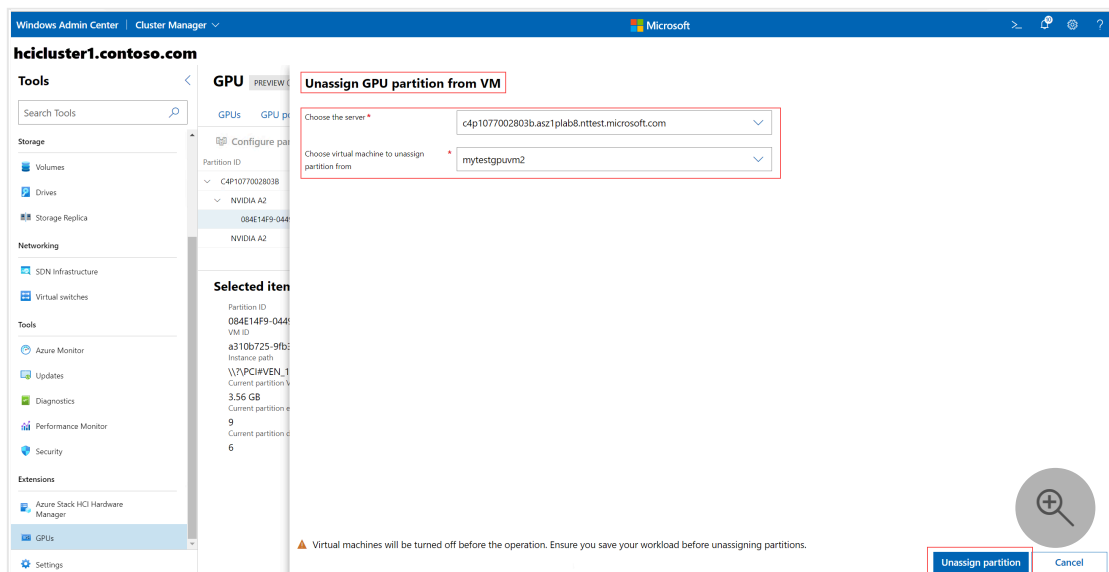
Follow these steps to unassign a partition from a VM:

1. On the **GPU partitions** tab, select the GPU partition that you want to unassign.
2. Select - **Unassign partition**.



The Unassign GPU partition from VM page is displayed.

- From **Choose the server** list, select the server that has the GPU partition that you want to unassign.
- From **Choose virtual machine to unassign partition from** list, search, or select the VM to unassign the partition from.
- Select **Unassign partition**.



After the partition is unassigned, Windows Admin Center notifies you that the partition is successfully unassigned and displays the **GPU partitions** tab again. On the **GPU partitions** tab, the VM from which the partition is unassigned no longer shows on the GPU partition row.

Was this page helpful?



Yes



No

Deploy graphics devices by using Discrete Device Assignment

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Learn how to use Discrete Device Assignment (DDA) to pass an entire PCIe device into a virtual machine (VM) on standalone (non-clustered) Hyper-V hosts with PowerShell. Doing so allows high performance access to devices like [NVMe storage](#) or graphics cards from within a VM while being able to apply the device's native drivers. For more information on devices that work and possible security implications, see [Plan for Deploying Devices using Discrete Device Assignment](#).

This article takes you through the steps to use a device with DDA:

1. [Configure the VM for DDA](#)
2. [Dismount the device from the host partition](#)
3. [Assign the device to the guest VM](#)

Important

Deploying graphics devices to clustered virtual machines? Learn about using DDA with GPU Pools in clustered environments. See [Use GPUs with Discrete Device Assignment in clustered VMs](#).

Prerequisites

Before you can use DDA to deploy graphics devices, you need to have the following.

- A Hyper-V host running Windows Server 2016 or later.
- A VM running one of the following operating systems:
 - Windows Server 2016 or later.
 - Windows 10 or later.
- Review [Plan for Deploying Devices using Discrete Device Assignment](#) to ensure your hardware is compatible with DDA.
 - Run the [SurveyDDA.ps1](#). PowerShell script to identify if the server is configured correctly. The script also displays which devices can be passed through by using Discrete Device Assignment.

- Administrative rights to the Hyper-V host.
- (Optional) Though not required, if [Single Root I/O Virtualization \(SR-IOV\)](#) isn't enabled or supported, you might encounter issues when you use DDA to deploy graphics devices.

Configure the VM for DDA

The first step in the solution is to address DDA restrictions to the VMs.

1. Sign in to the Hyper-V host as an administrator.
2. Open an elevated PowerShell prompt.
3. Configure the **Automatic Stop Action** of a VM to enable **TurnOff** with the following PowerShell cmdlet:

PowerShell

```
Set-VM -Name VMName -AutomaticStopAction TurnOff
```

VM preparation for graphics devices

Some hardware performs better if the VM is configured in a certain way. For details on whether you need the following configurations for your hardware, reach out to the hardware vendor. For more information, see [Plan for Deploying Devices using Discrete Device Assignment](#) and on this [blog post](#).

1. Enable Write-Combining on the CPU by using the following cmdlet:

PowerShell

```
Set-VM -GuestControlledCacheTypes $true -VMName VMName
```

2. Configure the 32-bit memory mapped IO (MMIO) space by using the following cmdlet:

PowerShell

```
Set-VM -LowMemoryMappedIoSpace 3Gb -VMName VMName
```

3. Configure greater than 32-bit MMIO space by using the following cmdlet:

PowerShell

```
Set-VM -HighMemoryMappedIoSpace 33280Mb -VMName VMName
```

Tip

The MMIO space values shown are reasonable values to set for experimenting with a single GPU. If after starting the VM the device is reporting an error relating to not enough resources, you'll likely need to modify these values. For more information about how to precisely calculate MMIO requirements, see [Plan for Deploying Devices using Discrete Device Assignment](#).

Dismount the device from the host partition

Follow the instructions in this section to dismount the device from the host partition.

Install the partitioning driver (optional)

DDA gives hardware vendors the ability to provide a security mitigation driver with their devices. This driver isn't the same as the device driver installed in the guest VM. It's up to the hardware vendor's discretion to provide this driver. But if they do provide a driver, install it before dismounting the device from the host partition. Reach out to the hardware vendor to see if they have a mitigation driver.

If no partitioning driver is provided, during dismount you must use the `-Force` option to bypass the security warning. For more information about the security implications, see [Plan for Deploying Devices using Discrete Device Assignment](#).

Locate the device's location path

The PCI location path is required to dismount and mount the device from the host. An example location path looks like this: `PCIROOT(20)#PCI(0300)#PCI(0000)#PCI(0800)#PCI(0000)`. For more information about locating the location path, see [Plan for Deploying Devices using Discrete Device Assignment](#).

Disable the device

Use Device Manager or PowerShell to ensure the device is **Disabled**.

Dismount the device

Depending on whether the vendor provided a mitigation driver, you must either use the `-Force` option or not, as shown here:

- If a mitigation driver was installed, use the following cmdlet:

PowerShell

```
Dismount-VMHostAssignableDevice -LocationPath $locationPath
```

- If a mitigation driver wasn't installed, use the following cmdlet:

PowerShell

```
Dismount-VMHostAssignableDevice -Force -LocationPath $locationPath
```

Assign the device to the guest VM

The final step is to tell Hyper-V that a VM should have access to the device. Specify the location path and the name of the VM.

PowerShell

```
Add-VMAssignableDevice -LocationPath $locationPath -VMName VMName
```

Complete tasks on the VM

After a device is successfully mounted in a VM, you're now able to start that VM and interact with the device as though you were running on a bare metal system. You're now able to install the hardware vendor's drivers in the VM, and applications are able to see the hardware. You can verify it by opening Device Manager in the guest VM and seeing that the hardware is available.

Remove a device and return it to the host

If you want to return the device back to its original state, you must stop the VM and issue this command:

PowerShell

```
# Remove the device from the VM
Remove-VMAssignableDevice -LocationPath $locationPath -VMName VMName

# Mount the device back in the host
Mount-VMHostAssignableDevice -LocationPath $locationPath
```

You can then re-enable the device in Device Manager, and the host operating system is able to interact with the device again.

Example - Mount a GPU to a VM

This example uses PowerShell to configure a VM named **ddatest1** to take the first GPU available by the manufacturer NVIDIA and assign it into the VM.

PowerShell

```
# Configure the VM for a Discrete Device Assignment
$vm = "ddatest1"
# Set automatic stop action to TurnOff
Set-VM -Name $vm -AutomaticStopAction TurnOff
# Enable Write-Combining on the CPU
Set-VM -GuestControlledCacheTypes $true -VMName $vm
# Configure 32 bit MMIO space
Set-VM -LowMemoryMappedIoSpace 3Gb -VMName $vm
# Configure Greater than 32 bit MMIO space
Set-VM -HighMemoryMappedIoSpace 33280Mb -VMName $vm

# Find the Location Path and disable the Device
# Enumerate all PNP Devices on the system
$pnpdevs = Get-PnpDevice -presentOnly
# Select only those devices that are Display devices manufactured by NVIDIA
$gpudev = $pnpdevs | Where-Object {$_.Class -like "Display" -and $_.Manufacturer -like "NVIDIA"}
# Select the location path of the first device that's available to be dismounted by the host.
$locationPath = ($gpudev | Get-PnpDeviceProperty DEVPKEY_Device_LocationPaths).data[0]
# Disable the PNP Device
Disable-PnpDevice -InstanceId $gpudev[0].InstanceId

# Dismount the Device from the Host
Dismount-VMHostAssignableDevice -Force -LocationPath $locationPath

# Assign the device to the guest VM.
Add-VMAssignableDevice -LocationPath $locationPath -VMName $vm
```

Troubleshoot issues with mounting a GPU

If you pass a GPU into a VM but Remote Desktop Services or an application isn't recognizing the GPU, check for the following common issues.

- Make sure you install the most recent version of the GPU vendor's supported driver, and that the driver isn't reporting errors. You can do so by checking the device state in Device Manager.

- Make sure your device has enough MMIO space allocated within the VM. For more information, see [MMIO Space](#).
- Make sure you use a GPU that the vendor supports being used in this configuration. For example, some vendors prevent their consumer cards from working when passed through to a VM.
- Make sure the application supports running inside a VM, and that the application supports both the GPU and its associated drivers. Some applications have allowlists of GPUs and environments.
- If you use the Remote Desktop Session Host role or Windows Multipoint Services on the guest, you must make sure that a specific Group Policy entry is set to allow use of the default GPU. Use a Group Policy Object applied to the guest (or the Local Group Policy Editor on the guest). Navigate to the following Group Policy item:

Computer Configuration\Administrator Templates\Windows Components\Remote Desktop Services\Remote Desktop Session Host\Remote Session Environment\Use hardware graphics adapters for all Remote Desktop Services sessions.

Set the Group Policy value to **Enabled**, then reboot the VM after you apply the policy.

Deploy NVMe Storage Devices using Discrete Device Assignment

Article • 11/01/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local](#), versions 23H2 and 22H2

Starting with Windows Server 2016, you can use Discrete Device Assignment, or DDA, to pass an entire PCIe Device into a VM. This will allow high performance access to devices like NVMe storage or Graphics Cards from within a VM while being able to leverage the devices native drivers. Please visit the [Plan for Deploying Devices using Discrete Device Assignment](#) for more details on which devices work, what are the possible security implications, etc. There are three steps to using a device with DDA:

- Configure the VM for DDA
- Dismount the Device from the Host Partition
- Assigning the Device to the Guest VM

All command can be executed on the Host on a Windows PowerShell console as an Administrator.

Configure the VM for DDA

Discrete Device Assignment imposes some restrictions to the VMs and the following step needs to be taken.

1. Configure the "Automatic Stop Action" of a VM to TurnOff by executing

```
Set-VM -Name VMName -AutomaticStopAction TurnOff
```

Dismount the Device from the Host Partition

Locating the Device's Location Path

The PCI Location path is required to dismount and mount the device from the Host. An example location path looks like the following:

```
"PCIROOT(20)#PCI(0300)#PCI(0000)#PCI(0800)#PCI(0000)". More details on located the
```

Location Path can be found here: [Plan for Deploying Devices using Discrete Device Assignment](#).

Disable the Device

Using Device Manager or PowerShell, ensure the device is “disabled.”

Dismount the Device

```
Dismount-VMHostAssignableDevice -LocationPath $locationPath
```

Assigning the Device to the Guest VM

The final step is to tell Hyper-V that a VM should have access to the device. In addition to the location path found above, you'll need to know the name of the vm.

```
Add-VMAssignableDevice -LocationPath $locationPath -VMName VMName
```

What's Next

After a device is successfully mounted in a VM, you're now able to start that VM and interact with the device as you normally would if you were running on a bare metal system. You can verify this by opening device manager in the Guest VM and seeing that the hardware now shows up.

Removing a Device and Returning it to the Host

If you want to return the device back to its original state, you will need to stop the VM and issue the following:

```
#Remove the device from the VM  
Remove-VMAssignableDevice -LocationPath $locationPath -VMName VMName
```



```
#Mount the device back in the host  
Mount-VMHostAssignableDevice -LocationPath $locationPath
```

You can then re-enable the device in device manager and the host operating system will be able to interact with the device again.

Feedback

Was this page helpful?

 Yes

 No

Run Hyper-V in a Virtual Machine with Nested Virtualization

07/25/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Nested virtualization enables you to run Hyper-V inside a virtual machine, allowing you to emulate complex environments without needing multiple physical hosts. This article explains how to configure and use nested virtualization on supported Windows platforms, including prerequisites, setup steps, and networking options. Use this article to test scenarios, run emulators, or develop solutions that require multiple layers of virtualization.

To learn more about Nested Virtualization and supported scenarios, see [What is Nested Virtualization for Hyper-V?](#).

Prerequisites

Intel processor with VT-x and EPT technology

- The Hyper-V host must be either Windows Server 2016 or later, or Windows 10 or later.
- VM configuration version 8.0 or higher.

AMD EPYC / Ryzen processor or later

- The Hyper-V host must be either Windows Server 2022 or later, or Windows 11 or later.
- VM configuration version 9.3 or higher.

Note

The guest can be any Windows supported guest operating system. Some newer versions of Windows can use extra CPU features that improve performance. To enable Nested Virtualization in an Azure VM, make sure to set Security Type as "**Standard**".

Enable Nested Virtualization

To enable nested virtualization, follow these steps:

1. Create a virtual machine. See the prerequisites for the required OS and VM versions.

2. While the virtual machine is in the OFF state, run the following command on the physical Hyper-V host to enable nested virtualization for the virtual machine.

PowerShell

```
Set-VMProcessor -VMName <VMName> -ExposeVirtualizationExtensions $true
```

3. Start the virtual machine.
4. Install Hyper-V within the virtual machine, just like you would for a physical server. For more information on installing Hyper-V, see, [Install Hyper-V](#).

ⓘ Note

With Windows Server 2019 and earlier as the first level VM, the number of vCPUs should be 225 or less. To learn more about virtual machine limits, see [Maximums for virtual machines](#).

Disable Nested Virtualization

You can disable nested virtualization for a stopped virtual machine using the following PowerShell command:

PowerShell

```
Set-VMProcessor -VMName <VMName> -ExposeVirtualizationExtensions $false
```

Networking options

There are two options for networking with nested virtual machines:

1. MAC address spoofing
2. NAT networking

MAC address spoofing

In order for network packets to be routed through two virtual switches, MAC address spoofing must be enabled on the first (L1) level of virtual switch. To enable MAC address spoofing, run the following PowerShell command.

PowerShell

```
Get-VMNetworkAdapter -VMName <VMName> | Set-VMNetworkAdapter -MacAddressSpoofing On
```

Network Address Translation (NAT)

The second option relies on network address translation (NAT). This approach is best suited for cases where MAC address spoofing isn't possible, like in a public cloud environment.

First, a virtual NAT switch must be created in the host virtual machine (the "middle" VM). The following example creates a new internal switch named `VmNAT` and creates a NAT object for all IP addresses in the `192.168.100.0/24` subnet.

PowerShell

```
New-VMSwitch -Name VmNAT -SwitchType Internal  
New-NetNat -Name LocalNAT -InternalIPInterfaceAddressPrefix "192.168.100.0/24"
```

Next, assign an IP address to the net adapter:

PowerShell

```
Get-NetAdapter "vEthernet (VmNat)" | New-NetIPAddress -IPAddress 192.168.100.1 -  
AddressFamily IPv4 -PrefixLength 24
```

Each nested virtual machine must have an IP address and gateway assigned to it. The gateway IP must point to the NAT adapter from the previous step. You might also want to assign a DNS server:

PowerShell

```
Get-NetAdapter "vEthernet (VmNat)" | New-NetIPAddress -IPAddress 192.168.100.2 -  
DefaultGateway 192.168.100.1 -AddressFamily IPv4 -PrefixLength 24  
Netsh interface ip add dnsserver "vEthernet (VmNat)" address=<my DNS server>
```

Next steps

- [Remotely manage Hyper-V hosts with Hyper-V Manager](#)

Windows Server storage architectures with Hyper-V

Article • 01/17/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2 and 22H2

Learn how you can use Windows Server with SAN, NAS, fiber channel, iSCSI, SMB storage for Hyper-V. Windows Server Hyper-V is a mature hypervisor platform that supports a wide range of physical storage architectures, from standalone systems with no resiliency to clustered systems with complex resiliency requirements. You'll also learn about common configurations using hyperconverged and disaggregated storage architectures.

This article doesn't describe every possible storage architecture configuration. It also doesn't describe less common types of configuration or edge cases.

SAN or NAS storage

With Windows Server in a disaggregated configuration using SAN or NAS storage, you install Hyper-V in a cluster where VMs access their storage over the network.

When using this configuration:

- VMs are highly available to any node in the same compute cluster.
- SAN and NAS systems provide their own availability guarantees.

In this model, the compute and storage scales independently from one another. For example, if you need more processing or memory resources to host your VMs (CPU or RAM), you can add more compute nodes or clusters without adding more storage. Each compute cluster can contain between 1 and 64 nodes.

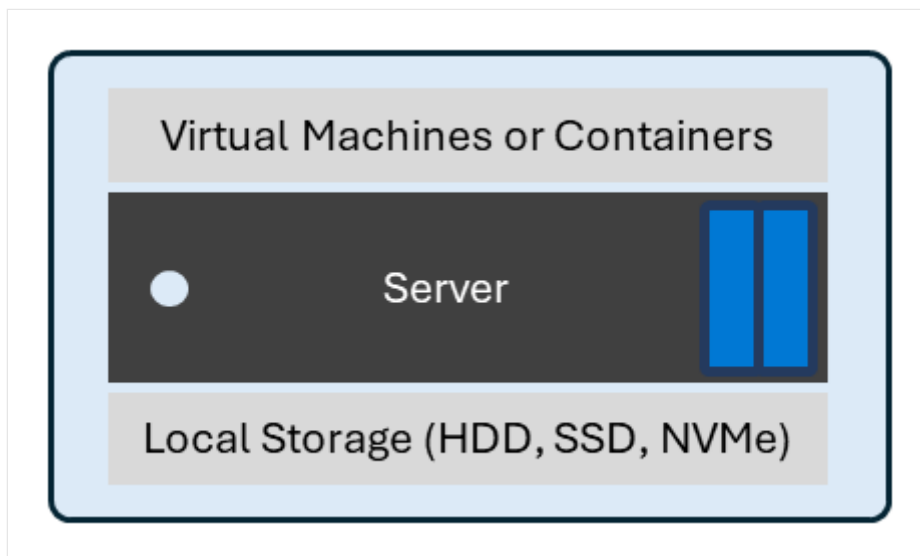
You can scale storage independently from the compute cluster. Storage cluster scaling is vendor-specific. Contact your vendors to understand how the storage solution they offer can scale.

Standalone with local disks

Windows Server supports several nonclustered or standalone configurations where you install Hyper-V on a standalone server and use local storage.

When using this configuration:

- Standalone Hyper-V with local disks doesn't support automatic failover of virtual machine (VM) workloads due to the lack of shared storage and the compute (VM processing and memory) systems being nonclustered. For example, if the physical host loses power, the VMs running on it restart when you power the physical host back on. You can also live migrate VMs to other nodes or clusters using [shared-nothing](#) migration.
- You can optionally configure local disk resiliency for local drives using hardware or software RAID solutions. If you need more information, we recommend you contact your storage vendor.



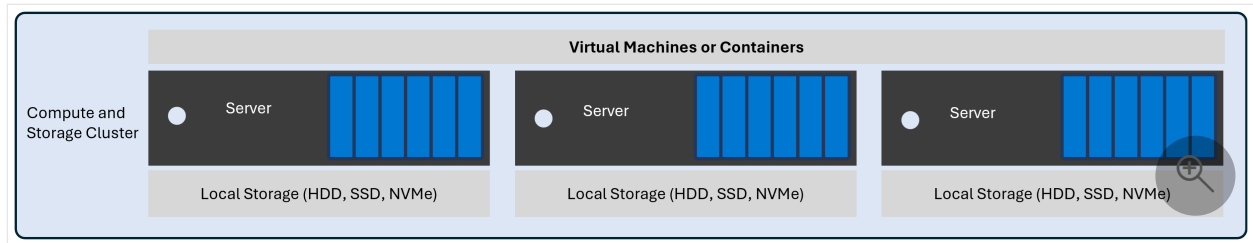
Hyperconverged and Storage Spaces Direct

In this configuration, you install Hyper-V and Storage Spaces Direct on each node in a cluster. Each node contains local disks with data replicated to other nodes in the same cluster.

When using this configuration:

- You can move VMs to any node in the cluster using Live Migrate or automatically restart after a failure (failover).
- Data gets replicated to other nodes in the cluster to increase storage resilience. Storage Spaces Direct supports several storage resiliency models. For more information, see [Fault tolerance and storage efficiency on Azure Local and Windows Server clusters](#).

- In this model, each physical host contains storage and compute resources. As a result, those physical resources scale symmetrically. Every new host automatically adds both compute and storage resources. Each cluster can contain between 1 and 16 nodes.



Disaggregated Storage Spaces Direct

In this configuration, you install Hyper-V and Storage Spaces Direct in separate clusters and access the VMs' configuration and storage over the network.

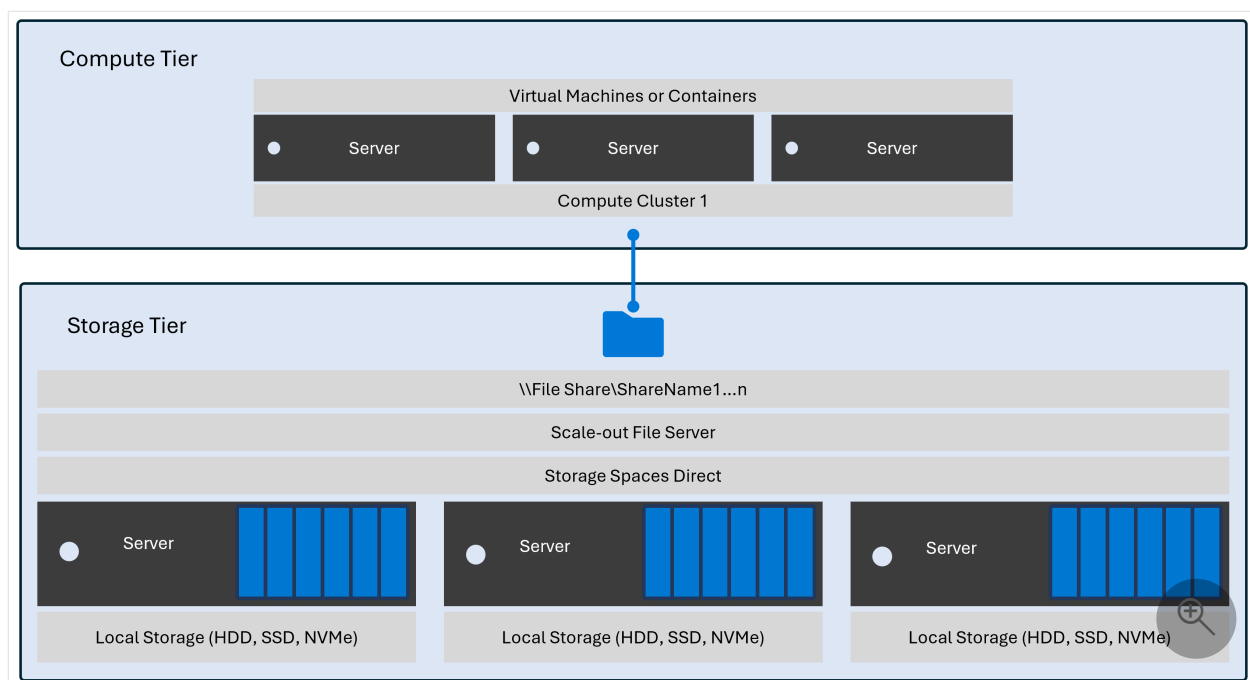
When using this configuration:

- VMs are highly available to any node in the same compute cluster.
- Data gets replicated to other nodes in the storage cluster to increase storage resilience. Storage Spaces Direct supports several storage resiliency models. For more information, see [Fault tolerance and storage efficiency on Azure Local and Windows Server clusters](#).

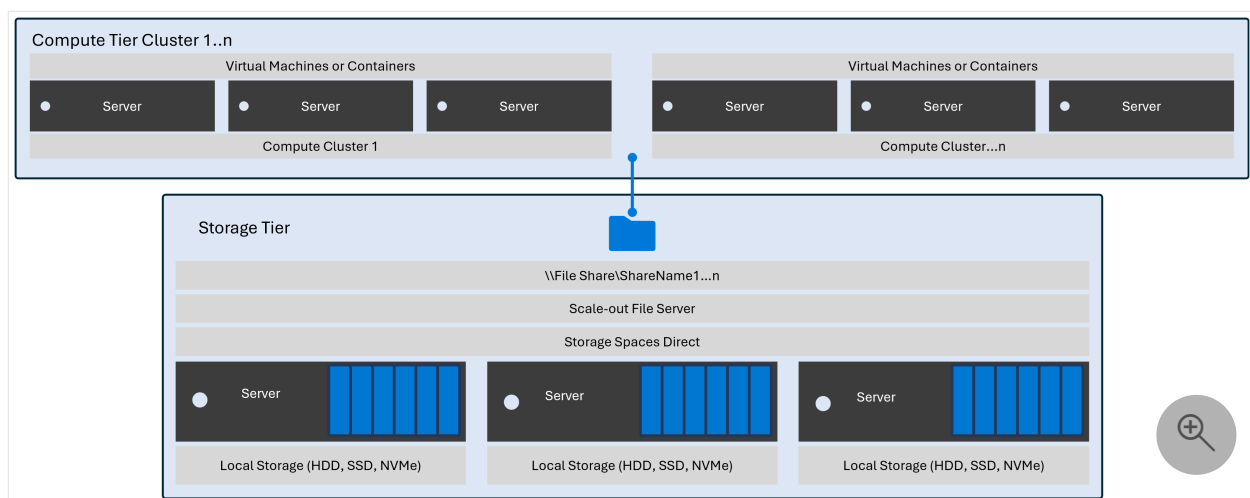
In this model, compute and storage scale independently from each other. This architecture is ideal for scenarios where your storage and compute requirements grow at different rates. For example, if you need more CPU or memory resources to host VMs, you can add extra cluster nodes without also adding more storage. If you have VMs that consume a lot of storage but don't consume many CPU resources, you can add another storage node or cluster without adding more compute resources.

- Each compute cluster can contain between 1 and 64 nodes.
- Each storage cluster can contain between 1 and 16 nodes.

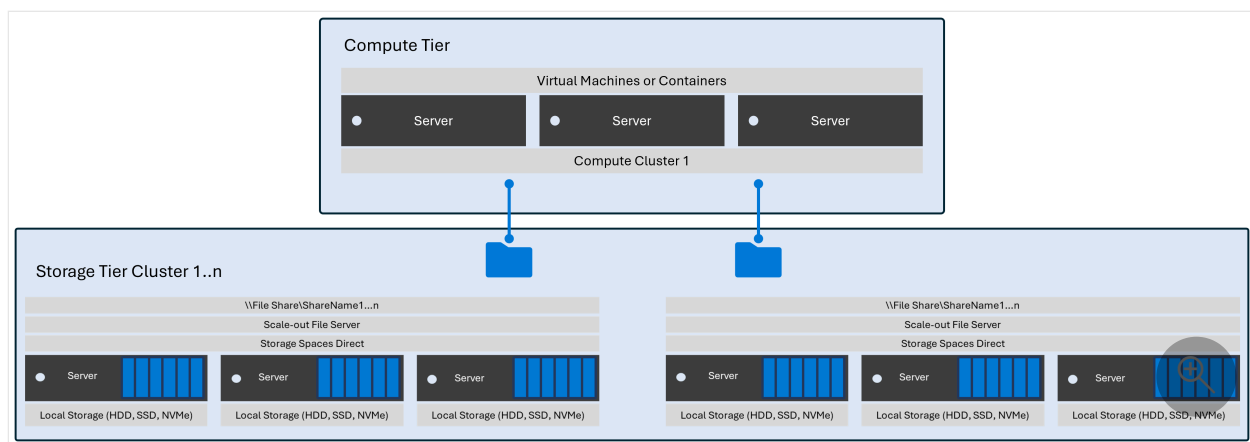
The following diagram shows a simple example deployment with one compute cluster and one storage cluster.



When you add more compute resources to run your VMs, you can either add a new node to the existing compute cluster or add a new cluster. The following diagram shows what happens to the simple deployment if you add a new cluster to it without adding more storage.



When you add more storage resources for hosting data without adding more compute resources, you can either add a new node to the existing storage cluster or add a new cluster. The following diagram shows what the simple deployment looks like when you add a new cluster without adding compute resources like CPU or RAM.

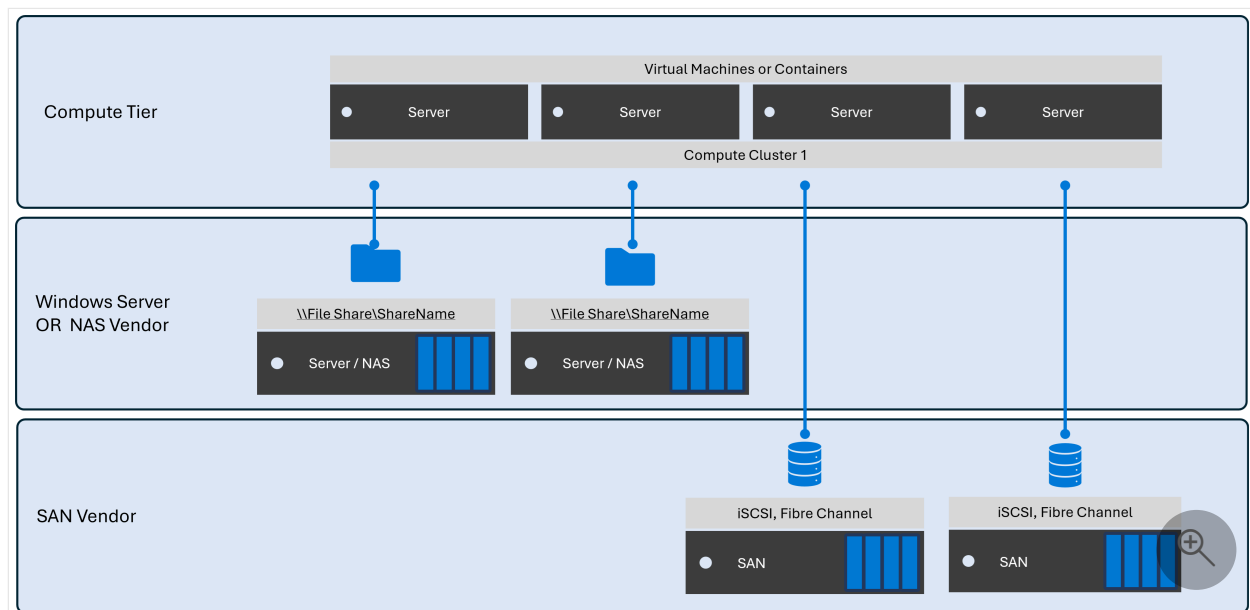


Mixed architecture support

Hyper-V supports combining the following architectures in the same compute cluster:

- Hyper-V with Disaggregated Storage Spaces Direct
- Disaggregated Hyper-V with SAN
- Disaggregated Hyper-V with NAS

The following diagram shows an example of a deployment with a compute cluster containing a mix of disaggregated SAN and NAS storage.



Network Storage Protocols

Windows Server supports the following network file storage protocols:

- [SMB over TCP](#)
- [SMB over QUIC](#)

- [SMB over RDMA \(SMB Direct\)](#)

Windows Server also supports the following network block storage protocols:

- iSCSI
- Fibre Channel
- InfiniBand

Note

Configuration specifics ultimately determine whether your deployment can support these protocols. For example, deployments that use the Hyper-V virtual switch don't support InfiniBand. However, they can support InfiniBand devices when they aren't bound to the virtual switch.

Microsoft also provides an in-box software-based iSCSI initiator for network block storage.

You can also use a storage vendor client for any device available in the Windows Server catalog.

Feedback

Was this page helpful?

 Yes

 No

Use GPUs with Discrete Device Assignment in clustered VMs

Applies to:  [Windows Server 2025](#),  [Azure Local 2311.2 and later](#)

You can include graphics processing units (GPUs) in your clusters to provide GPU acceleration to workloads running in clustered VMs. GPU acceleration can be provided via Discrete Device Assignment (DDA), which allows you to dedicate one or more physical GPUs to a VM, or through GPU Partitioning. Clustered VMs can take advantage of GPU acceleration, and clustering capabilities such as high availability via failover.

In this article, you'll learn how to use GPUs with clustered VMs to provide GPU acceleration to workloads using Discrete Device Assignment. This article guides you through preparing the cluster, assigning a GPU to a cluster VM, and failing over that VM using Windows Admin Center and PowerShell.

Tip

Live migration of virtual machines (VMs) using GPUs provided by DDA isn't currently supported, but VMs can be automatically restarted and placed where GPU resources are available if there's a failure. Looking to use Live Migration in clustered VMs? Consider using GPU partitioning. GPU partitioning allows you to share a fraction of the GPU instead of the entire GPU. To learn more about when to use GPU partition and support for live migration, see [Partition and assign GPUs to a virtual machine](#).

Prerequisites

There are several requirements and things to consider before you begin to use GPUs with clustered VMs:

- You need a Windows Server Failover cluster running Windows Server 2025 or later.
- Windows Server 2025 Datacenter edition is required for using DDA with GPU Pools in clustered VMs.
- You must have a familiarity with Failover clustering and Hyper-V.
- You must install the same make and model of the GPUs across all the servers in your cluster.

- Review and follow the instructions from your GPU manufacturer to install the necessary drivers and software on each server in the cluster.
- Depending on your hardware vendor, you might also need to configure any GPU licensing requirements.
- You need a machine with Windows Admin Center installed. This machine could be one of your cluster nodes.
- Create a VM to assign the GPU to. Prepare that VM for DDA by setting its cache behavior, stop action, and memory-mapped I/O (MMIO) properties according to the instructions in [Deploy graphics devices using Discrete Device Assignment](#).
- Prepare the GPUs in each server by installing security mitigation drivers on each server, disabling the GPUs, and dismounting them from the host. To learn more about this process, see [Deploy graphics devices by using Discrete Device Assignment](#).

Prepare the cluster

When the [prerequisites](#) are complete, you can prepare the cluster to use GPUs with clustered VMs.

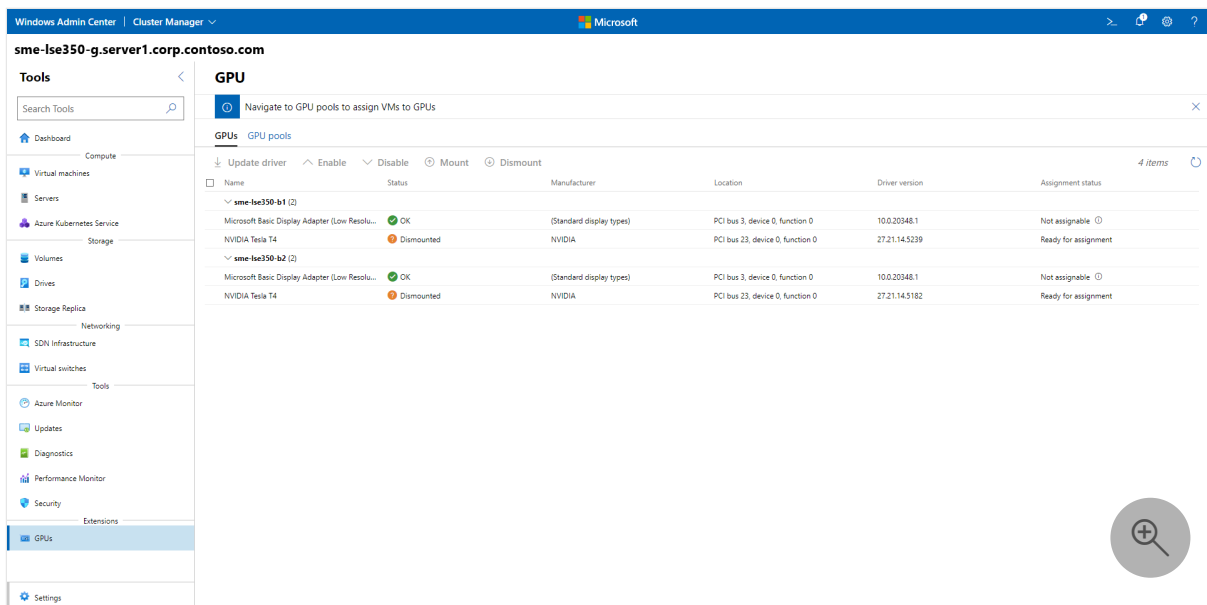
Preparing the cluster involves creating a resource pool that contains the GPUs that are available for assignment to VMs. The cluster uses this pool to determine VM placement for any started or moved VMs that are assigned to the GPU resource pool.

Windows Admin Center

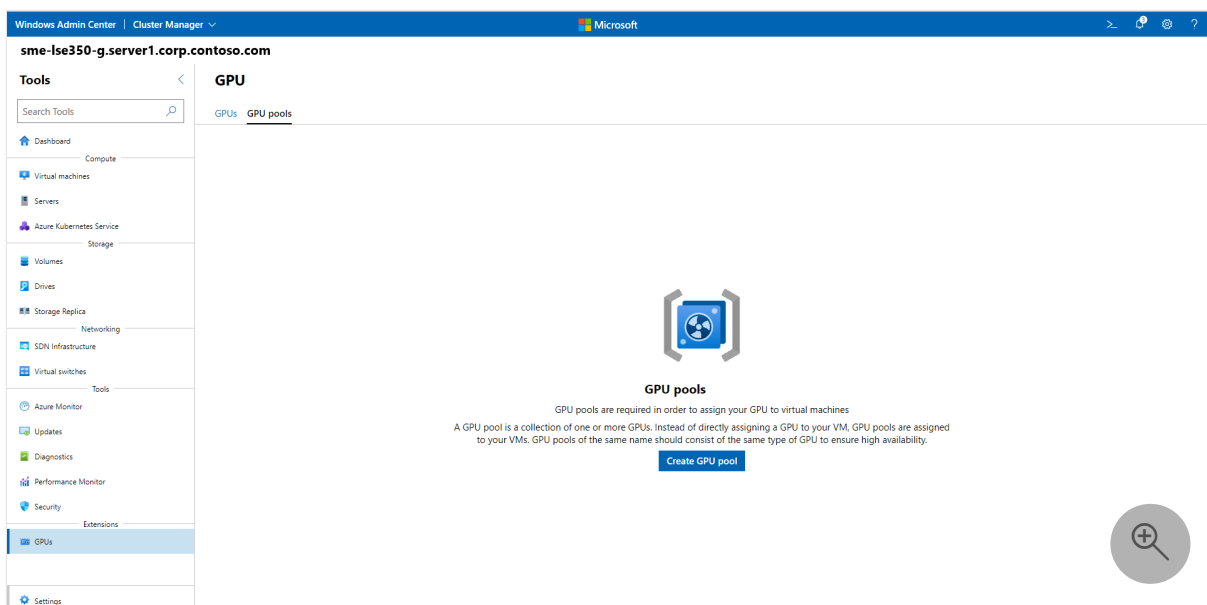
Using Windows Admin Center, follow these steps to prepare the cluster to use GPUs with clustered VMs.

To prepare the cluster and assign a VM to a GPU resource pool:

1. Launch Windows Admin Center and make sure the **GPUs** extension is already installed.
2. Select **Cluster Manager** from the top dropdown menu and connect to your cluster.
3. From the **Settings** menu, select **Extensions > GPUs**.
4. On the **Tools** menu, under **Extensions**, select **GPUs** to open the tool.

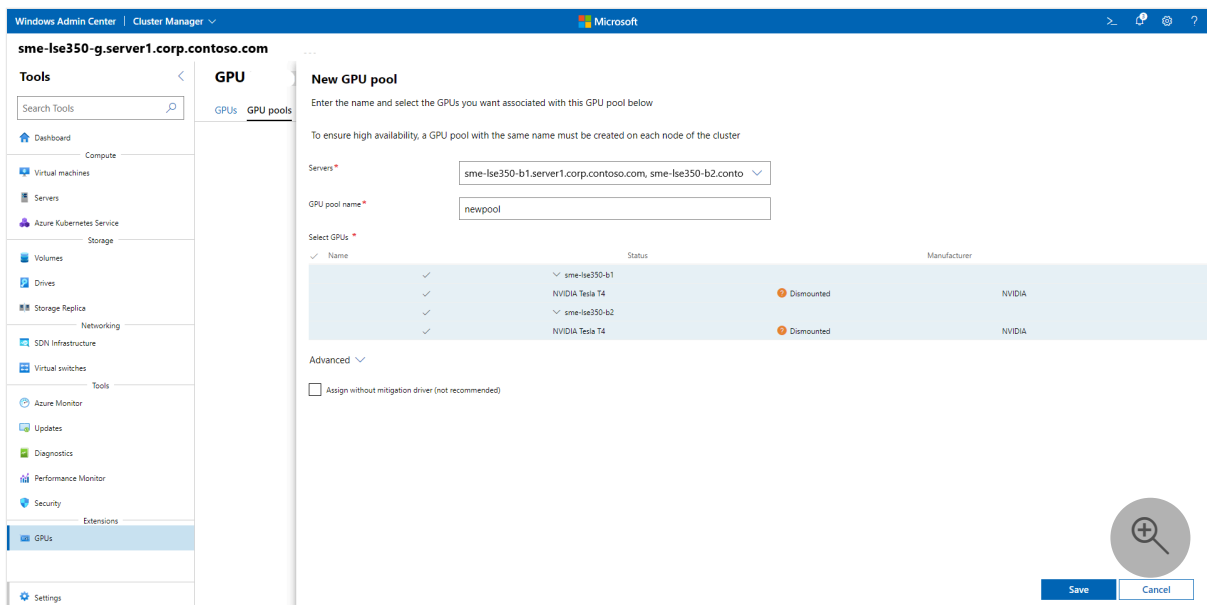


5. On tool's main page, select the **GPU pools** tab, and then select **Create GPU pool**.



6. On the **New GPU pool** page, specify the following and then select **Save**:

- Server name
- GPU pool name
- GPUs that you want to add to the pool



After the process completes, you'll receive a success prompt that shows the name of the new GPU pool and the host server.

Assign a VM to a GPU resource pool

You can now assign a VM to a GPU resource pool. You can assign one or more VMs to a clustered GPU resource pool, and remove a VM from a clustered GPU resource pool.

Windows Admin Center

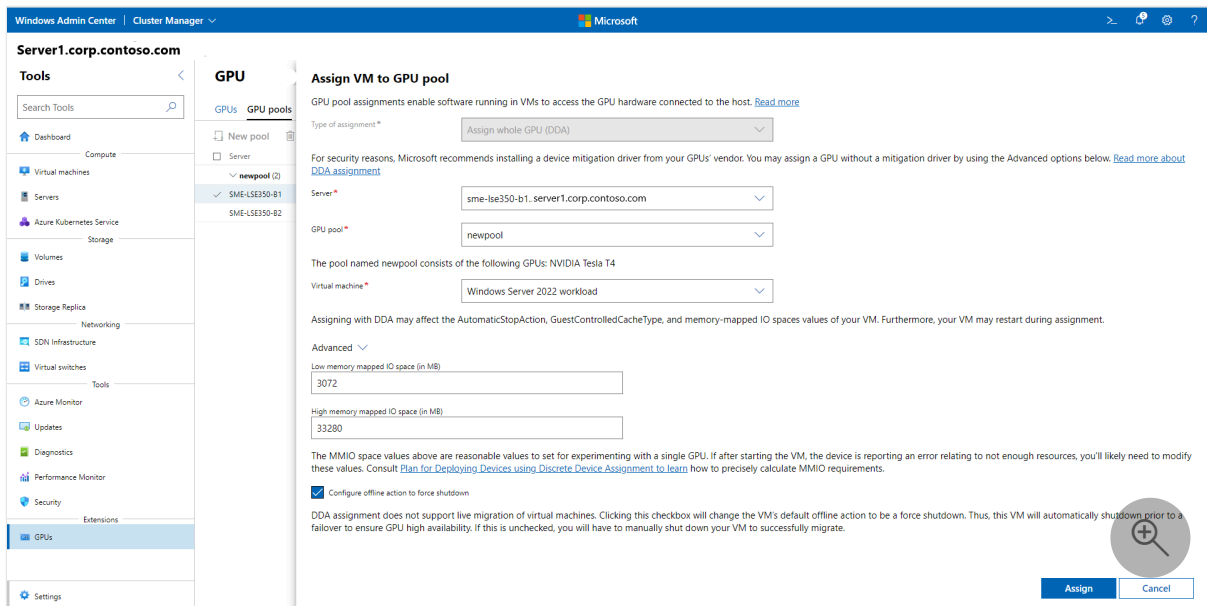
Follow these steps to assign an existing VM to a GPU resource pool using Windows Admin Center.

ⓘ Note

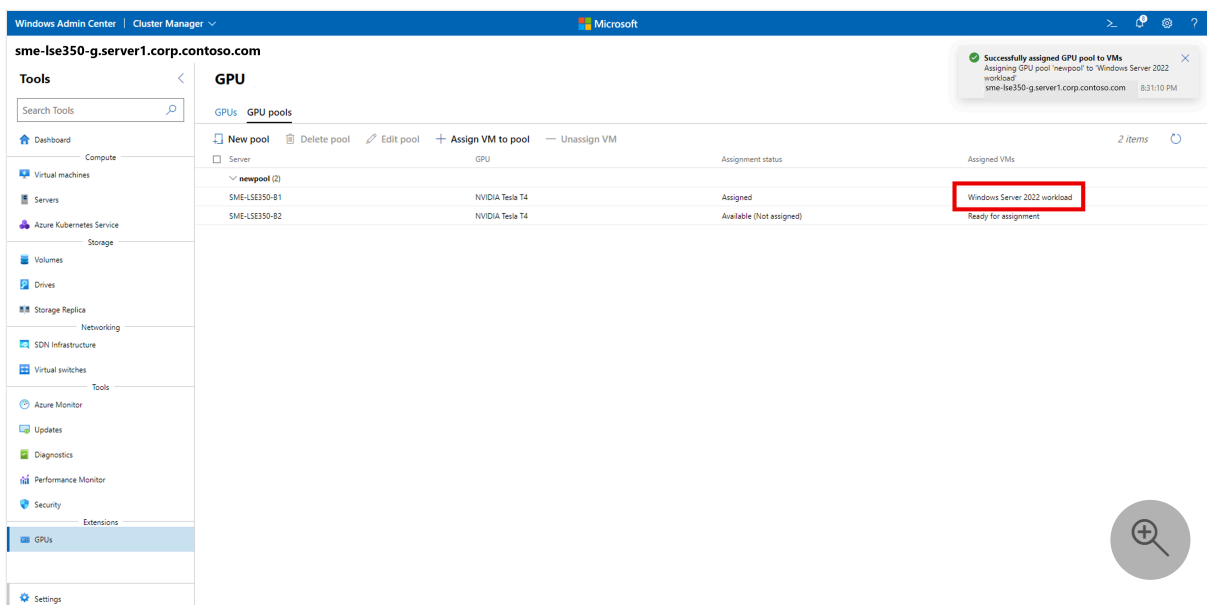
You also need to install drivers from your GPU manufacturer inside the VM so that apps in the VM can take advantage of the GPU assigned to them.

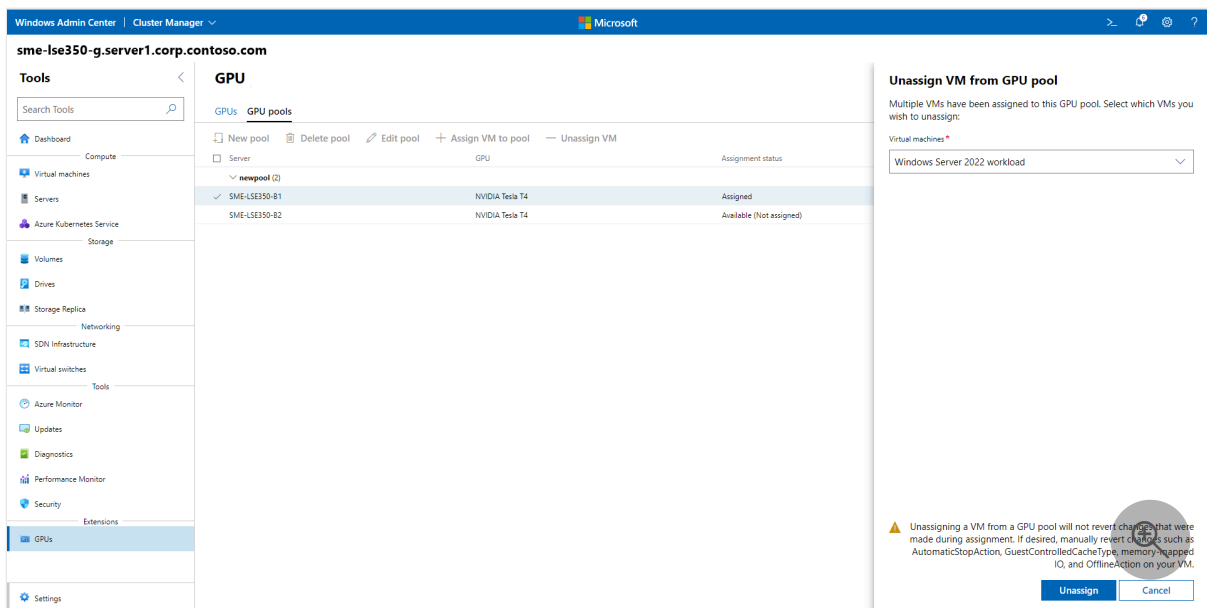
1. On the **Assign VM to GPU pool** page, specify the following, then select **Assign**:
 - a. **Server name**
 - b. **GPU pool name**
 - c. **Virtual machine** that you want to assign the GPU to from the GPU pool.

You can also define advanced setting values for memory-mapped IO (MMIO) spaces to determine resource requirements for a single GPU.



After the process completes, you'll receive a confirmation prompt that shows you successfully assigned the GPU from the GPU resource pool to the VM, which displays under **Assigned VMs**.





After the process completes, you receive a success prompt that the VM has been unassigned from the GPU pool, and under **Assignment status** the GPU shows **Available (Not assigned)**.

When you start the VM, the cluster ensures that the VM is placed on a server with available GPU resources from this cluster-wide pool. The cluster also assigns the GPU to the VM through DDA, which allows the GPU to be accessed from workloads inside the VM.

Fail over a VM with an assigned GPU

To test the cluster's ability to fail over your GPU workload, perform a drain operation on the server where the VM is running with an assigned GPU. Performing a drain operation on the server causes the cluster to restart the VM on another server in the cluster, as long as another server has sufficient available resources in the pool that you created.

To drain the server, follow the instructions in [Failover cluster maintenance procedures](#). The cluster restarts the VM on another server in the cluster, as long as another server has sufficient available GPU resources in the pool that you created.

Related content

For more information on using GPUs with your clustered VMs, see:

- [Manage VMs with Windows Admin Center](#)
- [Plan for deploying devices by using Discrete Device Assignment](#)

Deploy devices by using Discrete Device Assignment

Article • 02/19/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local](#), versions 23H2 and 22H2

Discrete Device Assignment allows physical Peripheral Component Interconnect Express (PCIe) hardware to be directly accessible from within a virtual machine (VM). This article discusses the type of devices that can be used, host system requirements, limitations imposed on the VMs, and security implications.

For Discrete Device Assignment, Microsoft supports two device classes: Graphics Adapters and NVMe Storage devices. Other devices are likely to work, and hardware vendors are able to offer statements of support for those devices. For other devices, contact specific hardware vendors for support.

To learn about other methods of GPU virtualization, see [Plan for GPU acceleration in Windows Server](#). If you're ready to try Discrete Device Assignment, you can go to [Deploy graphics devices using Discrete Device Assignment](#) or [Deploy NVMe Storage Devices using Discrete Device Assignment](#).

Supported VMs and guest operating systems

Discrete Device Assignment is supported for Generation 1 or 2 VMs. The guests supported include:

- Windows 10 or later
- Windows Server 2016 or later

For more information, see [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows Server and Windows](#).

System requirements

Your system must meet the [Hardware Requirements for Windows Server](#) and [System Requirements for Hyper-V on Windows Server](#). Discrete Device Assignment also requires server class hardware that's capable of granting the operating system control over configuring the PCIe fabric (Native PCI Express Control). In addition, the PCIe Root Complex has to support Access Control Services (ACS), which enables Hyper-V to force all PCIe traffic through the Input-Output Memory Management Unit.

These capabilities usually aren't exposed directly in the BIOS of the server and are often hidden behind other settings. If the same capabilities are required for SR-IOV support and in the BIOS, you might need to set "Enable SR-IOV." Reach out to your system vendor if you're unable to identify the correct setting in your BIOS.

To help ensure the hardware is capable of Discrete Device Assignment, you can run the [machine profile script](#) on a Hyper-V enabled host. The script tests if your server is correctly set up and what devices are capable of Discrete Device Assignment.

Device requirements

Not every PCIe device can be used with Discrete Device Assignment. Older devices that use legacy (INTx) PCI Interrupts aren't supported. For more information, see [Discrete Device Assignment - Machines and devices](#) [↗](#). You can also run the [Machine Profile Script](#) to display which devices are capable of being used for Discrete Device Assignment.

Device manufacturers can reach out to their Microsoft representative for more details.

Device driver

Discrete Device Assignment passes the entire PCIe device into the Guest VM. A host driver isn't required to be installed prior to the device being mounted within the VM. The only requirement on the host is that the device's [PCIe Location Path](#) can be determined. The device's driver can be installed to help in identifying the device. A GPU without its device driver installed on the host might appear as a Microsoft Basic Render Device. If the device driver is installed, its manufacturer and model is likely to be displayed.

When the device is mounted inside the guest, the Manufacturer's device driver can be installed like normal inside the guest VM.

VM limitations

Due to the nature of how Discrete Device Assignment is implemented, some features of a VM are restricted while a device is attached. The following features aren't available:

- VM Save/Restore
- Live migration of a VM
- The use of dynamic memory
- Adding the VM to a high availability (HA) cluster

Security

Discrete Device Assignment passes the entire device into the VM. This pass means all capabilities of that device are accessible from the guest operating system. Some capabilities, like firmware updating, might adversely affect the stability of the system. Numerous warnings are presented to the admin when dismounting the device from the host. You should only use Discrete Device Assignment where the tenants of the VMs are trusted.

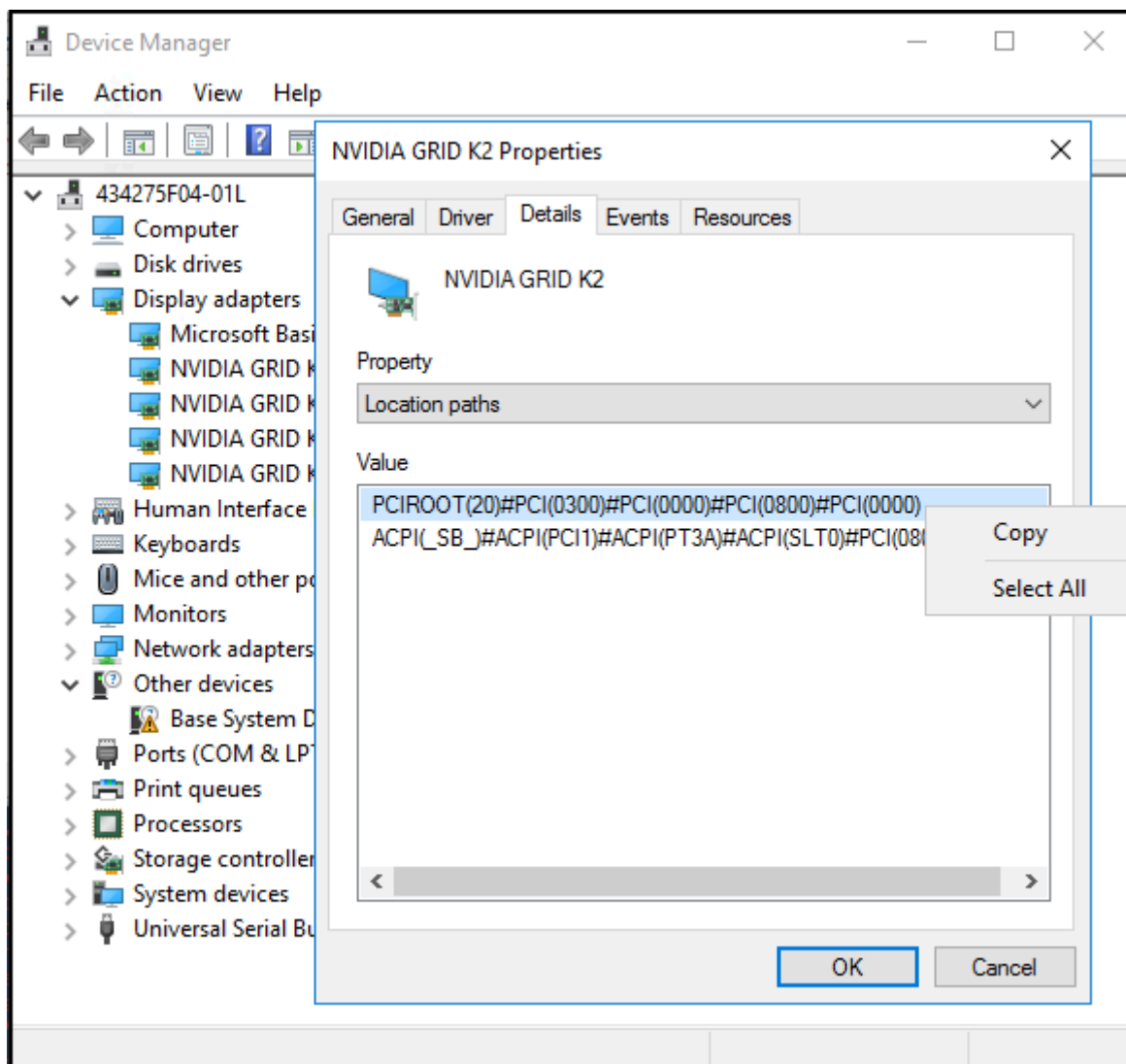
If the admin desires to use a device with an untrusted tenant, device manufacturers can create a Device Mitigation driver that can be installed on the host. Contact the device manufacturer for details on whether they provide a Device Mitigation Driver.

If you would like to bypass the security checks for a device that doesn't have a Device Mitigation Driver, you have to pass the `-Force` parameter to the `Dismount-VMHostAssignableDevice` cmdlet. When you make this pass, you have changed the security profile of that system. You should only make this change during prototyping or trusted environments.

PCIe location path

The PCIe location path is required to dismount and mount the device from the Host. An example location path is `PCIROOT(20)#PCI(0300)#PCI(0000)#PCI(0800)#PCI(0000)`. The [Machine Profile Script](#) also returns the location path of the PCIe device.

Get the location path by using Device Manager



1. Open Device Manager and locate the device.
2. Right-click the device and select **Properties**.
3. On the **Details** tab, expand the **Property** drop-down menu and select **Location Paths**.
4. Right-click the entry that begins with **PCIROOT** and select **Copy** to get the location path for the device.

MMIO space

Some devices, especially GPUs, require more MMIO space to be allocated to the VM for the memory of that device to be accessible. By default, each VM starts off with 128 MB of low MMIO space and 512 MB of high MMIO space allocated to it. However, a device might require more MMIO space, or multiple devices might be passed through such that the combined requirements exceed these values. Changing MMIO Space is straightforward and can be performed in PowerShell by using the following commands:

PowerShell

```
Set-VM -LowMemoryMappedIoSpace 3Gb -VMName $vm  
Set-VM -HighMemoryMappedIoSpace 33280Mb -VMName $vm
```

The easiest way to determine how much MMIO space to allocate is to use the [Machine Profile Script](#). To download and run the Machine Profile Script, run the following commands in a PowerShell console:

PowerShell

```
curl -o SurveyDDA.ps1  
https://raw.githubusercontent.com/MicrosoftDocs/Virtualization-  
Documentation/live/hyperv-tools/DiscreteDeviceAssignment/SurveyDDA.ps1  
.\SurveyDDA.ps1
```

For devices that can be assigned, the script displays the MMIO requirements of a given device. The following script output is an example:

PowerShell

```
NVIDIA GRID K520  
Express Endpoint -- more secure.  
...  
    And it requires at least: 176 MB of MMIO gap space  
...
```

The low MMIO space is used only by 32-bit operating systems and devices that use 32-bit addresses. In most circumstances, setting the high MMIO space of a VM is enough since 32-bit configurations aren't common.

Important

When you assign MMIO space to a VM, be sure to specify sufficient MMIO space. The MMIO space should be the sum of the requested MMIO space for all desired assigned devices plus a buffer for other virtual devices that require a few MB of MMIO space. Use the default MMIO values previously described as the buffer for low and high MMIO (128 MB and 512 MB, respectively).

Consider the previous example. If you assign a single K520 GPU, set the MMIO space of the VM to the value outputted by the machine profile script plus a buffer: 176 MB + 512 MB. If you assign three K520 GPUs, set the MMIO space to three times the base amount of 176 MB plus a buffer, or 528 MB + 512 MB.

For a more in-depth look at MMIO space, see [Discrete Device Assignment - GPUs](#) on the Tech Community blog.

Machine profile script

To identify if the server is configured correctly, and what devices can be passed through by using Discrete Device Assignment, run the [SurveyDDA.ps1](#) PowerShell script.

Before you use the script, ensure you have the Hyper-V role installed and you run the script from a PowerShell command window that has Administrator privileges.

If the system is incorrectly configured to support Discrete Device Assignment, the tool displays an error message with details about the issue. If the system is correctly configured, the tool enumerates all devices located on the PCIe Bus.

For each device it finds, the tool displays whether it's able to be used with Discrete Device Assignment. If a device is identified as being incompatible with Discrete Device Assignment, the script provides a reason. When a device is successfully identified as being compatible, the device's Location Path is displayed. Additionally, if that device requires [MMIO space](#), it's displayed as well.

```
Intel(R) C600/X79 series chipset USB2 Enhanced Host Controller #1 - 1D26
Old-style PCI device, switch port, etc. Not assignable.

Intel(R) Xeon(R)A E7 v2/Xeon(R) E5 v2/Core i7 IOAPIC - 0E2C
Embedded Endpoint -- less secure.
    And it has no interrupts at all -- assignment can work.
    And it requires at least: 1 MB of MMIO gap space
PCIR00T(0)#PCI(0504)

NVIDIA GRID K520
Express Endpoint -- more secure.
    And its interrupts are message-based, assignment can work.
    And it requires at least: 176 MB of MMIO gap space
PCIR00T(20)#PCI(0200)#PCI(0000)#PCI(0800)#PCI(0000)

Intel(R) Xeon(R)A E7 v2/Xeon(R) E5 v2/Core i7 Crystal Beach DMA Channel 0 - 0E20
Embedded Endpoint -- less secure.
All of the interrupts are line-based, no assignment can work.
```

Feedback

Was this page helpful?

☐ Yes

☐ No

Performance Tuning Hyper-V Servers

Article • 12/05/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#)

Hyper-V is the virtualization server role in Windows Server and Azure Local. Virtualization servers can host multiple virtual machines that are isolated from each other but share the underlying hardware resources by virtualizing the processors, memory, and I/O devices. By consolidating servers onto a single machine, virtualization can improve resource usage and energy efficiency and reduce the operational and maintenance costs of servers. In addition, virtual machines and the management APIs offer more flexibility for managing resources, balancing load, and provisioning systems.

Additional References

- [Hyper-V terminology](#)
- [Hyper-V architecture](#)
- [Hyper-V server configuration](#)
- [Hyper-V processor performance](#)
- [Hyper-V memory performance](#)
- [Hyper-V storage I/O performance](#)
- [Hyper-V network I/O performance](#)
- [Detecting bottlenecks in a virtualized environment](#)
- [Linux Virtual Machines](#)

Feedback

Was this page helpful?

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Create Security Policies with Extended Port Access Control Lists

07/25/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This topic provides information about extended port Access Control Lists (ACLs) in Windows Server 2016. You can configure extended ACLs on the Hyper-V Virtual Switch to allow and block network traffic to and from the virtual machines (VMs) that are connected to the switch via virtual network adapters.

This topic contains the following sections.

- [Detailed ACL rules](#)
- [Stateful ACL rules](#)

Detailed ACL rules

Hyper-V Virtual Switch extended ACLs allow you to create detailed rules that you can apply to individual VM network adapters that are connected to the Hyper-V Virtual Switch. The ability to create detailed rules allows Enterprises and Cloud Service Providers (CSPs) to address network-based security threats in a multitenant shared server environment.

With extended ACLs, rather than having to create broad rules that block or allow all traffic from all protocols to or from a VM, you can now block or allow the network traffic of individual protocols that are running on VMs. You can create extended ACL rules in Windows Server 2016 that include the following 5-tuple set of parameters: source IP address, destination IP address, protocol, source port, and destination port. In addition, each rule can specify network traffic direction (in or out), and the action the rule supports (block or allow traffic).

For example, you can configure port ACLs for a VM to allow all incoming and outgoing HTTP and HTTPS traffic on port 80, while blocking the network traffic of all other protocols on all ports.

This ability to designate the protocol traffic that can or cannot be received by tenant VMs provides flexibility when you configure security policies.

Configuring ACL rules with Windows PowerShell

To configure an extended ACL, you must use the Windows PowerShell command **Add-VMNetworkAdapterExtendedAcl**. This command has four different syntaxes, with a distinct use for each syntax:

1. Add an extended ACL to all of the network adapters of a named VM - which is specified by the first parameter, -VMName. Syntax:

❗ **Note**

If you want to add an extended ACL to one network adapter rather than all, you can specify the network adapter with the parameter -VMNetworkAdapterName.

```
Add-VMNetworkAdapterExtendedAcl [-VMName] <string[]> [-Action]
<VMNetworkAdapterExtendedAclAction> {Allow | Deny}
    [-Direction] <VMNetworkAdapterExtendedAclDirection> {Inbound | Outbound}
    [[-LocalIPAddress] <string>]
    [[-RemoteIPAddress] <string>] [[-LocalPort] <string>] [[-RemotePort]
<string>] [[-Protocol] <string>] [-Weight]
    <int> [-Stateful <bool>] [-IdleSessionTimeout <int>] [-IsolationID <int>]
    [-Passthru] [-VMNetworkAdapterName
    <string>] [-ComputerName <string[]>] [-WhatIf] [-Confirm]
    [<CommonParameters>]
```

2. Add an extended ACL to a specific virtual network adapter on a specific VM. Syntax:

```
Add-VMNetworkAdapterExtendedAcl [-VMNetworkAdapter] <VMNetworkAdapterBase[]>
[-Action]
    <VMNetworkAdapterExtendedAclAction> {Allow | Deny} [-Direction]
<VMNetworkAdapterExtendedAclDirection> {Inbound |
    Outbound} [[-LocalIPAddress] <string>] [[-RemoteIPAddress] <string>] [[-
LocalPort] <string>] [[-RemotePort]
    <string>] [[-Protocol] <string>] [-Weight] <int> [-Stateful <bool>] [-
IdleSessionTimeout <int>] [-IsolationID
    <int>] [-Passthru] [-WhatIf] [-Confirm]    [<CommonParameters>]
```

3. Add an extended ACL to all virtual network adapters that are reserved for use by the Hyper-V host management operating system.

❗ **Note**

If you want to add an extended ACL to one network adapter rather than all, you can specify the network adapter with the parameter `-VMNetworkAdapterName`.

```
Add-VMNetworkAdapterExtendedAcl [-Action] <VMNetworkAdapterExtendedAclAction>
{Allow | Deny} [-Direction]
    <VMNetworkAdapterExtendedAclDirection> {Inbound | Outbound} [[-
LocalIPAddress] <string>] [[-RemoteIPAddress]
    <string>] [[-LocalPort] <string>] [[-RemotePort] <string>] [[-Protocol]
<string>] [-Weight] <int> -ManagementOS
    [-Stateful <bool>] [-IdleSessionTimeout <int>] [-IsolationID <int>] [-
Passthru] [-VMNetworkAdapterName <string>]
    [-ComputerName <string[]>] [-WhatIf] [-Confirm] [<CommonParameters>]
```

4. Add an extended ACL to a VM object that you have created in Windows PowerShell, such as `$vm = get-vm "my_vm"`. In the next line of code you can run this command to create an extended ACL with the following syntax:

```
Add-VMNetworkAdapterExtendedAcl [-VM] <VirtualMachine[]> [-Action]
<VMNetworkAdapterExtendedAclAction> {Allow |
    Deny} [-Direction] <VMNetworkAdapterExtendedAclDirection> {Inbound |
Outbound} [[-LocalIPAddress] <string>]
    [[-RemoteIPAddress] <string>] [[-LocalPort] <string>] [[-RemotePort]
<string>] [[-Protocol] <string>] [-Weight]
    <int> [-Stateful <bool>] [-IdleSessionTimeout <int>] [-IsolationID <int>]
[-Passthru] [-VMNetworkAdapterName
    <string>] [-WhatIf] [-Confirm] [<CommonParameters>]
```

Detailed ACL rule examples

Following are several examples of how you can use the **Add-VMNetworkAdapterExtendedAcl** command to configure extended port ACLs and to create security policies for VMs.

- [Enforce application-level security](#)
- [Enforce both user-level and application-level security](#)
- [Provide security support to a non-TCP/UDP application](#)

ⓘ Note

The values for the rule parameter **Direction** in the tables below are based on traffic flow to or from the VM for which you are creating the rule. If the VM is receiving traffic, the traffic

is inbound; if the VM is sending traffic, the traffic is outbound. For example, if you apply a rule to a VM that blocks inbound traffic, the direction of inbound traffic is from external resources to the VM. If you apply a rule that blocks outbound traffic, the direction of outbound traffic is from the local VM to external resources.

Enforce application-level security

Because many application servers use standardized TCP/UDP ports to communicate with client computers, it is easy to create rules that block or allow access to an application server by filtering traffic going to and coming from the port designated to the application.

For example, you might want to allow a user to login to an application server in your datacenter by using Remote Desktop Connection (RDP). Because RDP uses TCP port 3389, you can quickly set up the following rule:

[Expand table](#)

Source IP	Destination IP	Protocol	Source Port	Destination Port	Direction	Action
*	*	TCP	*	3389	In	Allow

Following are two examples of how you can create rules with Windows PowerShell commands. The first example rule blocks all traffic to the VM named "ApplicationServer." The second example rule, which is applied to the network adapter of the VM named "ApplicationServer," allows only inbound RDP traffic to the VM.

ⓘ Note

When you create rules, you can use the **-Weight** parameter to determine the order in which the Hyper-V Virtual Switch processes the rules. Values for **-Weight** are expressed as integers; rules with a higher integer are processed before rules with lower integers. For example, if you have applied two rules to a VM network adapter, one with a weight of 1 and one with a weight of 10, the rule with the weight of 10 is applied first.

```
Add-VMNetworkAdapterExtendedAcl -VMName "ApplicationServer" -Action "Deny" -
Direction "Inbound" -Weight 1
Add-VMNetworkAdapterExtendedAcl -VMName "ApplicationServer" -Action "Allow" -
Direction "Inbound" -LocalPort 3389 -Protocol "TCP" -Weight 10
```

Enforce both user-level and application-level security

Because a rule can match a 5-tuple IP packet (Source IP, Destination IP, Protocol, Source Port, and Destination Port), the rule can enforce a more detailed security policy than a Port ACL.

For example, if you want to provide DHCP service to a limited number of client computers using a specific set of DHCP servers, you can configure the following rules on the Windows Server 2016 computer that is running Hyper-V, where the user VMs are hosted:

[Expand table](#)

Source IP	Destination IP	Protocol	Source Port	Destination Port	Direction	Action
*	255.255.255.255	UDP	*	67	Out	Allow
*	10.175.124.0/25	UDP	*	67	Out	Allow
10.175.124.0/25	*	UDP	*	68	In	Allow

Following are examples of how you can create these rules with Windows PowerShell commands.

```
Add-VMNetworkAdapterExtendedAcl -VMName "ServerName" -Action "Deny" -Direction "Outbound" -Weight 1
Add-VMNetworkAdapterExtendedAcl -VMName "ServerName" -Action "Allow" -Direction "Outbound" -RemoteIPAddress 255.255.255.255 -RemotePort 67 -Protocol "UDP"-Weight 10
Add-VMNetworkAdapterExtendedAcl -VMName "ServerName" -Action "Allow" -Direction "Outbound" -RemoteIPAddress 10.175.124.0/25 -RemotePort 67 -Protocol "UDP"-Weight 20
Add-VMNetworkAdapterExtendedAcl -VMName "ServerName" -Action "Allow" -Direction "Inbound" -RemoteIPAddress 10.175.124.0/25 -RemotePort 68 -Protocol "UDP"-Weight 20
```

Provide security support to a non-TCP/UDP application

While most network traffic in a datacenter is TCP and UDP, there is still some traffic that utilizes other protocols. For example, if you want to permit a group of servers to run an IP-multicast application that relies on Internet Group Management Protocol (IGMP), you can create the following rule.

 **Note**

IGMP has a designated IP protocol number of 0x02.

 Expand table

Source IP	Destination IP	Protocol	Source Port	Destination Port	Direction	Action
*	*	0x02	*	*	In	Allow
*	*	0x02	*	*	Out	Allow

Following is an example of how you can create these rules with Windows PowerShell commands.

```
Add-VMNetworkAdapterExtendedAcl -VMName "ServerName" -Action "Allow" -Direction "Inbound" -Protocol 2 -Weight 20
Add-VMNetworkAdapterExtendedAcl -VMName "ServerName" -Action "Allow" -Direction "Outbound" -Protocol 2 -Weight 20
```

Stateful ACL rules

Another new capability of extended ACLs allows you to configure stateful rules. A stateful rule filters packets based on five attributes in a packet - Source IP, Destination IP, Protocol, Source Port, and Destination Port.

Stateful rules have the following capabilities:

- They always allow traffic and are not used to block traffic.
- If you specify that the value for the parameter **Direction** is inbound and traffic matches the rule, Hyper-V Virtual Switch dynamically creates a matching rule that allows the VM to send outbound traffic in response to the external resource.
- If you specify that the value for the parameter **Direction** is outbound and traffic matches the rule, Hyper-V Virtual Switch dynamically creates a matching rule that allows the external resource inbound traffic to be received by the VM.
- They include a timeout attribute that is measured in seconds. When a network packet arrives at the switch and the packet matches a stateful rule, Hyper-V Virtual Switch creates a state so that all subsequent packets in both directions of the same flow are allowed. The state expires if there is no traffic in either direction in the period of time that is specified by the timeout value.

Following is an example of how you can use stateful rules.

Allow inbound remote server traffic only after it is contacted by the local server

In some cases, a stateful rule must be employed because only a stateful rule can keep track of a known, established connection, and distinguish the connection from other connections.

For example, if you want to allow a VM application server to initiate connections on port 80 to web services on the Internet, and you want the remote Web servers to be able to respond to the VM traffic, you can configure a stateful rule that allows initial outbound traffic from the VM to the Web services; because the rule is stateful, return traffic to the VM from the Web servers is also allowed. For security reasons, you can block all other inbound network traffic to the VM.

To achieve this rule configuration, you can use the settings in the table below.

 Note

Due to formatting restrictions and the amount of information in the table below, the information is displayed differently than in previous tables in this document.

 Expand table

Parameter	Rule 1	Rule 2	Rule 3
Source IP	*	*	*
Destination IP	*	*	*
Protocol	*	*	TCP
Source Port	*	*	*
Destination Port	*	*	80
Direction	In	Out	Out
Action	Deny	Deny	Allow
Stateful	No	No	Yes
Timeout (in seconds)	N/A	N/A	3600

The stateful rule allows the VM application server to connect to a remote Web server. When the first packet is sent out, Hyper-V Virtual switch dynamically creates two flow states to allow

all packets sent to and all returning packets from the remote Web server. When the flow of packets between the servers stops, the flow states time out in the designated timeout value of 3600 seconds, or one hour.

Following is an example of how you can create these rules with Windows PowerShell commands.

```
Add-VMNetworkAdapterExtendedAcl -VMName "ApplicationServer" -Action "Deny" -  
Direction "Inbound" -Weight 1  
Add-VMNetworkAdapterExtendedAcl -VMName "ApplicationServer" -Action "Deny" -  
Direction "Outbound" -Weight 1  
Add-VMNetworkAdapterExtendedAcl -VMName "ApplicationServer" -Action "Allow" -  
Direction "Outbound" 80 "TCP" -Weight 100 -Stateful -Timeout 3600
```


About dump encryption

Article • 09/17/2020 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local, versions 23H2 and 22H2](#)

Dump encryption can be used to encrypt crash dumps and live dumps generated for a system. The dumps are encrypted using a symmetric encryption key which is generated for each dump. This key itself is then encrypted using the public key specified by the trusted administrator of the host (crash dump encryption key protector). This ensures that only someone having the matching private key can decrypt and therefore access contents of the dump. This capability is leveraged in a guarded fabric. Note: If you configure dump encryption, also disable Windows Error Reporting. WER cannot read encrypted crash dumps.

Configuring dump encryption

Manual configuration

To turn on dump encryption using the registry, configure the following registry values under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\CrashControl`

 Expand table

Value Name	Type	Value
DumpEncryptionEnabled	DWORD	1 to enable dump encryption, 0 to disable dump encryption
EncryptionCertificates\Certificate.1::PublicKey	Binary	Public key (RSA, 2048 Bit) that should be used for encrypting dumps. This has to be formatted as BCRYPT_RSAKEY_BLOB .
EncryptionCertificates\Certificate.1::Thumbprint	String	Certificate thumbprint to allow automatic lookup of private key in the local certificate store when decrypting a crash dump.

Configuration using script

To simplify configuration, a [sample script](#) is available to enable dump encryption based on a public key from a certificate.

1. In a trusted environment: Create a certificate with a 2048 Bit RSA key and export the public certificate
2. On target hosts: Import the public certificate to the local certificate store
3. Run the sample configuration script

```
.\Set-DumpEncryptionConfiguration.ps1 -Certificate  
(Cert:\CurrentUser\My\093568AB328DF385544FAFD57EE53D73EFAAF519) -Force
```

Decrypting encrypted dumps

To decrypt an existing encrypted dump file, you need to download and install the Debugging Tools for Windows. This tool set contains KernelDumpDecrypt.exe which can be used to decrypt an encrypted dump file. If the certificate including the private key is present in the current user's certificate store, the dump file can be decrypted by calling

```
KernelDumpDecrypt.exe memory.dmp memory_decr.dmp
```

After decryption, tools like WinDbg can open the decrypted dump file.

Troubleshooting dump encryption

If dump encryption is enabled on a system but no dumps are being generated, please check the system's **System** event log for **Kernel-IO** event 1207. When dump encryption cannot be initialized, this event is created and dumps are disabled.

[Expand table](#)

Detailed error message	Steps to mitigate
Public Key or Thumbprint registry missing	Check if both registry values exist in the expected location
Invalid Public Key	Make sure that the public key stored in the PublicKey registry value is stored as BCRYPT_RSAKEY_BLOB .

Detailed error message	Steps to mitigate
Unsupported Public Key Size	Currently, only 2048 Bit RSA keys are supported. Configure a key that matches this requirement

Also check if the value `GuardedHost` under

`HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\CrashControl\ForceDumpsDisable`

`d` is set to a value other than 0. This disables crash dumps completely. If this is the case, set it to 0.

Feedback

Was this page helpful?

 Yes

 No

Cmdlets for configuring persistent memory devices for Hyper-V VMs

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

This article provides system administrators and IT Pros with information about configuring Hyper-V VMs with persistent memory (also known as storage class memory or NVDIMM). JEDEC-compliant NVDIMM-N persistent memory devices are supported in Windows Server 2016 and Windows 10 and provide byte-level access to very low latency nonvolatile devices. VM persistent memory devices are supported in Windows Server 2019.

Create a persistent memory device for a VM

Use the [New-VHD](#) cmdlet to create a persistent memory device for a VM. The device must be created on an existing NTFS DAX volume. The new filename extension (.vhdpmem) is used to specify that the device is a persistent memory device. Only the fixed VHD file format is supported.

Example: `New-VHD D:\VMPMEMDevice1.vhdpmem -Fixed -SizeBytes 4GB`

Create a VM with a persistent memory controller

Use the [New-VM](#) cmdlet to create a Generation 2 VM with specified memory size and path to a VHDX image. Then, use [Add-VMPmemController](#) to add a persistent memory controller to a VM.

Example:

PowerShell

```
New-VM -Name "ProductionVM1" -MemoryStartupBytes 1GB -VHDPath c:\vhd\BaseImage.vhdx  
  
Add-VMPmemController -VMName "ProductionVM1"
```

Attach a persistent memory device to a VM

Use [Add-VMHardDiskDrive](#) to attach a persistent memory device to a VM

Example: `Add-VMHardDiskDrive -VMName "ProductionVM1" -ControllerType PMEM -
ControllerLocation 1 -Path D:\VMPMEMDevice1.vhdpmem`

Persistent memory devices within a Hyper-V VM appear as a persistent memory device to be consumed and managed by the guest operating system. Guest operating systems can use the device as a block or DAX volume. When persistent memory devices within a VM are used as a DAX volume, they benefit from low latency byte-level address-ability of the host device (no I/O virtualization on the code path).

ⓘ Note

Persistent memory is only supported for Hyper-V Gen2 VMs. Live Migration and Storage Migration are not supported for VMs with persistent memory. Production checkpoints of VMs do not include persistent memory state.

Last updated on 01/21/2026

Using checkpoints to revert virtual machines to a previous state

08/14/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

One of the great benefits to virtualization is the ability to easily save the state of a virtual machine. In Hyper-V this is done through the use of virtual machine checkpoints. You may want to create a virtual machine checkpoint before making software configuration changes, applying a software update, or installing new software. If a system change were to cause an issue, the virtual machine can be reverted to the state at which it was when then checkpoint was taken.

Windows 10 and 11 Hyper-V includes two types of checkpoints:

- **Standard Checkpoints:** takes a snapshot of the virtual machine and virtual machine memory state at the time the checkpoint is initiated. A snapshot is not a full backup and can cause data consistency issues with systems that replicate data between different nodes such as Active Directory. Hyper-V only offered standard checkpoints (formerly called snapshots) prior to Windows 10.
- **Production Checkpoints:** uses Volume Shadow Copy Service or File System Freeze on a Linux virtual machine to create a data-consistent backup of the virtual machine. No snapshot of the virtual machine memory state is taken.

Production checkpoints are selected by default however this can be changed using either Hyper-V manager or PowerShell.

Note

The Hyper-V PowerShell module has several aliases so that **checkpoint** and **snapshot** can be used interchangeably.

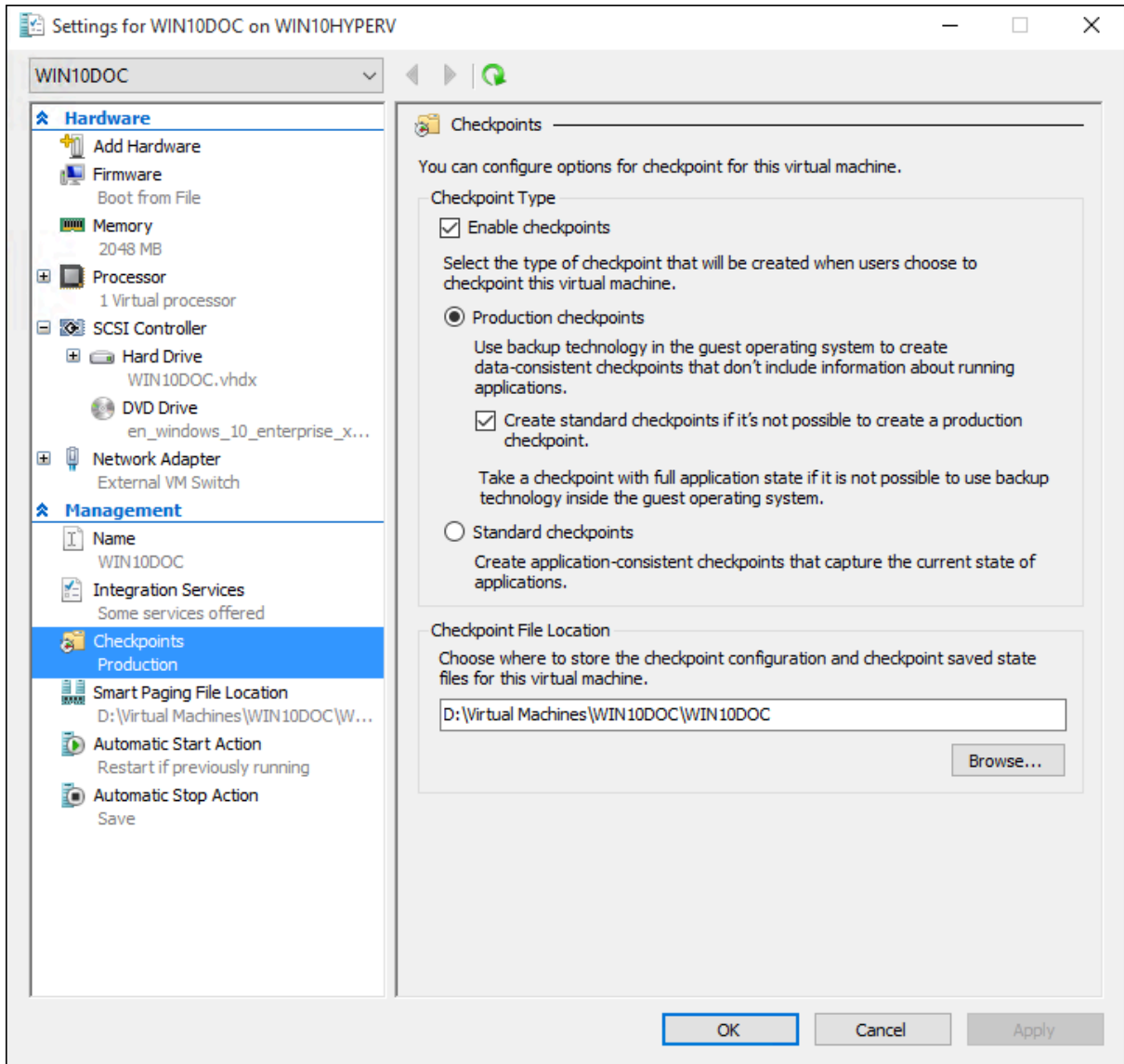
This document uses **checkpoint**, however be aware that you may see similar commands using the term **snapshot**.

Changing the Checkpoint Type

Hyper-V manager

1. Open Hyper-V Manager.

2. Right-click on a virtual machine and select **settings**.
3. Under Management select **Checkpoints**.
4. Select the desired checkpoint type.



Creating checkpoints

Creates a checkpoint of the type configured for the virtual machine. See the [Configuring Checkpoint Type](#) section earlier in this document for instructions on how to change this type.

Hyper-V manager

To create a checkpoint:

1. In Hyper-V Manager, select the virtual machine.

2. Right-click the name of the virtual machine, and then click **Checkpoint**.
3. When the process is complete, the checkpoint will appear under **Checkpoints** in the **Hyper-V Manager**.

Applying checkpoints

If you want to revert your virtual machine to a previous point-in-time, you can apply an existing checkpoint.

Hyper-V manager

1. In **Hyper-V Manager**, under **Virtual Machines**, select the virtual machine.
2. In the Checkpoints section, right-click the checkpoint that you want to use and click **Apply**.
3. A dialog box appears with the following options:
 - **Create Checkpoint and Apply**: Creates a new checkpoint of the virtual machine before it applies the earlier checkpoint.
 - **Apply**: Applies only the checkpoint that you have chosen. You cannot undo this action.
 - **Cancel**: Closes the dialog box without doing anything.

Select either Apply option to create apply the checkpoint.

Renaming checkpoints

Many checkpoints are created at a specific point. Giving them an identifiable name makes it easier to remember details about the system state when the checkpoint was created.

By default, the name of a checkpoint is the name of the virtual machine combined with the date and time the checkpoint was taken. This is the standard format:

Output

```
virtual_machine_name (MM/DD/YYYY -hh:mm:ss AM\PM)
```

Names are limited to 100 characters, and the name can't be blank.

1. In **Hyper-V Manager**, select the virtual machine.
2. Right-click the checkpoint, and then select **Rename**.
3. Enter in the new name for the checkpoint. It must be less than 100 characters, and the field cannot be empty.
4. Select **ENTER** when you are done.

Deleting checkpoints

Deleting checkpoints can help create space on your Hyper-V host.

Behind the scenes, checkpoints are stored as .avhdx files in the same location as the .vhd files for the virtual machine. When you delete a checkpoint, Hyper-V merges the .avhdx and .vhd files for you. Once completed, the checkpoint's .avhdx file will be deleted from the file system.

You should not delete the .avhdx files directly.

To cleanly delete a checkpoint:

1. In **Hyper-V Manager**, select the virtual machine.
2. In the **Checkpoints** section, right-click the checkpoint that you want to delete and select **Delete**. You can also delete a checkpoint and all subsequent checkpoints. To do so, right-click the earliest checkpoint that you want to delete, and then select **Delete Checkpoint Subtree**.
3. You might be asked to verify that you want to delete the checkpoint. Confirm that it is the correct checkpoint, and then select **Delete**.

Exporting checkpoints

Export bundles the checkpoint as a virtual machine so the checkpoint can be moved to a new location. Once imported, the checkpoint is restored as a virtual machine. Exported checkpoints can be used for backup.

To export a checkpoint:

1. In **Hyper-V Manager**, select the virtual machine.
2. In the **Checkpoints** section, right-click the checkpoint that you want to export and select **Export**.
3. Enter the location where you want to save the exported checkpoint, and then select **Export**.

The export process can take some time, depending on the size of the checkpoint. Check the status in the **Status** column in the **Hyper-V Manager** for the VM.

Enable or disable checkpoints

1. In **Hyper-V Manager**, right-click the name of the virtual machine, and select **Settings**.
2. In the **Management** section, select **Checkpoints**.
3. To allow checkpoints to be taken off this virtual machine, make sure **Enable Checkpoints** is selected -- this is the default behavior. To disable checkpoints, deselect the **Enable Checkpoints** check box.
4. Select **Apply** to apply your changes. If you are done, select **OK** to close the dialog box.

Configure checkpoint location

If the virtual machine has no checkpoints, you can change where the checkpoint configuration and saved state files are stored.

1. In **Hyper-V Manager**, right-click the name of the virtual machine, and select **Settings**.
2. In the **Management** section, select **Checkpoints** or **Checkpoint File Location**.
3. In **Checkpoint File Location**, enter the path to the folder where you would like to store the files.
4. Select **Apply** to apply your changes. If you are done, select **OK** to close the dialog box.

The default location for storing checkpoint configuration files is:

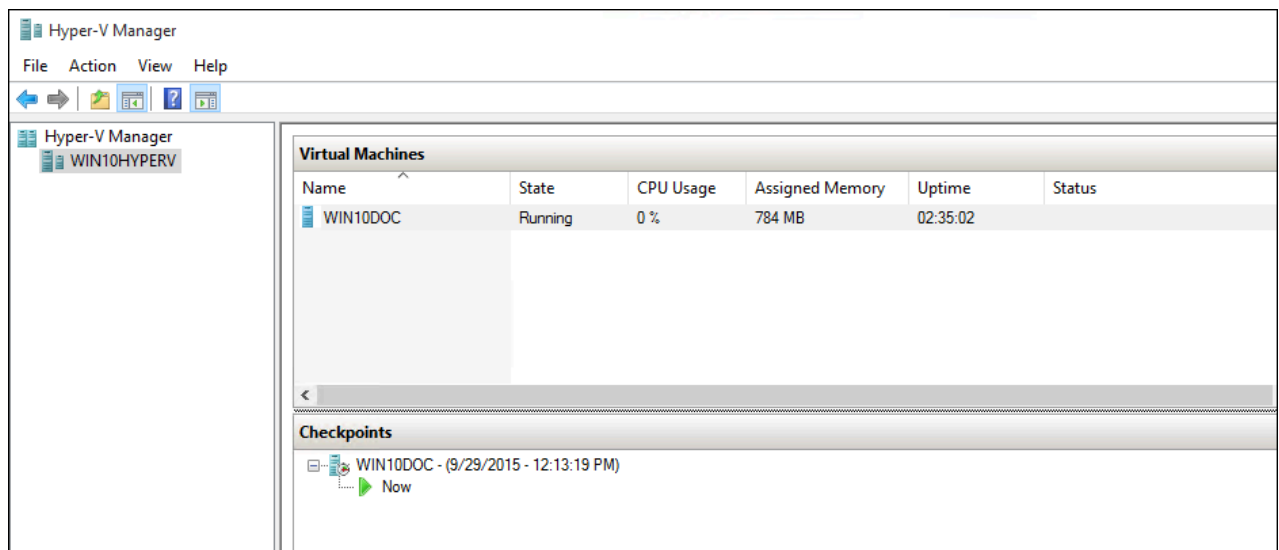
```
%systemroot%\ProgramData\Microsoft\Windows\Hyper-V\Snapshots.
```

Using standard checkpoints

This exercise walks through creating and applying a standard checkpoint versus a production checkpoint. For this example, you'll make a simple change to the virtual machine and observe the different behavior.

Create a standard checkpoint

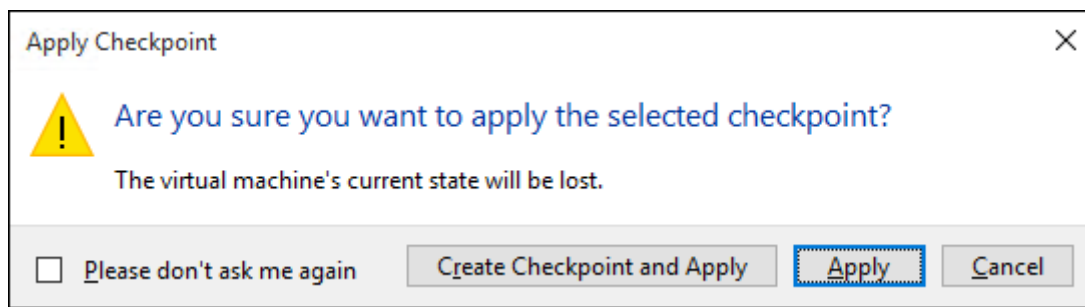
1. Log into your virtual machine and create a text file on the desktop.
2. Open the file with Notepad and enter the text 'This is a Standard Checkpoint.' **Do not save the file or close Notepad.**
3. Change the checkpoint to standard. Follow the instructions in [changing the checkpoints](#).
4. Create a new checkpoint.



Apply the standard checkpoint with Hyper-V Manager

Now that a checkpoint exists, make a modification to the virtual machine and then apply the checkpoint to revert the virtual machine back to the saved state.

1. Close the text file if it is still open and delete it from the virtual machine's desktop.
2. Open Hyper-V Manager, right-click on the standard checkpoint, and select Apply.
3. Select Apply on the Apply Checkpoint notification window.



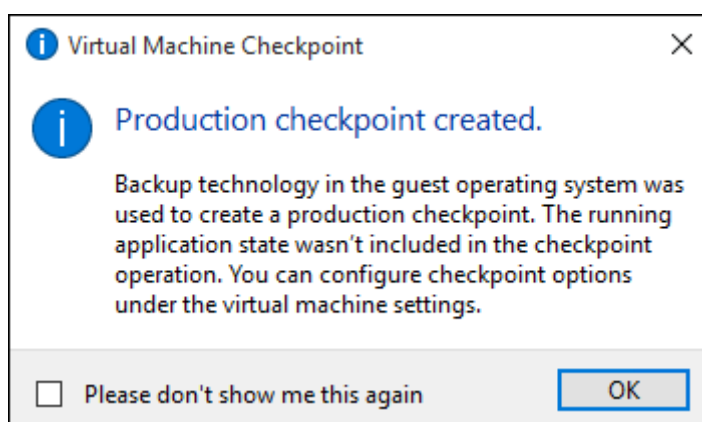
Once the checkpoint has been applied, notice that not only is the text file present, but the system is in the exact state that it was when the checkpoint was created. In this case Notepad is open and the text file loaded.

Using production checkpoints

Let's now examine production checkpoints. This process is almost identical to working with a standard checkpoint, however will have slightly different results. Before beginning make sure you have a virtual machine and that you have changes the checkpoint type to Production checkpoints.

Modify the virtual machine and Create a Production Checkpoint

1. Log into the virtual machine and create a new text file. If you followed the previous exercise, you can use the existing text file.
2. Enter 'This is a Production Checkpoint.' into the text file, save the file but **do not close Notepad**.
3. Open Hyper-V Manager, right-click on the virtual machine, and select **Checkpoint**.
4. Select **OK** on the Production Checkpoint Created Window.



Apply the Production Checkpoint with Hyper-V Manager

Now that a checkpoint exists make a modification to the system and then apply the checkpoint to revert the virtual machine back to the saved state.

1. Close the text file if it is still open and delete it from the virtual machine's desktop.
2. Open Hyper-V Manager, right click on the production checkpoint, and select **Apply**.
3. Select **Apply** on the Apply Checkpoint notification window.

Once the production checkpoint has been applied, noticed that the virtual machine is in an off state.

1. Start and log into the virtual machine.
2. Take note that the text file has been restored. But unlike the standard checkpoint, Notepad is not open.

Create Hyper-V VHD Set files

Article • 12/08/2020 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local, versions 23H2 and 22H2](#)

VHD Set files are a new shared Virtual Disk model for guest clusters in Windows Server 2016. VHD Set files support online resizing of shared virtual disks, support Hyper-V Replica, and can be included in application-consistent checkpoints.

VHD Set files use a new VHD file type, .VHDS. VHD Set files store checkpoint information about the group virtual disk used in guest clusters, in the form of metadata.

Hyper-V handles all aspects of managing the checkpoint chains and merging the shared VHD set. Management software can run disk operations like online resizing on VHD Set files in the same way it does for .VHDX files. This means that management software doesn't need to know about the VHD Set file format.

Note

It's important to evaluate the impact of VHD Set files before deployment into production. Make sure that there is no performance or functional degradation in your environment, such as disk latency.

Create a VHD Set file from Hyper-V Manager

1. Open Hyper-V Manager. Click **Start**, point to **Administrative Tools**, and then click **Hyper-V Manager**.
2. In the Action pane, click **New**, and then click **Hard Disk**.
3. On the **Choose Disk Format** page, select **VHD Set** as the format of the virtual hard disk.
4. Continue through the pages of the wizard to customize the virtual hard disk. You can click **Next** to move through each page of the wizard, or you can click the name of a page in the left pane to move directly to that page.
5. After you have finished configuring the virtual hard disk, click **Finish**.

Create a VHD Set file from Windows PowerShell

Use the [New-VHD](#) cmdlet, with the file type .VHDS in the file path. This example creates a VHD Set file named base.vhds that is 10 Gigabytes in size.

PowerShell

```
PS c:\>New-VHD -Path c:\base.vhds -SizeBytes 10GB
```

Migrate a shared VHDX file to a VHD Set file

Migrating an existing shared VHDX to a VHDS requires taking the VM offline. This is the recommended process using Windows PowerShell:

1. Remove the VHDX from the VM. For example, run:

PowerShell

```
PS c:\>Remove-VMHardDiskDrive existing.vhdx
```

2. Convert the VHDX to a VHDS. For example, run:

PowerShell

```
PS c:\>Convert-VHD existing.vhdx new.vhds
```

3. Add the VHDS to the VM. For example, run:

PowerShell

```
PS c:\>Add-VMHardDiskDrive new.vhds
```

Feedback

Was this page helpful?

 Yes

 No

Remotely manage Hyper-V hosts with Hyper-V Manager

07/08/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

This article lists the supported combinations of Hyper-V hosts and Hyper-V Manager versions and describes how to connect to remote and local Hyper-V hosts so you can manage them.

Hyper-V Manager lets you manage a small number of Hyper-V hosts, both remote and local. It's installed when you install the Hyper-V Management Tools, which you can do either through a full Hyper-V installation or a tools-only installation. If you do a tools-only installation, you can use the tools on computers that don't meet the hardware requirements to host Hyper-V. For details about hardware for Hyper-V hosts, see [System requirements](#). If Hyper-V Manager isn't installed, follow the [instructions to Install Hyper-V Manager](#).

Connect to a Hyper-V host

To connect to a Hyper-V host from Hyper-V Manager, right-click **Hyper-V Manager** in the left pane, and then select **Connect to Server**.

Manage Hyper-V on a local computer

Hyper-V Manager doesn't list any computers that host Hyper-V until you add the computer, including a local computer. To do this:

1. In the left pane, right-click **Hyper-V Manager**.
2. Select **Connect to Server**.
3. From **Select Computer**, select **Local computer** and then select **OK**.

If you can't connect:

- It's possible that only the Hyper-V tools are installed. To check whether Hyper-V platform is installed, look for the Virtual Machine Management service. To do so, open the Services desktop app: select **Start**, select the **Start Search** box, type **services.msc**, and then select **Enter**. If the Virtual Machine Management service isn't listed, install the Hyper-V platform by following the instructions in [Install Hyper-V](#).
- Check whether your hardware meets the requirements. See [System requirements](#).
- Ensure that your user account belongs to the Administrators group or the Hyper-V Administrators group.

Manage Hyper-V hosts remotely

To manage remote Hyper-V hosts, enable remote management on both the local computer and the remote host.

On Windows Server, open Server Manager > **Local Server** > **Remote management** and then select **Allow remote connections to this computer**.

Or, from either operating system, open Windows PowerShell as Administrator and run:

```
PowerShell
```

```
Enable-PSRemoting
```

Connect to hosts in the same domain

To add a remote Hyper-V host to Hyper-V Manager, select **Another computer** in the **Select Computer** dialog and enter the remote host's hostname, NetBIOS name, or fully qualified domain name (FQDN).

Hyper-V Manager offers more types of remote connections, which are described in the following sections.

Connect to a remote host as a different user

This lets you connect to the Hyper-V host when you're not running on the local computer as a user that's a member of either the Hyper-V Administrators group or the Administrators group on the Hyper-V host. To do this:

1. In the left pane, right-click **Hyper-V Manager**.
2. Select **Connect to Server**.
3. Select **Connect as another user** in the **Select Computer** dialog.
4. Select **Set User**.

Connect to a remote host by using an IP address

To do this:

1. In the left pane, right-click **Hyper-V Manager**.
2. Select **Connect to Server**.
3. Enter the IP address into the **Another Computer** box.

Connect to a remote host outside your domain, or with no domain

To do this:

1. On the Hyper-V host to be managed, open a Windows PowerShell session as Administrator.
2. Create the necessary firewall rules for private network zones:

PowerShell

```
Enable-PSRemoting
```

3. To allow remote access on public zones, enable firewall rules for CredSSP and WinRM:

PowerShell

```
Enable-WSManCredSSP -Role server
```

For details, see [Enable-PSRemoting](#) and [Enable-WSManCredSSP](#).

Next, configure the computer you'll use to manage the Hyper-V host.

1. Open a Windows PowerShell session as Administrator.
2. Run these commands:

PowerShell

```
Set-Item WSMan:\localhost\Client\TrustedHosts -Value "fqdn-of-hyper-v-host"
```

PowerShell

```
Enable-WSManCredSSP -Role client -DelegateComputer "fqdn-of-hyper-v-host"
```

3. You might also need to configure the following group policy:
 - **Computer Configuration > Administrative Templates > System > Credentials Delegation > Allow delegating fresh credentials with NTLM-only server authentication**
 - Select **Enable** and add *wsman/fqdn-of-hyper-v-host*.
4. Open Hyper-V Manager.

5. In the left pane, right-click **Hyper-V Manager**.

6. Select **Connect to Server**.

For cmdlet details, see [Set-Item](#) and [Enable-WSManCredSSP](#).

Install Hyper-V Manager

To use a UI tool, choose the one that's appropriate for the operating system on the computer where you'll run Hyper-V Manager:

On Windows Server, open Server Manager > **Manage** > **Add roles and features**. Move to the **Features** page and expand **Remote server administration tools** > **Role administration tools** > **Hyper-V management tools**.

On Windows, Hyper-V Manager is available on [any Windows operating system that includes Hyper-V](#).

1. On the Windows desktop, select the Start button and begin typing **Programs and features**.
2. In the search results, select **Programs and Features**.
3. In the left pane, select **Turn Windows features on or off**.
4. Expand the Hyper-V folder and select **Hyper-V Management Tools**.
5. To install Hyper-V Manager, select **Hyper-V Management Tools**. If you want to also install the Hyper-V module, select that option.

To use Windows PowerShell, run the following command as Administrator:

PowerShell

```
add-windowsfeature rsat-hyper-v-tools
```

Related content

- [Install Hyper-V](#)

Hyper-V Host CPU Resource Management

Article • 09/17/2020 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2
and 22H2

Hyper-V host CPU resource controls introduced in Windows Server 2016 or later allow Hyper-V administrators to better manage and allocate host server CPU resources between the "root", or management partition, and guest VMs. Using these controls, administrators can dedicate a subset of the processors of a host system to the root partition. This can segregate the work done in a Hyper-V host from the workloads running in guest virtual machines by running them on separate subsets of the system processors.

For details about hardware for Hyper-V hosts, see [Windows 10 Hyper-V System Requirements](#).

Background

Before setting controls for Hyper-V host CPU resources, it's helpful to review the basics of the Hyper-V architecture. You can find a general summary in the [Hyper-V Architecture](#) section. These are important concepts for this article:

- Hyper-V creates and manages virtual machine partitions, across which compute resources are allocated and shared, under control of the hypervisor. Partitions provide strong isolation boundaries between all guest virtual machines, and between guest VMs and the root partition.
- The root partition is itself a virtual machine partition, although it has unique properties and much greater privileges than guest virtual machines. The root partition provides the management services that control all guest virtual machines, provides virtual device support for guests, and manages all device I/O for guest virtual machines. Microsoft strongly recommends not running any application workloads in a host partition.
- Each virtual processor (VP) of the root partition is mapped 1:1 to an underlying logical processor (LP). A host VP will always run on the same underlying LP – there is no migration of the root partition's VPs.

- By default, the LPs on which host VPs run can also run guest VPs.
- A guest VP may be scheduled by the hypervisor to run on any available logical processor. While the hypervisor scheduler takes care to consider temporal cache locality, NUMA topology, and many other factors when scheduling a guest VP, ultimately the VP could be scheduled on any host LP.

The Minimum Root, or "Minroot" Configuration

Early versions of Hyper-V had an architectural maximum limit of 64 VPs per partition. This applied to both the root and guest partitions. As systems with more than 64 logical processors appeared on high-end servers, Hyper-V also evolved its host scale limits to support these larger systems, at one point supporting a host with up to 320 LPs. However, breaking the 64 VP per partition limit at that time presented several challenges and introduced complexities that made supporting more than 64 VPs per partition prohibitive. To address this, Hyper-V limited the number of VPs given to the root partition to 64, even if the underlying machine had many more logical processors available. The hypervisor would continue to utilize all available LPs for running guest VPs, but artificially capped the root partition at 64. This configuration became known as the "minimum root", or "minroot" configuration. Performance testing confirmed that, even on large scale systems with more than 64 LPs, the root did not need more than 64 root VPs to provide sufficient support to a large number of guest VMs and guest VPs – in fact, much less than 64 root VPs was often adequate, depending of course on the number and size of the guest VMs, the specific workloads being run, etc.

This "minroot" concept continues to be utilized today. In fact, even as Windows Server 2016 Hyper-V increased its maximum architectural support limit for host LPs to 512 LPs, the root partition will still be limited to a maximum of 320 LPs.

Using Minroot to Constrain and Isolate Host Compute Resources

With the high default threshold of 320 LPs in Windows Server 2016 Hyper-V, the minroot configuration will only be utilized on the very largest server systems. However, this capability can be configured to a much lower threshold by the Hyper-V host administrator, and thus leveraged to greatly restrict the amount of host CPU resources available to the root partition. The specific number of root LPs to utilize must of course be chosen carefully to support the maximum demands of the VMs and workloads allocated to the host. However, reasonable values for the number of host LPs can be

determined through careful assessment and monitoring of production workloads, and validated in non-production environments before broad deployment.

Enabling and Configuring Minroot

The minroot configuration is controlled via hypervisor BCD entries. To enable minroot, from a cmd prompt with administrator privileges:

```
bcdedit /set hypervisorrootproc n
```

Where n is the number of root VPs.

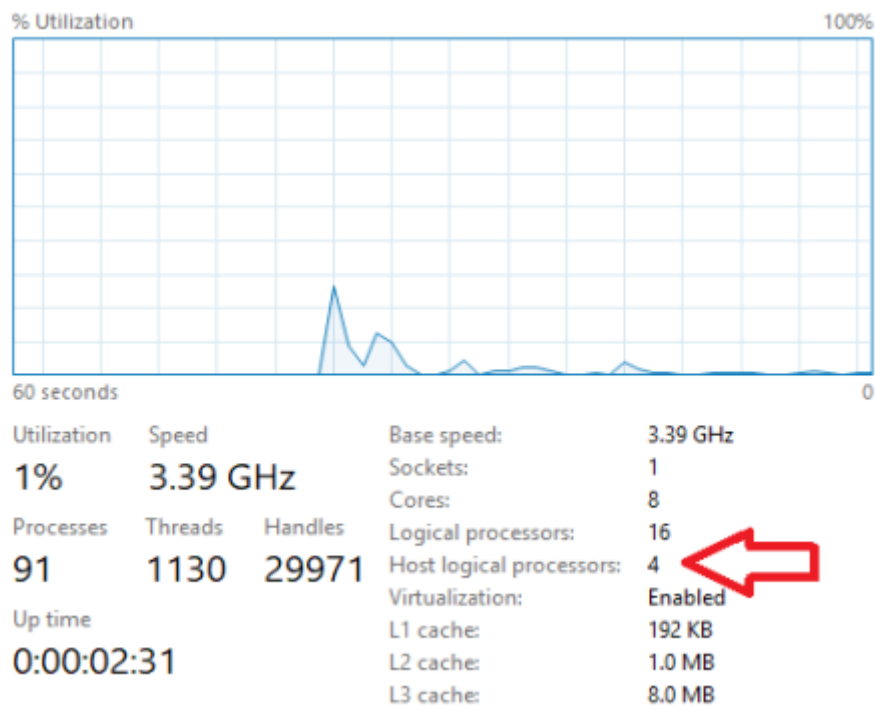
The system must be rebooted, and the new number of root processors will persist for the lifetime of the OS boot. The minroot configuration cannot be changed dynamically at runtime.

If there are multiple NUMA nodes, each node will get `n/NumaNodeCount` processors.

Note that with multiple NUMA nodes, you must ensure the VM's topology is such that there are enough free LPs (i.e., LPs without root VPs) on each NUMA node to run the corresponding VM's NUMA node VPs.

Verifying the Minroot Configuration

You can verify the host's minroot configuration using Task Manager, as shown below.



When Minroot is active, Task Manager will display the number of logical processors currently allotted to the host, in addition to the total number of logical processors in the system.

Feedback

Was this page helpful?

☐ Yes

☐ No

Virtual Machine Resource Controls

Article • 07/29/2021 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local, versions 23H2 and 22H2](#)

This article describes Hyper-V resource and isolation controls for virtual machines. These capabilities, which we'll refer to as Virtual Machine CPU Groups, or just "CPU groups", were introduced in Windows Server 2016. CPU groups allow Hyper-V administrators to better manage and allocate the host's CPU resources across guest virtual machines.

Using CPU groups, Hyper-V administrators can:

- Create groups of virtual machines, with each group having different allocations of the virtualization host's total CPU resources, shared across the entire group. This allows the host administrator to implement classes of service for different types of VMs.
- Set CPU resource limits to specific groups. This "group cap" sets the upper bound for host CPU resources that the entire group may consume, effectively enforcing the desired class of service for that group.
- Constrain a CPU group to run only on a specific set of the host system's processors. This can be used to isolate VMs belonging to different CPU groups from each other.

Managing CPU Groups

CPU groups are managed through the Hyper-V Host Compute Service, or HCS. A great description of the HCS, its genesis, links to the HCS APIs, and more is available on the Microsoft Virtualization team's blog in the posting [Introducing the Host Compute Service \(HCS\)](#) .

Note

Only the HCS may be used to create and manage CPU groups; the Hyper-V Manager applet, WMI and PowerShell management interfaces don't support CPU groups.

Microsoft provides a command line utility, `cpugroups.exe`, on the [Microsoft Download Center](#)  which uses the HCS interface to manage CPU groups. This utility can also

display the CPU topology of a host.

How CPU Groups Work

Allocation of host compute resources across CPU groups is enforced by the Hyper-V hypervisor, using a computed CPU group cap. The CPU group cap is a fraction of the total CPU capacity for a CPU group. The value of the group cap depends on the group class, or priority level assigned. The computed group cap can be thought of as "a number of LP's worth of CPU time". This group budget is shared, so if only a single VM were active, it could use the entire group's CPU allocation for itself.

The CPU group cap is calculated as $G = n \times C$, where:

- G is the amount of host LP we'd like to assign to the group
- n is the total number of logical processors (LPs) in the group
- C is the maximum CPU allocation — that is, the class of service desired for the group, expressed as a percentage of the system's total compute capacity

For example, consider a CPU group configured with 4 logical processors (LPs), and a cap of 50%.

- $G = n \times C$
- $G = 4 \times 50\%$
- $G = 2$ LP's worth of CPU time for the entire group

In this example, the CPU group G is allocated 2 LP's worth of CPU time.

Note that the group cap applies regardless of the number of virtual machines or virtual processors bound to the group, and regardless of the state (e.g., shutdown or started) of the virtual machines assigned to the CPU group. Therefore, each VM bound to the same CPU group will receive a fraction of the group's total CPU allocation, and this will change with the number of VMs bound to the CPU group. Therefore, as VMs are bound or unbound VMs from a CPU group, the overall CPU group cap must be readjusted and set to maintain the resulting per-VM cap desired. The VM host administrator or virtualization management software layer is responsible for managing group caps as necessary to achieve the desired per-VM CPU resource allocation.

Example Classes of Service

Let's look at some simple examples. To start with, assume the Hyper-V host administrator would like to support two tiers of service for guest VMs:

1. A low-end "C" tier. We'll give this tier 10% of the entire host's compute resources.
2. A mid-range "B" tier. This tier is allocated 50% of the entire host's compute resources.

At this point in our example we'll assert that no other CPU resource controls are in use, such as individual VM caps, weights, and reserves. However, individual VM caps are important, as we'll see a bit later.

For simplicity's sake, let's assume each VM has 1 VP, and that our host has 8 LPs. We'll start with an empty host.

To create the "B" tier, the host administrator sets the group cap to 50%:

- $G = n * C$
- $G = 8 * 50\%$
- $G = 4$ LP's worth of CPU time for the entire group

The host administrator adds a single "B" tier VM. At this point, our "B" tier VM can use at most 50% worth of the host's CPU, or the equivalent of 4 LPs in our example system.

Now, the admin adds a second "Tier B" VM. The CPU group's allocation—is divided evenly among all the VMs. We've got a total of 2 VMs in Group B, so each VM now gets half of Group B's total of 50%, 25% each, or the equivalent of 2 LPs worth of compute time.

Setting CPU Caps on Individual VMs

In addition to the group cap, each VM can also have an individual "VM cap". Per-VM CPU resource controls, including a CPU cap, weight, and reserve, have been a part of Hyper-V since its introduction. When combined with a group cap, a VM cap specifies the maximum amount of CPU that each VP can get, even if the group has CPU resources available.

For example, the host administrator might want to place a 10% VM cap on "C" VMs. That way, even if most "C" VPs are idle, each VP could never get more than 10%. Without a VM cap, "C" VMs could opportunistically achieve performance beyond levels allowed by their tier.

Isolating VM Groups to Specific Host Processors

Hyper-V host administrators may also want the ability to dedicate compute resources to a VM. For example, imagine the administrator wanted to offer a premium "A" VM that has a class cap of 100%. These premium VMs also require the lowest scheduling latency and jitter possible; that is, they may not be de-scheduled by any other VM. To achieve this separation, a CPU group can also be configured with a specific LP affinity mapping.

For example, to fit an "A" VM on the host in our example, the administrator would create a new CPU group, and set the group's processor affinity to a subset of the host's LPs. Groups B and C would be affinitized to the remaining LPs. The administrator could create a single VM in Group A, which would then have exclusive access to all LPs in Group A, while the presumably lower tier groups B and C would share the remaining LPs.

Segregating Root VPs from Guest VPs

By default, Hyper-V will create a root VP on each underlying physical LP. These root VPs are strictly mapped 1:1 with the system LPs, and do not migrate — that is, each root VP will always execute on the same physical LP. Guest VPs may be run on any available LP, and will share execution with root VPs.

However, it may be desirable to completely separate root VP activity from guest VPs. Consider our example above where we implement a premium "A" tier VM. To ensure our "A" VM's VPs have the lowest possible latency and "jitter", or scheduling variation, we'd like to run them on a dedicated set of LPs and ensure the root does not run on these LPs.

This can be accomplished using a combination of the "minroot" configuration, which limits the host OS partition to running on a subset of the total system logical processors, along with one or more affinitized CPU groups.

The virtualization host can be configured to restrict the host partition to specific LPs, with one or more CPU groups affinitized to the remaining LPs. In this manner, the root and guest partitions can run on dedicated CPU resources, and completely isolated, with no CPU sharing.

For more information about the "minroot" configuration, see [Hyper-V Host CPU Resource Management](#).

Using the CpuGroups Tool

Let's look at some examples of how to use the CpuGroups tool.

ⓘ Note

Command line parameters for the CpuGroups tool are passed using only spaces as delimiters. No '/' or '-' characters should proceed the desired command line switch.

Discovering the CPU Topology

Executing CpuGroups with the GetCpuTopology returns information about the current system, as shown below, including the LP Index, the NUMA node to which the LP belongs, the Package and Core IDs, and the ROOT VP index.

The following example shows a system with 2 CPU sockets and NUMA nodes, a total of 32 LPs, and multi threading enabled, and configured to enable Minroot with 8 root VPs, 4 from each NUMA node. The LPs that have root VPs have a RootVpIndex ≥ 0 ; LPs with a RootVpIndex of -1 are not available to the root partition, but are still managed by the hypervisor and will run guest VPs as allowed by other configuration settings.

Console

```
C:\vm\tools>CpuGroups.exe GetCpuTopology
```

LpIndex	NodeNumber	PackageId	CoreId	RootVpIndex
0	0	0	0	0
1	0	0	0	1
2	0	0	1	2
3	0	0	1	3
4	0	0	2	-1
5	0	0	2	-1
6	0	0	3	-1
7	0	0	3	-1
8	0	0	4	-1
9	0	0	4	-1
10	0	0	5	-1
11	0	0	5	-1
12	0	0	6	-1
13	0	0	6	-1
14	0	0	7	-1
15	0	0	7	-1
16	1	1	16	4
17	1	1	16	5
18	1	1	17	6
19	1	1	17	7
20	1	1	18	-1
21	1	1	18	-1
22	1	1	19	-1
23	1	1	19	-1
24	1	1	20	-1

25	1	1	20	-1
26	1	1	21	-1
27	1	1	21	-1
28	1	1	22	-1
29	1	1	22	-1
30	1	1	23	-1
31	1	1	23	-1

Example 2 – Print all CPU groups on the host

Here, we'll list all CPU groups on the current host, their GroupId, the group's CPU cap, and the indices of LPs assigned to that group.

Note that valid CPU cap values are in the range [0, 65536], and these values express the group cap in percent (e.g., 32768 = 50%).

Console

```
C:\vm\tools>CpuGroups.exe GetGroups
```

CpuGroupId	CpuCap	LpIndexes
-----	-----	-----
36AB08CB-3A76-4B38-992E-000000000002	32768	4,5,6,7,8,9,10,11,20,21,22,23
36AB08CB-3A76-4B38-992E-000000000003	65536	12,13,14,15
36AB08CB-3A76-4B38-992E-000000000004	65536	24,25,26,27,28,29,30,31

Example 3 – Print a single CPU group

In this example, we'll query a single CPU Group using the GroupId as a filter.

Console

```
C:\vm\tools>CpuGroups.exe GetGroups /GroupId:36AB08CB-3A76-4B38-992E-000000000003
```

CpuGroupId	CpuCap	LpIndexes
-----	-----	-----
36AB08CB-3A76-4B38-992E-000000000003	65536	12,13,14,15

Example 4 – Create a new CPU group

Here, we'll create a new CPU group, specifying the Group ID and the set of LPs to assign to the group.

Console

```
C:\vm\tools>CpuGroups.exe CreateGroup /GroupId:36AB08CB-3A76-4B38-992E-000000000001 /GroupAffinity:0,1,16,17
```

Now display our newly added group.

Console

```
C:\vm\tools>CpuGroups.exe GetGroups
CpuGroupId                      CpuCap  LpIndexes
-----
36AB08CB-3A76-4B38-992E-000000000001 65536 0,1,16,17
36AB08CB-3A76-4B38-992E-000000000002 32768 4,5,6,7,8,9,10,11,20,21,22,23
36AB08CB-3A76-4B38-992E-000000000003 65536 12,13,14,15
36AB08CB-3A76-4B38-992E-000000000004 65536 24,25,26,27,28,29,30,31
```

Example 5 – Set the CPU group cap to 50%

Here, we'll set the CPU group cap to 50%.

Console

```
C:\vm\tools>CpuGroups.exe SetGroupProperty /GroupId:36AB08CB-3A76-4B38-992E-000000000001 /CpuCap:32768
```

Now let's confirm our setting by displaying the group we just updated.

Console

```
C:\vm\tools>CpuGroups.exe GetGroups /GroupId:36AB08CB-3A76-4B38-992E-000000000001

CpuGroupId                      CpuCap  LpIndexes
-----
36AB08CB-3A76-4B38-992E-000000000001 32768 0,1,16,17
```

Example 6 – Print CPU group ids for all VMs on the host

Console

```
C:\vm\tools>CpuGroups.exe GetVmGroup
```

```
VmName                      VmId
CpuGroupId
-----
```

```

G2 4ABCFC2F-6C22-498C-BB38-7151CE678758 36ab08cb-3a76-4b38-992e-
000000000002
P1 973B9426-0711-4742-AD3B-D8C39D6A0DEC 36ab08cb-3a76-4b38-992e-
000000000003
P2 A593D93A-3A5F-48AB-8862-A4350E3459E8 36ab08cb-3a76-4b38-992e-
000000000004
G3 B0F3FCD5-FECF-4A21-A4A2-DE4102787200 36ab08cb-3a76-4b38-992e-
000000000002
G1 F699B50F-86F2-4E48-8BA5-EB06883C1FDC 36ab08cb-3a76-4b38-992e-
000000000002

```

Example 7 – Unbind a VM from the CPU group

To remove a VM from a CPU group, set to VM's CpuGroupId to the NULL GUID. This unbinds the VM from the CPU group.

Console

```

C:\vm\tools>CpuGroups.exe SetVmGroup /VmName:g1 /GroupId:00000000-0000-0000-
0000-000000000000

C:\vm\tools>CpuGroups.exe GetVmGroup
VmName                               VmId
CpuGroupId
-----
-----
G2 4ABCFC2F-6C22-498C-BB38-7151CE678758 36ab08cb-3a76-4b38-992e-
000000000002
P1 973B9426-0711-4742-AD3B-D8C39D6A0DEC 36ab08cb-3a76-4b38-992e-
000000000003
P2 A593D93A-3A5F-48AB-8862-A4350E3459E8 36ab08cb-3a76-4b38-992e-
000000000004
G3 B0F3FCD5-FECF-4A21-A4A2-DE4102787200 36ab08cb-3a76-4b38-992e-
000000000002
G1 F699B50F-86F2-4E48-8BA5-EB06883C1FDC 00000000-0000-0000-0000-
000000000000

```

Example 8 – Bind a VM to an existing CPU group

Here, we'll add a VM to an existing CPU group. Note that the VM must not be bound to any existing CPU group, or setting CPU group id will fail.

Console

```

C:\vm\tools>CpuGroups.exe SetVmGroup /VmName:g1 /GroupId:36AB08CB-3A76-4B38-
992E-000000000001

```

Now, confirm the VM G1 is in the desired CPU group.

Console

```
C:\vm\tools>CpuGroups.exe GetVmGroup
```

VmName	VmId
--------	------

CpuGroupId

G2	4ABCFC2F-6C22-498C-BB38-7151CE678758	36ab08cb-3a76-4b38-992e-000000000002
----	--------------------------------------	--------------------------------------

P1	973B9426-0711-4742-AD3B-D8C39D6A0DEC	36ab08cb-3a76-4b38-992e-000000000003
----	--------------------------------------	--------------------------------------

P2	A593D93A-3A5F-48AB-8862-A4350E3459E8	36ab08cb-3a76-4b38-992e-000000000004
----	--------------------------------------	--------------------------------------

G3	B0F3FCD5-FECF-4A21-A4A2-DE4102787200	36ab08cb-3a76-4b38-992e-000000000002
----	--------------------------------------	--------------------------------------

G1	F699B50F-86F2-4E48-8BA5-EB06883C1FDC	36AB08CB-3A76-4B38-992E-000000000001
----	--------------------------------------	--------------------------------------

Example 9 – Print all VMs grouped by CPU group id

Console

```
C:\vm\tools>CpuGroups.exe GetGroupVms
```

CpuGroupId	VmName
------------	--------

VmId

36AB08CB-3A76-4B38-992E-000000000001	G1 F699B50F-86F2-4E48-8BA5-EB06883C1FDC
--------------------------------------	---

36ab08cb-3a76-4b38-992e-000000000002	G2 4ABCFC2F-6C22-498C-BB38-7151CE678758
--------------------------------------	---

36ab08cb-3a76-4b38-992e-000000000002	G3 B0F3FCD5-FECF-4A21-A4A2-DE4102787200
--------------------------------------	---

36ab08cb-3a76-4b38-992e-000000000003	P1 973B9426-0711-4742-AD3B-D8C39D6A0DEC
--------------------------------------	---

36ab08cb-3a76-4b38-992e-000000000004	P2 A593D93A-3A5F-48AB-8862-A4350E3459E8
--------------------------------------	---

Example 10 – Print all VMs for a single CPU group

Console

```
C:\vm\tools>CpuGroups.exe GetGroupVms /GroupId:36ab08cb-3a76-4b38-992e-000000000002
```

CpuGroupId

VmName

VmId

```
-----  
----  
36ab08cb-3a76-4b38-992e-000000000002      G2 4ABCFC2F-6C22-498C-BB38-  
7151CE678758  
36ab08cb-3a76-4b38-992e-000000000002      G3 B0F3FCD5-FECF-4A21-A4A2-  
DE4102787200
```

Example 11 – Attempting to delete a non-empty CPU Group

Only empty CPU groups—that is, CPU groups with no bound VMs—can be deleted. Attempting to delete a non-empty CPU group will fail.

Console

```
C:\vm\tools>CpuGroups.exe DeleteGroup /GroupId:36ab08cb-3a76-4b38-992e-  
000000000001  
(null)  
Failed with error 0xc0350070
```

Example 12 – Unbind the only VM from a CPU group and delete the group

In this example, we'll use several commands to examine a CPU group, remove the single VM belonging to that group, then delete the group.

First, let's enumerate the VMs in our group.

Console

```
C:\vm\tools>CpuGroups.exe GetGroupVms /GroupId:36AB08CB-3A76-4B38-992E-  
000000000001  
CpuGroupId      VmName  
VmId  
-----  
----  
36AB08CB-3A76-4B38-992E-000000000001      G1 F699B50F-86F2-4E48-8BA5-  
EB06883C1FDC
```

We see that only a single VM, named G1, belongs to this group. Let's remove the G1 VM from our group by setting the VM's group ID to NULL.

Console

```
C:\vm\tools>CpuGroups.exe SetVmGroup /VmName:g1 /GroupId:00000000-0000-0000-0000-000000000000
```

And verify our change...

Console

```
C:\vm\tools>CpuGroups.exe GetVmGroup /VmName:g1
VmName                               VmId
CpuGroupId
-----
-----
      G1 F699B50F-86F2-4E48-8BA5-EB06883C1FDC 00000000-0000-0000-0000-000000000000
```

Now that the group is empty, we can safely delete it.

Console

```
C:\vm\tools>CpuGroups.exe DeleteGroup /GroupId:36ab08cb-3a76-4b38-992e-000000000001
```

And confirm our group is gone.

Console

```
C:\vm\tools>CpuGroups.exe GetGroups
CpuGroupId                               CpuCap                               LpIndexes
-----
36AB08CB-3A76-4B38-992E-000000000002 32768 4,5,6,7,8,9,10,11,20,21,22,23
36AB08CB-3A76-4B38-992E-000000000003 65536 12,13,14,15
36AB08CB-3A76-4B38-992E-000000000004 65536 24,25,26,27,28,29,30,31
```

Example 13 – Bind a VM back to its original CPU group

Console

```
C:\vm\tools>CpuGroups.exe SetVmGroup /VmName:g1 /GroupId:36AB08CB-3A76-4B38-992E-000000000002
```

```
C:\vm\tools>CpuGroups.exe GetGroupVms
```

```
CpuGroupId VmName VmId
-----
-----
36ab08cb-3a76-4b38-992e-000000000002 G2 4ABCFC2F-6C22-498C-BB38-7151CE678758
36ab08cb-3a76-4b38-992e-000000000002 G3 B0F3FCD5-FECF-4A21-A4A2-DE4102787200
```

36AB08CB-3A76-4B38-992E-000000000002 G1 F699B50F-86F2-4E48-8BA5-EB06883C1FDC
36ab08cb-3a76-4b38-992e-000000000003 P1 973B9426-0711-4742-AD3B-D8C39D6A0DEC
36ab08cb-3a76-4b38-992e-000000000004 P2 A593D93A-3A5F-48AB-8862-A4350E3459E8

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Manage Hyper-V hypervisor scheduler types

Article • 04/14/2025 •

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local, versions 23H2 and 22H2](#)

This article describes the modes of virtual processor scheduling logic in Windows Server and Windows. These modes, or *scheduler types*, determine how the Hyper-V hypervisor allocates and manages work across guest virtual processors. A Hyper-V host administrator can:

- Select hypervisor scheduler types best suited for guest virtual machines (VMs).
- Configure VMs to take advantage of scheduling logic.

Background

Before you consider the logic and controls behind Hyper-V virtual processor scheduling, it's important to understand certain concepts like simultaneous multithreading (SMT) and how Hyper-V virtualizes processors.

Understand SMT

SMT is a technique in modern processor design that provides a way for separate, independent execution threads to share processor resources. SMT usually gives a modest performance boost to workloads. It parallelizes computations when possible, increasing instruction throughput. However, when threads compete with each other for shared processor resources, there's sometimes no noticeable improvement in performance. There might even be a slight decrease in performance.

In order to use SMT with Windows Server, you must have a compatible processor. Processors with Intel Hyper-Threading Technology (Intel HT Technology) or Advanced Micro Devices (AMD) SMT are examples of compatible processors.

For the purposes of this article, the descriptions of SMT and how it's used by Hyper-V apply equally to both Intel and AMD systems.

- For more information on Intel HT Technology, see [Intel Hyper-Threading Technology](#) .
- For more information on AMD SMT, see [AMD "Zen" Core Architecture](#) .

Understand how Hyper-V virtualizes processors

Before you consider hypervisor scheduler types, you should understand Hyper-V architecture. For a detailed summary of how this architecture works, see [Hyper-V Technology overview](#). For now, keep the following concepts in mind:

- Hyper-V creates and manages VM partitions, allocating and sharing compute resources across them, under control of the hypervisor. Partitions provide strong isolation boundaries between all guest VMs and between guest VMs and the root partition.
- The root partition is itself a VM partition, although it has unique properties and greater privileges than guest VMs. The root partition:
 - Provides management services that control all guest VMs.
 - Provides virtual device support for guests.
 - Manages all device input and output for guest VMs.

We recommend not running any application workloads in the root partition.

- A one-to-one mapping exists between the virtual processors of the root partition and the underlying logical processors. A host virtual processor always runs on the same underlying logical processor. There's no migration of root partition virtual processors.
- By default, the logical processors that host the root partition virtual processors can also run guest virtual processors.
- The hypervisor can schedule a guest virtual processor to run on any available logical processor. The hypervisor scheduler tries to consider temporal cache locality, nonuniform memory access (NUMA) topology, and many other factors when it schedules a guest virtual processor. But ultimately, the virtual processor can be scheduled on any host logical processor.

Hypervisor scheduler types

The Hyper-V hypervisor supports several modes of scheduler logic, which determine how the hypervisor schedules virtual processors on the underlying logical processors. These scheduler types are:

- [The classic scheduler](#)
- [The core scheduler](#)
- [The root scheduler](#)

The classic scheduler

The classic scheduler is the default scheduler that's used on all versions of Hyper-V earlier than Windows Server 2019. The classic scheduler provides a fair-share, preemptive, round-robin

scheduling model for guest virtual processors.

The classic scheduler type is appropriate for most traditional Hyper-V uses, such as private clouds, hosting providers, and so on. The performance characteristics of the classic scheduler type are optimized to support a wide range of virtualization scenarios, such as:

- Over-subscribing of virtual processors to logical processors.
- Running many heterogeneous VMs and workloads at the same time.
- Running larger-scale high-performance VMs.
- Supporting the full feature set of Hyper-V without restrictions and other scenarios.

The core scheduler

The hypervisor core scheduler is an alternative to the classic scheduler logic. The core scheduler is available starting with Windows Server 2016 and Windows 10 version 1607. The core scheduler offers a strong security boundary for guest workload isolation. It also reduces performance variability for workloads inside VMs that run on an SMT-enabled virtualization host. The core scheduler supports running both SMT and non-SMT VMs at the same time on the same SMT-enabled virtualization host.

The core scheduler:

- Uses the virtualization host's SMT topology.
- Optionally exposes SMT pairs to guest VMs.
- Schedules groups of guest virtual processors from the same VM onto groups of SMT logical processors.

This work happens symmetrically. If logical processors are in groups of two, virtual processors are scheduled in groups of two, and a core is never shared between VMs. When you schedule a virtual processor for a VM without SMT enabled, that virtual processor consumes the entire core when it runs. When you use the core scheduler:

- It creates a strong security boundary for guest workload isolation. Guest virtual processors can only run on underlying physical core pairs, reducing vulnerability to side-channel snooping attacks.
- It reduces variability in throughput.
- It can potentially reduce performance. If only one virtual processor in a group can run, only one of the instruction streams in the core launches. The other is left idle.
- The operating system (OS) and applications running in the guest VM can use SMT behavior and programming interfaces (APIs) to control and distribute work across SMT threads, just like they do with a physical machine.

As of Windows Server 2019, Hyper-V uses the core scheduler by default. In earlier versions like Windows Server 2016, the core scheduler is optional, and the classic scheduler is the default one.

Core scheduler behavior with host SMT disabled

In some cases, you might configure the hypervisor to use the core scheduler type, but the SMT capability is disabled or isn't present on the virtualization host. In these cases, Hyper-V uses the classic scheduler behavior regardless of the hypervisor scheduler type setting.

The root scheduler

The root scheduler arrived with Windows 10, version 1803. When you enable the root scheduler type, the hypervisor gives the root partition control of work scheduling. In the root partition OS instance, the NT scheduler manages all aspects of assigning work to system logical processors.

The root scheduler addresses the unique requirements to support a utility partition and provide strong workload isolation, as used with Windows Defender Application Guard (WDAG). In this scenario, leaving scheduling responsibilities to the root OS offers several advantages:

- You can use CPU resource controls applicable to container scenarios with the utility partition, simplifying management and deployment.
- The root OS scheduler can readily gather metrics about workload CPU use inside the container. It can use this data as input to the scheduling policy and apply it to all other workloads in the system.
- These same metrics also help attribute work done in an application container to the host system. Tracking these metrics is more difficult with traditional VM workloads, where some work on behalf of all running VMs takes place in the root partition.

Root scheduler use on client systems

Starting with Windows 10, version 1803, the root scheduler is used by default on client systems only, which means:

- You can enable the hypervisor to support virtualization-based security and WDAG workload isolation.
- It's important to properly operate future systems with heterogeneous core architectures.

This configuration is the only supported hypervisor scheduler configuration for client systems. Administrators shouldn't attempt to override the default hypervisor scheduler type on Windows client systems.

VM CPU resource controls and the root scheduler

The resource controls that the Hyper-V hypervisor uses on individual VM processors aren't supported when you enable the hypervisor root scheduler. On the root OS, the scheduler logic manages host resources on a global basis and doesn't manage the guest resources of individual VMs. The Hyper-V per-VM processor resource controls, such as caps, weights, and reserves, only apply where the hypervisor directly controls virtual processor scheduling, such as with the classic and core scheduler types.

Root scheduler use on server systems

We don't recommend using the root scheduler with Hyper-V on servers. Its performance characteristics aren't yet fully characterized and tuned to accommodate the wide range of workloads typical of many server virtualization deployments.

Enable SMT in guest VMs

After you configure the virtualization host's hypervisor to use the core scheduler type, you can also configure guest VMs to use SMT. Specifically, you can specify the number of virtual SMT threads to expose to a guest VM. The guest OS scheduler and VM workloads can then detect and use the SMT topology in their own work scheduling.

- In Windows Server 2016, guest SMT isn't configured by default. A Hyper-V host administrator must explicitly enable it.
- In Windows Server 2019 and later versions, new VMs that you create on the host inherit the host SMT topology by default. For example, a version 9.0 VM that you create on a host with two SMT threads per core also has two SMT threads per core.

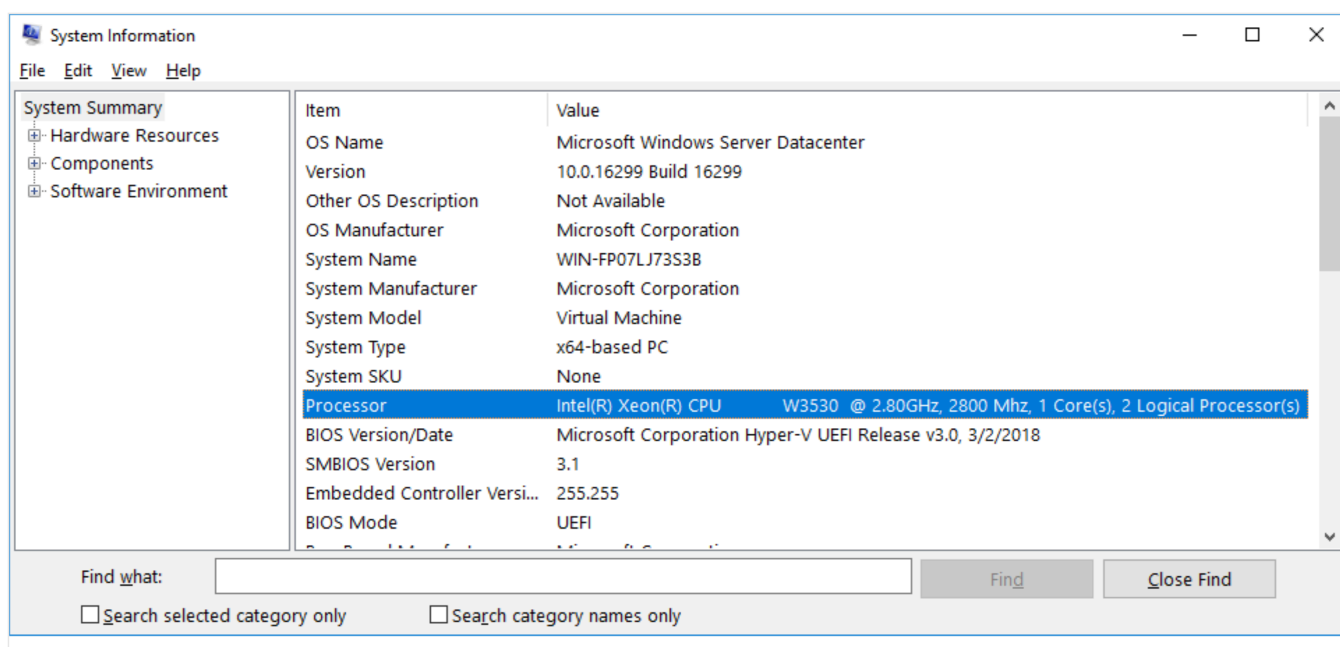
You must use PowerShell to enable SMT in a guest VM. There's no user interface provided in Hyper-V Manager. To enable SMT in a guest VM:

1. Open a PowerShell window by using an account that's a member of the Hyper-V Administrators or equivalent group.
2. Run `Set-VMProcessor -VMName <VM-name> -HwThreadCountPerCore <n>`, where `<n>` is the number of SMT threads per core the guest VM sees. If you use a value of `0` for `<n>`, the `HwThreadCountPerCore` value is set to the host's SMT thread count per core.

ⓘ Note

Windows Server 2016 doesn't support setting `HwThreadCountPerCore` to `0`.

The following screenshot shows system information taken from the guest OS running in a VM. There are two virtual processors, and SMT is enabled. The guest OS detects two logical processors that belong to the same core.



Configure the hypervisor scheduler type

The type of hypervisor scheduler that Hyper-V uses by default depends on the OS version. No matter which scheduler is the default one on your OS, you can configure the hypervisor to use the core scheduler. The core scheduler increases security by restricting guest virtual processors to run on corresponding physical SMT pairs. This configuration supports the use of VMs with SMT scheduling for their guest virtual processors.

ⓘ Note

On Windows Server 2016, Hyper-V uses the classic scheduler by default. If you use Hyper-V on Windows Server 2016 or later, we recommend that you select the core scheduler. Making this switch ensures your virtualization hosts are optimally protected against potentially malicious guest VMs.

Consider impacts on security and performance

To help ensure Hyper-V hosts are deployed in an optimal security configuration, Hyper-V uses the core hypervisor scheduler model by default starting with Windows Server 2019. The host administrator might optionally configure the host to use the legacy classic scheduler. Before overriding the default settings, administrators should carefully read, understand, and consider

the impacts each scheduler type has on the security and performance of virtualization hosts. For more information, see [About Hyper-V hypervisor scheduler type selection](#).

Select the hypervisor scheduler type on Windows Server

The hypervisor scheduler configuration is controlled by the `hypervisorschedulertype` boot configuration data (BCD) entry.

To select a scheduler type:

1. Open Command Prompt with administrator privileges.
2. Enter `bcdedit /set hypervisorschedulertype <type>`, where `<type>` is one of the following options:
 - `Classic`
 - `Core`
 - `Root`

You must restart the system for any changes you make to the hypervisor scheduler type to take effect.

 Note

The hypervisor root scheduler isn't supported on Hyper-V on Windows Server at this time. Hyper-V administrators shouldn't attempt to configure the root scheduler for use with server virtualization scenarios.

Determine the current scheduler type

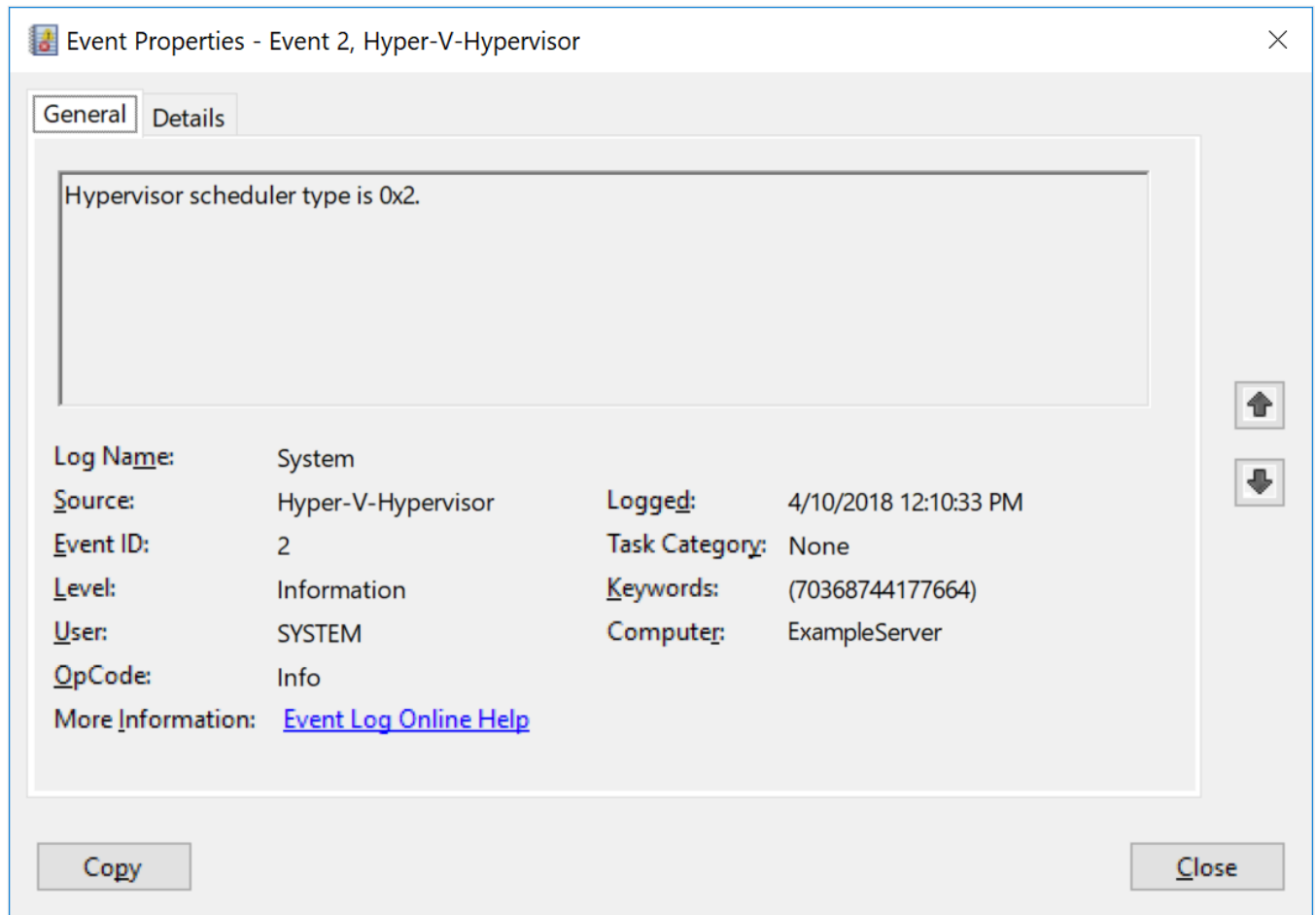
You can determine the hypervisor scheduler type that Hyper-V is currently using by examining the Windows system log. Look for the most recent hypervisor launch event that has an ID of 2. That event reports the hypervisor scheduler type configured at the hypervisor launch. The following values are possible for the hypervisor scheduler type:

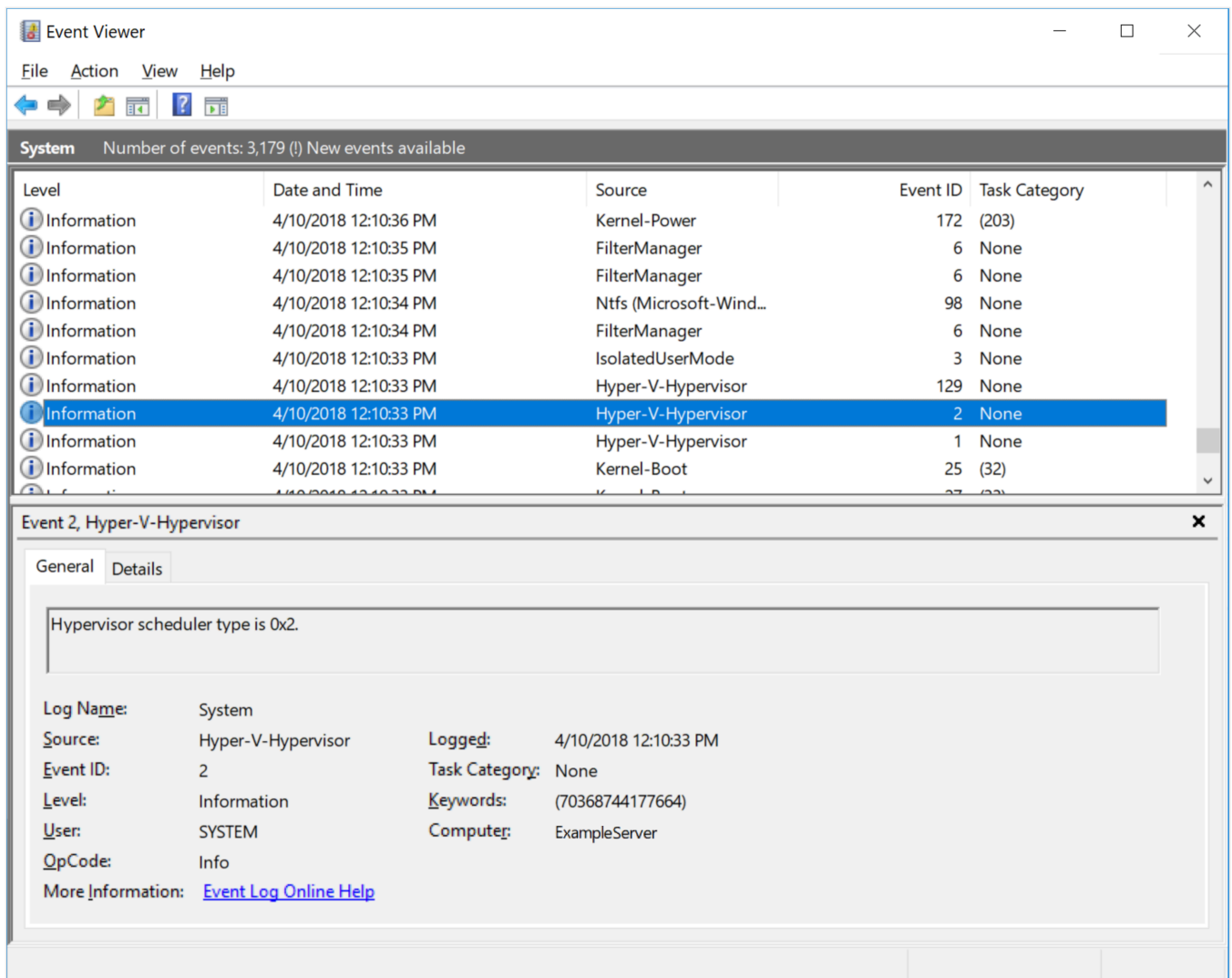
 Expand table

Value	Scheduler type
1	Classic scheduler, SMT disabled

Value	Scheduler type
2	Classic scheduler
3	Core scheduler
4	Root scheduler

You can use Event Viewer or PowerShell to view hypervisor launch event logs.



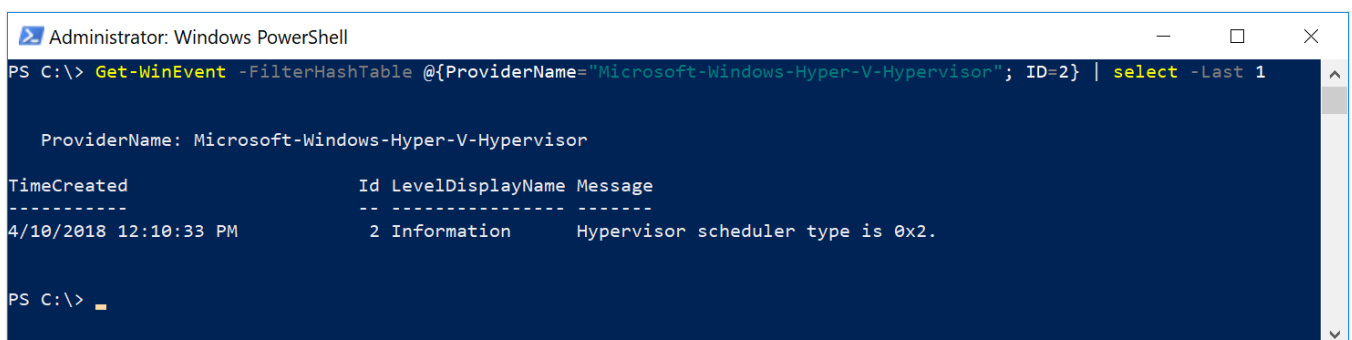


Use PowerShell to query system logs for a hypervisor launch event

To query the system logs for a hypervisor event with an ID of 2 by using PowerShell, run the following command:

PowerShell

```
Get-WinEvent -FilterHashTable @{ProviderName="Microsoft-Windows-Hyper-V-Hypervisor"; ID=2} -MaxEvents 1
```



About Hyper-V hypervisor scheduler type selection

Article • 12/23/2021 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2
and 22H2

This document describes important changes to Hyper-V's default and recommended use of hypervisor scheduler types. These changes impact both system security and virtualization performance. Virtualization host administrators should review and understand the changes and implications described in this document, and carefully evaluate the impacts, suggested deployment guidance and risk factors involved to best understand how to deploy and manage Hyper-V hosts in the face of the rapidly changing security landscape.

Important

Currently known side-channel security vulnerabilities sighted in multiple processor architectures could be exploited by a malicious guest VM through the scheduling behavior of the legacy hypervisor classic scheduler type when run on hosts with Simultaneous Multithreading (SMT) enabled. If successfully exploited, a malicious workload could observe data outside its partition boundary. This class of attacks can be mitigated by configuring the Hyper-V hypervisor to utilize the hypervisor core scheduler type and reconfiguring guest VMs. With the core scheduler, the hypervisor restricts a guest VM's VPs to run on the same physical processor core, therefore strongly isolating the VM's ability to access data to the boundaries of the physical core on which it runs. This is a highly effective mitigation against these side-channel attacks, which prevents the VM from observing any artifacts from other partitions, whether the root or another guest partition. Therefore, Microsoft is changing the default and recommended configuration settings for virtualization hosts and guest VMs.

Background

Starting with Windows Server 2016, Hyper-V supports several methods of scheduling and managing virtual processors, referred to as hypervisor scheduler types. A detailed description of all hypervisor scheduler types can be found in [Understanding and using Hyper-V hypervisor scheduler types](#).

ⓘ Note

New hypervisor scheduler types were first introduced with Windows Server 2016, and are not available in prior releases. All versions of Hyper-V prior to Windows Server 2016 support only the classic scheduler. Support for the core scheduler was only recently published.

About hypervisor scheduler types

This article focuses specifically on the use of the new hypervisor core scheduler type versus the legacy "classic" scheduler, and how these scheduler types intersect with the use of Symmetric Multi-Threading, or SMT. It is important to understand the differences between the core and classic schedulers and how each places work from guest VMs on the underlying system processors.

The classic scheduler

The classic scheduler refers to the fair-share, round robin method of scheduling work on virtual processors (VPs) across the system - including root VPs as well as VPs belonging to guest VMs. The classic scheduler has been the default scheduler type used on all versions of Hyper-V (until Windows Server 2019, as described herein). The performance characteristics of the classic scheduler are well understood, and the classic scheduler is demonstrated to ably support over-subscription of workloads - that is, over-subscription of the host's VP:LP ratio by a reasonable margin (depending on the types of workloads being virtualized, overall resource utilization, etc.).

When run on a virtualization host with SMT enabled, the classic scheduler will schedule guest VPs from any VM on each SMT thread belonging to a core independently. Therefore, different VMs can be running on the same core at the same time (one VM running on one thread of a core while another VM is running on the other thread).

The core scheduler

The core scheduler leverages the properties of SMT to provide isolation of guest workloads, which impacts both security and system performance. The core scheduler ensures that VPs from a VM are scheduled on sibling SMT threads. This is done symmetrically so that if LPs are in groups of two, VPs are scheduled in groups of two, and a system CPU core is never shared between VMs.

By scheduling guest VPs on underlying SMT pairs, the core scheduler offers a strong security boundary for workload isolation, and can also be used to reduce performance variability for latency sensitive workloads.

Note that when the VP is scheduled for a virtual machine without SMT enabled, that VP will consume the entire core when it runs, and the core's sibling SMT thread will be left idle. This is necessary to provide the correct workload isolation, but impacts overall system performance, especially as the system LPs become over-subscribed - that is, when total VP:LP ratio exceeds 1:1. Therefore, running guest VMs configured without multiple threads per core is a sub-optimal configuration.

Benefits of the using the core scheduler

The core scheduler offers the following benefits:

- A strong security boundary for guest workload isolation - Guest VPs are constrained to run on underlying physical core pairs, reducing vulnerability to side-channel snooping attacks.
- Reduced workload variability - Guest workload throughput variability is significantly reduced, offering greater workload consistency.
- Use of SMT in guest VMs - The OS and applications running in the guest virtual machine can utilize SMT behavior and programming interfaces (APIs) to control and distribute work across SMT threads, just as they would when run non-virtualized.

The core scheduler is currently used on Azure virtualization hosts, specifically to take advantage of the strong security boundary and low workload variability. Microsoft believes that the core scheduler type should be and will continue to be the default hypervisor scheduling type for the majority of virtualization scenarios. Therefore, to ensure our customers are secure by default, Microsoft is making this change for Windows Server 2019 now.

Core scheduler performance impacts on guest workloads

While required to effectively mitigate certain classes of vulnerabilities, the core scheduler may also potentially reduce performance. Customers may see a difference in the performance characteristics with their VMs and impacts to the overall workload capacity of their virtualization hosts. In cases where the core scheduler must run a non-SMT VP, only one of the instruction streams in the underlying logical core executes

while the other must be left idle. This will limit the total host capacity for guest workloads.

These performance impacts can be minimized by following the deployment guidance in this document. Host administrators must carefully consider their specific virtualization deployment scenarios and balance their tolerance for security risk against the need for maximum workload density, over-consolidation of virtualization hosts, etc.

Changes to the default and recommended configurations for Windows Server 2016 and Windows Server 2019

Deploying Hyper-V hosts with the maximum security posture requires use of the hypervisor core scheduler type. To ensure our customers are secure by default, Microsoft is changing the following default and recommended settings.

Note

While the hypervisor's internal support for the scheduler types was included in the initial release of Windows Server 2016, Windows Server 1709, and Windows Server 1803, updates are required in order to access the configuration control which allows selecting the hypervisor scheduler type. Please refer to [Understanding and using Hyper-V hypervisor scheduler types](#) for details on these updates.

Virtualization host changes

- The hypervisor will use the core scheduler by default beginning with Windows Server 2019.
- Microsoft recommends configuring the core scheduler on Windows Server 2016. The hypervisor core scheduler type is supported in Windows Server 2016, however the default is the classic scheduler. The core scheduler is optional and must be explicitly enabled by the Hyper-V host administrator.

Virtual machine configuration changes

- On Windows Server 2019, new virtual machines created using the default VM version 9.0 will automatically inherit the SMT properties (enabled or disabled) of the virtualization host. That is, if SMT is enabled on the physical host, newly

created VMs will also have SMT enabled, and will inherit the SMT topology of the host by default, with the VM having the same number of hardware threads per core as the underlying system. This will be reflected in the VM's configuration with `HwThreadCountPerCore = 0`, where 0 indicates the VM should inherit the host's SMT settings.

- Existing virtual machines with a VM version of 8.2 or earlier will retain their original VM processor setting for `HwThreadCountPerCore`, and the default for 8.2 VM version guests is `HwThreadCountPerCore = 1`. When these guests run on a Windows Server 2019 host, they will be treated as follows:
 1. If the VM has a VP count that is less than or equal to the count of LP cores, the VM will be treated as a non-SMT VM by the core scheduler. When the guest VP runs on a single SMT thread, the core's sibling SMT thread will be idled. This is non-optimal, and will result in overall loss of performance.
 2. If the VM has more VPs than LP cores, the VM will be treated as an SMT VM by the core scheduler. However, the VM will not observe other indications that it is an SMT VM. For example, use of the `CPUID` instruction or Windows APIs to query CPU topology by the OS or applications will not indicate that SMT is enabled.
- When an existing VM is explicitly updated from earlier VM versions to version 9.0 through the Update-VM operation, the VM will retain its current value for `HwThreadCountPerCore`. The VM will not have SMT force-enabled.
- On Windows Server 2016, Microsoft recommends enabling SMT for guest VMs. By default, VMs created on Windows Server 2016 would have SMT disabled, that is `HwThreadCountPerCore` is set to 1, unless explicitly changed.

 Note

Windows Server 2016 does not support setting `HwThreadCountPerCore` to 0.

Managing virtual machine SMT configuration

The guest virtual machine SMT configuration is set on a per-VM basis. The host administrator can inspect and configure a VM's SMT configuration to select from the following options:

1. Configure VMs to run as SMT-enabled, optionally inheriting the host SMT topology automatically

2. Configure VMs to run as non-SMT

The SMT configuration for a VM is displayed in the Summary panes in the Hyper-V Manager console. Configuring a VM's SMT settings may be done by using the VM Settings or PowerShell.

Configuring VM SMT settings using PowerShell

To configure the SMT settings for a guest virtual machine, open a PowerShell window with sufficient permissions, and type:

PowerShell

```
Set-VMProcessor -VMName <VMName> -HwThreadCountPerCore <0, 1, 2>
```

Where:

- 0 = Inherit SMT topology from the host (this setting of HwThreadCountPerCore=0 is not supported on Windows Server 2016)
- 1 = Non-SMT
- Values > 1 = the desired number of SMT threads per core. May not exceed the number of physical SMT threads per core.

To read the SMT settings for a guest virtual machine, open a PowerShell window with sufficient permissions, and type:

PowerShell

```
(Get-VMProcessor -VMName <VMName>).HwThreadCountPerCore
```

Note that guest VMs configured with HwThreadCountPerCore = 0 indicates that SMT will be enabled for the guest, and will expose the same number of SMT threads to the guest as they are on the underlying virtualization host, typically 2.

Guest VMs may observe changes to CPU topology across VM mobility scenarios

The OS and applications in a VM may see changes to both host and VM settings before and after VM lifecycle events such as live migration or save and restore operations. During an operation in which VM state is saved and restored, both the VM's HwThreadCountPerCore setting and the realized value (that is, the computed

combination of the VM's setting and source host's configuration) are migrated. The VM will continue running with these settings on the destination host. At the point the VM is shutdown and re-started, it's possible that the realized value observed by the VM will change. This should be benign, as OS and application layer software should look for CPU topology information as part of their normal startup and initialization code flows. However, because these boot time initialization sequences are skipped during live migration or save/restore operations, VMs that undergo these state transitions could observe the originally computed realized value until they are shut down and re-started.

Alerts regarding non-optimal VM configurations

Virtual machines configured with more VPs than there are physical cores on the host result in a non-optimal configuration. The hypervisor scheduler will treat these VMs as if they are SMT-aware. However, OS and application software in the VM will be presented a CPU topology showing SMT is disabled. When this condition is detected, the Hyper-V Worker Process will log an event on the virtualization host warning the host administrator that the VM's configuration is non-optimal, and recommending SMT be enabled for the VM.

How to identify non-optimally configured VMs

You can identify non-SMT VMs by examining the System Log in Event Viewer for Hyper-V Worker Process event ID 3498, which will be triggered for a VM whenever the number of VPs in the VM is greater than the physical core count. Worker process events can be obtained from Event Viewer, or via PowerShell.

Querying the Hyper-V worker process VM event using PowerShell

To query for Hyper-V worker process event ID 3498 using PowerShell, enter the following commands from a PowerShell prompt.

PowerShell

```
Get-WinEvent -FilterHashTable @{ProviderName="Microsoft-Windows-Hyper-V-Worker"; ID=3498}
```

Impacts of guest SMT configuration on the use of hypervisor enlightenments for guest operating systems

The Microsoft hypervisor offers multiple enlightenments, or hints, which the OS running in a guest VM may query and use to trigger optimizations, such as those that might benefit performance or otherwise improve handling of various conditions when running virtualized. One recently introduced enlightenment concerns the handling of virtual processor scheduling and the use of OS mitigations for side-channel attacks that exploit SMT.

ⓘ Note

Microsoft recommends that host administrators enable SMT for guest VMs to optimize workload performance.

The details of this guest enlightenment are provided below, however the key takeaway for virtualization host administrators is that virtual machines should have `HwThreadCountPerCore` configured to match the host's physical SMT configuration. This allows the hypervisor to report that there is no non-architectural core sharing. Therefore, any guest OS supporting optimizations that require the enlightenment may be enabled. On Windows Server 2019, create new VMs and leave the default value of `HwThreadCountPerCore` (0). Older VMs migrated from Windows Server 2016 hosts can be updated to the Windows Server 2019 configuration version. After doing so, Microsoft recommends setting `HwThreadCountPerCore` = 0. On Windows Server 2016, Microsoft recommends setting `HwThreadCountPerCore` to match the host configuration (typically 2).

NoNonArchitecturalCoreSharing enlightenment details

Starting in Windows Server 2016, the hypervisor defines a new enlightenment to describe its handling of VP scheduling and placement to the guest OS. This enlightenment is defined in the [Hypervisor Top Level Functional Specification v5.0c](#).

Hypervisor synthetic CPUID leaf

`CPUID.0x40000004.EAX:18[NoNonArchitecturalCoreSharing = 1]` indicates that a virtual processor will never share a physical core with another virtual processor, except for virtual processors that are reported as sibling SMT threads. For example, a guest VP will never run on an SMT thread alongside a root VP running simultaneously on a sibling SMT thread on the same processor core. This condition is only possible when running virtualized, and so represents a non-architectural SMT behavior that also has serious security implications. The guest OS can use `NoNonArchitecturalCoreSharing` = 1 as an indication that it is safe to enable optimizations, which may help it avoid the performance overhead of setting STIBP.

In certain configurations, the hypervisor will not indicate that `NoNonArchitecturalCoreSharing` = 1. As an example, if a host has SMT enabled and is configured to use the hypervisor classic scheduler, `NoNonArchitecturalCoreSharing` will be 0. This may prevent enlightened guests from enabling certain optimizations. Therefore, Microsoft recommends that host administrators using SMT rely on the hypervisor core scheduler and ensure that virtual machines are configured to inherit their SMT configuration from the host to ensure optimal workload performance.

Summary

The security threat landscape continues to evolve. To ensure our customers are secure by default, Microsoft is changing the default configuration for the hypervisor and virtual machines starting in Windows Server 2019 Hyper-V, and providing updated guidance and recommendations for customers running Windows Server 2016 Hyper-V.

Virtualization host administrators should:

- Read and understand the guidance provided in this document
- Carefully evaluate and adjust their virtualization deployments to ensure they meet the security, performance, virtualization density, and workload responsiveness goals for their unique requirements
- Consider re-configuring existing Windows Server 2016 Hyper-V hosts to leverage the strong security benefits offered by the hypervisor core scheduler
- Update existing non-SMT VMs to reduce the performance impacts from scheduling constraints imposed by VP isolation that addresses hardware security vulnerabilities

Feedback

Was this page helpful?

 Yes

 No

Hyper-V integration services

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Integration services, often called integration components, are services that allow the virtual machine to communicate with the Hyper-V host. Many of these services are conveniences, while others are important to the virtual machine's ability to function correctly.

This article is a reference for each integration service available in Windows. It also acts as a starting point for information related to specific integration services or their history.

For more information, see the Windows Server guide on [Managing integration services](#).

Quick reference

 Expand table

Name	Windows Service Name	Linux Daemon Name	Description	Impact on VM when disabled
Hyper-V Heartbeat Service	<code>vmicheartbeat</code>	<code>hv_utils</code>	Reports that the virtual machine is running correctly.	Varies
Hyper-V Guest Shutdown Service	<code>vmicshutdown</code>	<code>hv_utils</code>	Allows the host to trigger virtual machines shutdown.	High
Hyper-V Time Synchronization Service	<code>vmictimesync</code>	<code>hv_utils</code>	Synchronizes the virtual machine's clock with the host computer's clock.	High
Hyper-V Data Exchange Service (KVP)	<code>vmickvpexchange</code>	<code>hv_kvp_daemon</code>	Provides a way to exchange basic metadata between the virtual machine and the host.	Medium
Hyper-V Volume Shadow Copy Requestor	<code>vmicvss</code>	<code>hv_vss_daemon</code>	Allows Volume Shadow Copy Service to back up the virtual machine without shutting it down.	Varies
Hyper-V Guest Service Interface	<code>vmicguestinterface</code>	<code>hv_fcopys_daemon</code>	Provides an interface for the Hyper-V host to	Low

Name	Windows Service Name	Linux Daemon Name	Description	Impact on VM when disabled
			copy files to or from the virtual machine.	
Hyper-V PowerShell Direct Service	<code>vmicvmsession</code>	not available	Provides a way to manage virtual machine with PowerShell without a network connection.	Low

Hyper-V heartbeat service

Windows Service Name: vmicheartbeat

Linux Daemon Name: hv_utils

Description: Tells the Hyper-V host that the virtual machine has an operating system installed and that it booted correctly.

Added In: Windows Server 2012, Windows 8

Impact: When disabled, the virtual machine can't report that the operating system inside of the virtual machine is operating correctly. This limitation might affect some kinds of monitoring and host-side diagnostics.

The heartbeat service makes it possible to answer basic questions like "did the virtual machine boot?".

When Hyper-V reports that a virtual machine state is "running" (see the example in the following section), it means Hyper-V set aside resources for a virtual machine; it doesn't mean that an operating system is installed or functioning. This limitation is where heartbeat becomes useful. The heartbeat service tells Hyper-V that the operating system inside the virtual machine has booted.

Check heartbeat with PowerShell

Run [Get-VM](#) as Administrator to see a virtual machine's heartbeat:

PowerShell

```
Get-VM -VMName $VMName | select Name, State, Status
```

Your output should look something like this:

Output

Name	State	Status
----	-----	-----
DemoVM	Running	Operating normally

The `Status` field is determined by the heartbeat service.

Hyper-V guest shutdown service

Windows Service Name: `vmicshutdown`

Linux Daemon Name: `hv_utils`

Description: Allows the Hyper-V host to request that the virtual machine shut down. The host can always force the virtual machine to turn off, but that action is like flipping the power switch as opposed to selecting shutdown.

Added In: Windows Server 2012, Windows 8

Impact: High Impact When disabled, the host can't trigger a friendly shutdown inside the virtual machine. All shutdowns are hard power-offs, which could cause data loss or data corruption.

Hyper-V time synchronization service

Windows Service Name: `vmictimesync`

Linux Daemon Name: `hv_utils`

Description: Synchronizes the virtual machine's system clock with the system clock of the physical computer.

Added In: Windows Server 2012, Windows 8

Impact: High Impact When disabled, the virtual machine's clock drifts erratically.

Hyper-V data exchange service (KVP)

Windows Service Name: `vmickvpexchange`

Linux Daemon Name: `hv_kvp_daemon`

Description: Provides a mechanism to exchange basic metadata between the virtual machine and the host.

Added In: Windows Server 2012, Windows 8

Impact: When disabled, virtual machines running Windows 8 or Windows Server 2012 or earlier don't receive updates to Hyper-V integration services. Disabling data exchange might also affect some kinds of monitoring and host-side diagnostics.

The data exchange service (sometimes called KVP) shares small amounts of machine information between virtual machine and the Hyper-V host by using key-value pairs (KVP)

through the Windows registry. You can also use the same mechanism to share customized data between the virtual machine and the host.

Key-value pairs consist of a “key” and a “value”. Both the key and the value are strings, and no other data types are supported. When you create or change a key-value pair, both the guest and the host can see it. The key-value pair information transfers across the Hyper-V VMbus and doesn't require any kind of network connection between the guest and the Hyper-V host.

The data exchange service is a great tool for preserving information about the virtual machine. For interactive data sharing or data transfer, use [PowerShell Direct](#).

Learn more about [Using key-value pairs to share information between the host and guest on Hyper-V](#).

Hyper-V Volume Shadow Copy Requestor

Windows Service Name: `vmicvss`

Linux Daemon Name: `hv_vss_daemon`

Description: Allows Volume Shadow Copy Service to back up applications and data on the virtual machine.

Added In: Windows Server 2012, Windows 8

Impact: When disabled, the virtual machine can't be backed up while running (using VSS).

The Volume Shadow Copy Requestor integration service is required for Volume Shadow Copy Service ([VSS](#)). The Volume Shadow Copy Service (VSS) captures and copies images for backup on running systems, particularly servers, without unduly degrading the performance and stability of the services they provide. This integration service makes that possible by coordinating the virtual machine's workloads with the host's backup process.

Read more about Volume Shadow Copy in [Backing Up and Restoring Virtual Machines](#).

Hyper-V Guest Service Interface

Windows Service Name: `vmicguestinterface`

Linux Daemon Name: `hv_fcopy_daemon`

Description: Provides an interface for the Hyper-V host to bidirectionally copy files to or from the virtual machine.

Added In: Windows Server 2012 R2, Windows 8.1

Impact: When disabled, the host can't copy files to and from the guest using `Copy-VMFile`.

Read more about the [Copy-VMFile cmdlet](#).

Notes:

Disabled by default. See [PowerShell Direct using Copy-Item](#).

Hyper-V PowerShell Direct Service

Windows Service Name: `vmicvmsession`

Linux Daemon Name: n/a

Description: Provides a mechanism to manage a virtual machine with PowerShell through a VM session without a virtual network. **Added In:** Windows Server TP3, Windows 10

Impact: If you disable this service, the host can't connect to the virtual machine with PowerShell Direct.

Notes:

The service name was originally Hyper-V VM Session Service.

PowerShell Direct is under active development and is only available on Windows 10/Windows Server Technical Preview 3 or later hosts and guests.

PowerShell Direct enables PowerShell management inside a virtual machine from the Hyper-V host, regardless of any network configuration or remote management settings on either the Hyper-V host or the virtual machine. This feature makes it easier for Hyper-V administrators to automate and script management and configuration tasks.

The following resources might be helpful for managing PowerShell Direct:

- [PowerShell Direct](#).
 - [Running script in a virtual machine](#)
 - [Copying files to and from a virtual machine](#)

Last updated on 10/28/2025

Manage Hyper-V Integration Services

Article • 02/21/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local](#), versions 23H2
and 22H2

Hyper-V Integration Services enhance virtual machine performance and provide convenience features by leveraging two-way communication with the Hyper-V host. Many of these services are conveniences, such as guest file copy, while others are important to the virtual machine's functionality, such as synthetic device drivers. This set of services and drivers are sometimes referred to as *integration components*. You can control whether or not individual convenience services operate for any given virtual machine. The driver components aren't intended to be serviced manually.

For details about each integration service, see [Hyper-V Integration Services](#).

Important

Each service you want to use must be enabled in both the host and guest in order to function. When you enable a service in the host, it automatically starts running in the guest. All integration services except *Hyper-V Guest Service Interface* are on by default on Windows guest operating systems. The services can be turned on and off individually. The next sections show you how.

Turn an integration service on or off using Hyper-V Manager

1. From the center pane, right-click the virtual machine and select **Settings**.
2. From the left pane of the **Settings** window, under **Management**, select **Integration Services**.

The Integration Services pane lists all integration services available on the Hyper-V host, and whether the host has enabled the virtual machine to use them.

Turn an integration service on or off using PowerShell

To do this in PowerShell, use [Enable-VMIntegrationService](#) and [Disable-VMIntegrationService](#).

The following examples demonstrate turning the guest file copy integration service on and off for a virtual machine named *DemoVM*.

1. Get a list of running integration services:

PowerShell

```
Get-VMIntegrationService -VMName "DemoVM"
```

2. The output should look like this:

PowerShell

VMName	Name	Enabled	PrimaryStatusDescription	SecondaryStatusDescription
DemoVM	Guest Service Interface	False	OK	
DemoVM	Heartbeat	True	OK	OK
DemoVM	Key-Value Pair Exchange	True	OK	
DemoVM	Shutdown	True	OK	
DemoVM	Time Synchronization	True	OK	
DemoVM	VSS	True	OK	

3. Turn on Guest Service Interface:

PowerShell

```
Enable-VMIntegrationService -VMName "DemoVM" -Name "Guest Service Interface"
```

4. Verify that Guest Service Interface is enabled:

PowerShell

```
Get-VMIntegrationService -VMName "DemoVM"
```

5. Turn off Guest Service Interface:

PowerShell

```
Disable-VMIntegrationService -VMName "DemoVM" -Name "Guest Service Interface"
```

Checking the guest's integration services version

Some features may not work correctly or at all if the guest's integration services aren't current. To get the version information for Windows, sign in to the guest operating system, open a command prompt, and run this command:

PowerShell

```
REG QUERY "HKLM\Software\Microsoft\Virtual Machine\Auto" /v  
IntegrationServicesVersion
```

Start and stop an integration service from a Windows guest

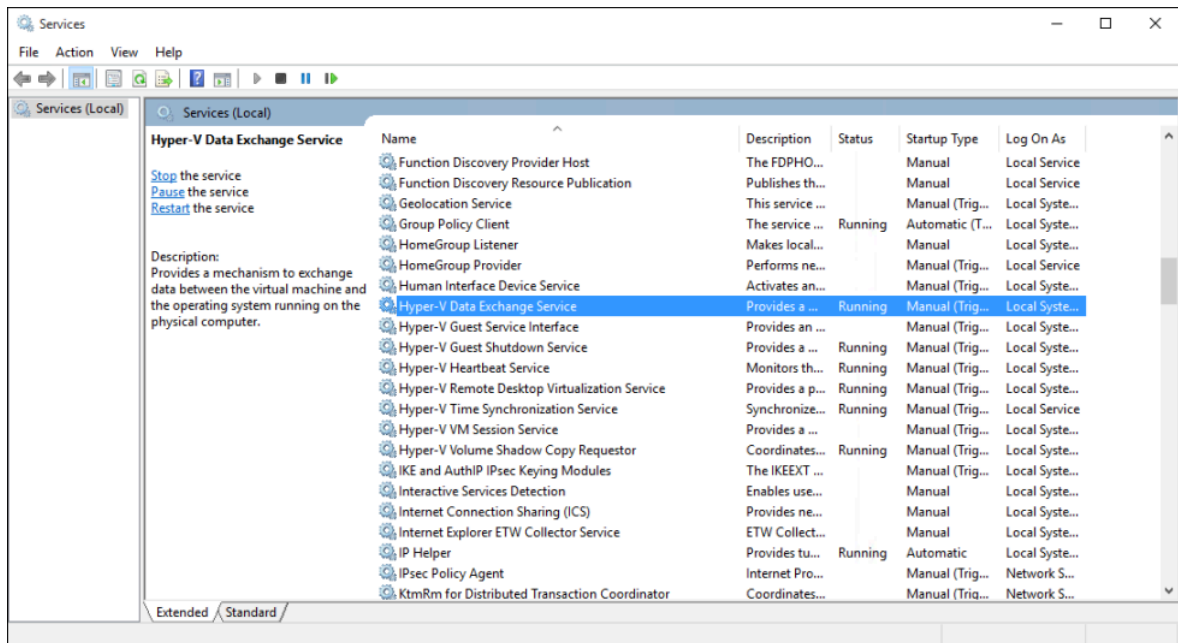
In order for an integration service to be fully functional, its corresponding service must be running within the guest in addition to being enabled on the host. In Windows guests, each integration service is listed as a standard Windows service. You can use the Services applet in Control Panel or PowerShell to stop and start these services.

Important

Stopping an integration service may severely affect the host's ability to manage your virtual machine. To work correctly, each integration service you want to use must be enabled on both the host and guest. As a best practice, you should only control integration services from Hyper-V using the instructions in this article. The matching service in the guest operating system stops or starts automatically when you change its status in Hyper-V. If you start a service in the guest operating system but it's disabled in Hyper-V, the service stops. If you stop a service in the guest operating system that is enabled in Hyper-V, Hyper-V starts it again. If you disable the service in the guest, Hyper-V is unable to start it.

Use Windows Services to start or stop an integration service within a Windows guest

1. Open Services manager by running `services.msc` as an Administrator or by double-clicking the Services icon in Control Panel.



2. Find the services that start with **Hyper-V**.
3. Right-click the service you want start or stop. Select the desired action.

Use PowerShell to start or stop an integration service within a Windows guest

1. To get a list of integration services, run:

PowerShell

```
Get-Service -Name vmic* | FT -AutoSize
```

2. The output should look similar to this:

PowerShell

Status	Name	DisplayName
Running	vmicguestinterface	Hyper-V Guest Service Interface
Running	vmicheartbeat	Hyper-V Heartbeat Service
Running	vmickvpexchange	Hyper-V Data Exchange Service
Running	vmicrdv	Hyper-V Remote Desktop Virtualization Service
Running	vmicshutdown	Hyper-V Guest Shutdown Service
Running	vmictimesync	Hyper-V Time Synchronization Service
Stopped	vmicvmsession	Hyper-V PowerShell Direct Service
Running	vmicvss	Hyper-V Volume Shadow Copy Requestor

3. Run either **Start-Service** or **Stop-Service**. For example, to turn off Windows PowerShell Direct, run:

PowerShell

```
Stop-Service -Name vmicvmsession
```

Start and stop an integration service from a Linux guest

Linux integration services are generally provided through the Linux kernel. The Linux integration services driver is named **hv_utils**.

1. To find out if **hv_utils** is loaded, use this command:

Bash

```
lsmod | grep hv_utils
```

2. The output should look similar to this:

Bash

Module	Size	Used by
hv_utils	20480	0
hv_vmbus	61440	8
hv_balloon,hyperv_keyboard,hv_netvsc,hid_hyperv,hv_utils,hyperv_fb,hv_s torvsc		

3. To find out if the required daemons are running, use this command.

Bash

```
ps -ef | grep hv
```

4. The output should look similar to this:

Bash

root	236	2	0	Jul11	?	00:00:00	[hv_vmbus_con]
root	237	2	0	Jul11	?	00:00:00	[hv_vmbus_ctl]
...							
root	252	2	0	Jul11	?	00:00:00	[hv_vmbus_ctl]
root	1286	1	0	Jul11	?	00:01:11	/usr/lib/linux- tools/3.13.0-32-generic/hv_kvp_daemon
root	9333	1	0	Oct12	?	00:00:00	/usr/lib/linux- tools/3.13.0-32-generic/hv_kvp_daemon
root	9365	1	0	Oct12	?	00:00:00	/usr/lib/linux-

```
tools/3.13.0-32-generic/hv_vss_daemon
user      43774 43755  0 21:20 pts/0    00:00:00 grep --color=auto hv
```

5. To see what daemons are available, run:

```
Bash

compgen -c hv_
```

6. The output should look similar to this:

```
Bash

hv_vss_daemon
hv_get_dhcp_info
hv_get_dns_info
hv_set_ifconfig
hv_kvp_daemon
hv_fcopy_daemon
```

Integration service daemons that might be listed include the following. If any are missing, they might not be supported on your system or they might not be installed. Find details, see [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows](#).

- **hv_vss_daemon:** This daemon is required to create live Linux virtual machine backups.
- **hv_kvp_daemon:** This daemon allows setting and querying intrinsic and extrinsic key value pairs.
- **hv_fcopy_daemon:** This daemon implements a file copying service between the host and guest.

Examples

These examples demonstrate stopping and starting the KVP daemon, named `hv_kvp_daemon`.

1. Use the process ID (PID) to stop the daemon's process. To find the PID, look at the second column of the output, or use `pidof`. Hyper-V daemons run as root, so you'll need root permissions.

```
Bash
```



```
sudo kill -15 `pidof hv_kvp_daemon`
```

2. To verify that all `hv_kvp_daemon` processes are gone, run:

```
Bash
```

```
ps -ef | hv
```

3. To start the daemon again, run the daemon as root:

```
Bash
```

```
sudo hv_kvp_daemon
```

4. To verify that the `hv_kvp_daemon` process is listed with a new process ID, run:

```
Bash
```

```
ps -ef | hv
```

Keep integration services up to date

We recommend that you keep integration services up to date to get the best performance and most recent features for your virtual machines. This happens for Windows guests by default if they are set up to get important updates from Windows Update. Linux guests using current kernels contain integration services built in, but there may be optional updates available. You will receive the latest integration components when you update the kernel. For more information about Linux guests, see [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows](#).

Install or update integration services for Hyper-V hosts earlier than Windows Server 2016 and Windows 10

For Hyper-V hosts earlier than Windows Server 2016 and Windows 10, you'll need to **manually install or update** the integration services in the guest operating systems.

To manually install or update the integration services:

1. Open Hyper-V Manager.

2. Connect to the virtual machine. Right-click the virtual machine and select **Connect**.
 3. From the Action menu of Virtual Machine Connection, select **Insert Integration Services Setup Disk**. This action loads the setup disk in the virtual DVD drive.
Depending on the guest operating system, you might need to start the installation manually from File Explorer.
 4. After the installation finishes, integration services are available for use.
-

Feedback

Was this page helpful?

 Yes

 No

Hyper-V Replica overview

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Hyper-V Replica is a built-in feature of Hyper-V in Windows Server that enables the replication of virtual machines (VMs) between Hyper-V hosts for business continuity and disaster recovery (BCDR) purposes. It enables you to create and maintain offline copies of VMs on a secondary host, and even extend to a third host, which you can use for failover if a primary host fails. Hyper-V Replica isn't available in Hyper-V in Windows client operating systems.

Hyper-V failover clustering and Hyper-V Replica solve related but distinct availability scenarios. A Hyper-V failover cluster provides local high availability for running VMs by using multiple hosts (nodes) that typically access the same clustered storage. If a node fails, the cluster service restarts the affected VM on another node, with no (or minimal) data loss because the VM's virtual hard disks remain on shared storage. Hyper-V Replica, by contrast, is a disaster recovery technology that maintains an asynchronous copy of a VM's storage on another host or cluster - often at a different site - so you can manually initiate failover if the primary host, cluster, or site becomes unavailable. Replica doesn't require shared storage, introduces a recovery point objective (potential data loss up to the replication interval), and complements rather than replaces failover clustering. Many organizations use both: clustering for intra-site high availability and Hyper-V Replica for inter-site resilience and recovery.

This article provides an overview of Hyper-V Replica, its key features, how it works, and how you can use it to enhance your BCDR strategy.

Key features of Hyper-V Replica

Hyper-V Replica offers several key features that make it a powerful and affordable tool for VM replication and disaster recovery:

- **Workload and application agnostic:** Hyper-V Replica works at the hypervisor level, making it suitable for replicating a wide range of workloads and applications without requiring application-specific agents or configurations.
- **Asynchronous replication:** changes made to the primary VM are collated by using resilient change tracking (RCT) at the block level and can be sent to the replica VM at regular intervals of 30 seconds, 5 minutes, or 15 minutes, depending on your recovery point objective (RPO) requirements. RCT decreases the need for time-consuming consistency checks and provides great resiliency.

- **Planned and unplanned failover:** In a planned failover, the primary VM shuts down gracefully and replicates remaining changes before switching to the replica. In an unplanned failover, the replica VM starts without shutting down the primary VM, which might result in some data loss. Once you recover the primary host, you can perform a reverse replication to synchronize changes back to the original primary VM. For more information, see [Failover and recovery](#).
- **Flexible topologies:** replicate between single hosts, clustered hosts, or a combination of both, allowing you to design a replication strategy that fits your infrastructure. Hosts can be located in the same site or across different geographic locations.
- **Encryption and authentication:** support for both Kerberos (for domain-joined hosts) and certificate-based authentication (for non-domain-joined hosts), providing flexibility in securing replication traffic. Encrypt replication traffic with a certificate to protect data in transit. You can restrict replication between specific Hyper-V hosts to enhance security.
- **Flexible storage options:** file-based replication that is storage and workload agnostic, allowing flexibility in your infrastructure design to use different types of storage solutions for the primary and replica hosts. You can choose to replicate all virtual hard disks for a VM or only a subset of them, enabling you to optimize storage usage and replication performance.
- **Network and compression:** replication is TCP-based by using HTTP or HTTPS. You can enable compression to reduce bandwidth usage during replication.
- **Recovery history:** up to 24 hourly recovery points enable you to revert to a previous state of the VM if needed. Volume Shadow Copy Service (VSS) integration can provide application-consistent recovery points for VMs running VSS-aware applications, such as Microsoft SQL Server.
- **Test failover:** perform test failovers to validate your disaster recovery plan without impacting the production environment. For more information, see [Failover and recovery](#).
- **Extended replication:** You can extend replication from the secondary host to a third Hyper-V host, creating a three-tier replication topology. This approach provides an extra layer of redundancy and caters for more complex BCDR strategies. You can use a different replication interval for the extended replication to balance recovery point objectives (RPO) and bandwidth usage across different sites. Replication doesn't happen from the primary server to two other replica servers. Instead, the primary server replicates to the replica server, which in turn replicates to the extended replica server.
- **No additional licensing costs:** Hyper-V Replica is included with Windows Server at no extra cost, making it a cost-effective solution for VM replication and disaster recovery.

Hyper-V Replica components

Hyper-V Replica includes the components described in the following table:

 Expand table

Component	Description
Replication engine	Manages initial replication, replication configuration details, replication of delta changes, and failover and test failover operations. Tracks VM and storage mobility events and takes appropriate actions when necessary.
Change tracking module	Tracks changes that occur to the VM on a source Hyper-V host by monitoring write operations to the virtual hard disks (VHDs), regardless of storage location (local, SAN, NAS, SMB 3 or newer share, or Cluster Shared Volume).
Network module	Provides a secure and efficient way to transfer VM data between Hyper-V hosts. Minimizes traffic by compressing data by default and can encrypt data when HTTPS and certificate-based authentication are used.
Hyper-V Replica Broker	Used only when a Hyper-V host is a node in a failover cluster. Enables use of Hyper-V Replica with highly available VMs that can move between cluster nodes by querying the cluster database and redirecting requests to the node where the VM is running.
Management tools	Configure and manage Hyper-V Replica with Hyper-V Manager and Windows PowerShell. Use Failover Cluster Manager for all VM management and Hyper-V Replica configurations when source or replica hosts are part of a failover cluster.

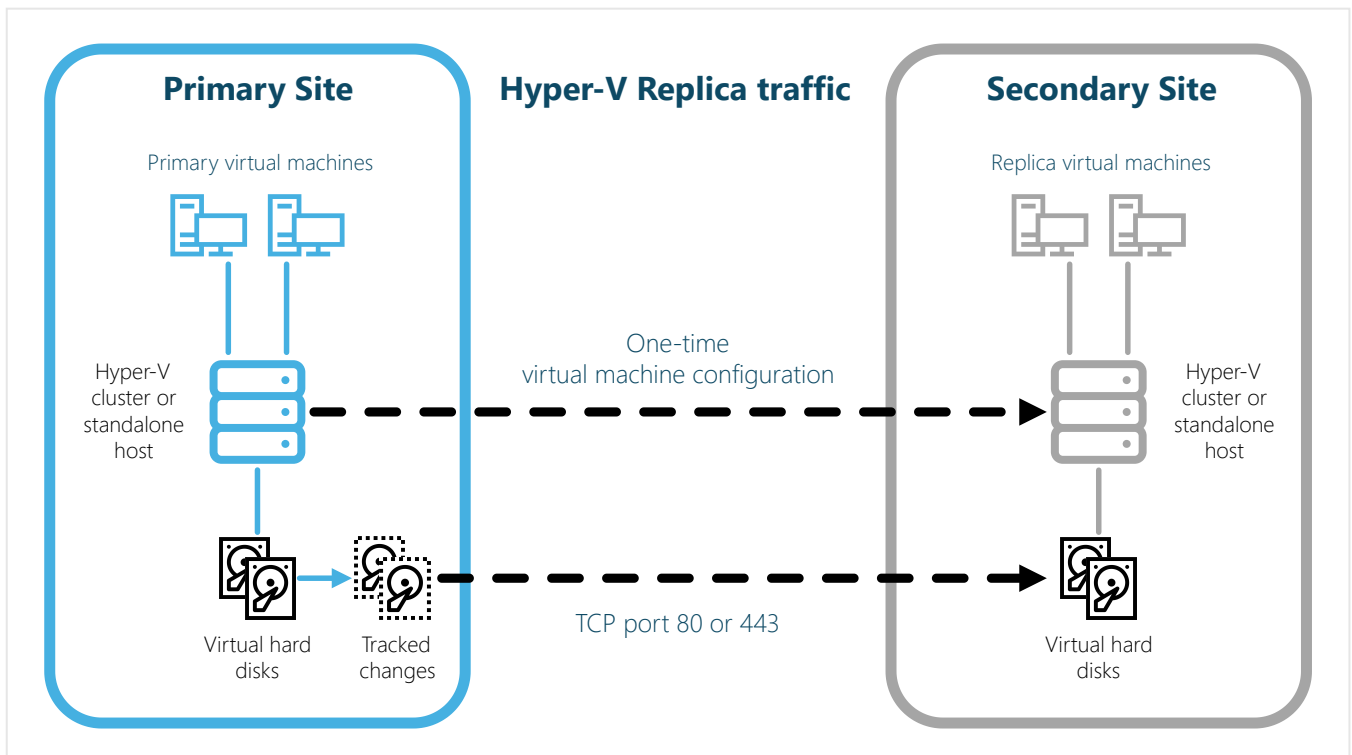
How Hyper-V Replica works

Hyper-V Replica works by replicating the changes you make to a primary VM to a replica VM on a secondary Hyper-V host. The approach you take depends on whether you're replicating to a Hyper-V cluster or a single host. You configure the receiving cluster or host, not the primary cluster or host. Configure both the primary and secondary hosts for replication to ensure that you can fail back to the original primary host after a failover.

Hyper-V Replica can have the following two instances of a single VM residing on different Hyper-V hosts:

- The main, actively running VM, which is called a *primary VM*.
- An offline copy of the primary VM, which is called a *replica VM*.

The replication process involves several key components and steps, which are outlined in the following diagram:



1. When you enable Hyper-V Replica for a VM, you create an initial copy of the VM on the secondary host. You can send this copy over the network or by using external media.
2. Hyper-V uses a change tracking mechanism to monitor changes you make to the primary VM's virtual hard disks (VHDs). It stores changes in `.hr1` files in the same location. This approach enables Hyper-V to identify which blocks of data changed since the last replication cycle.
3. At the configured replication interval (30 seconds, 5 minutes, or 15 minutes), Hyper-V sends the changes to the secondary host. The replication process is asynchronous, so the primary VM continues to operate while Hyper-V replicates changes.
4. The secondary host receives the changes and applies them to the replica VM's VHDs, ensuring the replica VM stays up-to-date with the primary VM.
5. If a primary host fails, you initiate a failover to the replica VM. Depending on whether it's a planned or unplanned failover, the process differs slightly, but the end result is that the replica VM becomes the new primary VM. You can initiate failover manually or through automation scripts.
6. Once you restore the primary host, you can perform a reverse replication to synchronize any changes made to the replica VM back to the original primary VM, allowing it to resume its role as the primary host.

Plan for Hyper-V Replica

When planning to implement Hyper-V Replica as part of your BCDR strategy, consider and make decisions on the following design points:

[Expand table](#)

Decision point	Supporting details
Which workloads do you want to replicate?	List target VMs and their workloads. Default replication protects OS state, not in-flight application state. If you need to recover application state, enable (and schedule) application-consistent recovery points.
Which VHDs need to be replicated?	Exclude disks whose changing data isn't required after failover (for example, page file or temp data disks) to conserve bandwidth and storage. Document exclusions.
How often do you need to synchronize data?	Choose 30 seconds, 5 minutes, or 15 minutes based on RPO, criticality, and bandwidth. Higher criticality and lower RPO needs favor shorter intervals; validate available network capacity.
What's the rate of change of the data on each VM?	High churn increases bandwidth and replica storage consumption. Consider compression or a longer interval if saturation occurs. Include every VM in sizing calculations.
What authentication method will you use and do you require encryption?	Use Kerberos when both hosts are domain-joined and you don't require encryption. Use certificate-based authentication for encrypting replica traffic and/or if either host is non-domain-joined; provision and trust the required certificates beforehand.
Do you need to fail over to an earlier point in time?	Default is a single (latest) recovery point. Configure up to 24 hourly points for point-in-time recovery; more points increase storage and I/O overhead.
How will you seed the initial replication of VM data?	Options: - Send over the network immediately. - Schedule network transfer for a later time window. - Use an existing restored VM on the replica host. - Export to external media, ship, and import at the replica site.

Failover and recovery

It's important to regularly test your BCDR plan to ensure that Hyper-V Replica functions as expected and that you can successfully fail over and recover VMs. You should perform test failovers, validate recovery points, and ensure that all components of your replication strategy are working correctly. Regular testing helps identify potential issues and ensures that your team is familiar with the failover and recovery processes.

There are three types of failover scenarios in Hyper-V Replica:

- You can run a test failover at any point without impacting the production environment. A test failover creates a temporary copy of the replica VM on the secondary host, allowing you to validate the failover process and ensure that the VM starts correctly. The test VM is isolated from the production network to prevent any conflicts. The primary VM continues to run during a test failover, so there's no downtime, and the replication process isn't affected. After completing the test, including other components of your BCDR plan, you can delete the test VM without impacting the replica VM or the primary VM.
- In a planned failover, the primary VM shuts down gracefully, and any remaining changes are replicated to the replica VM before switching roles. This approach ensures that there's no data loss during the failover process. No duplicate VM is created during a planned failover. The failover action is initiated from the primary host. You can reverse the replication after the primary host is available again to synchronize any changes made to the replica VM back to the original primary VM, then perform another planned failover to switch back.
- In an unplanned failover, the replica VM starts without shutting down the primary VM, which can result in data loss. The replica VM becomes the new primary VM, and you can later perform a reverse replication to synchronize changes back to the original primary VM once it's restored. No duplicate VM is created during an unplanned failover. The failover action is initiated from the secondary host.

For more information about performing failover and recovery operations with Hyper-V Replica, see [Fail over a replicated virtual machine with Hyper-V Replica](#).

Next step

To configure Hyper-V Replica, select one of the following articles based on your environment:

[Enable Hyper-V Replica on a failover cluster](#)

[Enable Hyper-V Replica on a single host](#)

Note: The author created this article with assistance from AI. [Learn more](#)

Last updated on 11/17/2025

Enable Hyper-V Replica on a failover cluster

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Hyper-V Replica helps you protect your workloads by replicating virtual machines (VMs) between Hyper-V hosts running Windows Server. This article explains how to enable Hyper-V Replica on a failover cluster.

You can replicate between clusters, single hosts, or a combination of both. If you use a certificate for authentication, there's no Active Directory dependency between the hosts. Single hosts can either be domain members or be in a workgroup.

If you want to learn how to enable Hyper-V Replica on a single host, see [Enable Hyper-V Replica on a single host](#). For more information about Hyper-V Replica and how it works, see [Hyper-V Replica overview](#).

Prerequisites

Before you begin, make sure you meet the following prerequisites:

- You have a Hyper-V cluster to receive the replicated VM. You also have another cluster or single host that runs the VM you want to replicate.
- Decide on an authentication method:
 - If your clusters and hosts are joined to the same or trusted Active Directory domains, you can use Kerberos (HTTP) authentication.
 - If your clusters and hosts aren't domain joined or are in untrusted domains, or you also want to use encryption, you must use certificate-based (HTTPS) authentication. You need to have a valid certificate installed on each host, both sending and receiving. The certificate must meet the following requirements:
 - Not be expired
 - Have both Client and Server Authentication Enhanced Key Usage (EKU) attributes and an associated private key.
 - Terminate at a valid root certificate.
 - The subject common name (CN) or subject alternative name (SAN) must match the fully qualified domain name (FQDN) of the Hyper-V Replica Broker role you provide for the receiving cluster. If you're sending a VM from a cluster, you also need a certificate for the Hyper-V Replica Broker role FQDN on each primary host.

- Network connectivity between the clusters and hosts. By default, if you use Kerberos authentication, replication uses HTTP over port 80. If you use certificate-based authentication, replication uses HTTPS over port 443.
- A storage location on the receiving cluster that's available to all nodes to store the replicated VMs.
- The Hyper-V Replica Broker role requires an IP address. Make sure that for the cluster network used for management, you either have a static IP address available in the subnet or you have DHCP available on the subnet.
- A user account that has administrative privileges on both the primary and replica clusters or hosts.
- Either a user account that is a domain administrator, or prestage a computer account in Active Directory for the Hyper-V Replica Broker role for the receiving cluster. For more information, see [Prestage an account for a clustered service or application](#).

Enable the Hyper-V Replica Broker role

Before you can replicate VMs to a Hyper-V cluster, you need to enable the Hyper-V Replica Broker role. You configure the receiving cluster, not the primary cluster or host.

The Hyper-V Replica Broker is a clustered role that acts as the target for replication traffic. It provides a single point of contact for the primary server to connect to, even if the replica VM moves between nodes in the cluster.

To ensure you can fail a VM back to the original primary cluster or host after a failover event, configure both primary and replica clusters and hosts for replication.

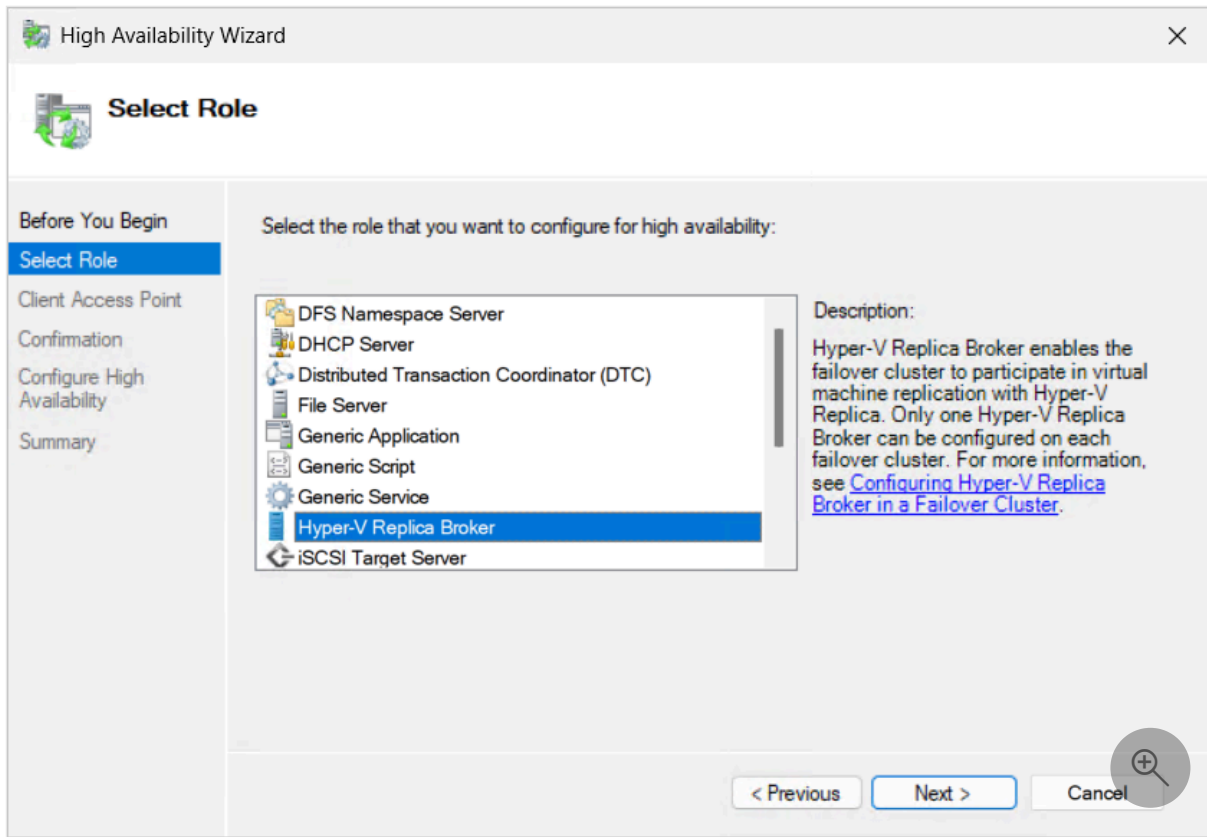
Use Failover Cluster Manager or PowerShell to enable and configure the Hyper-V Replica Broker role. Select the relevant tab for instructions.

Failover Cluster Manager

To enable and configure the Hyper-V Replica Broker cluster role by using Failover Cluster Manager:

1. Open **Failover Cluster Manager** on a device you use to manage the cluster that you want to replicate to, or on one of the nodes in the cluster.
2. In the left pane, expand the name of your cluster you want to replicate to, then select **Roles**.

3. In the right **Actions** pane, select **Configure Role**. In the **High Availability Wizard**, if you see **Before You Begin**, select **Next**.
4. For **Select Role**, select **Hyper-V Replica Broker**, then select **Next**.



5. For **Client Access Point**, enter the name you want to use for the Hyper-V Replica Broker, then select **Next**. The name is used as the NetBIOS name for the role and is limited to 15 characters. This name must be unique in your Active Directory domain.

High Availability Wizard

Client Access Point

Before You Begin

Select Role

Client Access Point

Confirmation

Configure High Availability

Summary

Type the name that clients will use when accessing this clustered role:

Name:

The NetBIOS name is limited to 15 characters. One or more DHCP IPv4 addresses were configured automatically. All networks were configured automatically.

< Previous Next > Cancel

6. For **Confirmation**, review the information, then select **Next**. The Hyper-V Replica Broker role is configured. Initially, the role attempts to obtain an IP address from DHCP. You can change this setting later if you want to use a static IP address. If DHCP isn't available on the subnet, the role doesn't come online until you assign a static IP address.
7. View the **Summary**, then select **Finish**. You can also view a log of the actions performed by selecting **View Report**, which opens in a browser window.
8. Back in the **Roles** pane, right-click the new **Hyper-V Replica Broker** role, then select **Replication Settings**.
9. In the **Hyper-V Replica Broker Configuration** dialog box, make the following changes:
 - a. Check the box **Enable this cluster as a Replica server**.
 - b. Check the box for the authentication method you want to use from **Use Kerberos (HTTP)** or **Use certificate-based Authentication (HTTPS)**. Change the port if you don't want to use the default ports. If you're using certificate-based authentication, select **Select Certificate**, then you're prompted to select the certificate that matches the requirements.
 - c. For **Authorization and storage**, select either **Allow replication from any authenticated server** to allow the replica server to accept VM replication traffic from any primary server that authenticates successfully, or **Allow replication from**

the specified servers to accept traffic only from the primary servers you specifically select. For both options, you need to specify where the replicated VHDs should be stored on the replica Hyper-V cluster. For a cluster, this location needs to be accessible by all nodes in the cluster, such as a Cluster Shared Volume (CSV) at `C:\ClusterStorage\Volume1\Replica`.

If you select **Allow replication from the specified servers**, select **Add**. In **Add Authorization Entry**, specify the FQDN of a primary host, a location to store replica files, and a trust group. A trust group is a free-form text field that you can use to group primary servers. Select **OK**.

d. Select **OK** to save your settings.

The image shows the 'Hyper-V Replica Broker Configuration' window. The 'Replication Configuration' section has 'Enable this cluster as a Replica server.' checked. Under 'Authentication and ports', 'Use Kerberos (HTTP):' is checked, with 'Data sent over the network will not be encrypted.' and 'Specify the port:' set to 80. 'Use certificate-based Authentication (HTTPS):' is unchecked, with 'Data sent over the network will be encrypted.' and 'Specify the port:' set to 443. There is a 'Specify the certificate:' section with fields for 'Issued To:', 'Issued By:', 'Expiration Date:', and 'Intended Purpose:', and a 'Select Certificate...' button. The 'Authorization and storage' section has 'Allow replication from any authenticated server' selected. 'Specify the default location to store Replica files:' is set to 'C:\ClusterStorage\Volume1\'. There is a 'Browse...' button. 'Allow replication from the specified servers:' is unselected. Below it is a table with columns 'Primary Server', 'Storage Location', and 'Trust Group'. At the bottom are 'OK', 'Cancel', and 'Apply' buttons. A circular icon with a plus sign is in the bottom right corner.

Primary Server	Storage Location	Trust Group
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Enable Windows Firewall rules for Hyper-V Replica

To allow replication between the primary and replica clusters and hosts, traffic must get through the Windows Firewall (or any other third-party firewalls). When you install the Hyper-V role on each host, the Windows Firewall creates exceptions for HTTP (80) and HTTPS (443), but doesn't enable them by default. You need to enable the appropriate rules for all receiving hosts.

You can enable the rules by using your preferred method of managing the Windows Firewall, such as centrally using Group Policy, or locally on each node by using the Windows Firewall with Advanced Security console or PowerShell. For more information about managing the Windows Firewall and how-to guides, see [Windows Firewall tools](#).

The rules you need to enable depend on the authentication method you chose when you configured the Hyper-V Replica Broker role:

- Enable `Hyper-V Replica HTTP Listener (TCP-In)` for Kerberos (HTTP) authentication.
- Enable `Hyper-V Replica HTTPS Listener (TCP-In)` for certificate-based (HTTPS) authentication.

Test replication configuration

After you enable and configure Hyper-V Replica and enable the appropriate Windows Firewall rules, test the configuration to ensure that the primary cluster or host can connect to the replica host as expected.

You can use the PowerShell cmdlet [Test-VMReplicationConnection](#) to test the connection, as shown in the following examples. Be sure to replace placeholder `<values>` with your own.

1. Open a PowerShell session as an administrator on one of the hosts that you want to replicate from, or connect remotely by using the [Enter-PSSession](#) cmdlet on a device you use to manage the cluster or host.
2. Use one of the following example commands:
 - To test a connection to a replica cluster by using Kerberos authentication, run the following command:

PowerShell

```
Test-VMReplicationConnection -ReplicaServerName '<Hyper-V Broker role FQDN>' -ReplicaServerPort 80 -AuthenticationType Kerberos
```

- To test a connection to a replica cluster by using certificate-based authentication, run the following command. You need the thumbprint of a valid certificate for

Hyper-V Replica on the primary host or cluster.

PowerShell

```
Test-VMReplicationConnection -ReplicaServerName '<Hyper-V Broker role FQDN>' -ReplicaServerPort 443 -AuthenticationType Certificate -CertificateThumbprint AA11BB22CC33DD44EE55FF66AA77BB88CC99DD00
```

For either of these example commands, here's the output you should see if the test is successful:

Output

The connection to the specified Replica server with the specified parameters was successful.

 **Note:** The author created this article with assistance from AI. [Learn more](#)

Last updated on 11/17/2025

Enable Hyper-V Replica on a single host

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Hyper-V Replica helps you protect your workloads by replicating virtual machines (VMs) between Hyper-V hosts running Windows Server. This article explains how to enable Hyper-V Replica on a single host by using Hyper-V Manager, PowerShell, or Windows Admin Center - Virtualization mode.

You can replicate between clusters, single hosts, or a combination of both. If you use a certificate for authentication, there's no Active Directory dependency between the hosts. Single hosts can either be domain members or be in a workgroup.

If you want to learn how to enable Hyper-V Replica on a failover cluster instead, see [Enable Hyper-V Replica on a failover cluster](#). For more information about Hyper-V Replica and how it works, see [Hyper-V Replica overview](#).

Prerequisites

Before you begin, make sure you meet the following prerequisites:

- You have a single Hyper-V host to receive the replicated VM. You also have another cluster or single host that runs the VM you want to replicate.
- Decide on an authentication method:
 - If your hosts are joined to the same or trusted Active Directory domains, you can use Kerberos (HTTP) authentication.
 - If your hosts aren't domain joined or are in untrusted domains, or you also want to use encryption, you must use certificate-based (HTTPS) authentication. You need to have a valid certificate installed on each host, both sending and receiving. The certificate must meet the following requirements:
 - Not be expired
 - Have both Client and Server Authentication Enhanced Key Usage (EKU) attributes and an associated private key.
 - Terminate at a valid root certificate.
 - The subject common name (CN) or subject alternative name (SAN) must match the fully qualified domain name (FQDN) of the host. If you're sending a VM from a cluster, you also need a certificate for the Hyper-V Replica Broker role FQDN on each primary host.

- Network connectivity between the hosts. By default, if you use Kerberos authentication, replication uses HTTP over port 80. If you use certificate-based authentication, replication uses HTTPS over port 443.
- A storage location on the receiving host to store the replicated VMs.
- A user account that has administrative privileges on both the primary and replica clusters or hosts.

Enable Hyper-V Replica

Before you can replicate VMs to a single Hyper-V host, you need to enable Hyper-V Replica. You configure the receiving host, not the primary cluster or host.

To ensure that you can fail a VM back to the original primary host or cluster after a failover event, configure both primary and replica hosts and clusters for replication.

Use Hyper-V Manager, PowerShell, or Windows Admin Center - Virtualization mode to enable and configure Hyper-V Replica. Select the relevant tab for instructions.

Hyper-V Manager

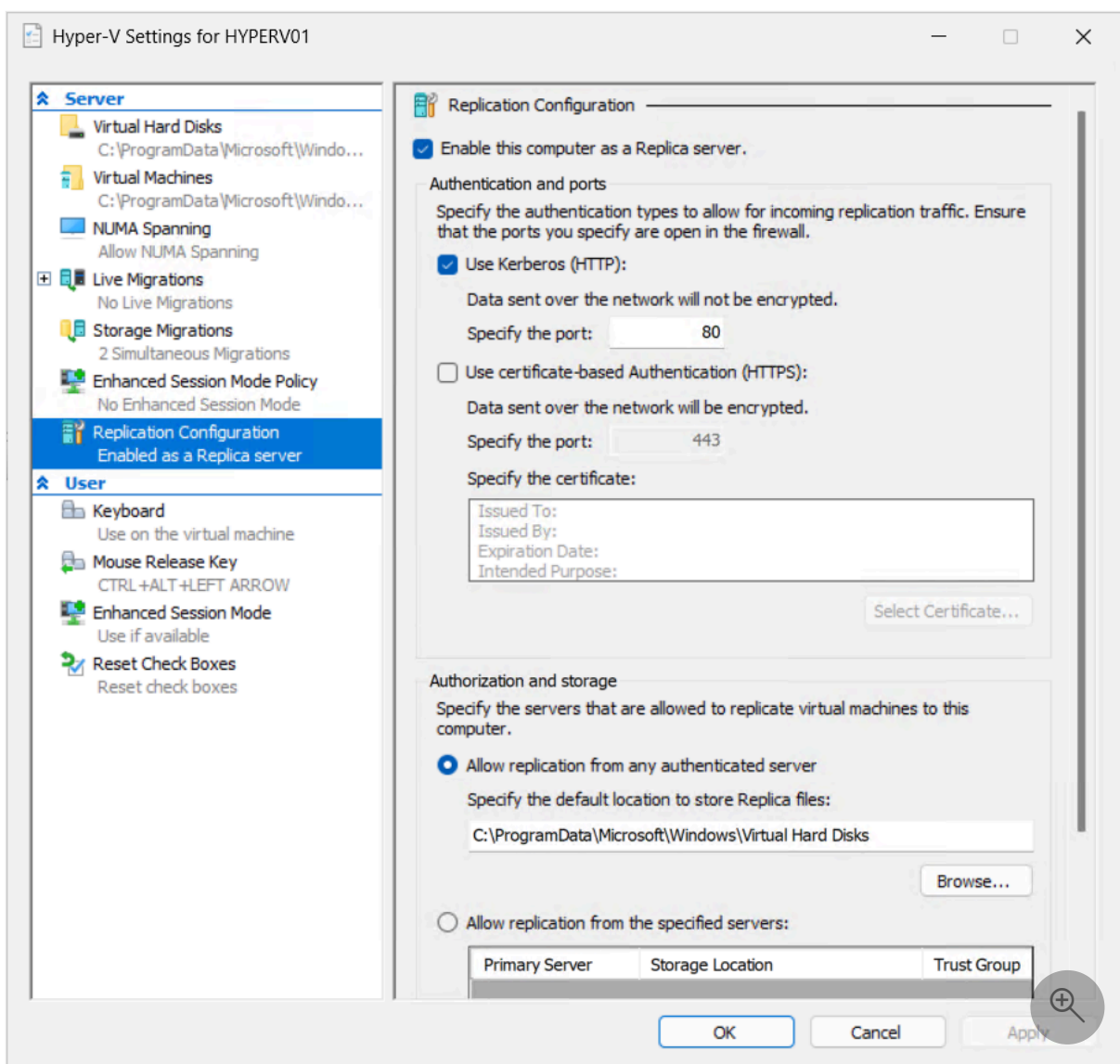
To enable and configure Hyper-V Replica by using Hyper-V Manager:

1. Open **Hyper-V Manager** on a device you use to manage the host that you want to replicate to, or on the host itself where Windows Server with Desktop Experience is installed.
2. In the left pane, select the host you want to configure as the replica server.
3. In the right pane, under **Actions**, select **Hyper-V Settings**.
4. Under the **Server** section, select **Replication Configuration** in the left pane and make the following changes:
 - a. Check the box **Enable this computer as a Replica server**.
 - b. For **Authentication and ports**, check the box for the authentication method you want to use from **Use Kerberos (HTTP)** or **Use certificate-based authentication (HTTPS)**. Change the port if you don't want to use the default ports. If you're using certificate-based authentication, select **Select Certificate**, then you're prompted to select the certificate that matches the requirements.

- c. For **Authorization and storage**, select either **Allow replication from any authenticated server** to allow the replica server to accept VM replication traffic from any primary server that authenticates successfully, or **Allow replication from the specified servers** to accept traffic only from the primary servers you specifically select. For both options, you need to specify where the replicated VHDs should be stored on the replica host.

If you select **Allow replication from the specified servers**, select **Add**. In **Add Authorization Entry**, specify the FQDN of a primary host, a location to store replica files, and a trust group. A trust group is a free-form text field that you can use to group primary servers. Select **OK**.

- d. Select **OK** to save your settings.



Enable Windows Firewall rules for Hyper-V Replica

To allow replication between the primary and replica clusters and hosts, traffic must get through the Windows Firewall (or any other third-party firewalls). When you install the Hyper-V role on each host, the Windows Firewall creates exceptions for HTTP (80) and HTTPS (443), but it doesn't enable them by default. You need to enable the appropriate rules for all receiving hosts.

You can enable the rules by using your preferred method of managing the Windows Firewall, such as centrally using Group Policy, or locally on each node by using the Windows Firewall with Advanced Security console or PowerShell. For more information about managing the Windows Firewall and how-to guides, see [Windows Firewall tools](#).

The rules you need to enable depend on the authentication method you chose when you configured the Hyper-V Replica Broker role:

- Enable `Hyper-V Replica HTTP Listener (TCP-In)` for Kerberos (HTTP) authentication.
- Enable `Hyper-V Replica HTTPS Listener (TCP-In)` for certificate-based (HTTPS) authentication.

Test replication configuration

After you enable and configure Hyper-V Replica and enable the appropriate Windows Firewall rules, test the configuration to ensure that the primary cluster or host can connect to the replica host as expected.

You can use the PowerShell cmdlet [Test-VMReplicationConnection](#) to test the connection, as shown in the following examples. Be sure to replace placeholder `<values>` with your own.

1. Open a PowerShell session as an administrator on one of the hosts that you want to replicate from, or connect remotely by using the [Enter-PSSession](#) cmdlet on a device you use to manage the cluster or host.
2. Use one of the following example commands:
 - To test a connection to a replica cluster by using Kerberos authentication, run the following command:

PowerShell

```
Test-VMReplicationConnection -ReplicaServerName '<Replica host FQDN>' -  
ReplicaServerPort 80 -AuthenticationType Kerberos
```

- To test a connection to a replica cluster by using certificate-based authentication, run the following command. You need the thumbprint of a valid certificate for

Hyper-V Replica on the primary host or cluster.

PowerShell

```
Test-VMReplicationConnection -ReplicaServerName '<Replica host FQDN>' -  
ReplicaServerPort 443 -AuthenticationType Certificate -  
CertificateThumbprint AA11BB22CC33DD44EE55FF66AA77BB88CC99DD00
```

For either of these example commands, here's the output you should see if the test is successful:

Output

The connection to the specified Replica server with the specified parameters was successful.

 **Note:** The author created this article with assistance from AI. [Learn more](#)

Last updated on 11/17/2025

Replicate a virtual machine with Hyper-V Replica

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Hyper-V Replica helps you protect your workloads by replicating virtual machines (VMs) between Hyper-V hosts running Windows Server. This article explains how to replicate a virtual machine after you enable replication on the receiving host or cluster by using Hyper-V Manager, Failover Cluster Manager, PowerShell, or Windows Admin Center - Virtualization mode.

You can replicate between clusters, single hosts, or a combination of both. If you use a certificate for authentication, there's no Active Directory dependency between the hosts. Single hosts can either be domain members or be in a workgroup.

For more information about Hyper-V Replica and how it works, see [Hyper-V Replica overview](#). If you need to enable Hyper-V Replica, see:

- [Enable Hyper-V Replica on a failover cluster](#).
- [Enable Hyper-V Replica on a single host](#).

Prerequisites

Before you begin, make sure you meet the following prerequisites:

- You have a Hyper-V cluster to receive the replicated VM on which you already configured Hyper-V Replica. If you need to configure Hyper-V Replica, see [Enable Hyper-V Replica on a failover cluster](#) or [Enable Hyper-V Replica on a single host](#). You also have another cluster or single host that runs the VM you want to replicate.
- Sufficient network connectivity and bandwidth between primary and replica sites to meet your Recovery Point Objective (RPO) and initial replication method. Consider dedicating or QoS-throttling replication traffic to avoid impacting production workloads.
- If you're using certificate-based authentication, you need to have a valid certificate installed on each host in each cluster or single host, both sending and receiving. The certificate must meet the following requirements:
 - Not be expired
 - Have both Client and Server Authentication Enhanced Key Usage (EKU) attributes and an associated private key.
 - Terminate at a valid root certificate.

- The subject common name (CN) or subject alternative name (SAN) must match the fully qualified domain name (FQDN) of the Hyper-V Replica Broker role you provide for the receiving cluster.
- Adequate storage capacity on the replica site for the VM, including its configuration files, virtual hard disk (VHDX or VHD) files, and optional extra recovery points (snapshots).
- A user account that's a member of the **Hyper-V Administrators** security group on each host. In an Active Directory domain, you can add users or groups to this group by using Group Policy Preferences. Alternatively, the account can be a local administrator on each host. You can find more information about the Hyper-V Administrators group in [Active Directory Security Groups](#).

Replicate a virtual machine

You need to enable replication for each VM you want to replicate. You can replicate a VM using the following combinations:

- Hyper-V Manager on clusters or single hosts.
- Failover Cluster Manager on clusters only.
- PowerShell on clusters or single hosts.
- Windows Admin Center - Virtualization mode on single hosts only.

Select the relevant tab for instructions.

Hyper-V Manager

To replicate a virtual machine by using Hyper-V Manager:

1. Open **Hyper-V Manager** on a device you use to manage a host in the **primary** cluster or a **primary** single host with the VM, or on the host itself where Windows Server with Desktop Experience is installed.
2. In the resources pane, select the host that contains the VM you want to replicate.
3. Right-click the VM you want to replicate, then select **Enable Replication**. In the **Enable Replication** screen, if you see **Before You Begin**, select **Next**.
4. For **Specify Replica Server**, in the **Replica Server** box, enter either the NetBIOS or FQDN of the Replica server. If the Replica server is part of a failover cluster, enter the name of the Hyper-V Replica Broker. Then select **Next**. The connection to the replica server is verified to retrieve the authentication and port settings you configured for the replica server.

The screenshot shows a Windows-style window titled "Enable Replication for contoso-vm-01". Inside, the "Specify Replica Server" step is active, highlighted in blue in the left-hand navigation pane. The main area contains instructions: "Specify the Replica server name to use to replicate this virtual machine. If the Replica server is on a failover cluster, specify the name of the Hyper-V Replica Broker as the Replica server. Use the Failover Cluster Manager on the Replica server to find the name of the Replica Broker." Below this, a text box labeled "Replica server:" contains the value "cluster02rep.contoso.com", followed by a "Browse..." button. At the bottom right, there are four buttons: "< Previous", "Next >" (which is highlighted with a blue border), "Finish", and "Cancel". A small circular icon with a plus sign and a magnifying glass is also visible near the "Cancel" button.

5. Review the replica server settings. If you're using certificate-based authentication, select **Select Certificate**, then you're prompted to select the certificate for the primary host that matches the requirements. If you want to compress the data that is transmitted over the network, check the box. Select **Next**.

Enable Replication for contoso-vm-01

Specify Connection Parameters

Before You Begin

Specify Replica Server

Specify Connection Parameters

Choose Replication VHDs

Configure Replication Frequency

Configure Additional Recovery Points

Choose Initial Replication Method

Summary

Replica server: cluster02rep.contoso.com

Replica server port: 80

Authentication Type

☒ Use Kerberos authentication (HTTP)
Data will not be encrypted while being transmitted over the network.

☐ Use certificate-based authentication (HTTPS)
Data will be encrypted while being transmitted over the network.

Issued To:
Issued By:
Expiration Date:
Intended Purpose:

Select Certificate...

☒ Compress the data that is transmitted over the network.

< Previous Next > Finish Cancel

6. For **Choose Replication VHDs**, make sure the checkboxes for the VHDs you want to replicate are selected, and clear the checkboxes for any VHDs that you want to exclude from replication. Then select **Next**.

Enable Replication for contoso-vm-01

Choose Replication VHDs

Before You Begin

Specify Replica Server

Specify Connection Parameters

Choose Replication VHDs

Configure Replication Frequency

Configure Additional Recovery Points

Choose Initial Replication Method

Summary

Clear the check boxes of any virtual hard disks (VHDs) that you do not want to replicate (for example, a VHD used for a dedicated paging file).

Not replicating certain VHDs, such as the operating system VHD, could result in the Replica virtual machine not starting up properly.

Virtual Hard Disks:

☒ C:\ClusterStorage\Volume1\contoso-vm-01\Virtual Hard Disks\contoso-vm-01.vhdx

< Previous Next > Finish Cancel

7. For **Configure Replication Frequency**, specify how often changes should be synchronized from the primary to replica clusters or hosts from **30 seconds**, **5 minutes**, or **15 minutes**. Then select **Next**.

The screenshot shows a Windows wizard window titled 'Enable Replication for contoso-vm-01'. The main heading is 'Configure Replication Frequency'. On the left, a list of steps includes 'Before You Begin', 'Specify Replica Server', 'Specify Connection Parameters', 'Choose Replication VHDs', 'Configure Replication Frequency' (which is highlighted), 'Configure Additional Recovery Points', 'Choose Initial Replication Method', and 'Summary'. The main area contains the instruction 'Choose the frequency at which changes will be sent to the Replica Server' and a dropdown menu with four options: '5 minutes', '30 seconds', '5 minutes', and '15 minutes'. The '5 minutes' option is selected. At the bottom right, there are four buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'. A circular icon with a plus sign is also visible next to the 'Cancel' button.

8. For **Configure Additional Recovery Points**, select **Maintain only the latest recovery point** or **Create additional hourly recovery points** and the number of extra recovery points you want to create in hours. Then select **Next**.

When you create extra hourly recovery points, you can also check the box **Volume Shadow Copy Service (VSS) snapshot frequency** if you want recovery points to provide application-consistent recovery points for VMs running VSS-aware applications.

Enable Replication for contoso-vm-01

Configure Additional Recovery Points

Before You Begin

- Specify Replica Server
- Specify Connection Parameters
- Choose Replication VHDs
- Configure Replication Frequency
- Configure Additional Recovery Points**
- Choose Initial Replication Method
- Summary

You can choose to store only the latest recovery point of the primary virtual machine on the Replica server or to add additional recovery points, allowing you to recover to an earlier point in time. Additional recovery points require more storage and processing resources.

Configure additional recovery points for this virtual machine

☒ Maintain only the latest recovery point

☐ Create additional hourly recovery points

Coverage provided by additional recovery points (in hours): 24

☐ Volume Shadow Copy Service (VSS) snapshot frequency (in hours): 4

This option requires the latest Integration Components be installed in the guest. For non-Windows guest operating systems, these snapshots are file-system consistent.

< Previous Next > Finish Cancel

9. For **Choose Initial Replication**, select the initial replication method. The total size of the initial copy is displayed. Select an option from the following list, then select **Next**.

- **Send initial copy over the network:** this method is the default option.
- **Send initial copy using external media:** specify a folder location on local or external media where the initial copy can be stored.
- **Use an existing virtual machine on the Replica server as the initial copy:** restore the VM on the replica server and use it as the basis for replication, then replicate only the changes. You can find the process to import the initial copy in the section [Send initial copy using external media](#).

If you select **Send initial copy over the network** or **Use an existing virtual machine on the Replica server as the initial copy**, you can optionally specify whether to start replication immediately, or start replication at a specified time and date up to seven days in the future.

Enable Replication for contoso-vm-01

Choose Initial Replication Method

Before You Begin

Specify Replica Server

Specify Connection Parameters

Choose Replication VHDs

Configure Replication Frequency

Configure Additional Recovery Points

Choose Initial Replication Method

Summary

Before replication can start, an initial copy of all virtual hard disks that you selected must be transferred to the Replica server.

Size of the initial copy of selected virtual hard disks: 4 MB

Initial Replication Method

☒ Send initial copy over the network

☐ Send initial copy using external media

Specify the location to export initial copy.

[Browse...](#)

☐ Use an existing virtual machine on the Replica server as the initial copy.

Choose this option if you have restored a copy of this virtual machine on the Replica server. The restored virtual machine will be used as the initial copy.

Schedule Initial Replication

☒ Start replication immediately

☐ Start replication on:

You can choose any time up to Monday, September 22, 2025.

< Previous **Next >** Finish Cancel

10. Review the summary information, then select **Finish**.

Enable Replication for contoso-vm-01

Completing the Enable Replication wizard

Before You Begin

Specify Replica Server

Specify Connection Parameters

Choose Replication VHDs

Configure Replication Frequency

Configure Additional Recovery Points

Choose Initial Replication Method

Summary

You have successfully completed the Enable Replication wizard. You are about to enable replication for "contoso-vm-01" with the following settings:

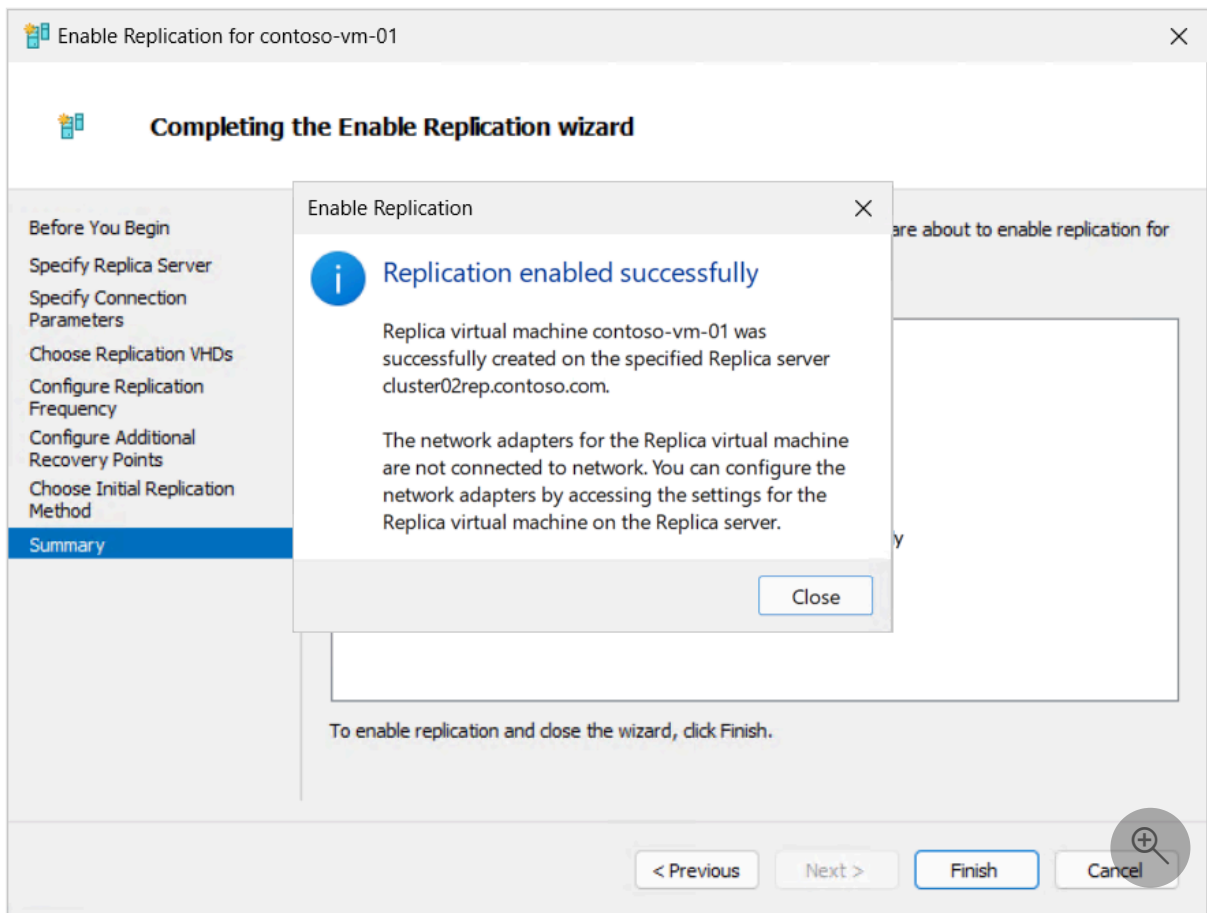
Description:

Replica server:	cluster02rep.contoso.com
Replica server port:	80
Compress data:	Yes
Authentication type:	Kerberos authentication
VHDs not selected for replication:	None
Replication Frequency:	5 minutes
Store additional recovery points:	No
Initial replication method:	Using network, start immediately

To enable replication and close the wizard, click Finish.

< Previous **Next >** **Finish** Cancel

11. A dialog box appears to indicate that replication was successfully enabled.



12. To monitor the initial replication progress, see the section [Monitor replication progress](#). You can change replication settings later if you need to in the **Replication** section of the VM settings.

If you chose to send the initial copy by using external media, see the section [Send initial copy using external media](#) for the process to import the initial copy. A placeholder VM is created on the replica site and its replication health state is **Warning**. The replica contents for the VM are stored on your external media in a subfolder for the VM. Once the export is complete, safely eject the external media and transport it to the replica server.

Important

The following behaviors apply to VMs that you replicate:

- When you replicate a VM, its network adapters aren't connected to a virtual switch on the replica host. After the initial replication is complete, connect any network adapters to a virtual switch on the replica host so that VMs are ready to bring online if you need to fail over.

- If you make changes to the VM configuration, such as changing memory or processor settings or modifying network adapters, those changes aren't automatically replicated to the replica VM. You must manually update the replica VM if needed. Alternatively, you can remove replication and re-enable it to create a new replica VM with the updated configuration.
- Hyper-V Replica points to a single default volume for VM replication. To prevent capacity or performance issues, move the VM's storage to a different volume than the default volume for replication.
- If a VM runs an application that saves data across virtual hard disks, you can ensure that all virtual hard disks selected for replication are replicated to the same point in time. An example of where this approach is useful is when an application has one virtual hard disk dedicated for application data, and another virtual hard disk dedicated for application log files. To use this feature, configure VM replication by using PowerShell with the `Enable-VMReplication` cmdlet and add the parameter `EnableWriteOrderPreservationAcrossDisks`. For more information, see [Enable-VMReplication](#).

Send initial copy using external media

If you send the initial copy by using external media, such as a hard disk or USB drive, the process creates a placeholder VM on the replica site with a checkpoint. The process stores the replica contents for the VM on your external media in a subfolder for the VM. You transport the initial copy to the replica server, then import it to the placeholder VM.

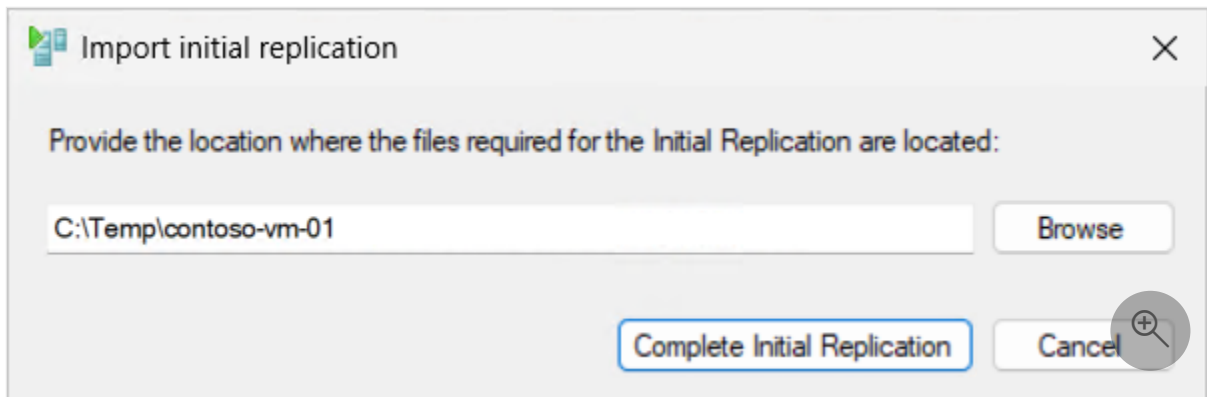
You can enable replication by using Hyper-V Manager, Failover Cluster Manager, PowerShell, or Windows Admin Center - Virtualization mode. Select the relevant tab for instructions.

Hyper-V Manager

To send the initial copy by using external media and import it by using Hyper-V Manager:

1. Follow the steps in the section [Replicate a virtual machine](#) using Hyper-V Manager. The process creates a placeholder VM on the replica site and sets its replication health state to **Warning**. The process stores the replica contents for the VM on your external media in a subfolder for the VM. Once the export is complete, safely eject the external media and transport it to the replica server.

2. Open **Hyper-V Manager** on a device you use to manage the replica host, or on the replica host itself where Windows Server with Desktop Experience is installed.
3. In the resources pane, select the host that contains the placeholder VM you're replicating.
4. Right-click the placeholder VM, then select **Import Initial Replica**.
5. Specify the location where the initial copy is stored, then select **Complete Initial Replication**.



6. The process applies the initial checkpoint to the placeholder VM and changes the replication health state to **Normal**. Replication continues at the specified frequency. See the section [Monitor replication progress](#) to monitor continued replication progress.

Monitor replication progress

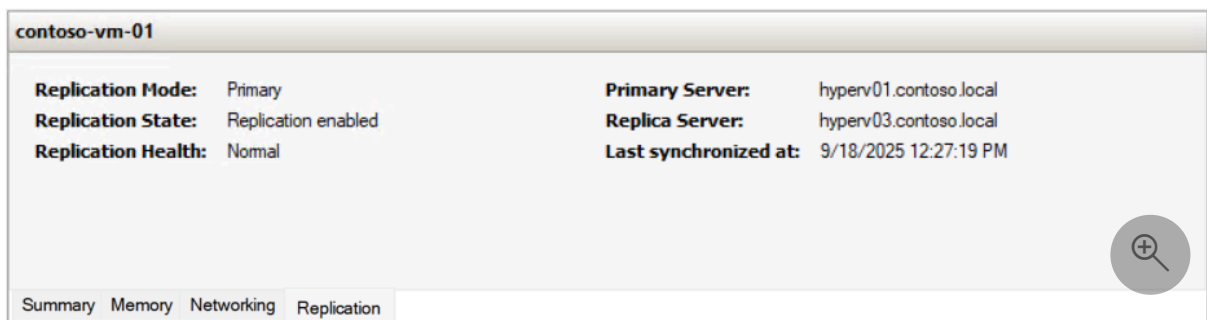
You can monitor the progress of the initial replication and ongoing replication by using Hyper-V Manager, Failover Cluster Manager, PowerShell, or Windows Admin Center - Virtualization mode. The replication health states are:

- **Normal:** Replication cycles complete on schedule (RPO met), backlog is minimal, and there are no recent replication errors. No action needed.
- **Warning:** Replication works but lags or backlog grows (often transient network or storage load). Monitor and investigate if it doesn't clear quickly.
- **Critical:** Replication stalls or fails. The RPO is at risk as changes aren't applying. Investigate immediately, such as check connectivity, storage capacity and performance, authentication and certificates, and event logs. Resume or resynchronize as required.

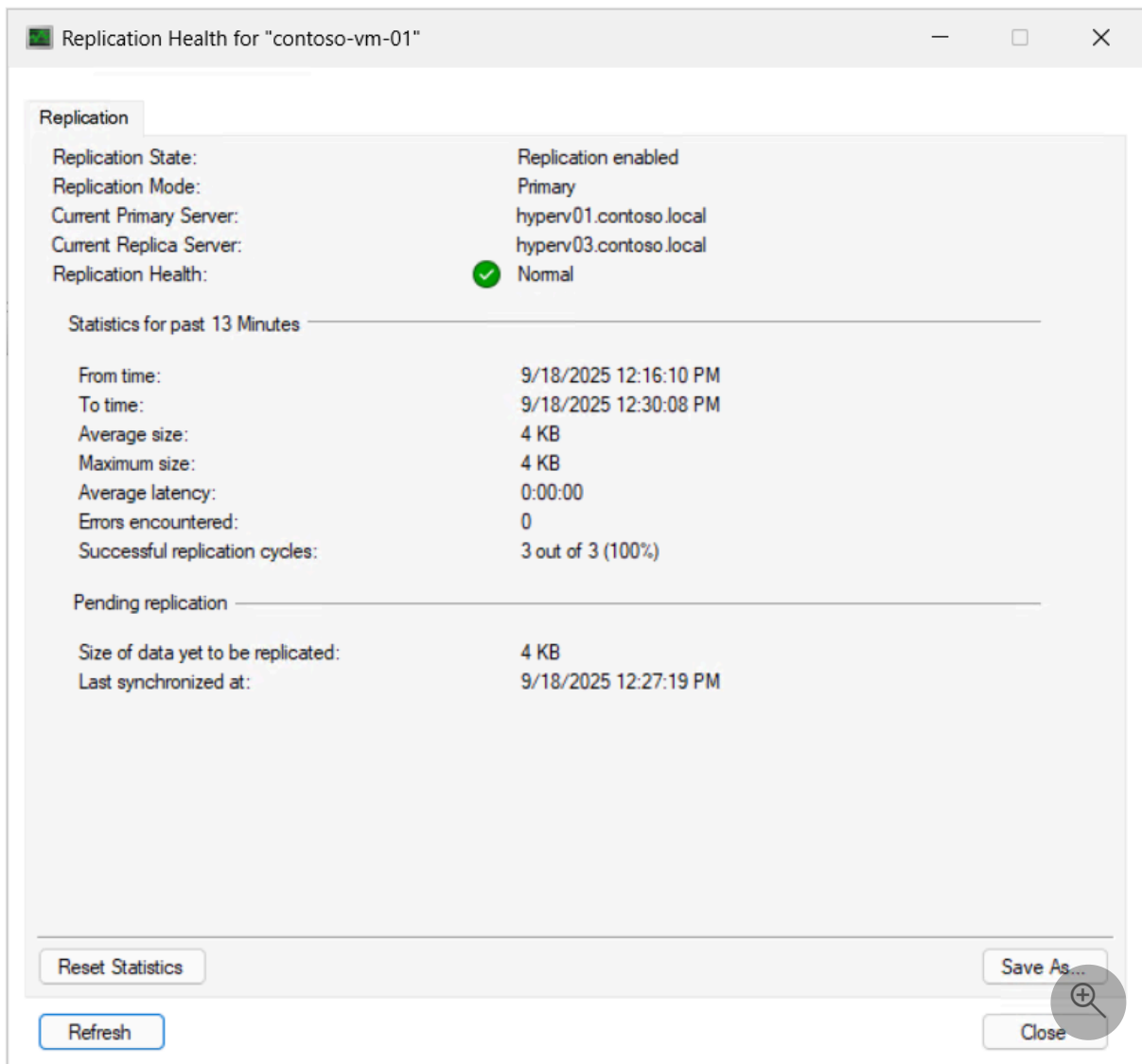
Select the relevant tab for instructions.

To monitor replication progress by using Hyper-V Manager:

1. Back in **Hyper-V Manager**, select a host.
2. In the central pane, add a column for **Replication Health** to show the high-level health status of replication for each VM. Right-click on any column heading, select **Add/Remove Columns**. In the list of **Available Columns**, select **Replication Health**, then select **Add** to move it to **Displayed Columns**. Select **OK**.
3. View summary replication information, such as the last synchronized time and primary and replica host FQDNs. Select the VM, then in the bottom pane select the **Replication** tab.



4. To view more detailed replication information, right-click the VM and select **Replication**, then **View Replication Health**. You can reset the replication statistics if needed by selecting **Reset Statistics**.



Configure extended replication (optional)

You can configure extended replication, which enables you to replicate a VM to a third location, from one replica server to another replica server. This approach can help you meet compliance requirements for offsite backups or provide an extra layer of protection for your workloads.

You can enable extended replication by using Hyper-V Manager, Failover Cluster Manager, or PowerShell. You can't use Windows Admin Center - Virtualization mode to configure extended replication at this time.

Select the relevant tab for instructions.

Hyper-V Manager

To configure extended replication by using Hyper-V Manager:

1. Open **Hyper-V Manager** on a device you use to manage the host with the VM, or on the host itself where Windows Server with Desktop Experience is installed.
2. In the resources pane, select the host that contains the VM you want to configure extended replication for.
3. Right-click the VM you want to configure extended replication for, then select **Replication > Extend Replication**.
4. The process to configure extended replication is similar to replicating a VM to the first replica location. Follow the process in the section [Replicate a virtual machine](#) with the following exceptions:
 - You can't specify which virtual hard disks to replicate. The virtual hard disks that are replicated to the first replica server are used for extended replication.
 - The replication frequency can only be 5 minutes or 15 minutes.
 - You can't change the authentication type.

Next step

After you enable replication for a VM, you can test failover to the replica server without impacting ongoing replication. For more information about performing a test failover and how to fail over during an outage, see [Failover a replicated virtual machine with Hyper-V Replica](#).

Note: The author created this article with assistance from AI. [Learn more](#)

Last updated on 11/17/2025

Fail over a replicated virtual machine with Hyper-V Replica

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

After you enable Hyper-V Replica and replicate a virtual machine (VM), you can perform failover operations to switch the VM to the replica host or cluster. This article explains how to perform different types of failover operations by using Hyper-V Replica with Hyper-V Manager, Failover Cluster Manager, PowerShell, or Windows Admin Center - Virtualization mode.

For more information about Hyper-V Replica, or if you need to enable replication or replicate a VM, see the following articles:

- [Hyper-V Replica overview](#)
- [Enable Hyper-V Replica on a single host](#)
- [Enable Hyper-V Replica on a failover cluster](#)
- [Replicate a virtual machine with Hyper-V Replica](#)

Failover scenarios

Hyper-V Replica supports the following three failover scenarios.

- **Test failover:** creates a test VM on the replica host or cluster without affecting the ongoing replication. It enables you to verify that the replica VM is functioning correctly. After testing, you can stop the test failover to remove the test VM.

By default, the test VM is created from the latest recovery point and isn't connected to a network. If you configured extra recovery points, you can choose to create the test VM from one of those points. You can only run one test failover on a VM at a time.

- **Planned failover:** use when you can gracefully shut down the primary VM. It helps you to perform an end-to-end validation of your recovery plan. It ensures that all changes are replicated to the replica before switching over, resulting in zero data loss. After the planned failover, you can optionally reverse replication back to the original primary host or cluster, then fail back the VM to the original location.

Planned failover isn't a substitute for high availability, but it can enable you to keep your workloads running with minimal downtime without data loss. It can be useful for events such as planned maintenance of an entire site or datacenter.

- **Unplanned failover:** use when the primary VM is unavailable due to a failure, such as a power outage. It enables you to switch to the replica VM by using the latest recovery point or previous recovery points, if configured. After the unplanned failover, you can complete the failover process and optionally reverse replication back to the original primary host or cluster, then fail back the VM to the original location.

Unplanned failover is designed for disaster recovery scenarios where the primary VM can't be gracefully shut down. It might result in some data loss, depending on the recovery point selected.

Prerequisites

Before you begin, make sure you meet the following prerequisites:

- You have a VM that you're replicating by using Hyper-V Replica.
- The VM's replication health state is **Normal**. To check the replication health, see [Monitor replication progress](#).
- A user account that's a member of the **Hyper-V Administrators** security group on each host. In an Active Directory domain, you can add users or groups to this group by using Group Policy Preferences. Alternatively, the account can be a local administrator on each host. For more information about the Hyper-V Administrators group, see [Active Directory Security Groups](#).

Fail over a virtual machine

You need to run a failover operation for each VM. You can only run one failover operation at a time for a VM. You can run each failover operation by using the following combinations:

- Hyper-V Manager on clusters or single hosts
- Failover Cluster Manager on clusters
- PowerShell on clusters or single hosts
- Windows Admin Center - Virtualization mode on single hosts only. You can only fail over by using Windows Admin Center - Virtualization mode from the primary host to the replica host. You can't perform a test failover or reverse replication at this time.

Select the relevant tab for instructions.

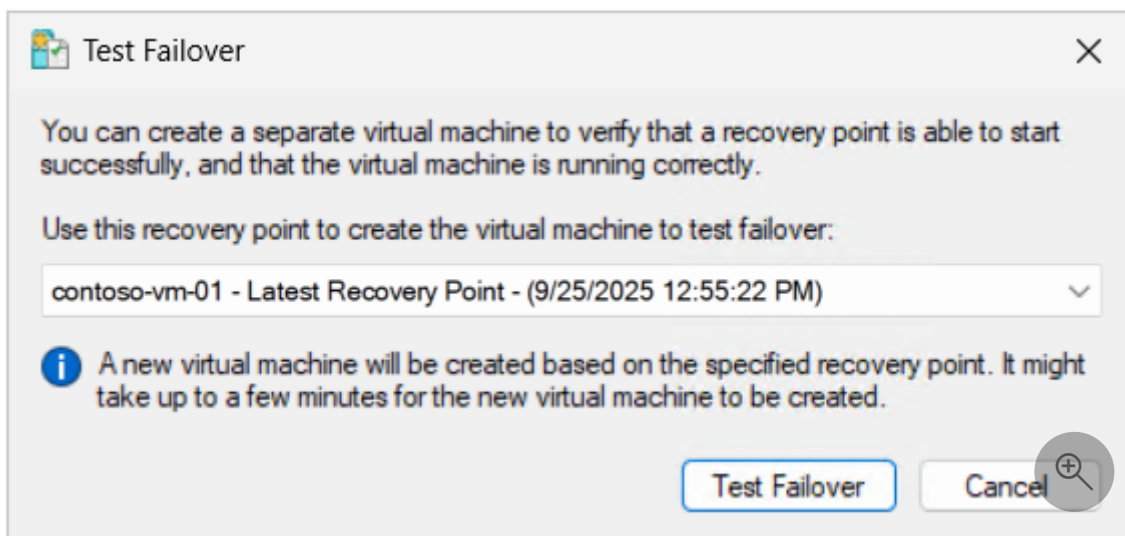
Hyper-V Manager

Failing over a virtual machine by using Hyper-V Manager depends on the type of failover you want to perform.

Test failover by using Hyper-V Manager

To run a test failover by using Hyper-V Manager, follow these steps:

1. Open **Hyper-V Manager** on a device you use to manage a host in the **replica** cluster or a **replica** single host with the VM, or on the host itself where Windows Server with Desktop Experience is installed.
2. In the resources pane, select the host that contains the replica VM for which you want to run a test failover.
3. Right-click the VM, select **Replication**, then select **Test Failover**.
4. In the **Test Failover** screen, select the recovery point you want to use for the test failover from the drop-down list, then select **Test Failover**.

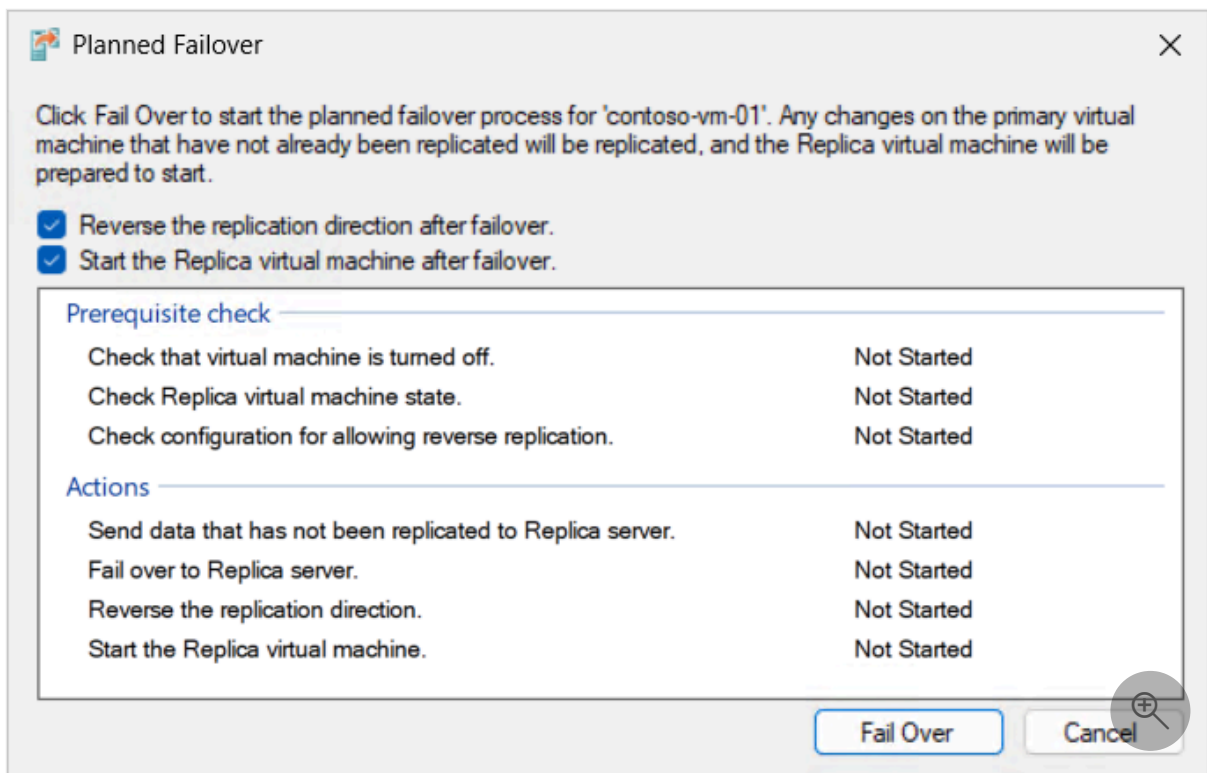


5. The host creates a duplicate VM. The VM name is the original VM name with **- Test** appended. The duplicate VM isn't connected to a network by default. At this point, you can start the VM and perform tests that you require to verify that it functions correctly.
6. When you finish testing, right-click the replica VM (not the test duplicate), select **Replication**, then select **Stop Test Failover**. In the confirmation dialog, select **Stop Test Failover** to delete the test VM and discard its data.

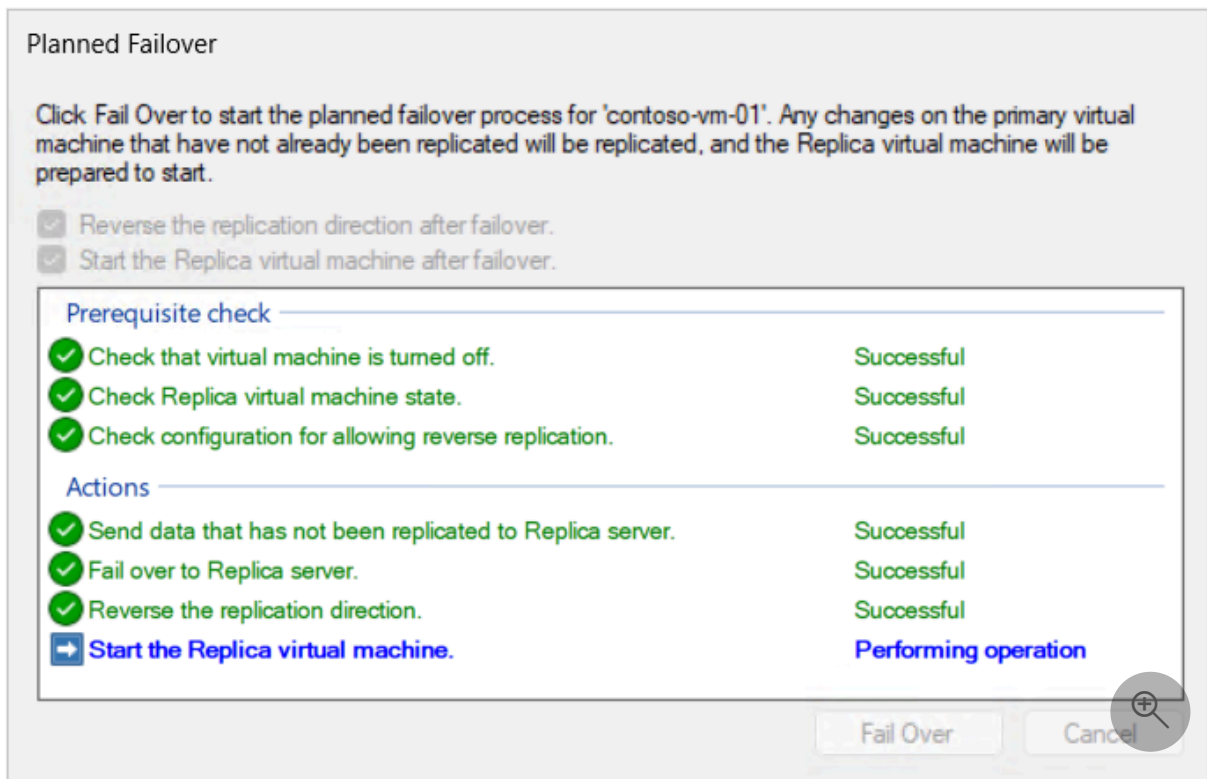
Planned failover using Hyper-V Manager

To run a planned failover using Hyper-V Manager:

1. Open **Hyper-V Manager** on a device you use to manage a host in the **primary** cluster or a **primary** single host with the VM, or on the host itself where Windows Server with Desktop Experience is installed.
2. In the resources pane, select the host that contains the VM for which you want to run a planned failover.
3. Shut down the VM if it's running. You must shut down the VM to perform a planned failover.
4. Right-click the VM, select **Replication**, then select **Planned Failover**.
5. In the **Planned Failover** screen, check the box to **Reverse the replication direction after failover** and optionally check the box to **Start the Replica virtual machine after failover**, depending on your requirements. The dialog shows a list of the prerequisite checks and actions the planned failover does. By default, the replica VM starts after failover. Don't forget to attach the VM to a network if necessary.



6. Select **Fail Over** to begin the process.



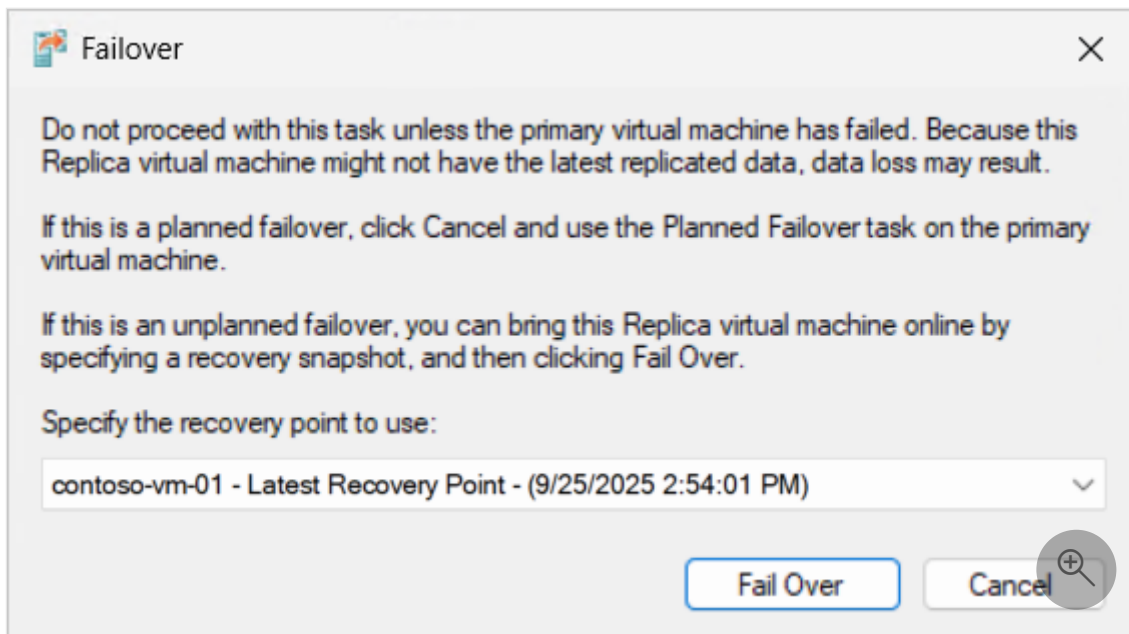
At this point, failover is complete. If you selected the option to reverse the replication direction, replication is automatically reconfigured to replicate changes back to the original primary host or cluster. Alternatively, right-click the replica VM, select **Replication**, then select **Reverse Replication** to manually configure replication back to the original primary host or cluster. Once reverse replication is configured, the replication health returns to **Normal**.

7. Start the VM if it isn't already running by right-clicking the VM, then selecting **Start**. Remember to attach the VM to a network if necessary.

Unplanned failover using Hyper-V Manager

To run an unplanned failover using Hyper-V Manager:

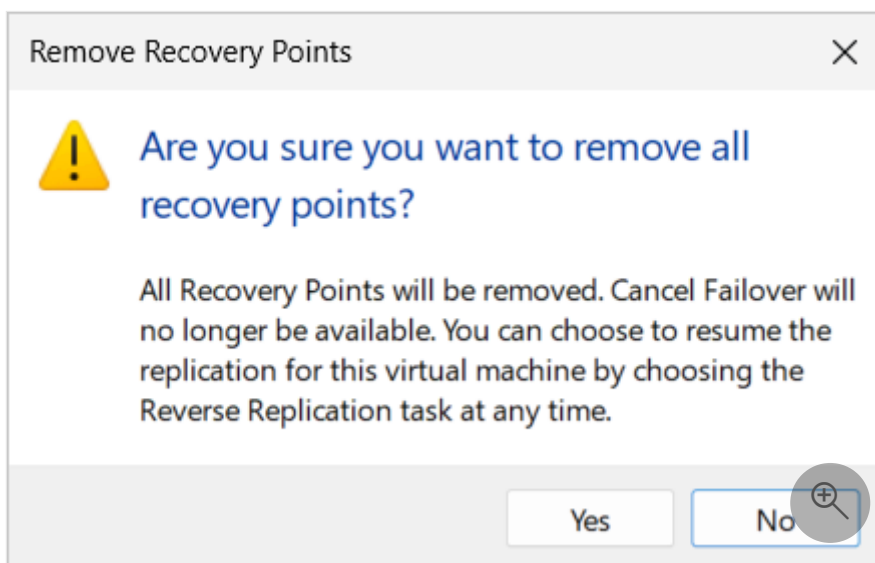
1. Open **Hyper-V Manager** on a device you use to manage a host in the **replica** cluster or a **replica** single host with the VM, or on the host itself where Windows Server with Desktop Experience is installed.
2. In the resources pane, select the host that contains the replica VM for which you want to run an unplanned failover.
3. Right-click the replica VM, select **Replication**, then select **Failover**.
4. In the **Failover** screen, select the recovery point you want to use for the failover from the drop-down list, then select **Fail Over**.



5. The process creates a checkpoint for the replica VM and then starts the VM. You should perform tests that you require to verify that the recovery point you chose functions correctly. Don't forget to attach the VM to a network if necessary.

If you need to select a different recovery point, you can cancel the failover by right-clicking the replica VM, select **Replication**, then select **Cancel Failover**. Then you can choose a different recovery point.

6. Once you test the VM and don't need to revert to a different recovery point, you need to complete the failover. Right-click the replica VM, select **Replication**, then select **Remove Recovery Points**. This action removes the recovery points and merges the checkpoint, which means you can't revert to an earlier recovery point. If you want to keep extra recovery points, you can first export the replica VM before completing failover.



At this point failover is complete, however the replication health shows as **Warning** because that replication direction isn't configured.

Reverse replication after unplanned failover using Hyper-V Manager

Once the primary VM is available again, you should replicate the changes made to the **replica** VM back to the original **primary** VM. The process is called **Reverse replication**. Before you can reverse the replication, you need to mark the original **primary** VM as able to be a target to receive replication. You can't perform this action in Hyper-V Manager.

1. On the original **primary** host that contains the original **primary** VM, open a PowerShell session as an administrator, or connect remotely using the [Enter-PSSession](#) cmdlet on a device you use to manage the host.
2. Mark the original **primary** VM to be a target to receive replication by running the following command. Be sure to replace placeholder `<VMName>` with the name of the original primary VM.

PowerShell

```
Set-VMReplication -VMName '<VM Name>' -AsReplica
```

3. Check the status of the original primary VM by running the following command on the primary host:

PowerShell

```
Get-VMReplication -VMName '<VM Name>' | FL VMName, ReplicationMode, ReplicationState
```

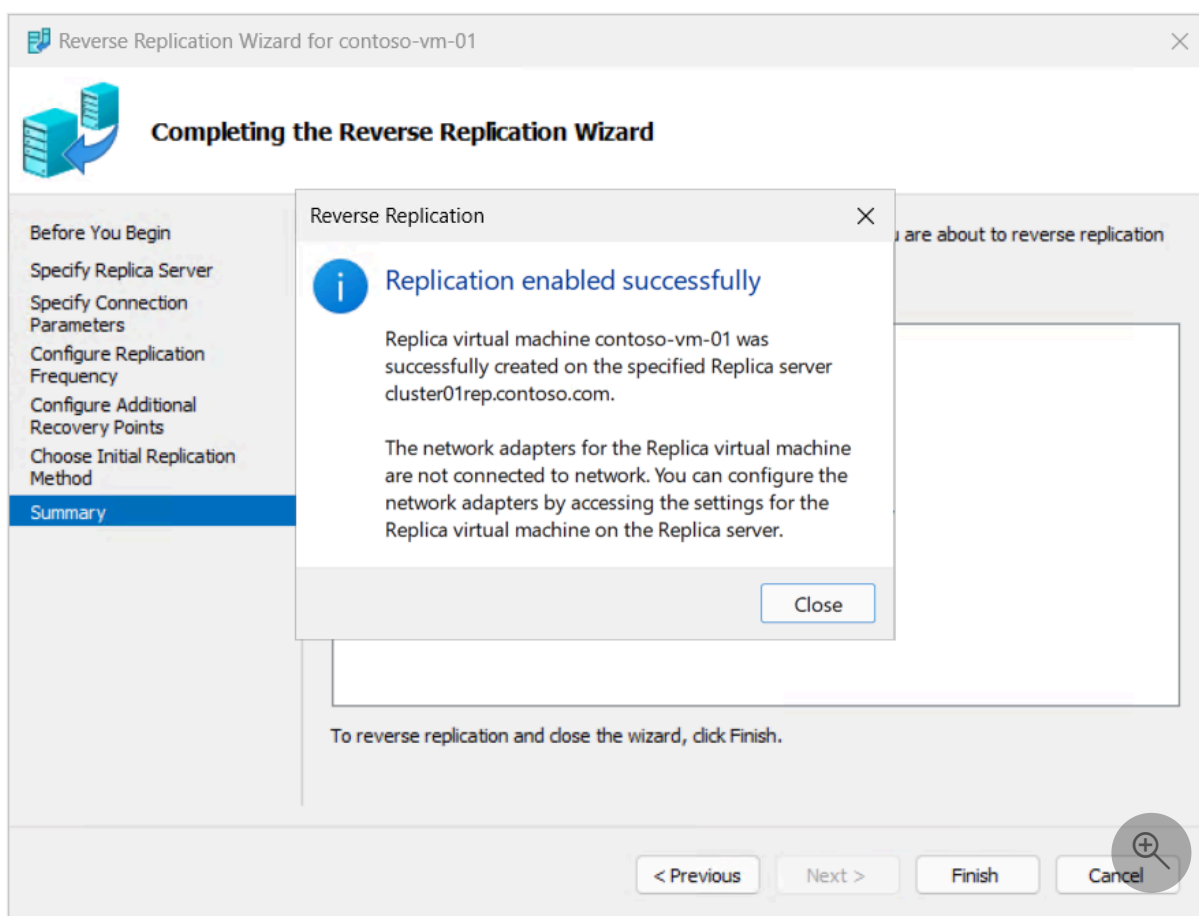
Here's an example of the output, which shows that the primary VM is now configured as a replica and is waiting for replication:

Output

```
VMName           : contoso-vm-01
ReplicationMode   : Replica
ReplicationState  : WaitingForInitialReplication
```

4. On the **replica** host that contains the **replica** VM, in Hyper-V Manager right-click the replica VM, select **Replication**, then select **Reverse Replication**. In the **Reverse Replication** screen, if you see **Before You Begin**, select **Next**.

5. The options in the wizard are prepopulated based on the settings of the original primary VM. If you don't need to change any settings, select **Next** on each screen until you reach the **Summary** screen, then select **Finish** to begin the reverse replication. Alternatively the options are the same as when you initially enabled replication for the VM.



At this point, any changes are replicated and the replication health returns to **Normal**. The primary and replica roles have now swapped. To return to the original replication direction, you can perform a planned failover, as described in the [Planned failover](#) section.

Note: The author created this article with assistance from AI. [Learn more](#)

Configure multiple CSV for VM replication

Article • 01/15/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Azure Local 23H2](#),  [Azure](#)
to: [Local 22H2](#)

When setting up Virtual Machine (VM) replication, you might encounter a limitation where only a single path for Cluster Shared Volumes (CSV) can be specified. This guide outlines how to overcome this limitation to configure multiple CSV for VM replication. This approach provides greater control over your VM by improving flexibility and storage efficiency of your resources.

Prerequisites

The following are required on your device before proceeding:

- The **Hyper-V** role must be installed. To learn more, see [Install the Hyper-V role on Windows Server](#).
- VM replication is enabled and inbound firewall rules are configured. To learn more, see [Set up Hyper-V Replica](#).
- Sufficient disk space for successful storage migration.

Note

During the replication process, the hardware and network you use affects the services that rely on them. Depending on the amount of data replicated between the source and target systems, this process consumes a large amount of system resources. Your device performance is impacted until this process completes.

Configure multiple CSV locations

When copying VMs through a replication process, it's possible to use multiple storage volumes to store the replicated data. There are two methods to configure new replication locations, through the Hyper-V Manager UI or via PowerShell.

Hyper-V

1. Open the **Hyper-V Manager**, under **Actions**, select **Hyper-V Settings**.

2. Under **Authorization and storage**, select either **Allow replication from any authenticated server** or **Allow replication from the specified servers**, then select **Browse**.
3. In the **Select Folder** window, choose a new default location, then select **Select Folder**.

When changes are made to the "default location" where replication is enabled for a VM, the new target volume is used for any new subsequent replication enablement. However, VMs that were already being replicated continues to use the original target volume that was set up when replication was first enabled.

Feedback

Was this page helpful?

 Yes

 No

Enable Intel Performance Monitoring Hardware in a Hyper-V virtual machine

Article • 05/17/2021 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 

to: [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local, versions 23H2 and 22H2](#)

Intel processors contain features collectively called performance monitoring hardware (e.g. PMU, PEBS, LBR). These features are used by performance tuning software like Intel VTune Amplifier to analyze software performance. Prior to Windows Server 2019 and Windows 10 Version 1809, neither the host operating system nor Hyper-V guest virtual machines could use performance monitoring hardware when Hyper-V was enabled. Starting with Windows Server 2019 and Windows 10 Version 1809, the host operating system has access to performance monitoring hardware by default. Hyper-V guest virtual machines do not have access by default, but Hyper-V administrators may choose to grant access to one or more guest virtual machines. This document describes the steps required to expose performance monitoring hardware to guest virtual machines.

Requirements

To enable performance monitoring hardware in a virtual machine, you'll need:

- An Intel processor with performance monitoring hardware (i.e. PMU, PEBS, LBR). Refer to [this document](#)  from Intel to determine which performance monitoring hardware your system supports.
- Windows Server 2019 or Windows 10 Version 1809 (October 2018 Update) or later
- A Hyper-V virtual machine *without* [nested virtualization](#) that is also in the stopped state

To enable upcoming Intel Processor Trace (IPT) performance monitoring hardware in a virtual machine, you'll need:

- An Intel processor that supports IPT and the PT2GPA feature. [^1] Refer to [this document](#)  from Intel to determine which performance monitoring hardware your system supports.
- Windows Server version 1903 (SAC) or Windows 10 Version 1903 (May 2019 Update) or later
- A Hyper-V virtual machine *without* [nested virtualization](#) that is also in the stopped state
- PMU needs to be enabled via the command line using the command seen below.

[^1]: PT2GPA refers to the “Intel PT uses guest physical addresses” bit. This is described in 25.5.4.1 of the Intel SDM.

Enabling performance monitoring components in a virtual machine

To enable different performance monitoring components for a specific guest virtual machine, use the `Set-VMProcessor` PowerShell cmdlet while running as Administrator:

ⓘ Note

The virtual machine generation must be 9.1 or higher. If nested virtualization is offered to the guest as well, then this needs 9.3 and up.

Powershell

```
# Enable IPT
Set-VMProcessor MyVMName -Perfmon @("ipt", "pmu")
```

Powershell

```
# Enable all components
Set-VMProcessor MyVMName -Perfmon @("ipt", "pmu", "lbr", "pebs")
```

Powershell

```
# Disable all components
Set-VMProcessor MyVMName -Perfmon @()
```

ⓘ Note

When enabling the performance monitoring components, if "pebs" is specified, then "pmu" must also be specified.

PEBS is only supported on hardware that has a PMU Version ≥ 4 .

Also, any command that attempts to enable "ipt" must also specify "pmu".

Enabling a component that is not supported by the host's physical processors will result in a virtual machine start failure.

Effects of enabling performance monitoring hardware on save/restore, export, and live migration

Microsoft does not recommend live migrating or saving/restoring virtual machines with performance monitoring hardware between systems with different Intel hardware. The specific behavior of performance monitoring hardware is often non-architectural and changes between Intel hardware systems. Moving a running virtual machine between different systems can result in unpredictable behavior of the non-architectural counters.

Feedback

Was this page helpful?

 Yes

 No

Processor compatibility for Hyper-V virtual machines

08/11/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Hyper-V processor compatibility mode enables virtual machine migration between hosts with different processor capabilities. A Hyper-V host passes through its processor capabilities to the virtual machines running on it, enabling the guest operating system and applications to take advantage of the same processor features as the host.

There are times when you might need to limit the processor capabilities that are passed through, such as when you want to move a live virtual machine (VM) or a saved VM between hosts or clusters with different processor capabilities, or developing applications that need to run on a wide range of hardware. In these cases, you can use processor compatibility mode. We recommend you enable processor compatibility mode only during the migration process, and then disable it to allow the VM to take advantage of the full capabilities of the host's processor.

Processor compatibility mode provides a VM with a common set of processor features. There are two modes of processor compatibility:

- **Dynamic processor compatibility mode:** for Hyper-V hosts in a cluster, it dynamically calculates the common set of processor features across all nodes, enabling VMs to take advantage of the maximum capabilities available across the cluster. Dynamic processor compatibility mode was introduced in Windows Server 2025.
- **Standard processor compatibility mode:** uses a fixed set of processor features, regardless of the capabilities of the host or cluster.

This article describes processor compatibility mode, why it's useful and how it works.

Why processor compatibility is useful

Processor manufacturers often introduce optimizations and capabilities in their processors. These capabilities often improve performance or security by using specialized hardware for a particular task. For example, many media applications use processor capabilities to speed up vector calculations. These features are rarely required for applications to run; they boost performance.

The capability set that's available on a processor varies depending on its make, model, and age. Operating systems and application software typically enumerate the system's processor capability set when they're first launched. Software doesn't expect the available processor capabilities on a device to change during its lifetime.

However, VM mobility features allow a running VM to be migrated to a new virtualization host. If software in the VM detects and starts using a particular processor capability, and then the VM is moved to a new virtualization host without that capability, the software can fail. Different processor features can result in the application or VM crashing.

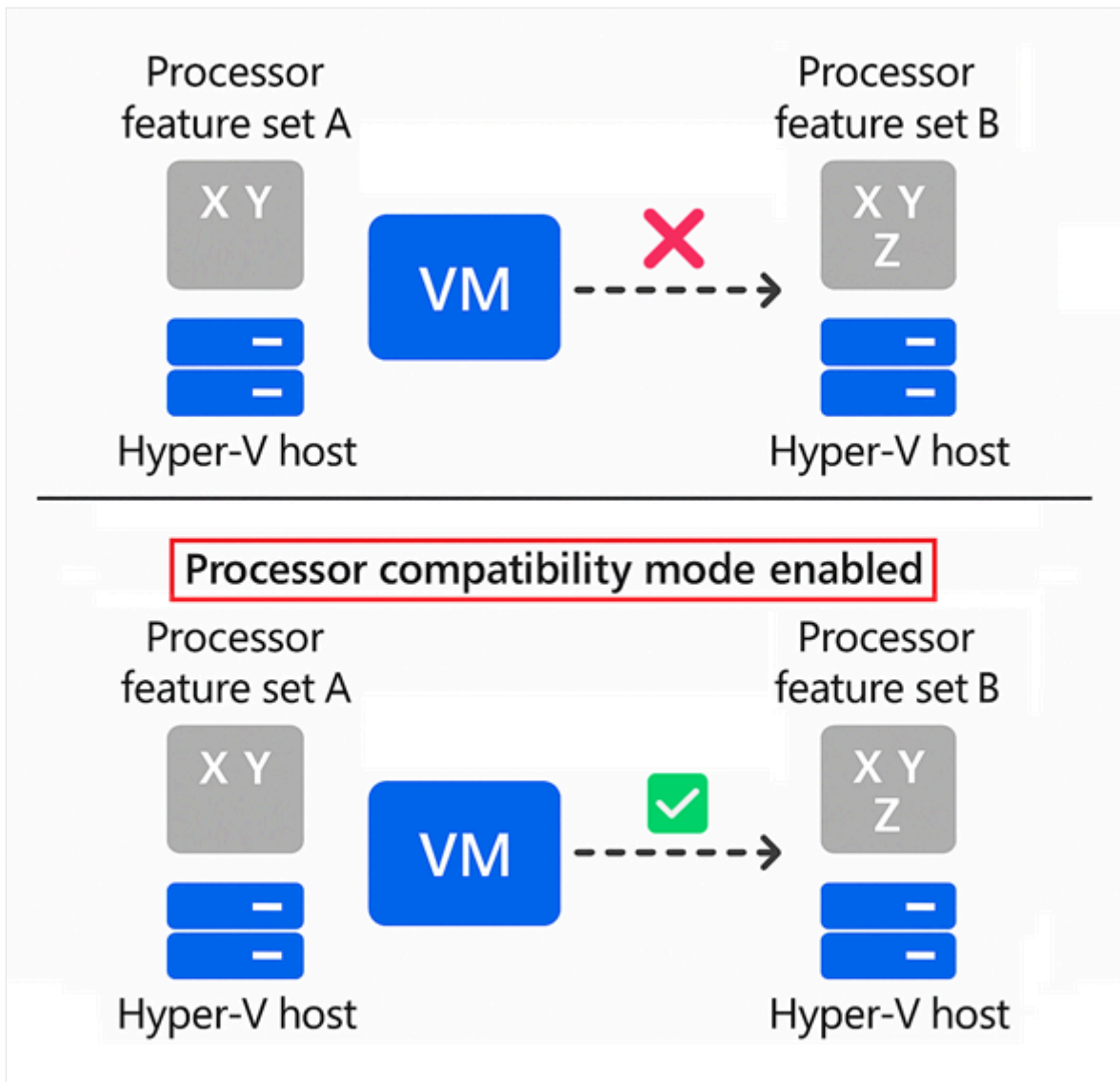
To avoid failures, Hyper-V performs initial checks whenever a VM live migration or save/restore operation is initiated. These checks compare the set of processor features that are available to the VM on the source host against the set of features that are available on the target host. If these feature sets don't match, the migration or restore operation is canceled.

You don't need to enable processor compatibility mode if you plan to stop and restart the VMs, as the guest operating system enumerates the processor capabilities available on the new host when the VM is restarted.

How processor compatibility mode works

Processor compatibility mode isn't enabled by default. You must explicitly enable it for each VM that you want to use it with.

When you enable processor compatibility mode on a VM, it ensures that the set of processor features available across a disparate set of virtualization hosts match by presenting only a limited set of processor features to the virtual machine. Processor compatibility mode hides newer processor instruction sets, typically those processors introduced within the last 10 years. However, hiding these features means that the guest operating system and application software can't take advantage of these processor instruction set enhancements.



Dynamic processor compatibility mode, introduced in Windows Server 2025 for VMs that use configuration version 10.0 or later, enhances the previous version of processor compatibility mode by enabling VMs to use a dynamically calculated set of processor features that are common across all nodes in a cluster. This means that VMs can take advantage of the maximum capabilities available across the cluster while still ensuring compatibility when moving VMs between hosts.

Each VM receives the maximum number of processor instruction sets that are present across all servers in the cluster. This process occurs automatically and is always enabled and replicated across the cluster, so there's no command to enable or disable the process.

It's difficult to quantify the overall performance effects of processor compatibility mode. The performance loss is primarily dependent on the workload running in the VM. Some workloads might be unaffected, while others show a noticeable difference. Software that heavily relies on hardware optimizations (such as encryption, compression, or intensive floating-point calculations) is impacted the most. If you're concerned about the performance impact of processor compatibility mode, it's best to compare VM workload performance with processor compatibility mode enabled and with it disabled.

Even when processor compatibility is enabled, you can't move VMs between hosts with different processor manufacturers. For example, you can't move running VMs or saved state VMs from a host with Intel processors to a host with AMD processors.

Next step

Learn how to [Configure a virtual machine to use processor compatibility mode](#).

 **Note:** The author created this article with assistance from AI. [Learn more](#)

Configure processor compatibility mode in Hyper-V virtual machines

08/11/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Processor compatibility mode enables virtual machine migration between hosts with different processor capabilities. It limits the processor features available to a virtual machine (VM) to ensure compatibility when moving VMs between hosts or clusters with different processors.

There are times when you might need to limit the processor capabilities that are passed through, such as when you want to move a live virtual machine (VM) or a saved VM between hosts or clusters with different processor capabilities, or developing applications that need to run on a wide range of hardware. In these cases, you can use processor compatibility mode. We recommend you enable processor compatibility mode only during the migration process, and then disable it to allow the VM to take advantage of the full capabilities of the host's processor. If you have a cluster with different processor generations, you need to leave processor compatibility mode enabled for live migration to work.

Dynamic processor compatibility mode, introduced in Windows Server 2025 for VMs that use configuration version 10.0 or later, enhances the previous version of processor compatibility mode by allowing VMs to use a dynamically calculated set of processor features that are common across all nodes in a cluster. This means that VMs can take advantage of the maximum capabilities available across the cluster while still ensuring compatibility when moving VMs between hosts.

To learn about processor compatibility mode, see [Processor compatibility in Hyper-V virtual machines](#).

This article explains how to configure processor compatibility mode for a VM using Hyper-V Manager, PowerShell, or Windows Admin Center.

Prerequisites

Before you begin, make sure you meet the following prerequisites:

- You have administrative access to the Hyper-V host or cluster where the VM is running and where you want to migrate it to.
- The VM you want to configure processor compatibility mode is powered off. Processor compatibility mode can't be enabled or disabled while the VM is running.

- Dynamic processor compatibility mode was introduced for VMs on a Hyper-V host running Windows Server 2025 or Azure Local (previously Azure Stack HCI) version 21H2 and use configuration version 10.0 or later. If your VM is using an earlier configuration version, upgrade its configuration version, otherwise it uses standard processor compatibility mode.

Configure a VM to use processor compatibility mode

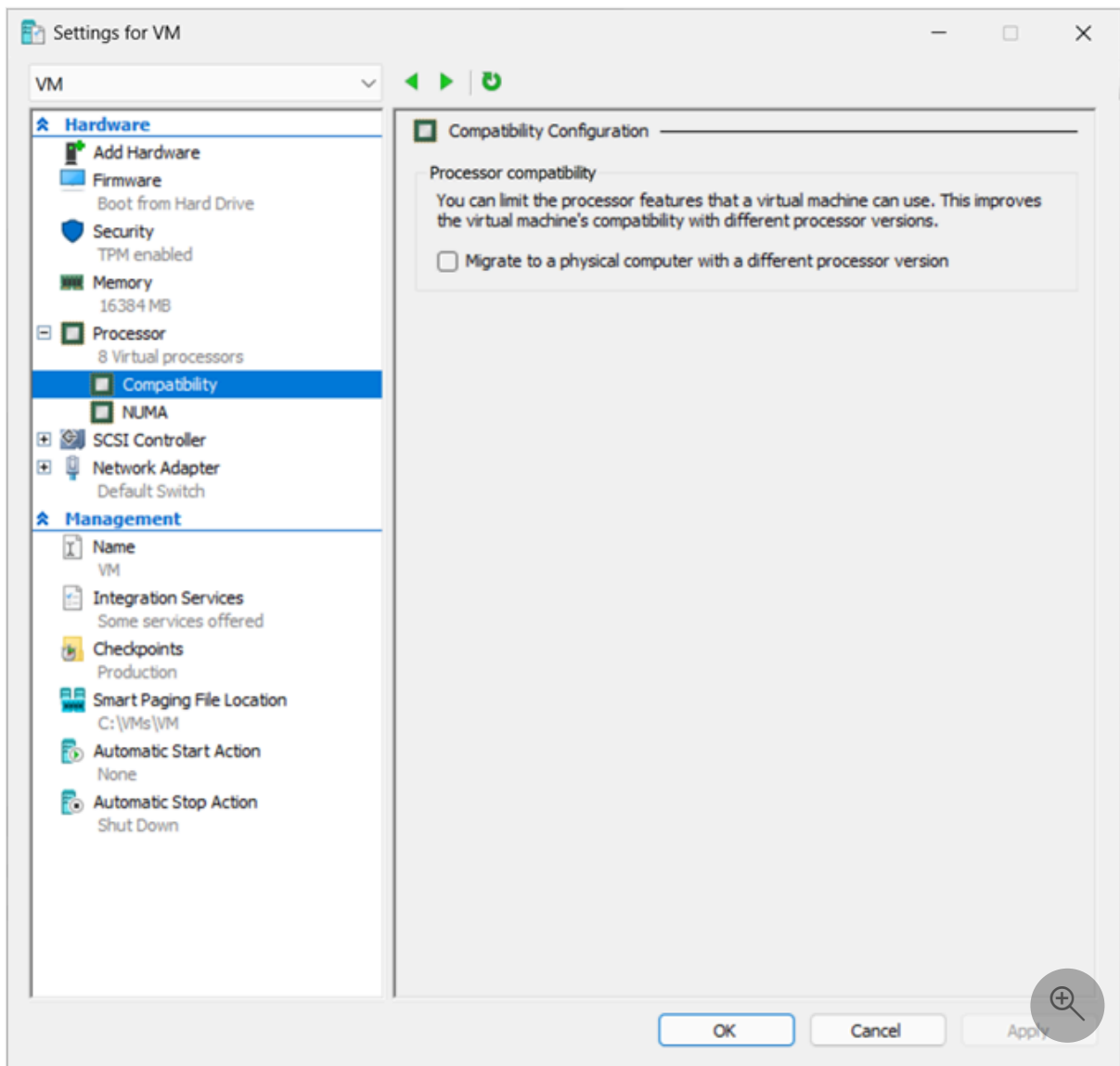
This section explains how to configure a VM to use processor compatibility mode using Hyper-V manager, PowerShell, or Windows Admin Center. It's possible to run VMs with and without compatibility mode in the same cluster. The two modes of processor compatibility are enabled and disabled the same way. Dynamic processor compatibility mode is transparent to the user and is enabled when the VM is running on a cluster.

Select the relevant tab for the method you want to use.

Hyper-V manager

To enable or disable processor compatibility mode for a VM using Hyper-V Manager, follow these steps. You can't configure dynamic processor compatibility mode in Hyper-V Manager; use PowerShell or Windows Admin Center instead.

1. Open **Hyper-V Manager**.
2. Select the server running Hyper-V and the desired VM.
3. If the VM is running, you must shut down the VM to change the processor compatibility mode setting.
4. In the **Actions** pane for the VM, select **Settings**.
5. In the **Hardware** pane, select **Processor**. Expand **Processor**, then select **Compatibility**.
6. To enable processor compatibility mode, check the box for **Migrate to a physical computer with a different processor**. To disable it, uncheck the box.



7. Select **OK** to save the change, then start the VM.

Verify processor features on a Hyper-V host and in a VM

Once you configure processor compatibility mode, you can verify the processor features available on a Hyper-V host and in the VM. This information can be useful if you want to check the VM has the expected set of features available.

To check the processor features on a Hyper-V host and a Windows VM, you can run [Coreinfo from Sysinternals](#):

1. Sign in to the host or the VM and download and extract the [Coreinfo tool](#).
2. Open PowerShell as an administrator and navigate to the directory where you extracted Coreinfo.

3. Run the following command. If it's the first time running Coreinfo on this host, you need to accept the license agreement.

PowerShell

```
.\Coreinfo.exe -f
```

Here's a truncated example of the output you might see, where each line represents a processor feature and an asterisk (*) indicates that the feature is available:

Output

HTT	*	Hyperthreading enabled
CET	-	Supports Control Flow Enforcement Technology
Kernel CET	-	Kernel-mode CET Enabled
User CET	-	User-mode CET Allowed
HYPervisor	*	Hypervisor is present
VMX	-	Supports Intel hardware-assisted virtualization
SVM	-	Supports AMD hardware-assisted virtualization
X64	*	Supports 64-bit mode
SMX	-	Supports Intel trusted execution
SKINIT	-	Supports AMD SKINIT
SGX	-	Supports Intel SGX
...		

When processor compatibility mode is enabled, you see that some features aren't available (indicated by a dash -) when compared to the output from the host. A reduced list indicates that the VM is running in processor compatibility mode and is limited to a common set of features.

If you're using dynamic processor compatibility mode, the features available in the VM are the maximum set of features that are common across all nodes in the cluster.

Linux VMs

Here's an example of how to check the processor features on a Debian-based or Red Hat-based Linux VM:

1. Sign in to the VM and open a terminal.
2. Run the following command to list the processor features:

Bash

```
lscpu | grep Flags
```

Here's a truncated example of the output you might see:

Output

```
Flags:                                fpu vme de pse tsc msr pae mce cx8 apic sep
mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm
constant_tsc rep_good nopl
xtopology aperfmperf pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic
movbe popcnt
tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm
3dnowprefetch invpcid_single
ssbd ibrs ibpb stibp ibrs_enhanced kaiser fsgsbase tsc_adjust bmi1 avx2 smep
bmi2 erms invpcid
avx512f avx512dq rdseed adx smap clflushopt avx512cd sha_ni avx512bw avx512vl
xsaveopt xsavec
xgetbv1 xsaves rdpid md_clear flush_l1d arch_capabilities
```

When processor compatibility mode is enabled, you see that some features aren't listed when compared to the output from the host. A reduced list indicates that the VM is running in processor compatibility mode and is limited to a common set of features.

If you're using dynamic processor compatibility mode, the features available in the VM are the maximum set of features that are common across all nodes in the cluster.

Live migration behavior of VMs between separate clusters

Assuming that all servers in each cluster are running the same hardware, it's possible to live migrate running VMs between clusters. There are three common scenarios.

- **Live migrating a VM from a cluster with new processors to a cluster with the same processors.** The VM capabilities are transferred to the destination cluster. This scenario doesn't require processor compatibility mode to be enabled; however, leaving it enabled shouldn't cause any migration issues.
- **Live migrating a VM from a cluster with older processors to a cluster with newer processors.** The VM capabilities are transferred to the destination cluster. In this scenario, when the VM is restarted, it receives the latest calculated capability of the destination cluster.

- **Live migrating a VM from a cluster with newer processors to a cluster with older processors.** You need to set the VM processor to use the `MinimumFeatureSet` for the `CompatibilityForMigrationMode` parameter in PowerShell, or select **Compatible across other hosts with the same CPU manufacturer** in Windows Admin Center. Once the VM is restarted, it receives the latest calculated capability of the destination cluster.

ⓘ **Note:** The author created this article with assistance from AI. [Learn more](#)

Live Migration Overview

Article • 09/17/2020 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local](#), versions 23H2 and 22H2

Live migration is a Hyper-V feature in Windows Server. It allows you to transparently move running Virtual Machines from one Hyper-V host to another without perceived downtime. The primary benefit of live migration is flexibility; running Virtual Machines are not tied to a single host machine. This allows actions like draining a specific host of Virtual Machines before decommissioning or upgrading it. When paired with Windows Failover Clustering, live migration allows the creation of highly available and fault tolerant systems.

Related Technologies and Documentation

Live migration is often used in conjunction with a few related technologies like Failover Clustering and System Center Virtual Machine Manager. If you're using Live Migration via these technologies, here are pointers to their latest documentation:

- [Failover Clustering](#) (Windows Server 2016)
- [System Center Virtual Machine Manager](#) (System Center 2016)

If you're using older versions of Windows Server, or need details on features introduced in older versions of Windows Server, here are pointers to historical documentation:

- [Live Migration](#) (Windows Server 2008 R2)
- [Live Migration](#) (Windows Server 2012 R2)
- [Failover Clustering](#) (Windows Server 2012 R2)
- [Failover Clustering](#) (Windows Server 2008 R2)
- [System Center Virtual Machine Manager](#) (System Center 2012 R2)
- [System Center Virtual Machine Manager](#)  (System Center 2008 R2)

Live Migration in Windows Server 2016

In Windows Server 2016, there are fewer restrictions on live migration deployment. It now works without Failover Clustering. Other functionality remains unchanged from previous releases of Live Migration. For more details on configuring and using live migration without Failover Clustering:

- [Set up hosts for live migration without Failover Clustering](#)

- [Use live migration without Failover Clustering to move a virtual machine](#)
-

Feedback

Was this page helpful?

 Yes

 No

Set up hosts for live migration without Failover Clustering

08/15/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Azure Local 2311.2 and later](#)

Live migration lets you move running virtual machines between servers with minimal downtime. This article shows you how to set up nonclustered hosts for live migration in Hyper-V, including authentication options, network configuration, and security best practices. Use these steps if you didn't configure live migration during Hyper-V installation, or need to update your settings.

Requirements for setting up live migration

To set up nonclustered hosts for live migration, you need:

- Membership in the local Hyper-V Administrators group or the Administrators group on both the source and destination computers meets this requirement, unless you're setting up constrained delegation. Membership in the Domain Administrators group is required to set up constrained delegation.
- The Hyper-V role in Windows Server installed on the source and destination servers. You can run a live migration between hosts running Windows Server 2012 R2 and later. For installation instructions, see [Install the Hyper-V role on Windows Server](#).
- Source and destination computers that belong to the same Active Directory domain, or to domains that trust each other.
- The Hyper-V management tools installed on a computer running at least Windows Server 2016 or Windows 10, unless the tools are installed on the source or destination server and you run the tools from the server.

Consider options for authentication and networking

Decide how you want to set up authentication, performance, and networking options.

- **Authentication:** Which protocol do you use to authenticate live migration traffic between the source and destination server? The protocol you choose determines whether you need to sign in to the source server before starting a live migration:

- Kerberos lets you avoid signing in to the server, but requires you to set up constrained delegation. See the following details for instructions.
- CredSSP lets you avoid setting up constrained delegation, but requires you to sign in to the source server. You can do this through a local console session, a Remote Desktop session, or a remote Windows PowerShell session.

CredSSP requires you to sign in for situations that aren't always obvious. For example, if you sign in to TestServer01 to move a virtual machine to TestServer02, and then want to move the virtual machine back to TestServer01, you need to sign in to TestServer02 before you try to move the virtual machine back to TestServer01. If you don't, the authentication attempt fails, an error occurs, and you see the following message:

"Virtual machine migration operation failed at migration Source. Failed to establish a connection with host *computer name*: No credentials are available in the security package 0x8009030E."

- **Performance:** Should you set up performance options? These options can reduce network and CPU usage, and make live migrations go faster. Check your requirements and infrastructure, and test different configurations to decide. The options are described at the end of step 2.
- **Network preference:** Do you allow live migration traffic through any available network, or do you isolate the traffic to specific networks? As a security best practice, isolate the traffic onto trusted, private networks because live migration traffic isn't encrypted when sent over the network. You can isolate the network by using a physically isolated network or another trusted networking technology like VLANs.

Upgrading to Windows Server 2025

Starting with Windows Server 2025, [Credential Guard is enabled by default](#) on all domain-joined servers that aren't domain controllers. Because of this, you can't use CredSSP-based live migration with Hyper-V after upgrading to Windows Server 2025. CredSSP-based delegation is the default for Windows Server 2022 and earlier for live migration. Instead, use Kerberos constrained delegation, as described in the following section. For more information, see [Live migration with Hyper-V breaks when upgrading to Windows Server 2025](#).

Step 1: Configure constrained delegation (optional)

If you decide to use Kerberos to authenticate live migration traffic, configure constrained delegation using an account that is a member of the Domain Administrators group.

Use the Users and Computers snap-in to set up constrained delegation

1. Open the Active Directory Users and Computers snap-in. In Server Manager, select the server, then select **Tools > Active Directory Users and Computers**.
2. In the navigation pane in **Active Directory Users and Computers**, select the domain and then double-click the **Computers** folder.
3. In the **Computers** folder, right-click the computer account of the source server, and then select **Properties**.
4. In **Properties**, select the **Delegation** tab.
5. On the **Delegation** tab, select **Trust this computer for delegation to the specified services only**, and then select **Use any authentication protocol**.
6. Select **Add**.
7. In **Add Services**, select **Users or Computers**.
8. From **Select Users or Computers**, type the name of the destination server. Select **Check Names** to verify it, and then select **OK**.
9. From **Add Services**, in the list of available services, do the following and then select **OK**:
 - To move virtual machine storage, select **cifs**. This is required if you want to move the storage along with the virtual machine, and if you want to move only a virtual machine's storage. If the server is configured to use SMB storage for Hyper-V, this should already be selected.
 - To move virtual machines, select **Microsoft Virtual System Migration Service**.
10. On the **Delegation** tab of the Properties dialog box, check that the services you selected in the previous step are listed as the services to which the destination computer can present delegated credentials. Select **OK**.
11. In the **Computers** folder, select the computer account of the destination server and repeat the process. In the **Select Users or Computers** dialog box, make sure to specify the name of the source server.

The configuration changes take effect after:

- The changes replicate to the domain controllers that the servers running Hyper-V are logged into.
- The domain controller issues a new Kerberos ticket.

Step 2: Set up the source and destination computers for live migration

This step covers options for authentication and networking. For better security, select specific networks for live migration traffic, as discussed earlier. You also choose a performance option in this step.

Use Hyper-V Manager to set up the source and destination computers for live migration

1. Open Hyper-V Manager. In Server Manager, select **Tools > Hyper-V Manager**.
2. In the navigation pane, select a server. If it isn't listed, right-click **Hyper-V Manager**, select **Connect to Server**, enter the server name, and select **OK**. Repeat to add more servers.
3. In the **Action** pane, select **Hyper-V Settings > Live Migrations**.
4. In the **Live Migrations** pane, select **Enable incoming and outgoing live migrations**.
5. Under **Simultaneous live migrations**, enter a different number if you don't want to use the default of 2.
6. Under **Incoming live migrations**, if you want to use specific network connections to accept live migration traffic, select **Add** to type the IP address. To set Kerberos and performance options, expand **Live Migrations** and select **Advanced Features**.
 - If you set up constrained delegation, under **Authentication protocol**, select **Kerberos**.
 - Under **Performance options**, review the details and select a different option if it fits your environment. **ce options****, review the details and choose a different option if it's appropriate for your environment.
7. Select **OK**.
8. Select the other server in Hyper-V Manager and repeat these steps.

Use Windows PowerShell to set up the source and destination computers for live migration

You can use three cmdlets to set up live migration on nonclustered hosts: [Enable-VMMigration](#), [Set-VMMigrationNetwork](#), and [Set-VMHost](#). This example uses all three to:

- Set up live migration on the local host

- Let incoming migration traffic only on a specific network
- Choose Kerberos as the authentication protocol

Each line is a separate command.

PowerShell

```
PS C:\> Enable-VMMigration
```

```
PS C:\> Set-VMMigrationNetwork 192.168.10.1
```

```
PS C:\> Set-VMHost -VirtualMachineMigrationAuthenticationType Kerberos
```

`Set-VMHost` also lets you choose a performance option and other host settings. For example, to choose SMB but keep the authentication protocol set to the default of CredSSP, enter:

PowerShell

```
PS C:\> Set-VMHost -VirtualMachineMigrationPerformanceOption SMB
```

This table describes how the performance options work.

[Expand table](#)

Option	Description
TCP/IP	Copies the memory of the virtual machine to the destination server over a TCP/IP connection.
Compression	Compresses the memory content of the virtual machine before copying it to the destination server over a TCP/IP connection. Compression is the default setting.
SMB	Copies the memory of the virtual machine to the destination server over an SMB 3.0 connection. - SMB Direct is used when the network adapters on the source and destination servers have Remote Direct Memory Access (RDMA) enabled. - SMB Multichannel automatically detects and uses multiple connections when a proper SMB Multichannel configuration is set up. For more information, see Improve Performance of a File Server with SMB Direct.

Next steps

After you set up the hosts, you're ready to do a live migration. For instructions, see [Use live migration without Failover Clustering to move a virtual machine](#).

Use live migration without Failover Clustering to move a virtual machine

Article • 02/21/2025 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local](#), versions 23H2 and 22H2

This article shows you how to move a virtual machine by doing a live migration without using Failover Clustering. A live migration moves running virtual machines between Hyper-V hosts without any noticeable downtime.

To be able to do this, you'll need:

- A user account that's a member of the local Hyper-V Administrators group or the Administrators group on both the source and destination computers.
- The Hyper-V role in Windows Server 2016 and Windows Server 2012 R2 installed on the source and destination servers and set up for live migrations. You can do a live migration between hosts running Windows Server 2016 and Windows Server 2012 R2 if the virtual machine is at least version 5.

For version upgrade instructions, see [Upgrade virtual machine version in Hyper-V on Windows 10 or Windows Server 2016](#). For installation instructions, see [Set up hosts for live migration](#).

- The Hyper-V management tools installed on a computer running at least Windows Server 2016 or Windows 10, unless the tools are installed on the source or destination server and you'll run them from there.

Use Hyper-V Manager to move a running virtual machine

1. Open Hyper-V Manager. (From Server Manager, select **Tools** > > **Hyper-V Manager**.)
2. In the navigation pane, select one of the servers. (If it isn't listed, right-click **Hyper-V Manager**, select **Connect to Server**, type the server name, and select **OK**. Repeat to add more servers.)
3. From the **Virtual Machines** pane, right-click the virtual machine and then select **Move**. This opens the Move Wizard.

4. Use the wizard pages to choose the type of move, destination server, and options.
5. On the **Summary** page, review your choices and then select **Finish**.

Use Windows PowerShell to move a running virtual machine

The following example uses the Move-VM cmdlet to move a virtual machine named *LMTest* to a destination server named *TestServer02* and moves the virtual hard disks and other file, such checkpoints and Smart Paging files, to the *D:\LMTest* directory on the destination server.

PowerShell

```
PS C:\> Move-VM LMTest TestServer02 -IncludeStorage -DestinationStoragePath  
D:\LMTest
```

Troubleshooting

Failed to establish a connection

If you haven't set up constrained delegation, you must sign in to source server before you can move a virtual machine. If you don't do this, the authentication attempt fails, an error occurs, and this message is displayed:

"Virtual machine migration operation failed at migration Source. Failed to establish a connection with host *computer name*: No credentials are available in the security package 0x8009030E."

To fix this problem, sign in to the source server and try the move again. To avoid having to sign in to a source server before doing a live migration, set up constrained delegation. You'll need domain administrator credentials to set up constrained delegation. For instructions, see [Set up hosts for live migration](#).

Failed because the host hardware isn't compatible

If a virtual machine doesn't have processor compatibility turned on and has one or more snapshots, the move fails if the hosts have different processor versions. An error occurs and this message is displayed:

The virtual machine cannot be moved to the destination computer. The hardware on the destination computer is not compatible with the hardware requirements of this virtual machine.

To fix this problem, shut down the virtual machine and turn on the processor compatibility setting.

1. From Hyper-V Manager, in the **Virtual Machines** pane, right-click the virtual machine and select **Settings**.
2. In the navigation pane, expand **Processors** and select **Compatibility**.
3. Check **Migrate to a computer with a different processor version**.
4. Select **OK**.

To use Windows PowerShell, use the [Set-VMProcessor](#) cmdlet:

PowerShell

```
PS C:\> Set-VMProcessor TestVM -CompatibilityForMigrationEnabled $true
```

Feedback

Was this page helpful?

 Yes

 No

Use live migration with workgroup clusters

Article • 07/29/2024 •

Applies  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#), 
to: [Windows Server 2016](#),  [Azure Local](#), versions 23H2 and 22H2

This article describes how to move a virtual machine by doing a live migration between hosts using workgroup clusters. Workgroup clusters are a type of Failover Cluster that doesn't use an on-premises domain controller or Active Directory forest. Instead, workgroup clusters are joined by workgroup. Workgroup clusters were introduced in Windows Server 2016. However, live migration for workgroup clusters wasn't supported until now, in Windows Server 2025. Benefit from the flexibility of live migration combined with the high availability of workgroup clusters.

Follow the steps in this article to perform your own live migration.

Prerequisites

The following prerequisites must be met in order to do a live migration of hosts using workgroup clusters:

- A workgroup cluster with two or more nodes is up and running. To learn more about creating a workgroup cluster, see [Create a workgroup cluster](#).
- A local user account exists on each server node with an identical username and password.

Consider options for authentication and performance

When setting up live migrations for workgroup clusters, it's important to consider how authentication and performance.

- **Authentication:** Local accounts with an identical username and password on each node are used to create and configure the workgroup cluster. The cluster uses self-signing PKU2U certificates to authenticate and be able to move a virtual machine from one host node to another host node without Kerberos. The local accounts on each node are the only way to successfully authenticate workgroup clusters and allow for live migration between source and destination servers.

- **Performance:** Configuring performance options can reduce network and CPU usage. Different options like simultaneous migrations can also make live migrations go faster. Consider your requirements and your infrastructure, and test different configurations to help you decide.

Do a live migration with Hyper-V workgroup clusters

In the next sections, you'll complete the following steps:

- **Install the Hyper-V role and Hyper-V management tools.** Each virtual machine needs Hyper-V installed in order to be able to connect to other hosts and do a live migration.
- **Create a new virtual machine and add it to the workgroup cluster.** Add in a virtual machine role to facilitate the live migration.
- **Set up source and destination servers.** Set up each server node to enable live migrations.
- **Move a running virtual machine with live migration.** Complete a live migration by moving a running virtual machine between Hyper-V hosts without any noticeable downtime.

Step 1: Install the Hyper-V role

The Hyper-V role must be installed on the source and destination servers and set up for live migrations. Install this role before continuing.

Hyper-V provides the services that you can use to create and manage virtual machines. [Hyper-V may be installed in many ways](#). This section describes how to install the role using either PowerShell or the Server Manager.

PowerShell

1. Open a PowerShell session as an Administrator.
2. Use the [Install-WindowsFeature cmdlet](#) to install the Hyper-V role with the following command:

PowerShell

```
Install-WindowsFeature -Name Hyper-V -IncludeManagementTools -Restart
```

3. Wait for the role to be installed, and then restart your virtual machine.
4. Repeat the installation for the other virtual machines in the cluster.

ⓘ Note

If you're unable to install the Hyper-V role, you might need to enable Nested Virtualization. See how to [enable Nested Virtualization](#) to allow Hyper-V to run inside of a Hyper-V virtual machine.

Step 2: Create a new virtual machine and add it to the workgroup cluster

Add a new Hyper-V virtual machine as a role to your workgroup cluster in order to do a live migration between hosts.

PowerShell

1. Connect to one of your server nodes.
2. Open a PowerShell session as an Administrator.
3. Run the following [New-VM](#) command to create a new virtual machine that has 10GB of memory and uses an existing VHDX image on the server node. Change the parameters and values as needed to customize your setup.

PowerShell

```
New-VM -Name "<VM_NAME>" -MemoryStartupBytes 10GB -VHDPath  
<PATH_TO_VHDX_FILE>
```

4. Add in the virtual machine as a **Virtual Machine** role in the workgroup cluster enable automatic failover.

PowerShell

```
Add-ClusterVirtualMachineRole -VirtualMachine <VM_NAME>
```

Step 3: Set up the source and destination computers for live migration

In this step, set up your source and host destination virtual machines to enable live migrations. Here, you can also specify live migration settings, such as how many live and storage migrations to allow at the same time.

PowerShell

1. Connect to one of your server nodes.
2. Open a PowerShell session as an Administrator.
3. First, use the [Enable-VMMigration](#) cmdlet to configure live migration on the Hyper-V virtual machine host.

PowerShell

```
Enable-VMMigration
```

4. Use the [Set-VMHost](#) cmdlet to configure the local Hyper-V host. The following command configures the virtual machine to allow 10 simultaneous live migrations and storage migrations. Change these values to the number of simultaneous live and storage migrations your setup allows. You might need to test different configurations to help you decide.

PowerShell

```
Set-VMHost -MaximumVirtualMachineMigrations 10 -  
MaximumStorageMigrations 10
```

Set-VMHost also lets you specify a performance option, and other host settings. Consider using a parameter like `-VMMigrationPerformance` to choose more settings for your virtual machine.

5. Repeat the steps for the other server node.

Step 4: Move a running virtual machine with live migration

Finally, do a live migration to move a running virtual machine.

1. Connect to the current owner node in your workgroup cluster.
2. Open Failover Cluster Manager.
3. In the **Roles** section, select the virtual machine role, and right-click.
4. Select **Move**, then **Live Migration**, and then **Best Possible Node**.
5. In the **Information** column, see a status appear with the message **Live Migrating, X% completed**.
6. Once complete, confirm that the Owner Node column updates with the other node in your workgroup cluster.

Next steps

After completing a live migration, it's important to confirm that the migration works, and all virtual machines migrated were migrated successfully. If you notice any issues during or after a migration, it might be necessary to revisit the simultaneous migrations allowed or configure live migration performance options.

To learn more about Live Migration performance options, see [Virtual Machine Live Migration Overview](#)

You can also read more about live migration performance in [Hyper-V Network I/O Performance](#).

Feedback

Was this page helpful?

 Yes

 No

Working with Hyper-V and Windows PowerShell

08/14/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Now that you have walked through the basics of deploying Hyper-V, creating virtual machines and managing these virtual machines, let's explore how you can automate many of these activities with PowerShell.

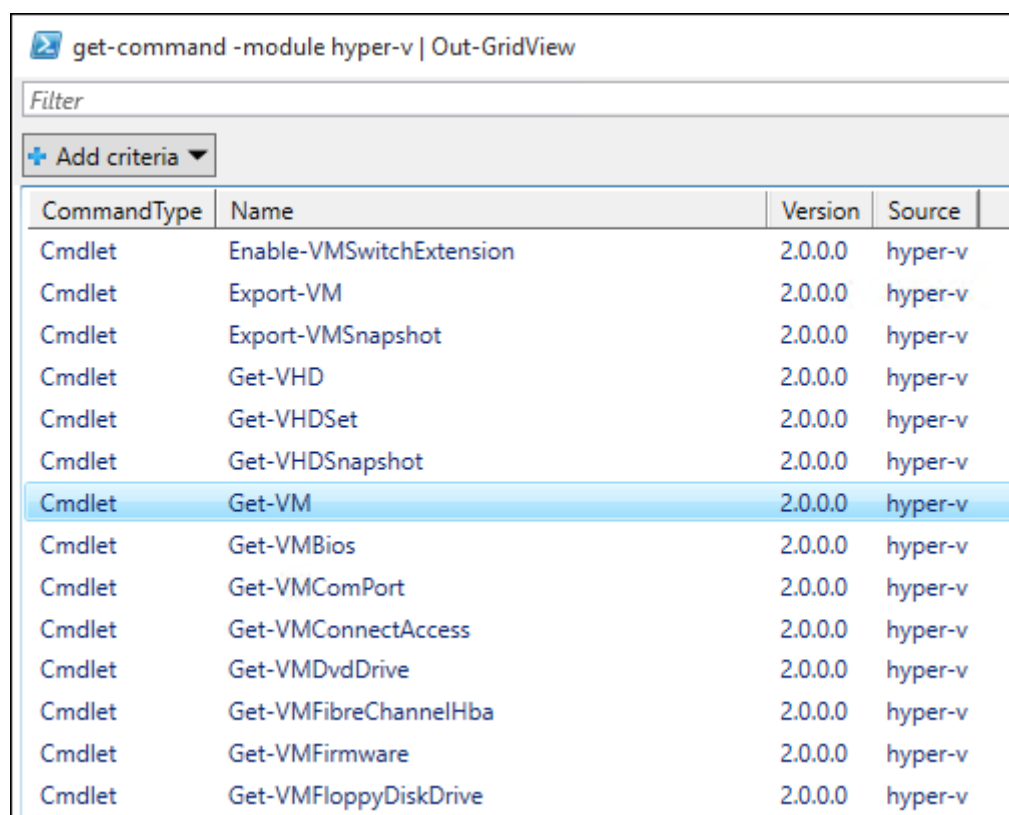
Return a list of Hyper-V commands

1. Select the Windows start button, and then type **PowerShell**.
2. Run the following command to display a searchable list of PowerShell commands available with the Hyper-V PowerShell Module.

PowerShell

```
Get-Command -Module hyper-v | Out-GridView
```

You get something like this:



The screenshot shows a PowerShell Out-GridView window titled "get-command -module hyper-v | Out-GridView". It contains a table with four columns: CommandType, Name, Version, and Source. The table lists various Hyper-V cmdlets, with "Get-VM" highlighted in blue.

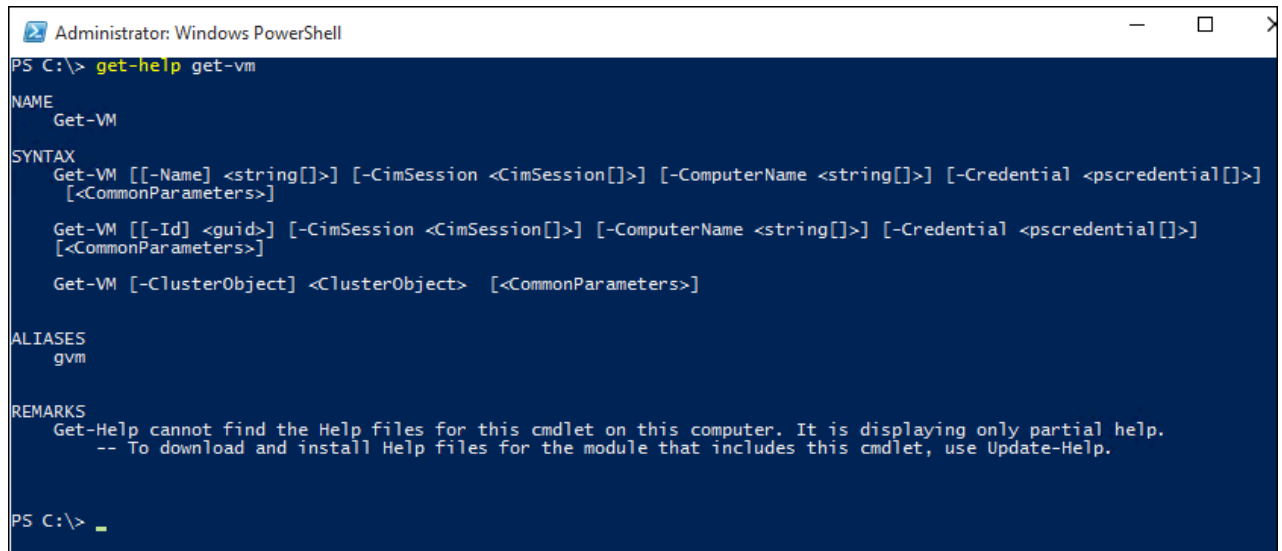
CommandType	Name	Version	Source
Cmdlet	Enable-VMSwitchExtension	2.0.0.0	hyper-v
Cmdlet	Export-VM	2.0.0.0	hyper-v
Cmdlet	Export-VMSnapshot	2.0.0.0	hyper-v
Cmdlet	Get-VHD	2.0.0.0	hyper-v
Cmdlet	Get-VHDSets	2.0.0.0	hyper-v
Cmdlet	Get-VHDSnapshot	2.0.0.0	hyper-v
Cmdlet	Get-VM	2.0.0.0	hyper-v
Cmdlet	Get-VM BIOS	2.0.0.0	hyper-v
Cmdlet	Get-VMComPort	2.0.0.0	hyper-v
Cmdlet	Get-VMConnectAccess	2.0.0.0	hyper-v
Cmdlet	Get-VM DVD Drive	2.0.0.0	hyper-v
Cmdlet	Get-VMFibreChannelHba	2.0.0.0	hyper-v
Cmdlet	Get-VMFirmware	2.0.0.0	hyper-v
Cmdlet	Get-VMFloppyDiskDrive	2.0.0.0	hyper-v

1. To learn more about a particular PowerShell command use `Get-Help`. For instance, running the following command returns information about the `Get-VM` Hyper-V command.

PowerShell

`Get-Help Get-VM`

The output shows you how to structure the command, what the required and optional parameters are, and the aliases that you can use.



```
Administrator: Windows PowerShell
PS C:\> get-help get-vm

NAME
    Get-VM

SYNTAX
    Get-VM [[-Name] <string[]>] [-CimSession <CimSession[]>] [-ComputerName <string[]>] [-Credential <pscredential[]>]
    [-<CommonParameters>]

    Get-VM [[-Id] <guid>] [-CimSession <CimSession[]>] [-ComputerName <string[]>] [-Credential <pscredential[]>]
    [-<CommonParameters>]

    Get-VM [-ClusterObject] <ClusterObject> [-<CommonParameters>]

ALIASES
    gvm

REMARKS
    Get-Help cannot find the Help files for this cmdlet on this computer. It is displaying only partial help.
    -- To download and install Help files for the module that includes this cmdlet, use Update-Help.

PS C:\> _
```

Return a list of virtual machines

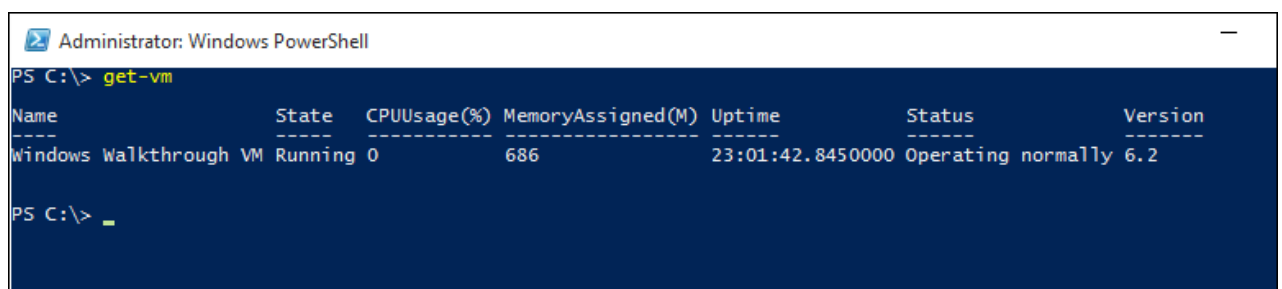
Use the `Get-VM` command to return a list of virtual machines.

1. In PowerShell, run the following command:

PowerShell

`Get-VM`

This displays something like this:



```
Administrator: Windows PowerShell
PS C:\> get-vm

Name                State      CPUUsage(%) MemoryAssigned(M) Uptime                Status              Version
-----
Windows Walkthrough VM Running 0           686                23:01:42.8450000 Operating normally 6.2

PS C:\> _
```

2. To return a list of only powered on virtual machines add a filter to the `Get-VM` command. A filter can be added by using the `Where-Object` command. For more information on filtering, see [Using the Where-Object](#) documentation.

PowerShell

```
Get-VM | where {$_.State -eq 'Running'}
```

3. To list all virtual machines in a powered off state, run the following command. This command is a copy of the command from step 2 with the filter changed from 'Running' to 'Off'.

PowerShell

```
Get-VM | where {$_.State -eq 'Off'}
```

Start and shut down virtual machines

1. To start a particular virtual machine, run the following command with name of the virtual machine:

PowerShell

```
Start-VM -Name <virtual machine name>
```

2. To start all currently powered off virtual machines, get a list of those machines and pipe the list to the `Start-VM` command:

PowerShell

```
Get-VM | where {$_.State -eq 'Off'} | Start-VM
```

3. To shut down all running virtual machines, run the following command:

PowerShell

```
Get-VM | where {$_.State -eq 'Running'} | Stop-VM
```

Create a virtual machine checkpoint

To create a checkpoint using PowerShell, select the virtual machine using the `Get-VM` command and pipe this to the `Checkpoint-VM` command. Finally give the checkpoint a name using `-SnapshotName`. The complete command looks like the following:

PowerShell

```
Get-VM -Name <VM Name> | Checkpoint-VM -SnapshotName <name for snapshot>
```

Create a new virtual machine

The following example shows how to create a new virtual machine in the PowerShell Integrated Scripting Environment (ISE). This is a simple example and could be expanded on to include additional PowerShell features and more advanced VM deployments.

1. To open the PowerShell ISE click on start, type **PowerShell ISE**.
2. Run the following code to create a virtual machine. See the [New-VM](#) documentation for detailed information on the `New-VM` command.

PowerShell

```
$VMName = "VMNAME"

$VM = @{
    Name = $VMName
    MemoryStartupBytes = 2147483648
    Generation = 2
    NewVHDPATH = "C:\Virtual Machines\$VMName\$VMName.vhdx"
    NewVHDSIZEBytes = 53687091200
    BootDevice = "VHD"
    Path = "C:\Virtual Machines\$VMName"
    SwitchName = (Get-VMSwitch).Name
}

New-VM @VM
```

Wrap up and References

This document has shown some simple steps to explore the Hyper-V PowerShell module as well as some sample scenarios. For more information on the Hyper-V PowerShell module, see the [Hyper-V Cmdlets in Windows PowerShell reference](#).

Hyper-V Dynamic Memory Overview

10/28/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

If you have idle or low-load virtual machines, such as in pooled Virtual Desktop Infrastructure (VDI) environments, Dynamic Memory enables you to increase consolidation and improve reliability for restart operations. You also gain agility in responding to requirement changes with these new capabilities.

Key benefits

With Dynamic Memory, you can attain higher consolidation numbers with improved reliability for restart operations. This capability can lead to lower costs, especially in environments that have many idle or low-load virtual machines, such as pooled VDI environments. Dynamic Memory run-time configuration changes can reduce downtime and provide increased agility to respond to requirement changes.

Hyper-V running on Windows Server automatically calculates an amount of memory to reserve for exclusive use by the management host operating system. This memory is used to run virtualization services. If the computer is part of a failover cluster, Hyper-V also reserves enough memory to run failover cluster services.

Technical overview

Dynamic Memory defines startup memory as the minimum amount of memory that a virtual machine can have. However, Windows requires more memory during startup than the steady state. As a result, administrators sometimes assign extra memory to a virtual machine because Hyper-V can't reclaim memory from these virtual machines after startup. Dynamic Memory introduces a minimum memory setting, which allows Hyper-V to reclaim the unused memory from the virtual machines. This feature is reflected as increased virtual machine consolidation numbers, especially in Virtual Desktop Infrastructure (VDI) environments.

Windows Server also introduced Smart Paging for reliable virtual machine restart operations. Although minimum memory increases virtual machine consolidation numbers, it also brings a challenge. If a virtual machine has a smaller amount of memory than its startup memory and if it's restarted, Hyper-V needs additional memory to restart the virtual machine. Due to host memory pressure or virtual machine states, Hyper-V might not always have additional memory available. This limitation can cause sporadic virtual machine restart failures. Smart Paging

bridges the memory gap between minimum memory and startup memory, and allows virtual machines to restart reliably.

Minimum memory configuration with reliable restart operation

As in the previous version of Dynamic Memory, you can configure a minimum memory amount for virtual machines, and Hyper-V continues to ensure that it always assigns this amount to running virtual machines.

To provide a reliable restart experience for virtual machines configured with less minimum memory than startup memory, Hyper-V uses Smart Paging. This memory management method uses disk resources as additional, temporary memory when more memory is required to restart a virtual machine. This approach has advantages and drawbacks. It provides a reliable way to keep the virtual machines running when there's no available physical memory. However, it can degrade virtual machine performance because disk access speeds are much slower than memory access speeds.

Important

When installing or upgrading the operating system of a virtual machine, the amount of memory that's available to the virtual machine during the installation and upgrade process is the value specified as **Startup RAM**. Even if you configure Dynamic Memory for the virtual machine, the virtual machine uses only the amount of memory configured in the **Startup RAM** setting. Ensure the **Startup RAM** value meets the minimum memory requirements of the operating system during the install or upgrade procedure.

To minimize the performance impact of Smart Paging, Hyper-V uses it only when all of the following conditions occur:

- The virtual machine is being restarted.
- There's no available physical memory.
- No memory can be reclaimed from other virtual machines running on the host.

Hyper-V doesn't use Smart Paging when:

- You start a virtual machine from an "off state" (instead of a restart).
- You oversubscribe memory for a running virtual machine.
- A virtual machine fails over in Hyper-V clusters.

When you oversubscribe host memory, Hyper-V continues to rely on the paging operation in the guest operating system because it's more effective than Smart Paging. The paging

operation in the guest operating system is performed by Windows Memory Manager. Windows Memory Manager has more information than the Hyper-V host about memory usage within the virtual machine, which means it can provide Hyper-V with better information to use when choosing the memory to be paged. Because of this, the system incurs less overhead compared to Smart Paging.

To further reduce the impact of Smart Paging, Hyper-V removes memory from the virtual machine after it completes the start process. It accomplishes this goal by coordinating with Dynamic Memory components inside the guest operating system ("ballooning"), so the virtual machine stops using Smart Paging. With this technique, the use of Smart Paging is temporary and isn't expected to last longer than 10 minutes.

Also note the following about how Hyper-V uses Smart Paging:

- It creates Smart Paging files only when needed for a virtual machine.
- It deletes Smart Paging files after it removes the additional amount of memory.
- It doesn't use Smart Paging for this virtual machine again until another restart occurs and there's not enough physical memory.

Run-time Dynamic Memory configuration changes

Hyper-V enables you to make the following configuration changes to Dynamic Memory when the virtual machine is running:

- Increase the maximum memory.
- Decrease the minimum memory.

About the Dynamic Memory settings

The following virtual machine settings in Hyper-V Manager allow you to configure Dynamic Memory. Review the following descriptions to understand how the features work and how they are related.

 Expand table

Setting	Description
Startup RAM	Specifies the amount of memory required to start the virtual machine. The value needs to be high enough to allow the guest operating system to start, but should be as low as possible to allow for optimal memory utilization and potentially higher consolidation ratios.
Minimum RAM	Specifies the minimum amount of memory that should be allocated to the virtual machine after the virtual machine starts. Set the value as low as 32 MB up to a maximum value equal

Setting	Description
	to the Startup RAM value.
Maximum RAM	Specifies the maximum amount of memory that this virtual machine can use. Set the value from as low as the value for Startup RAM up to as high as 1 TB. However, a virtual machine can use only as much memory as the maximum amount supported by the guest operating system. For example, if you specify 64 GB for a virtual machine running a guest operating system that supports a maximum of 32 GB, the virtual machine can't use more than 32 GB.
Memory buffer	Specifies how much memory Hyper-V attempts to assign to the virtual machine compared to the amount of memory actually needed by the applications and services running inside the virtual machine. Memory buffer is specified as a percentage because the actual amount of memory that represents the buffer changes in response to changes in memory usage while the virtual machine is running. Hyper-V uses performance counters in the virtual machine that identify committed memory to determine the current memory requirements of the virtual machine and then calculates the amount of memory to add as a buffer. The buffer is determined using the following formula: Amount of memory buffer = how much memory the virtual machine actually needs / (memory buffer value / 100). For example, if the memory committed to the guest operating system is 1,000 MB and the memory buffer is 20%, Hyper-V attempts to allocate an additional 20% (200 MB) for a total of 1,200 MB of physical memory allocated to the virtual machine. Note: The buffer isn't maintained when there's not enough physical memory available in the computer to give every virtual machine its requested memory buffer.
Memory weight	Provides Hyper-V with a way to determine how to distribute memory among virtual machines if there's not enough physical memory available in the computer to give every virtual machine its requested amount of memory.

Dynamic Memory / Hyper-V Memory Management FAQs

The following are some frequently asked questions regarding Dynamic Memory.

Do I need to manually configure the size of the page file after installing the Hyper-V role?

No, it's recommended to allow the system to manage the page file size when running Hyper-V.

How is the memory divided between the host operating system and running virtual machines?

Hyper-V monitors the runtime health of the host operating system to determine how much memory it can safely allocate to virtual machines.

My virtual machine failed to start with the following error message “Not enough memory in the system to start the virtual machine” or “Could not initialize memory: Ran out of memory (0x8007000E)”. How can I tell how much memory is available for virtual machines?

As described earlier, Hyper-V monitors the runtime health of the host operating system to determine how much memory it can safely allocate to virtual machines.

You can view the current amount of memory available to virtual machines in the following Performance Monitor counter: **Hyper-V Dynamic Memory Balancer – Available Memory**.

Hyper-V Virtual Machine Connection

07/25/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Virtual Machine Connection (VMConnect) is a tool you can use to connect to a virtual machine to install or interact with the guest operating system in a virtual machine. Some of the tasks you can perform by using VMConnect include the following:

- Start and shut down a virtual machine
- Connect to a DVD image (.iso file) or a USB flash drive
- Create a checkpoint
- Modify the settings of a virtual machine

Tips for using VMConnect

You might find the following information helpful for using VMConnect:

 [Expand table](#)

To do this...	Do this...
Send mouse clicks or keyboard input to the virtual machine	Click anywhere in the virtual machine window. The mouse pointer might appear as a small dot when you connect to a running virtual machine.
Return mouse clicks or keyboard input to the physical computer	Press CTRL+ALT+LEFT ARROW and then move the mouse pointer outside of the virtual machine window. This mouse release key combination can be changed in the Hyper-V settings in Hyper-V Manager.
Send the CTRL+ALT+DELETE key combination to a virtual machine	Select Action > Ctrl+Alt+Delete or use the key combination CTRL+ALT+END.
Switch from window mode to full-screen mode	Select View > Full Screen Mode . To switch back to window mode, press CTRL+ALT+BREAK.
Create a checkpoint to capture the current state of the machine for troubleshooting	Select Action > Checkpoint or use the key combination CTRL+N.

To do this...	Do this...
Change the settings of the virtual machine	Select File > Settings .
Connect to a DVD image (.iso file) or a virtual floppy disk (.vfd file)	Select Media. Virtual floppy disks aren't supported for generation 2 virtual machines. For more information, see Should I create a generation 1 or 2 virtual machine in Hyper-V?.
Use a host's local resources on Hyper-V virtual machine like a USB flash drive	Turn on enhanced session mode on the Hyper-V host, use VMConnect to connect to the virtual machine, and before you connect, choose the local resource that you want to use. For the specific steps, see Use local resources on Hyper-V virtual machine with VMConnect .
Change saved VMConnect settings for a virtual machine	Run the following command in Windows PowerShell or the command prompt: <pre>VMConnect.exe <ServerName> <VMName> /edit</pre>
Prevent a VMConnect user from taking over another user's VMConnect session	Not having enhanced session mode turned on might pose a security and privacy risk. If a user is connected and logged on to a virtual machine through VMConnect and another authorized user connects to the same virtual machine, the session is taken over by the second user and the first user loses the session. The second user will be able to view the first user's desktop, documents, and applications.
Manage integration services or components that allow the VM to communicate with the Hyper-V host	On Hyper-V hosts that run Windows 10 or Windows Server 2016, you can't manage integration services with VMConnect. See these articles: - Turn on/turn off integration services from the Hyper-V host - Turn on/turn off integration services from a Windows virtual machine - Turn on/turn off integration services from a Linux virtual machine - Keep integration services updated for the virtual machine For hosts that run Windows Server 2012 or Windows Server 2012 R2, see Integration Services.
Resize the VMConnect window	You can change the size of the VMConnect window for generation 2 virtual machines that run a Windows operating system. To do this, you might need to turn on enhanced session mode on the Hyper-V host. For virtual machines that run Ubuntu, see Changing Ubuntu Screen Resolution in a Hyper-V VM .

Keyboard shortcuts

By default, keyboard input and mouse clicks are sent to the virtual machine. So you might need to press CTRL + ALT + LEFT ARROW before you use the following shortcut keys.

Key combination	Description
CTRL+ALT+LEFT ARROW	Mouse release
CTRL+ALT+END	Equivalent of CTRL+ALT+DELETE in the virtual machine
CTRL+ALT+BREAK	Switch from full-screen mode back to windowed mode
CTRL+O	Opens the settings for the virtual machine
CTRL+S	Starts the virtual machine
CTRL+N	Creates a checkpoint
CTRL+E	Reverts to a checkpoint
CTRL+C	Does a screen capture

Related content

- [Hyper-V overview](#)
- [Use local resources on Hyper-V virtual machine with VMConnect](#)

Use local resources on Hyper-V virtual machine with VMConnect

07/25/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Virtual Machine Connection (VMConnect) lets you use a computer's local resources in a virtual machine, like a removable USB flash drive or a printer. Enhanced session mode also lets you resize the VMConnect window. This article shows you how to configure the host and then give the virtual machine access to a local resource.

Enhanced session mode and Type clipboard text are available only for virtual machines that run recent Windows operating systems. See [Requirements for using local resources](#).

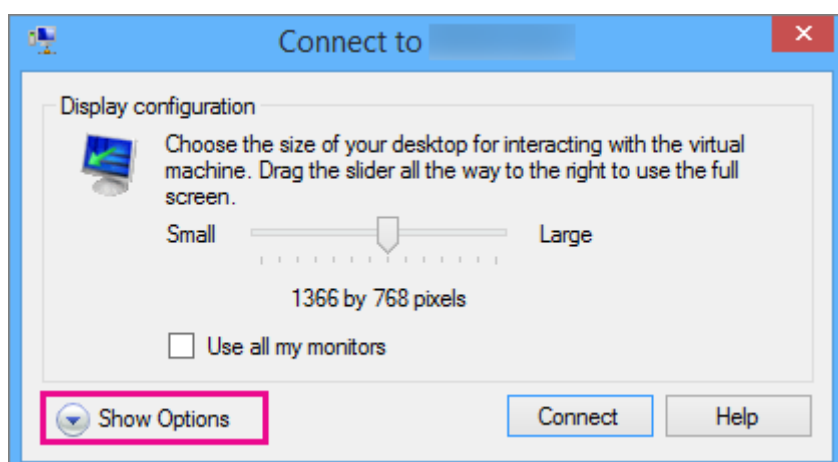
For virtual machines that run Ubuntu, see [Changing Ubuntu Screen Resolution in a Hyper-V VM](#).

Choose a local resource

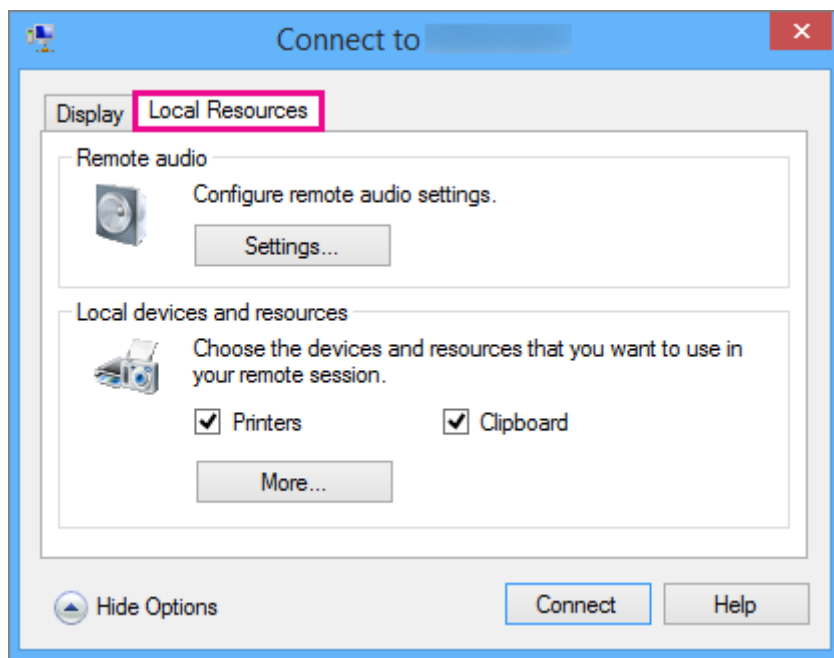
Local resources include printers, the clipboard, and local drive(s) on the computer where you're running VMConnect. For more information, see [Requirements for using local resources](#).

To choose a local resource:

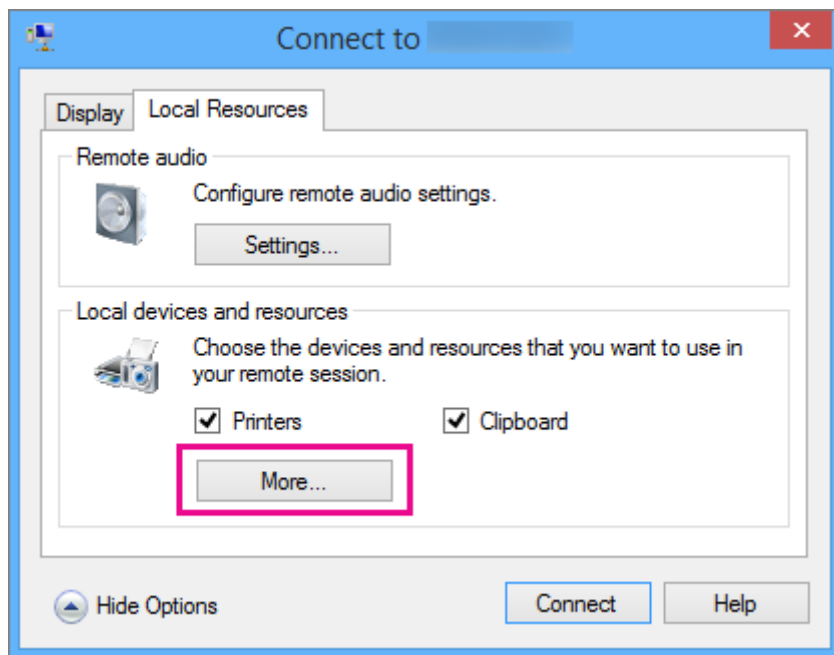
1. Open VMConnect.
2. Select the virtual machine that you want to connect to.
3. Select **Show options**.



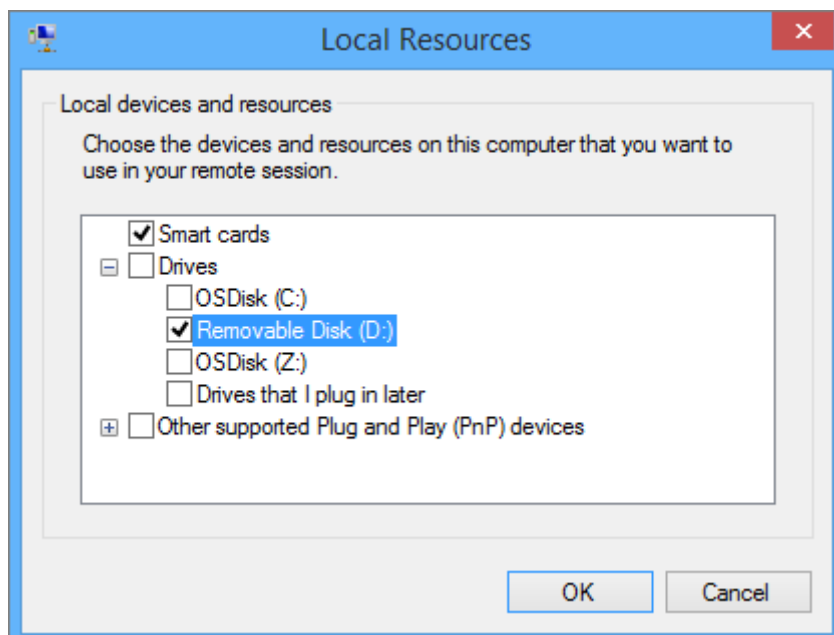
4. Select **Local resources**.



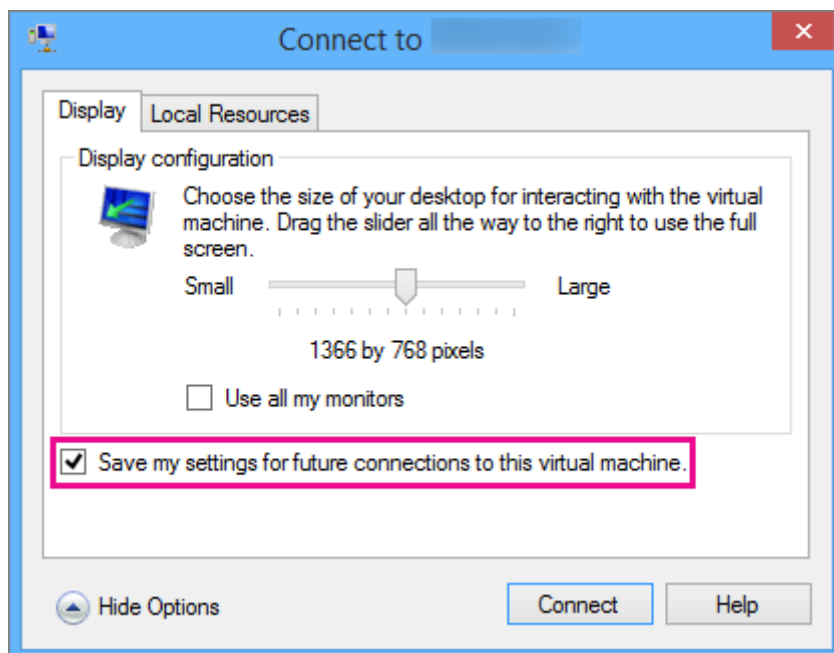
5. Select **More**.



6. Select the drive that you want to use on the virtual machine and click **Ok**.



7. Select **Save my settings for future connections to this virtual machine.**



8. Select **Connect**.

The path to the local drive shared to the virtual machine, in a Windows-based guest VM, is located at **This PC** under **Redirected drives and folders**. The path for a Linux-based guest VM is located at `/home/shared-drives`.

Edit VMConnect settings

You can easily edit your connection settings for VMConnect by running the following command in Windows PowerShell or the command prompt:

```
VMConnect.exe <ServerName> <VMName> /edit
```

ⓘ Note

An elevated command prompt may be required.

Requirements for using local resources

To be able to use a computer's local resources on a virtual machine:

- The Hyper-V host must have **Enhanced session mode policy** and **Enhanced session mode** settings turned on.
- The computer on which you use VMConnect must run at least Windows 10, or Windows Server 2016.
- The virtual machine must have Remote Desktop Services enabled, and run at least Windows 10, or Windows Server 2016.

If the computer running VMConnect and the virtual machine both meet the requirements, you can use any of the following local resources if they're available:

- Display configuration
- Audio
- Printers
- Clipboards for copy and paste
- Smart cards
- USB devices
- Drives
- Supported plug and play devices

Why use a computer's local resources?

You may want to use the computer's local resources to:

- Troubleshoot a virtual machine without a network connection to the virtual machine.
- Copy and paste files to and from the virtual machine in the same way you copy and paste using a Remote Desktop Connection (RDP).

- Sign in to the virtual machine by using a smart card.
- Print from a virtual machine to a local printer.
- Test and troubleshoot developer applications that require USB and sound redirection without using RDP.

See also

- [Hyper-V Virtual Machine Connection](#)
- [Should I create a generation 1 or 2 virtual machine in Hyper-V?](#)

Share devices with your Hyper-V virtual machine

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

When you use Enhanced Session Mode in the Virtual Machine Connection (VMConnect) tool, you can connect to Hyper-V virtual machines (VMs) by using Remote Desktop Protocol (RDP). Not only does this capability improve your general VM viewing experience, using RDP also provides a way for the VM and your computer to share devices. This functionality is only available for Windows VMs.

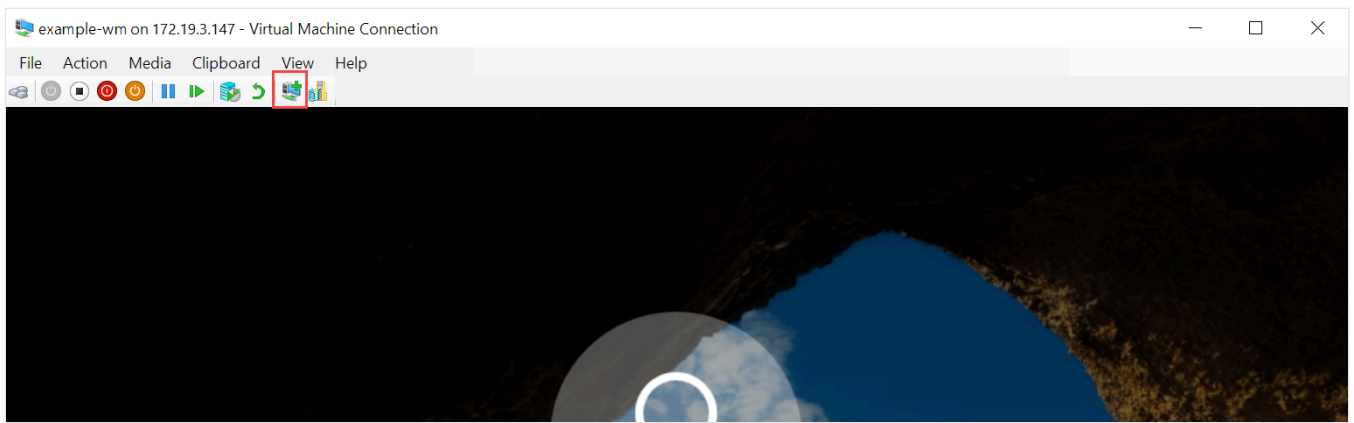
Using RDP in Enhanced Session Mode offers many benefits:

- Makes VMs resizable and high-DPI aware
- Improves VM integration:
 - Shared clipboard
 - File sharing via dragging or copying and pasting
- Makes device sharing possible:
 - Microphone and speakers
 - USB devices
 - Data disks, including drive C
 - Printers

Because RDP is turned on by default in Windows, you probably already use RDP to connect to your Windows VMs. This article highlights some of the benefits and hidden options in the connection settings dialog. It shows you how to see your session type, enter Enhanced Session Mode, and configure your session settings.

Check the session type

You can check your connection type by using the Enhanced Session Mode icon on the VMConnect toolbar. You can also use this button to switch between Basic Session Mode and Enhanced Session Mode.

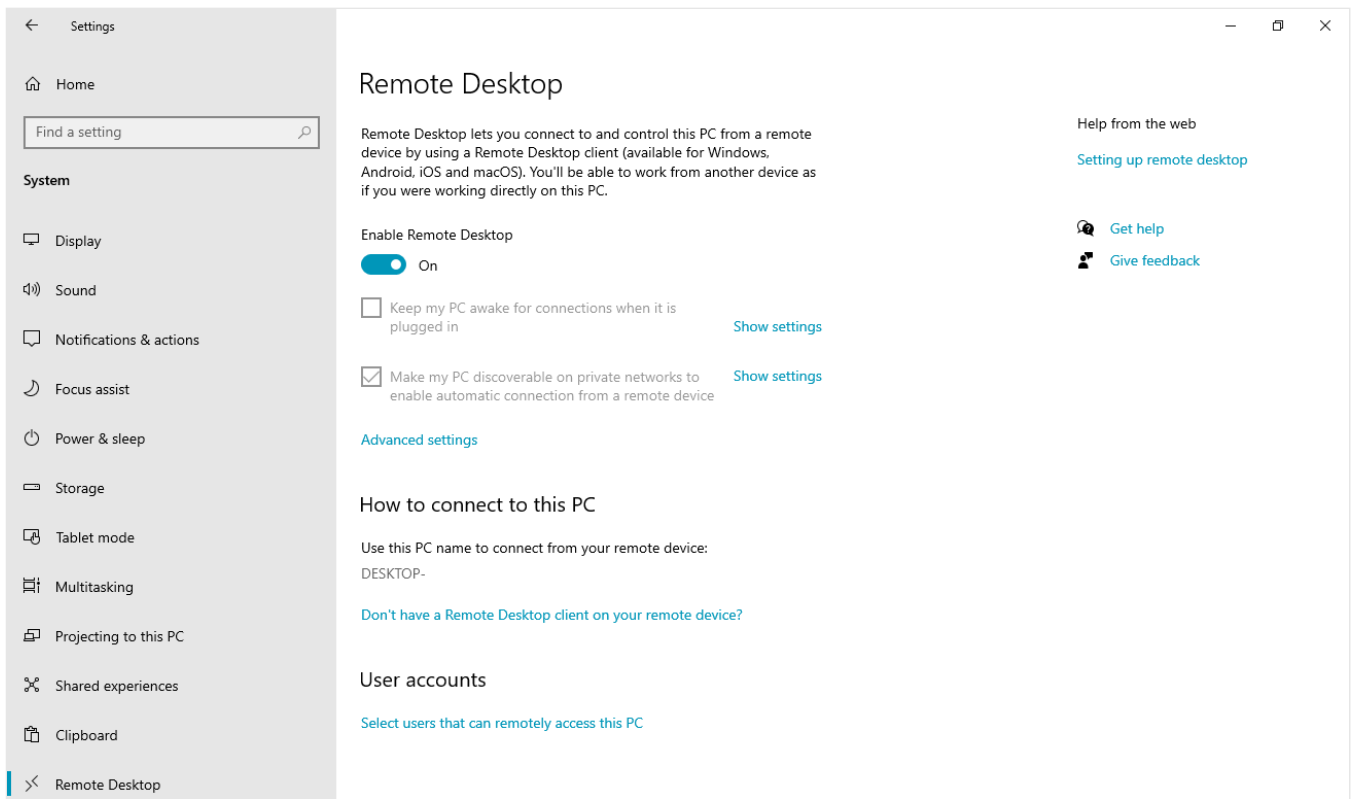


[Expand table](#)

Icon	Connection state
	You're currently running in Enhanced Session Mode. Selecting this icon reconnects you to your VM in Basic Mode.
	You're currently running in Basic Session Mode, but Enhanced Session Mode is available. Selecting this icon reconnects you to your VM in Enhanced Session Mode.
	You're currently running in Basic Session Mode. Enhanced Session Mode isn't available for this VM.

Configure a VM for Remote Desktop

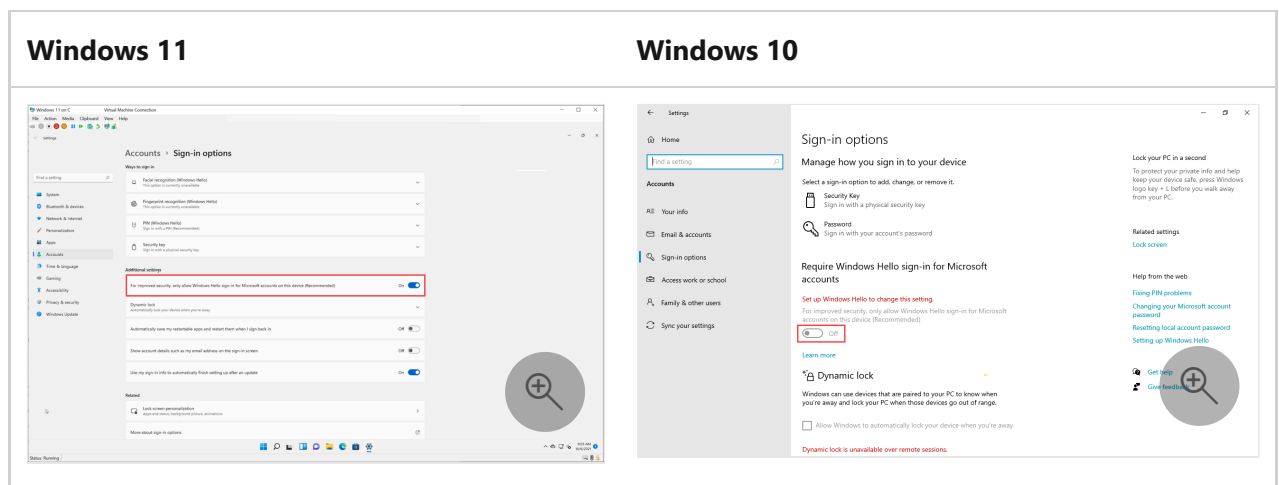
Enhanced Session Mode requires Remote Desktop to be turned on in the VM. In the Settings app or Start menu, search for **Remote Desktop settings**. Turn on the **Enable Remote Desktop** toggle.



Windows 10 (version 2004 and later) and Windows 11 require an extra setting. If the VMConnect window background doesn't display a sign-in prompt, disable Windows Hello sign-in:

1. Sign in to the VM in basic session mode.
2. Go to **Settings > Accounts > Sign-in options**.
3. Turn off **For improved security, only allow Windows Hello sign-in for Microsoft accounts on this device**.

 Expand table

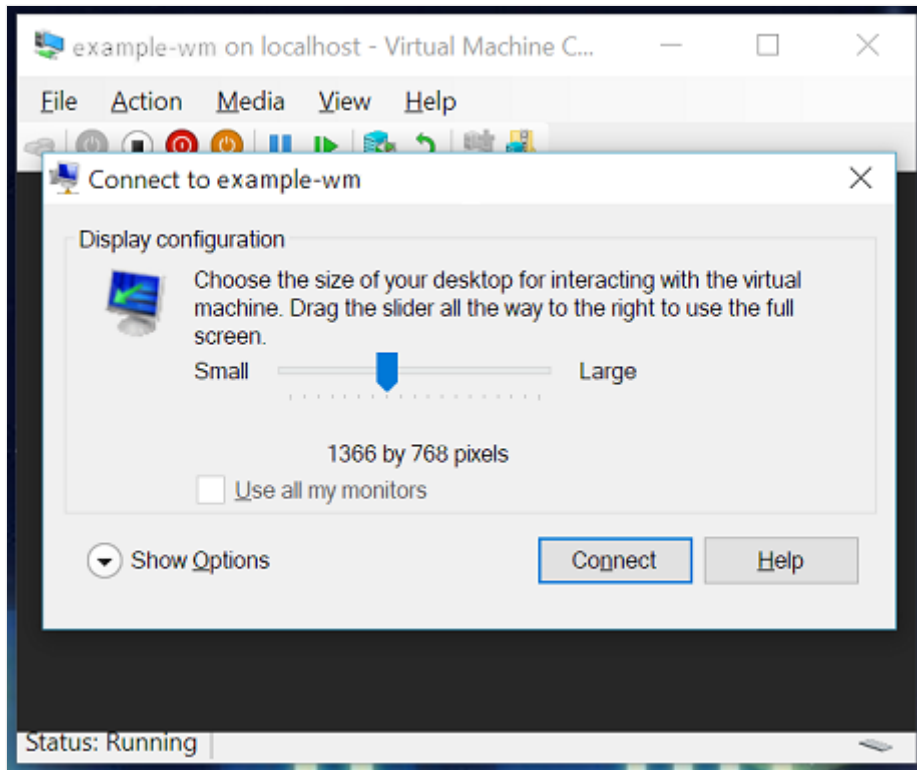


4. Sign out of the VM or reboot it before you close the VMConnect window.

To verify, check that the VMConnect window displays an Enhanced Session Mode sign-in prompt after reconnecting.

Share drives and devices

You can find the device sharing capabilities of Enhanced Session Mode in the connection dialog that opens when you connect to a VM:

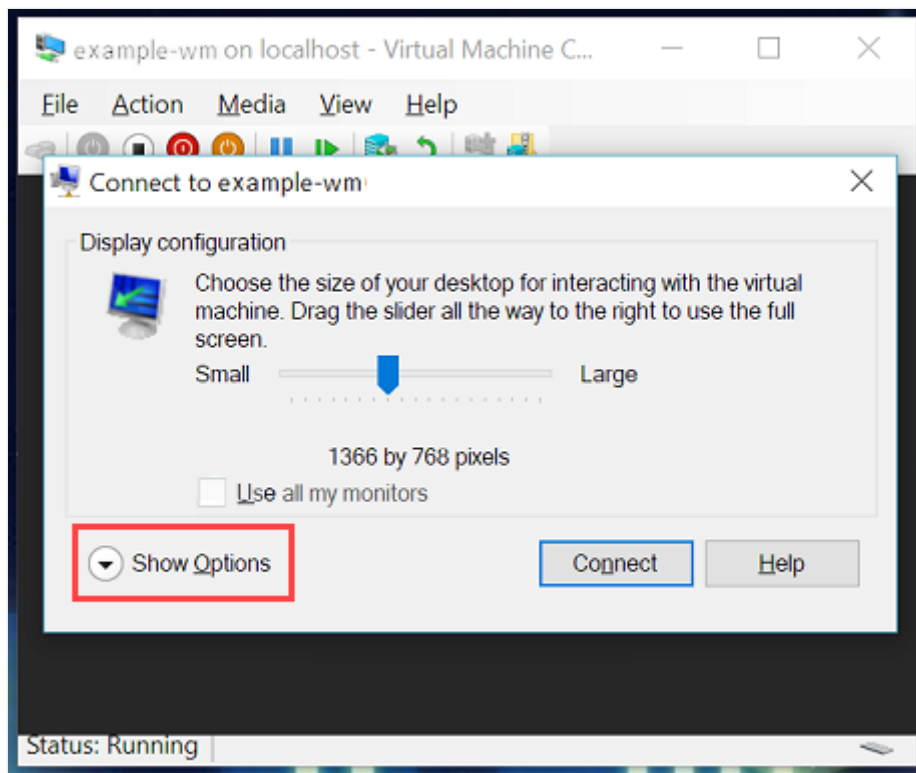


By default, Enhanced Session Mode provides VMs with access to the following resources:

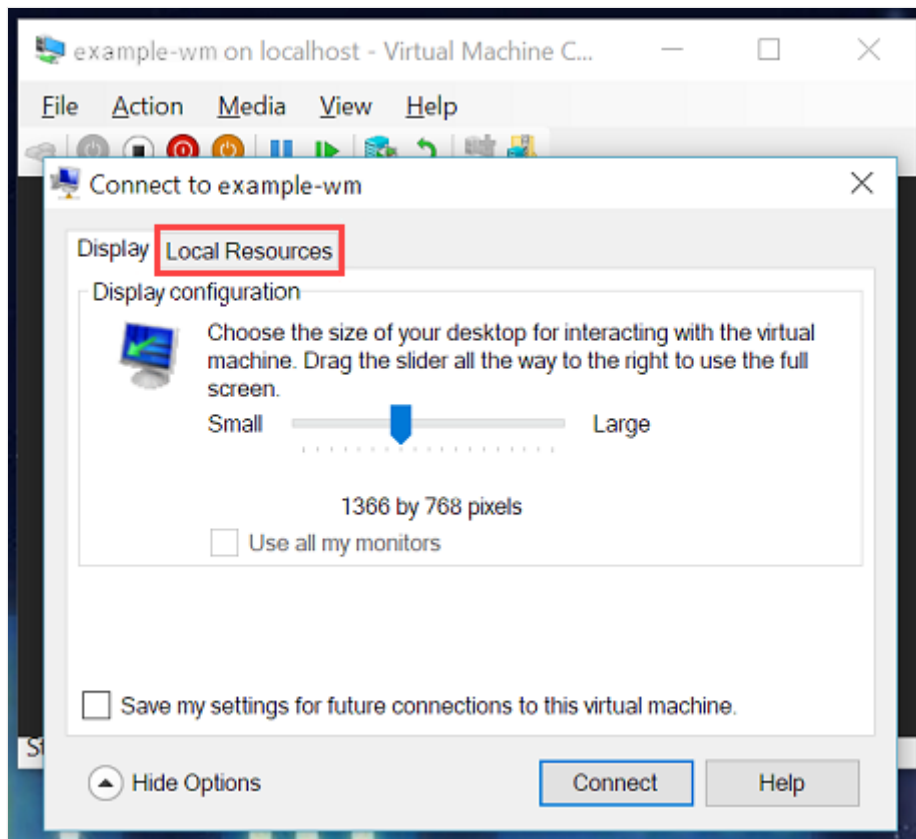
- Clipboard
- Printers
- Audio playback (to host speakers)

To share devices with your VM or to change the default settings:

1. In VMConnect, select the virtual machine that you want to connect to.
2. Select **Show Options**.



3. Select **Local Resources**, and then adjust the settings.



To verify, check that elected resources are available after connecting to the VM.

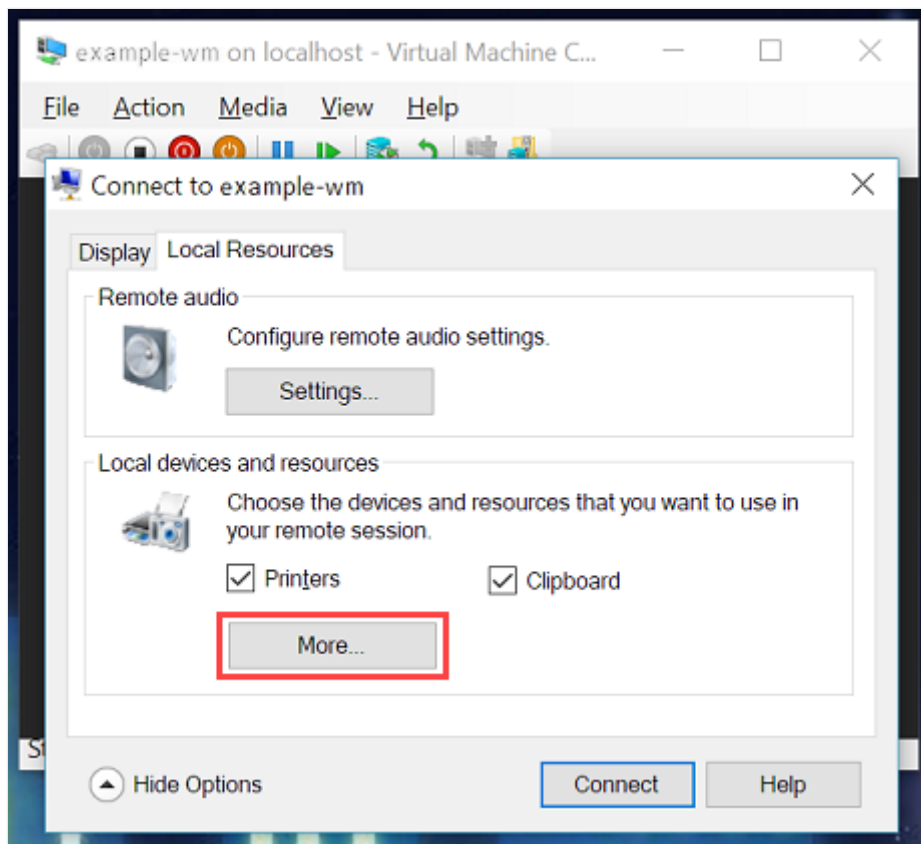
Share storage and USB devices

Enhanced Session Mode also supports sharing additional device types, such as:

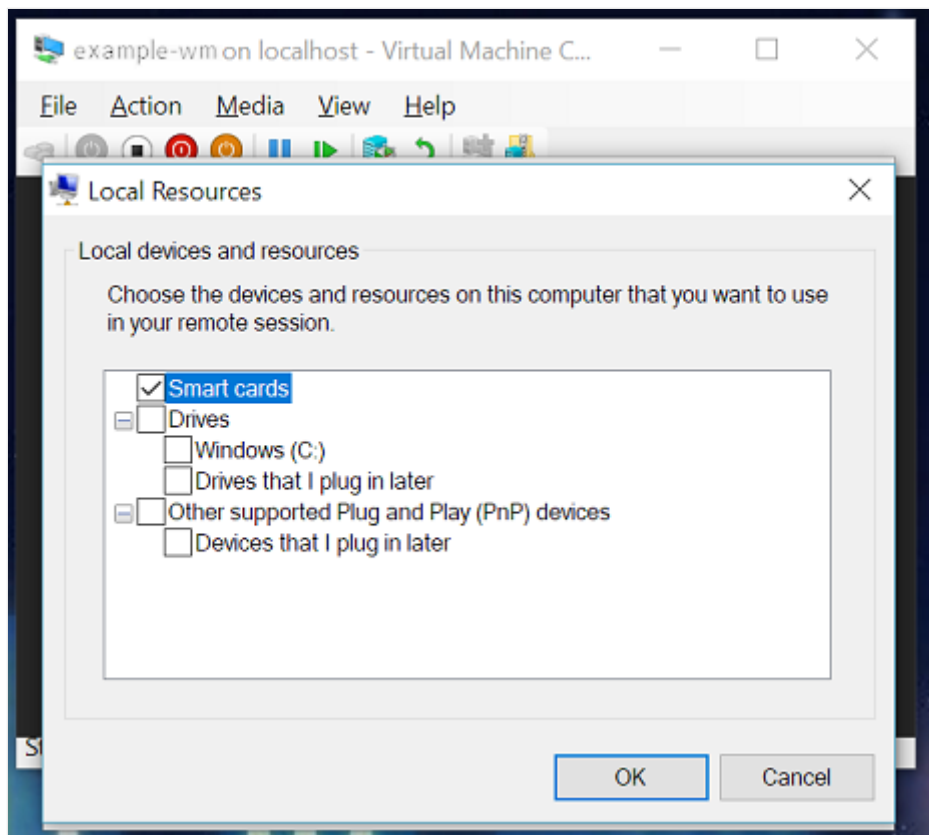
Device type	Examples
Storage	Local drives (including C:), removable drives
USB devices	Flash drives, external storage
Security devices	Smart cards, Windows Hello companion devices

To share these devices with a VM:

1. Under **Local devices and resources**, select **More**.



2. Select the devices that you want to share with the VM. The system drive, which is drive C on Windows, is especially helpful for file sharing.



To verify, check that shared drives appear under **This PC** in File Explorer within the VM.

Share audio devices (speakers and microphones)

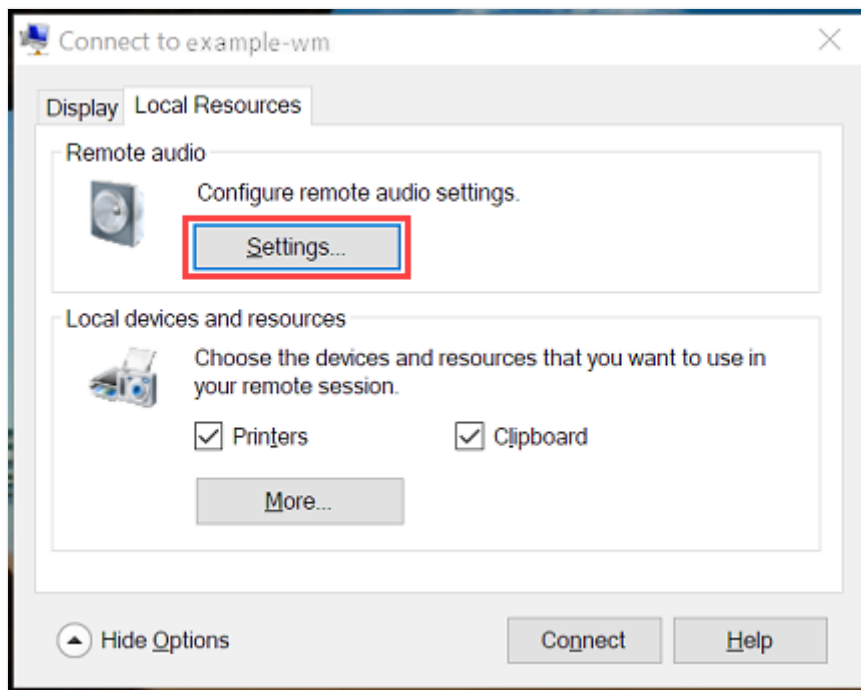
By default, VMs transfer audio to the host machine using the currently selected audio device. You can configure the following audio settings:

[Expand table](#)

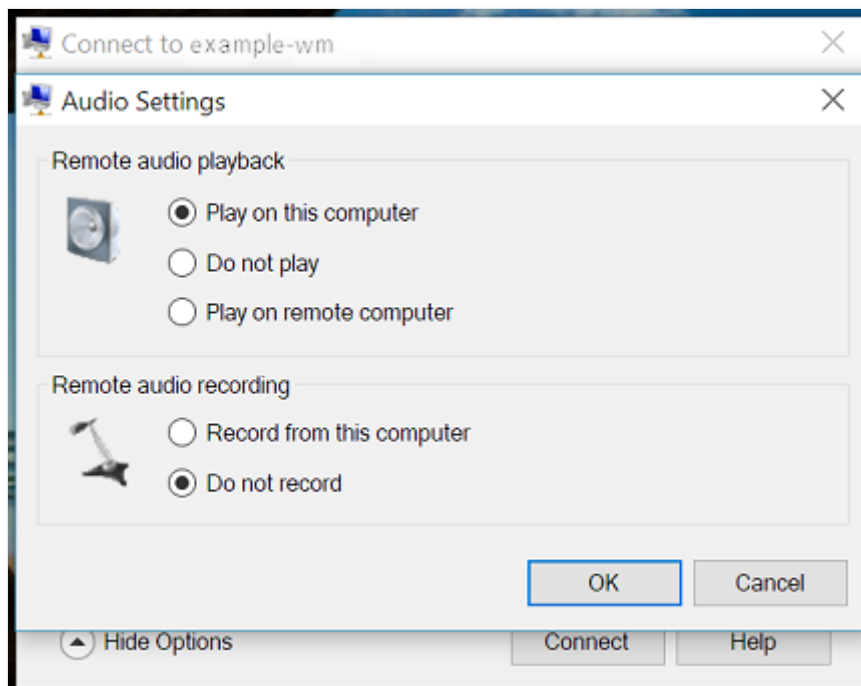
Setting	Options
Remote audio playback	Play on this computer, Play on remote computer, Do not play
Remote audio recording	Record from this computer (microphone pass-through), Do not record

To change audio settings:

1. Select **Settings** to configure remote audio settings.



2. Configure audio and microphone settings.



For local VMs, **Play on this computer** and **Play on remote computer** produce the same result.

To verify, check that audio plays through the configured device, and microphone input is detected if recording is enabled.

Reopen the connection settings

If you don't see the connection dialog or the **Show Options** button, try opening VMConnect independently from either the Windows menu or the command line as an administrator.

PowerShell

```
vmconnect.exe
```

VMConnect opens and displays the connection dialog with **Show Options** available.

Last updated on 01/21/2026

Virtual machine automation and management using PowerShell

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

You can use PowerShell Direct to run arbitrary PowerShell in a Windows 10 or greater, or Windows Server 2016 or newer, virtual machine from your Hyper-V host. Use PowerShell Direct regardless of network configuration or remote management settings.

Here are some ways you can run PowerShell Direct:

- [As an interactive session using the Enter-PSSession cmdlet](#)
- [As a single-use section to execute a single command or script using the Invoke-Command cmdlet](#)
- [As a persistent session \(build 14280 and later\) using the New-PSSession, Copy-Item, and Remove-PSSession cmdlets](#)

Requirements

Operating system requirements:

- Host: Windows 10, Windows Server 2016, or later running Hyper-V.
- Guest/Virtual Machine: Windows 10, Windows Server 2016, or later.

If you're managing older virtual machines, use Virtual Machine Connection (VMConnect) or connect over a network connection.

Configuration requirements:

- The virtual machine must run locally on the host.
- The virtual machine must be turned on and running with at least one configured user profile.
- You must be logged into the host computer as a Hyper-V administrator.
- You must supply valid user credentials for the virtual machine.

Create and exit an interactive PowerShell session

The easiest way to run PowerShell commands in a virtual machine is to start an interactive session.

When the session starts, the commands that you type run on the virtual machine, just as though you typed them directly into a PowerShell session on the virtual machine itself.

To start an interactive session:

1. On the Hyper-V host, open PowerShell as Administrator.
2. Run one of the following commands to create an interactive session using the virtual machine name or GUID:

PowerShell

```
Enter-PSSession -VMName <VMName>  
Enter-PSSession -VMId <VMId>
```

Enter credentials for the virtual machine when prompted.

3. Run commands on your virtual machine. You should see the VMName as the prefix for your PowerShell prompt as the following:

Output

```
[VMName]: PS C:\>
```

Any command you run executes on your virtual machine. To test, run `ipconfig` or `hostname` to make sure that these commands run in the virtual machine.

4. When you're done, run the following command to close the session:

PowerShell

```
Exit-PSSession
```

ⓘ Note

If your session won't connect, see the [troubleshooting](#) for potential causes.

To learn more about these cmdlets, see [Enter-PSSession](#) and [Exit-PSSession](#).

Run a script or command with Invoke-Command

PowerShell Direct with Invoke-Command is perfect for situations where you need to run one command or one script on a virtual machine but don't need to continue interacting with the virtual machine beyond that point.

To run a single command:

1. On the Hyper-V host, open PowerShell as Administrator.
2. Run one of the following commands to create a session using the virtual machine name or GUID:

PowerShell

```
Invoke-Command -VMName <VMName> -ScriptBlock { command }  
Invoke-Command -VMId <VMId> -ScriptBlock { command }
```

Enter credentials for the virtual machine when prompted.

The command runs on the virtual machine. If the command produces output, you see it in your console. The connection closes automatically as soon as the command runs.

To run a script:

1. On the Hyper-V host, open PowerShell as Administrator.
2. Run one of the following commands to create a session using the virtual machine name or GUID:

PowerShell

```
Invoke-Command -VMName <VMName> -FilePath C:\host\script_path\script.ps1  
Invoke-Command -VMId <VMId> -FilePath C:\host\script_path\script.ps1
```

Enter credentials for the virtual machine when prompted.

The script runs on the virtual machine. The connection closes automatically as soon as the command runs.

For more information about this cmdlet, see [Invoke-Command](#).

Copy files with New-PSSession and Copy-Item

ⓘ Note

PowerShell Direct only supports persistent sessions in Windows builds 14280 and later

Persistent PowerShell sessions are incredibly useful when writing scripts that coordinate actions across one or more remote machines. Once created, persistent sessions exist in the

background until you decide to delete them. This means you can reference the same session over and over again with `Invoke-Command` or `Enter-PSSession` without passing credentials.

By the same token, sessions hold state. Since persistent sessions persist, any variables you create in a session or pass to a session are preserved across multiple calls. You can use a number of tools to work with persistent sessions. For this example, use [New-PSSession](#) and [Copy-Item](#) to move data from the host to a virtual machine and from a virtual machine to the host.

To create a session then copy files:

1. On the Hyper-V host, open PowerShell as Administrator.
2. Run one of the following commands to create a persistent PowerShell session to the virtual machine by using `New-PSSession`.

PowerShell

```
$s = New-PSSession -VMName <VMName> -Credential (Get-Credential)
$s = New-PSSession -VMId <VMId> -Credential (Get-Credential)
```

Enter credentials for the virtual machine when prompted.

Warning

There's a bug in builds before 14500. If you don't explicitly specify credentials with the `-Credential` flag, the service in the guest crashes and needs to be restarted. If you encounter this issue, see the section [Error: A remote session might have ended](#) for workaround instructions.

3. Copy a file into the virtual machine.

To copy `C:\host_path\data.txt` to the virtual machine from the host machine, run:

PowerShell

```
Copy-Item -ToSession $s -Path C:\host_path\data.txt -Destination C:\guest_path\
```

4. Copy a file from the virtual machine (on to the host).

To copy `C:\guest_path\data.txt` to the host from the virtual machine, run:

PowerShell

```
Copy-Item -FromSession $s -Path C:\guest_path\data.txt -Destination  
C:\host_path\
```

5. Stop the persistent session by using `Remove-PSSession`.

```
PowerShell
```

```
Remove-PSSession $s
```

Troubleshooting

PowerShell Direct surfaces a small set of common error messages. The following sections describe the most common error messages, some causes, and tools for diagnosing issues.

-VMName or -VMID parameters don't exist

Problem:

`Enter-PSSession`, `Invoke-Command`, or `New-PSSession` don't have a `-VMName` or `-VMId` parameter.

Potential causes:

The most likely issue is that your host operating system doesn't support PowerShell Direct.

You can check your Windows build by running the following command:

```
PowerShell
```

```
[System.Environment]::OSVersion.Version
```

If you're running a supported build, it's also possible that your version of PowerShell doesn't support PowerShell Direct. For PowerShell Direct and JEA, the major version must be 5 or later.

You can check your PowerShell version build by running the following command:

```
PowerShell
```

```
$PSVersionTable.PSVersion
```

Error: A remote session might have ended

ⓘ Note

For Enter-PSSession between host builds 10240 and 12400, all errors report as "A remote session might have ended".

Error message:

Enter-PSSession : An error occurred that Windows PowerShell can't handle. A remote session might have ended.

Potential causes:

- The virtual machine exists but isn't running.
- The guest OS doesn't support PowerShell Direct. See [requirements](#).
- PowerShell isn't available in the guest yet
 - The operating system hasn't finished booting
 - The operating system can't boot correctly
 - Some boot time event needs user input

Use the [Get-VM](#) cmdlet to check which VMs are running on the host.

Error message:

New-PSSession : An error occurred that Windows PowerShell can't handle. A remote session might have ended.

Potential causes:

- One of the reasons listed earlier -- they all apply to `New-PSSession`
- A bug in current builds where you must explicitly pass credentials with `-Credential`.
When this bug occurs, the entire service hangs in the guest operating system and needs to be restarted. You can check if the session is still available with Enter-PSSession.

To work around the credential issue, sign in to the virtual machine using VMConnect, open PowerShell, and restart the vmicvmsession service with the following PowerShell command:

PowerShell

```
Restart-Service -Name vmicvmsession
```

Error: Parameter set can't be resolved

Error message:

Enter-PSSession : Parameter set can't be resolved by using the specified named parameters.

Potential causes:

- You can't use `-RunAsAdministrator` when connecting to virtual machines.

When connecting to a Windows container, the `-RunAsAdministrator` flag allows Administrator connections without explicit credentials. Since virtual machines don't give the host implied administrator access, you need to explicitly enter credentials.

You can pass administrator credentials to the virtual machine with the `-Credential` parameter or by entering them manually when prompted.

Error: The credential is invalid

Error message:

Enter-PSSession : The credential is invalid.

Potential causes:

- The guest credentials couldn't be validated
 - The supplied credentials were incorrect.
 - There are no user accounts in the guest (the OS didn't boot before)
 - If connecting as Administrator: Administrator isn't set as an active user. For more information, see [Enable and Disable the Built-in Administrator Account](#).

Error: The input VMName parameter doesn't resolve to any virtual machine.

Error message:

Enter-PSSession : The input VMName parameter doesn't resolve to any virtual machine.

Potential causes:

- You're not a Hyper-V Administrator.
- The virtual machine doesn't exist.

Use the [Get-VM](#) cmdlet to check that the credentials you're using have the Hyper-V administrator role and to see which VMs are running locally on the host and booted.

Samples and user guides

PowerShell Direct supports Just Enough Administration (JEA).

Check out samples on [GitHub](#) .

Last updated on 10/28/2025

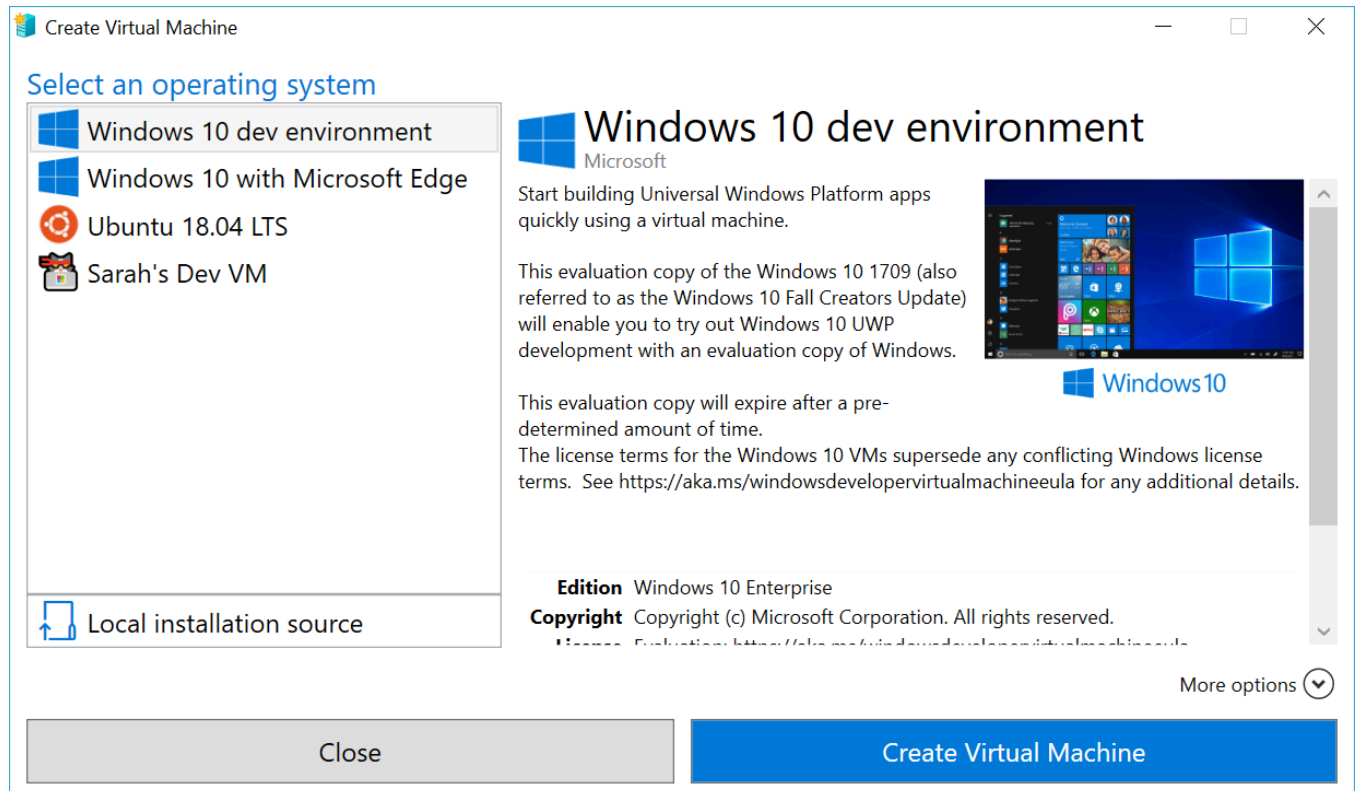
Create a custom virtual machine gallery

08/14/2025

Applies to:  Windows Server 2025,  Windows Server 2022,  Windows Server 2019,  Windows Server 2016,  Windows 11,  Windows 10,  Azure Local 2311.2 and later

Windows 10 Fall Creators Update and later.

In Fall Creators Update, Quick Create expanded to include a virtual machine gallery.



While there is a set of images provided by Microsoft and Microsoft partners, the gallery can also list your own images.

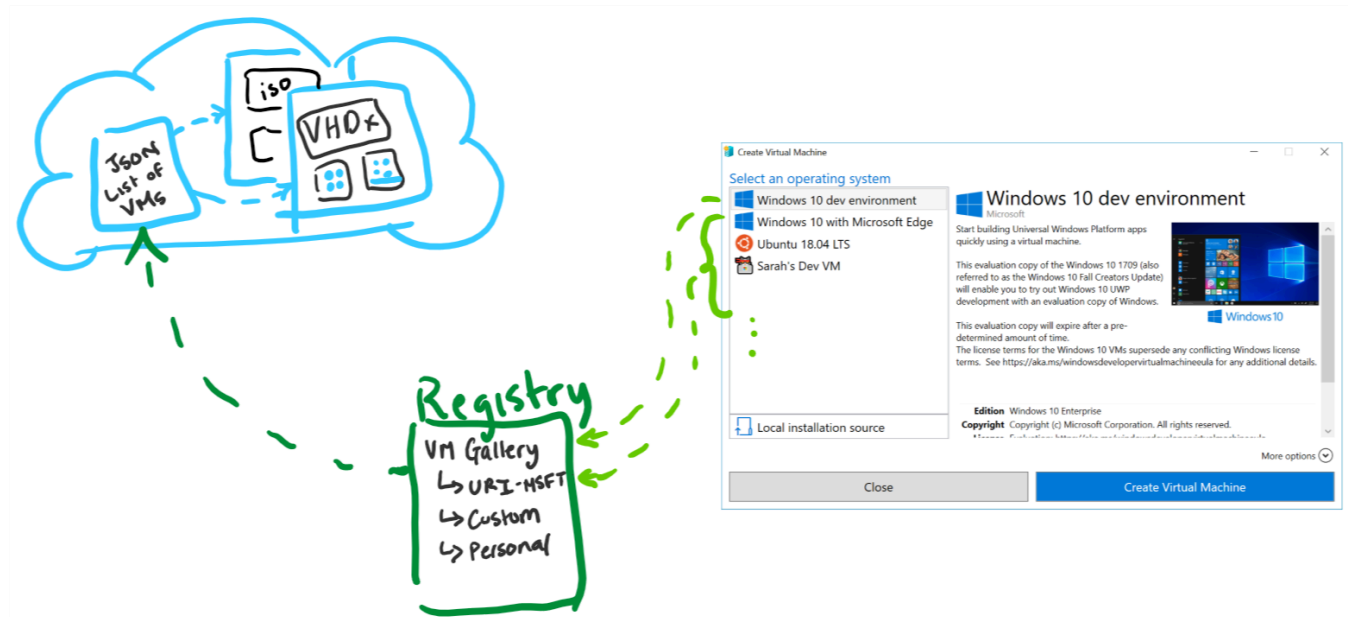
This article details:

- building virtual machines that are compatible with the gallery.
- creating a new gallery source.
- adding your custom gallery source to the gallery.

Gallery architecture

The virtual machine gallery is a graphical view for a set of virtual machine sources defined in the Windows registry. Each virtual machine source is a path (local path or URI) to a JSON file with virtual machines as list items.

The list of virtual machines you see in the gallery is the full contents of the first source, followed by the contents of the second source, so on and so forth until all of the available virtual machines have been listed. The list is dynamically created every time you launch the gallery.



Registry Key: `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Virtualization`

Value Name: `GalleryLocations`

Type: `REG_MULTI_SZ`

Create gallery-compatible virtual machines

Virtual machines in the gallery can be either a disk image (.iso) or virtual hard drive (.vhdx).

Virtual machines made from a virtual hard drive have a few configuration requirements:

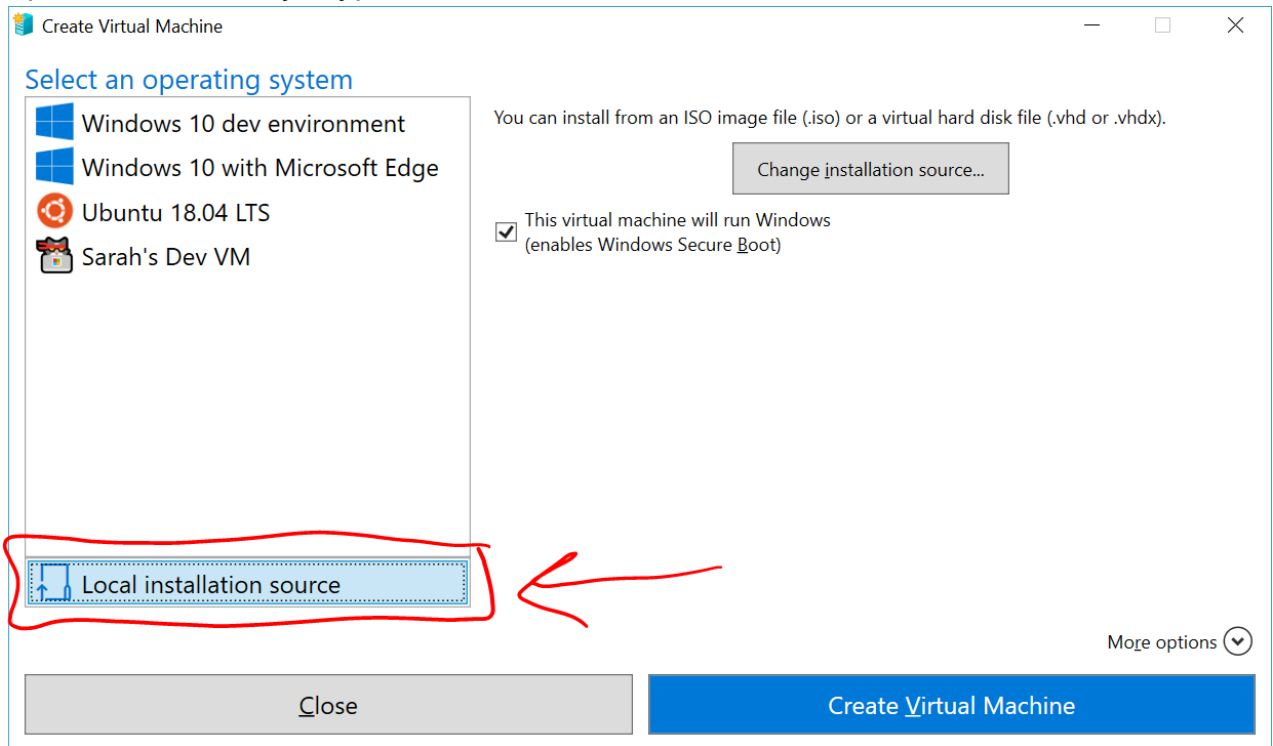
1. Built to support UEFI firmware. If they're created using Hyper-V, that's a Generation 2 VM.
2. The virtual hard drive should be at least 20GB - keep in mind, that's the max size. Hyper-V will not take space the VM isn't actively using.

Testing a new VM image

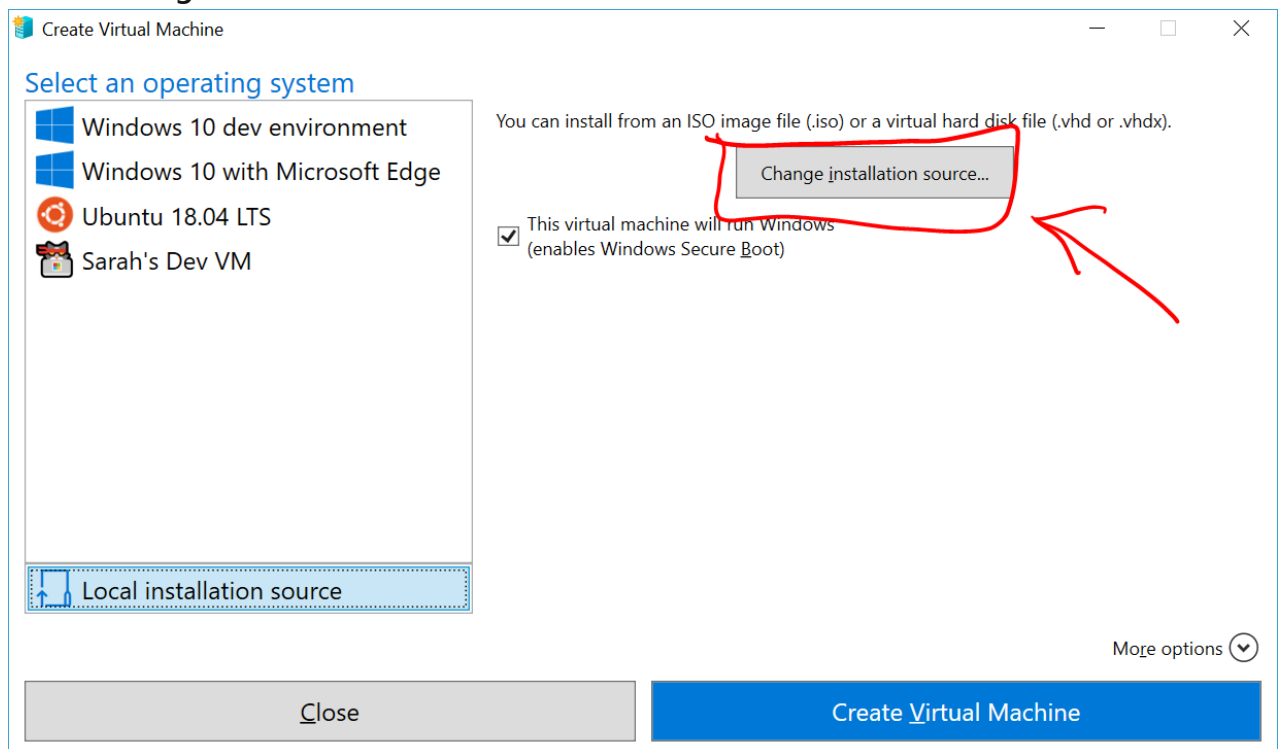
The virtual machine gallery creates virtual machines using the same mechanism as installing from a local installation source.

To validate a virtual machine image will boot and run:

1. Open the VM Gallery (Hyper-V Quick Create) and select **Local Installation Source**.

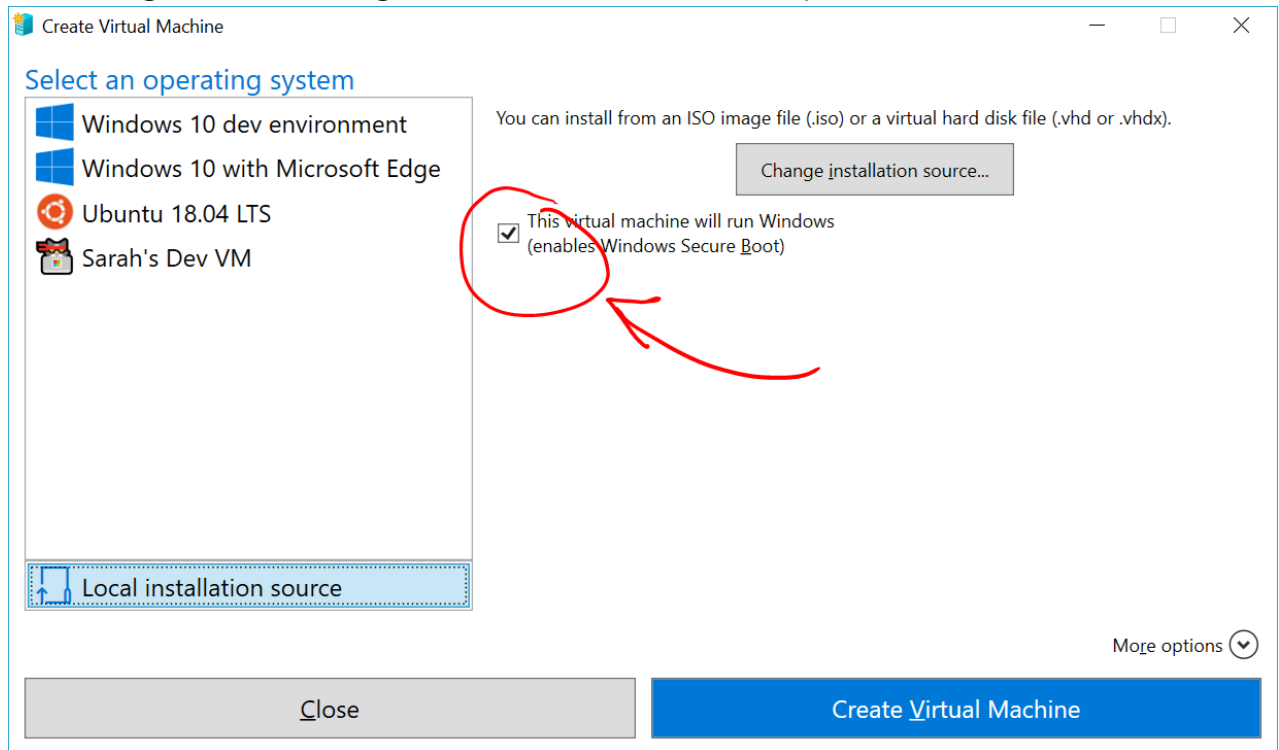


2. Select **Change Installation Source**.



3. Pick the .iso or .vhdx that will be used in the gallery.

4. If the image is a Linux image, deselect the Secure Boot option.

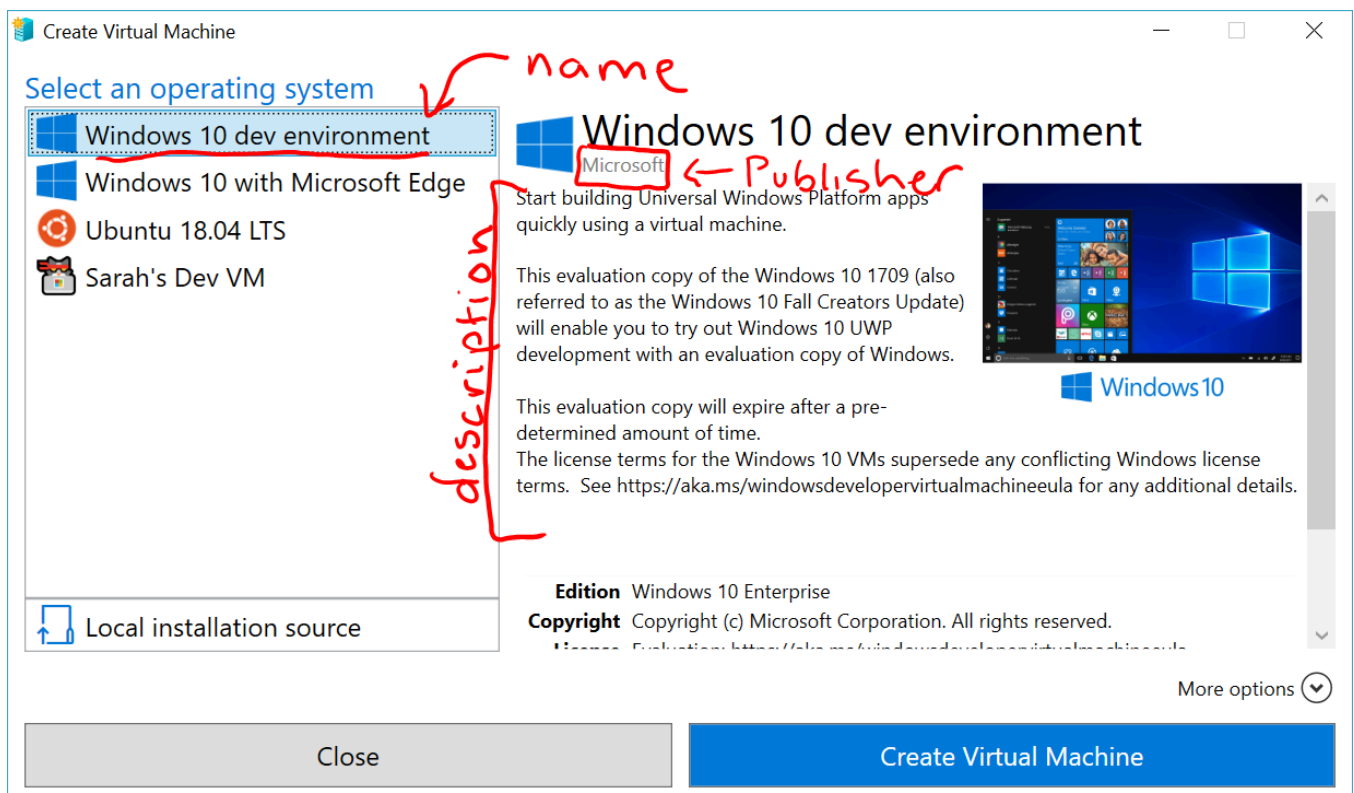


5. Create Virtual Machine. If the virtual machine boots correctly, it's ready for the gallery.

Build a new gallery source

The next step is to create a new gallery source. This is the JSON file that lists your virtual machines and adds all of the extra information you see in the gallery.

Text information:



- **name** - required - this is the name that appears in the left column and also at the top of the virtual machine view.
- **publisher** - required
- **description** - required - List of strings that describe the VM.
- **version** - required
- **lastUpdated** - defaults to Monday, January 1, 0001.

The format should be: yyyy-mm-ddThh:mm:ssZ

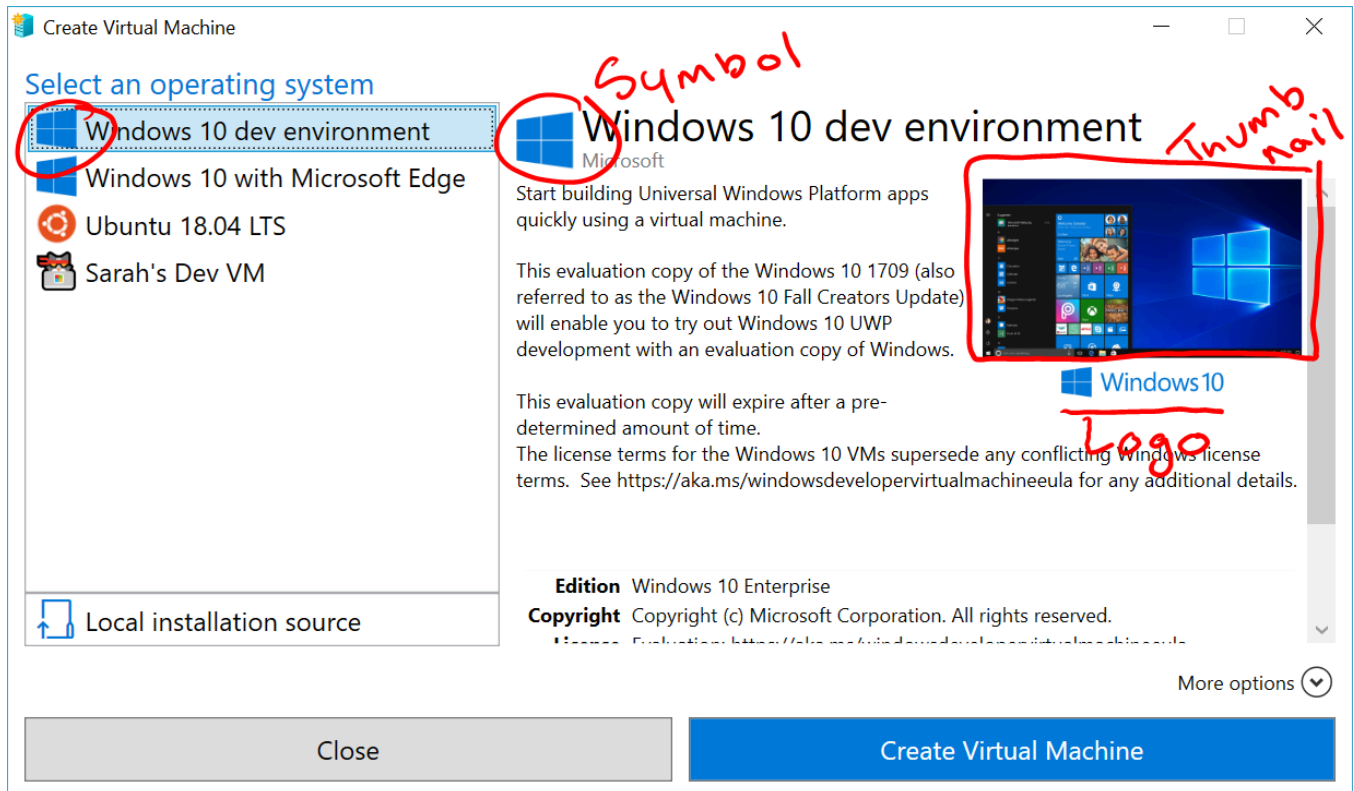
The following PowerShell command will provide today's date in the proper format and put it on the clipboard:

PowerShell

```
Get-Date -UFormat "%Y-%m-%dT%TZ" | clip.exe
```

- **locale** - defaults to blank.

Pictures:



- **logo** - required
- **symbol**
- **thumbnail**

And, of course, your virtual machine (.iso or .vhdx).

To generate the hashes, you can use the following powershell command:

PowerShell

```
Get-FileHash -Path .\TMLogo.jpg -Algorithm SHA256
```

The below JSON template has starter items and the gallery's schema. If you edit it in VSCode, it will automatically provide IntelliSense.

JSON

```
{
  "$schema": "https://raw.githubusercontent.com/MicrosoftDocs/Virtualization-Documentation/live/hyperv-tools/vmgallery/vm-gallery-schema.json",
  "images": [
    {
      "name": "",
      "version": "",
      "locale": "",
      "publisher": "",
      "lastUpdated": "",
      "description": [
        ""
      ],
      "disk": {
        "uri": "",
        "hash": ""
      },
      "logo": {
        "uri": "",
        "hash": ""
      },
      "symbol": {
        "uri": "",
        "hash": ""
      },
      "thumbnail": {
        "uri": "",
        "hash": ""
      }
    }
  ]
}
```

Connect your gallery to the VM Gallery UI

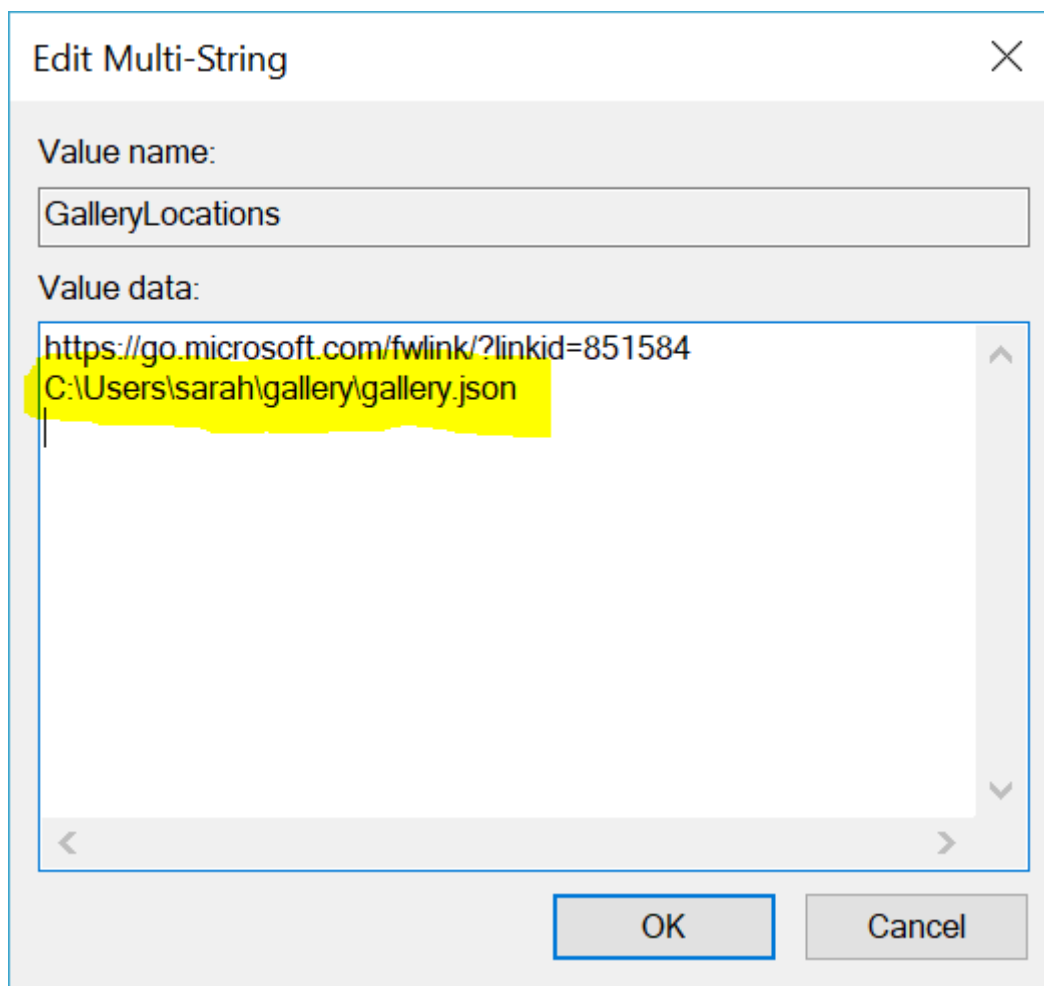
The easiest way to add your custom gallery source to the VM Gallery is to add it in regedit.

1. Open **regedit.exe**
2. Navigate to `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Virtualization\`
3. Look for the `GalleryLocations` item.

If it already exists, go to the **Edit** menu and **modify**.

If it doesn't already exist, go to the **Edit** menu, navigate through **New** to **Multi-String Value**

4. Add your gallery to the `GalleryLocations` registry key.



Troubleshooting

Check for errors loading Gallery

The virtual machine gallery does provide error reporting in the Windows Event Viewer. To check for errors:

1. Open Event Viewer
2. Navigate to **Windows Logs** -> **Application**
3. Look for events from Source VMCreate.

Resources

There are a handfull of gallery scripts and helpers in GitHub [link](#)↗.

See a sample gallery entry [here](#)↗. This is the JSON file that defines the in-box gallery.

Hyper-V Module

This reference provides cmdlet descriptions and syntax for all Hyper-V-specific cmdlets. It lists the cmdlets in alphabetical order based on the verb at the beginning of the cmdlet.

Hyper-V Cmdlets

 Expand table

Cmdlet	Description
Add-VMAssignableDevice	Adds an assignable device to a specific virtual machine.
Add-VMDVDDrive	Adds a DVD drive to a virtual machine.
Add-VMFibreChannelHba	Adds a virtual Fibre Channel host bus adapter to a virtual machine.
Add-VMGPUPartitionAdapter	Adds a GPU partition adapter to a virtual machine.
Add-VMGroupMember	Adds group members to a virtual machine group.
Add-VMHardDiskDrive	Adds a hard disk drive to a virtual machine.
Add-VMHostAssignableDevice	Adds an assignable device to a virtual machine (VM) host.
Add-VMMigrationNetwork	Adds a network for virtual machine migration on one or more virtual machine hosts.
Add-VMNetworkAdapter	Adds a virtual network adapter to a virtual machine.
Add-VMNetworkAdapterAcl	Creates an ACL to apply to the traffic through a virtual machine network adapter.
Add-VMNetworkAdapterExtendedAcl	Creates an extended ACL for a virtual network adapter.
Add-VMNetworkAdapterRoutingDomainMapping	Adds a routing domain and virtual subnets to a virtual network adapter.
Add-VMRemoteFx3dVideoAdapter	Adds a RemoteFX video adapter in a virtual machine.
Add-VMScsiController	Adds a SCSI controller in a virtual machine.
Add-VMStoragePath	Adds a path to a storage resource pool.
Add-VMSwitch	Adds a virtual switch to an Ethernet resource pool.

Cmdlet	Description
Add-VMSwitchExtensionPortFeature	Adds a feature to a virtual network adapter.
Add-VMSwitchExtensionSwitchFeature	Adds a feature to a virtual switch.
Add-VMSwitchTeamMember	Adds members to a virtual switch team.
Checkpoint-VM	Creates a checkpoint of a virtual machine.
Compare-VM	Compares a virtual machine and a virtual machine host for compatibility, returning a compatibility report.
Complete-VMFailover	Completes a virtual machine's failover process on the Replica server. Removes all recovery points on a failed over virtual machine.
Connect-VMNetworkAdapter	Connects a virtual network adapter to a virtual switch.
Connect-VMSan	Associates a host bus adapter with a virtual storage area network (SAN).
Convert-VHD	Converts the format, version type, and block size of a virtual hard disk file.
Copy-VMFile	Copies a file to a virtual machine.
Debug-VM	Debugs a virtual machine.
Disable-VMConsoleSupport	Disables keyboard, video, and mouse for a generation 2 virtual machine.
Disable-VMEventing	Disables virtual machine eventing.
Disable-VMIntegrationService	Disables an integration service on a virtual machine.
Disable-VMMigration	Disables migration on one or more virtual machine hosts.
Disable-VMRemoteFXPhysicalVideoAdapter	Disables one or more RemoteFX physical video adapters from use with RemoteFX-enabled virtual machines.
Disable-VMResourceMetering	Disables collection of resource utilization data for a virtual machine or resource pool.
Disable-VMSwitchExtension	Disables one or more extensions on one or more virtual switches.
Disable-VMTPM	Disables TPM functionality on a virtual machine.
Disconnect-VMNetworkAdapter	Disconnects a virtual network adapter from a virtual switch or Ethernet resource pool.

Cmdlet	Description
Disconnect-VMsSan	Removes a host bus adapter from a virtual storage area network (SAN).
Dismount-VHD	Dismounts a virtual hard disk.
Dismount-VMHostAssignableDevice	Dismounts a device from a virtual machine (VM) host.
Enable-VMConsoleSupport	Enables keyboard, video, and mouse for virtual machines.
Enable-VMEventing	Enables virtual machine eventing.
Enable-VMIntegrationService	Enables an integration service on a virtual machine.
Enable-VMMigration	Enables migration on one or more virtual machine hosts.
Enable-VMRemoteFXPhysicalVideoAdapter	Enables one or more RemoteFX physical video adapters for use with RemoteFX-enabled virtual machines.
Enable-VMReplication	Enables replication of a virtual machine.
Enable-VMResourceMetering	Collects resource utilization data for a virtual machine or resource pool.
Enable-VMSwitchExtension	Enables one or more extensions on one or more switches.
Enable-VMTPM	Enables TPM functionality on a virtual machine.
Export-VM	Exports a virtual machine to disk.
Export-VMSnapshot	Exports a virtual machine checkpoint to disk.
Get-VHD	Gets the virtual hard disk object associated with a virtual hard disk.
Get-VHDSets	Gets information about a VHD set.
Get-VHDSnapshot	Gets information about a checkpoint in a VHD set.
Get-VM	Gets the virtual machines from one or more Hyper-V hosts.
Get-VMAssignableDevice	Retrieves information about the assignable device from a specific virtual machine.
Get-VMbios	Gets the BIOS of a virtual machine or snapshot.
Get-VMComPort	Gets the COM ports of a virtual machine or snapshot.
Get-VMConnectAccess	Gets entries showing users and the virtual machines to which they can connect on one or more Hyper-V hosts.

Cmdlet	Description
Get-VMVhdDrive	Gets the VHD drives attached to a virtual machine or snapshot.
Get-VMFibreChannelHba	Gets the Fibre Channel host bus adapters associated with one or more virtual machines.
Get-VMFirmware	Gets the firmware configuration of a virtual machine.
Get-VMFloppyDiskDrive	Gets the floppy disk drives of a virtual machine or snapshot.
Get-VMGpuPartitionAdapter	Gets the information of assigned GPU partitions to a virtual machine.
Get-VMGroup	Gets virtual machine groups.
Get-VMHardDiskDrive	Gets the virtual hard disk drives attached to one or more virtual machines.
Get-VMHost	Gets a Hyper-V host.
Get-VMHostAssignableDevice	Retrieves device information assigned to a virtual machine (VM) host.
Get-VMHostCluster	Gets virtual machine host clusters.
Get-VMHostNumaNode	Gets the NUMA topology of a virtual machine host.
Get-VMHostNumaNodeStatus	Gets the status of the virtual machines on the non-uniform memory access (NUMA) nodes of a virtual machine host or hosts.
Get-VMHostPartitionableGpu	Gets the host machine's partitionable GPU.
Get-VMHostSupportedVersion	Returns a list of virtual machine configuration versions that are supported on a host.
Get-VMIdeController	Gets the IDE controllers of a virtual machine or snapshot.
Get-VMIntegrationService	Gets the integration services of a virtual machine or snapshot.
Get-VMKeyProtector	Retrieves a key protector for a virtual machine.
Get-VMMemory	Gets the memory of a virtual machine or snapshot.
Get-VMMigrationNetwork	Gets the networks added for migration to one or more virtual machine hosts.
Get-VMNetworkAdapter	Gets the virtual network adapters of a virtual machine, snapshot, management operating system, or of a virtual

Cmdlet	Description
	machine and management operating system.
Get-VMNetworkAdapterAcl	Gets the ACLs configured for a virtual machine network adapter.
Get-VMNetworkAdapterExtendedAcl	Gets extended ACLs configured for a virtual network adapter.
Get-VMNetworkAdapterFailoverConfiguration	Gets the IP address of a virtual network adapter configured to be used when a virtual machine fails over.
Get-VMNetworkAdapterIsolation	Gets isolation settings for a virtual network adapter.
Get-VMNetworkAdapterRoutingDomainMapping	Gets members of a routing domain.
Get-VMNetworkAdapterTeamMapping	
Get-VMNetworkAdapterVlan	Gets the virtual LAN settings configured on a virtual network adapter.
Get-VMProcessor	Gets the processor of a virtual machine or snapshot.
Get-VMRemoteFx3dVideoAdapter	Gets the RemoteFX video adapter of a virtual machine or snapshot.
Get-VMRemoteFXPhysicalVideoAdapter	Gets the RemoteFX physical graphics adapters on one or more Hyper-V hosts.
Get-VMReplication	Gets the replication settings for a virtual machine.
Get-VMReplicationAuthorizationEntry	Gets the authorization entries of a Replica server.
Get-VMReplicationServer	Gets the replication and authentication settings of a Replica server.
Get-VMResourcePool	Gets the resource pools on one or more virtual machine hosts.
Get-VMSan	Gets the available virtual machine storage area networks on a Hyper-V host or hosts.
Get-VMScsiController	Gets the SCSI controllers of a virtual machine or snapshot.
Get-VMSecurity	Gets security information about a virtual machine.
Get-VMSnapshot	Gets the checkpoints associated with a virtual machine or checkpoint.
Get-VMStoragePath	Gets the storage paths in a storage resource pool.

Cmdlet	Description
Get-VMSwitch	Gets virtual switches from one or more virtual Hyper-V hosts.
Get-VMSwitchExtension	Gets the extensions on one or more virtual switches.
Get-VMSwitchExtensionPortData	Retrieves the status of a virtual switch extension feature applied to a virtual network adapter.
Get-VMSwitchExtensionPortFeature	Gets the features configured on a virtual network adapter.
Get-VMSwitchExtensionSwitchData	Gets the status of a virtual switch extension feature applied on a virtual switch.
Get-VMSwitchExtensionSwitchFeature	Gets the features configured on a virtual switch.
Get-VMSwitchTeam	Gets virtual switch teams from Hyper-V hosts.
Get-VMSystemSwitchExtension	Gets the switch extensions installed on a virtual machine host.
Get-VMSystemSwitchExtensionPortFeature	Gets the port-level features supported by virtual switch extensions on one or more Hyper-V hosts.
Get-VMSystemSwitchExtensionSwitchFeature	Gets the switch-level features on one or more Hyper-V hosts.
Get-VMVideo	Gets video settings for virtual machines.
Grant-VMConnectAccess	Grants a user or users access to connect to a virtual machine or machines.
Import-VM	Imports a virtual machine from a file.
Import-VMInitialReplication	Imports initial replication files for a Replica virtual machine to complete the initial replication when using external media as the source.
Measure-VM	Reports resource utilization data for one or more virtual machines.
Measure-VMReplication	Gets replication statistics and information associated with a virtual machine.
Measure-VMResourcePool	Reports resource utilization data for one or more resource pools.
Merge-VHD	Merges virtual hard disks.
Mount-VHD	Mounts one or more virtual hard disks.

Cmdlet	Description
Mount-VMHostAssignableDevice	Mounts a device to a virtual machine (VM) host.
Move-VM	Moves a virtual machine to a new Hyper-V host.
Move-VMStorage	Moves the storage of a virtual machine.
New-VFD	Creates a virtual floppy disk.
New-VHD	Creates one or more new virtual hard disks.
New-VM	Creates a new virtual machine.
New-VMGroup	Creates a virtual machine group. With Hyper-V, there are two types of VMGroups: a VMCollectionType and a ManagementCollectionType. A VMCollectionType VMGroup contains VMs while the ManagementCollectionType VMGroup contains VMCollectionType VMGroups. For example, you could have two VMCollectionType VMGroups VMG1 (containing VMs VM1 and VM2) and a second VMG2 (containing VMs VM3 and VM4). You could then create a ManagementCollectionType VMGroup VM-All containing the two VMCollectionType VMGroups. You use the Add-VMGroupMember cmdlet to add VMs to VMCollectionType VMGroups and to add VMCollectionType groups to ManagementCollectionType VMGroups.
New-VMReplicationAuthorizationEntry	Creates a new authorization entry that allows one or more primary servers to replicate data to a specified Replica server.
New-VMResourcePool	Creates a resource pool.
New-VMSan	Creates a new virtual storage area network (SAN) on a Hyper-V host.
New-VMSwitch	Creates a new virtual switch on one or more virtual machine hosts.
Optimize-VHD	Optimizes the allocation of space used by virtual hard disk files, except for fixed virtual hard disks.
Optimize-VHDSets	Optimizes VHD set files.
Remove-VHDSnapshot	Removes a checkpoint from a VHD set file.
Remove-VM	Deletes a virtual machine.

Cmdlet	Description
Remove-VMAssignableDevice	Removes information about the assignable devices from a specific virtual machine.
Remove-VM dvdDrive	Deletes a DVD drive from a virtual machine.
Remove-VMFibreChannelHba	Removes a Fibre Channel host bus adapter from a virtual machine.
Remove-VMGpuPartitionAdapter	Removes an assigned GPU partition from a virtual machine.
Remove-VMGroup	Removes a virtual machine group.
Remove-VMGroupMember	Removes members from a virtual machine group.
Remove-VMHardDiskDrive	Deletes a hard disk drive from a virtual machine.
Remove-VMHostAssignableDevice	Removes a device assigned to a virtual machine (VM) host.
Remove-VMMigrationNetwork	Removes a network from use with migration.
Remove-VMNetworkAdapter	Removes one or more virtual network adapters from a virtual machine.
Remove-VMNetworkAdapterAcl	Removes an ACL applied to the traffic through a virtual network adapter.
Remove-VMNetworkAdapterExtendedAcl	Removes an extended ACL for a virtual network adapter.
Remove-VMNetworkAdapterRoutingDomainMapping	Removes a routing domain from a virtual network adapter.
Remove-VMNetworkAdapterTeamMapping	
Remove-VMRemoteFx3dVideoAdapter	Removes a RemoteFX 3D video adapter from a virtual machine.
Remove-VMReplication	Removes the replication relationship of a virtual machine.
Remove-VMReplicationAuthorizationEntry	Removes an authorization entry from a Replica server.
Remove-VMResourcePool	Deletes a resource pool from one or more virtual machine hosts.
Remove-VMsSan	Removes a virtual storage area network (SAN) from a Hyper-V host.
Remove-VMSavedState	Deletes the saved state of a saved virtual machine.
Remove-VMScsiController	Removes a SCSI controller from a virtual machine.

Cmdlet	Description
Remove-VMSnapshot	Deletes a virtual machine checkpoint.
Remove-VMStoragePath	Removes a path from a storage resource pool.
Remove-VMSwitch	Deletes a virtual switch.
Remove-VMSwitchExtensionPortFeature	Removes a feature from a virtual network adapter.
Remove-VMSwitchExtensionSwitchFeature	Removes a feature from a virtual switch.
Remove-VMSwitchTeamMember	Removes a member from a virtual machine switch team.
Rename-VM	Renames a virtual machine.
Rename-VMGroup	Renames virtual machine groups.
Rename-VMNetworkAdapter	Renames a virtual network adapter on a virtual machine or on the management operating system.
Rename-VMResourcePool	Renames a resource pool on one or more Hyper-V hosts.
Rename-VMSan	Renames a virtual storage area network (SAN).
Rename-VMSnapshot	Renames a virtual machine checkpoint.
Rename-VMSwitch	Renames a virtual switch.
Repair-VM	Repairs one or more virtual machines.
Reset-VMReplicationStatistics	Resets the replication statistics of a virtual machine.
Reset-VMResourceMetering	Resets the resource utilization data collected by Hyper-V resource metering.
Resize-VHD	Resizes a virtual hard disk.
Restart-VM	Restarts a virtual machine.
Restore-VMSnapshot	Restores a virtual machine checkpoint.
Resume-VM	Resumes a suspended (paused) virtual machine.
Resume-VMReplication	Resumes a virtual machine replication that is in a state of Paused, Error, Resynchronization Required, or Suspended.
Revoke-VMConnectAccess	Revokes access for one or more users to connect to a one or more virtual machines.
Save-VM	Saves a virtual machine.
Set-VHD	Sets properties associated with a virtual hard disk.

Cmdlet	Description
Set-VM	Configures a virtual machine.
Set-VMBios	Configures the BIOS of a Generation 1 virtual machine.
Set-VMComPort	Configures the COM port of a virtual machine.
Set-VMDVDDrive	Configures a virtual DVD drive.
Set-VMFibreChannelHba	Configures a Fibre Channel host bus adapter on a virtual machine.
Set-VMFirmware	Sets the firmware configuration of a virtual machine.
Set-VMFloppyDiskDrive	Configures a virtual floppy disk drive.
Set-VMGpuPartitionAdapter	Assigns a partition of a GPU to a virtual machine.
Set-VMHardDiskDrive	Configures a virtual hard disk.
Set-VMHost	Configures a Hyper-V host.
Set-VMHostCluster	Configures a virtual machine host cluster.
Set-VMHostPartitionableGpu	Configures the host partitionable GPU to the number of partitions supported by the manufacturer.
Set-VMKeyProtector	Configures a key protector for a virtual machine.
Set-VMMemory	Configures the memory of a virtual machine.
Set-VMMigrationNetwork	Sets the subnet, subnet mask, and/or priority of a migration network.
Set-VMNetworkAdapter	Configures features of the virtual network adapter in a virtual machine or the management operating system.
Set-VMNetworkAdapterFailoverConfiguration	Configures the IP address of a virtual network adapter to be used when a virtual machine fails over.
Set-VMNetworkAdapterIsolation	Modifies isolation settings for a virtual network adapter.
Set-VMNetworkAdapterRoutingDomainMapping	Sets virtual subnets on a routing domain.
Set-VMNetworkAdapterTeamMapping	
Set-VMNetworkAdapterVlan	Configures the virtual LAN settings for the traffic through a virtual network adapter.
Set-VMProcessor	Configures settings for the virtual processors of a virtual machine. Settings are applied uniformly to all virtual

Cmdlet	Description
	processors belonging to the virtual machine.
Set-VMRemoteFx3dVideoAdapter	Configures the RemoteFX 3D video adapter of a virtual machine.
Set-VMReplication	Modifies the replication settings of a virtual machine.
Set-VMReplicationAuthorizationEntry	Modifies an authorization entry on a Replica server.
Set-VMReplicationServer	Configures a host as a Replica server.
Set-VMResourcePool	Sets the parent resource pool for a selected resource pool.
Set-VMSan	Configures a virtual storage area network (SAN) on one or more Hyper-V hosts.
Set-VMSecurity	Configures security settings for a virtual machine.
Set-VMSecurityPolicy	Configures the security policy for a virtual machine.
Set-VMSwitch	Configures a virtual switch.
Set-VMSwitchExtensionPortFeature	Configures a feature on a virtual network adapter.
Set-VMSwitchExtensionSwitchFeature	Configures a feature on a virtual switch.
Set-VMSwitchTeam	Configures a virtual switch team.
Set-VMVideo	Configures video settings for virtual machines.
Start-VM	Starts a virtual machine.
Start-VMFailover	Starts failover on a virtual machine.
Start-VMInitialReplication	Starts replication of a virtual machine.
Start-VMTrace	Starts tracing to a file.
Stop-VM	Shuts down, turns off, or saves a virtual machine.
Stop-VMFailover	Stops failover of a virtual machine.
Stop-VMInitialReplication	Stops an ongoing initial replication.
Stop-VMReplication	Cancels an ongoing virtual machine resynchronization.
Stop-VMTrace	Stops tracing to file.
Suspend-VM	Suspends, or pauses, a virtual machine.
Suspend-VMReplication	Suspends replication of a virtual machine.

Cmdlet	Description
Test-VHD	Tests a virtual hard disk for any problems that would make it unusable.
Test-VMNetworkAdapter	Tests connectivity between virtual machines.
Test-VMReplicationConnection	Tests the connection between a primary server and a Replica server.
Update-VMVersion	Updates the version of virtual machines.

Hyper-V specification and APIs documentation

Documentation for the top level functional specification and APIs for Hyper-V for developers.

Hyper-V top level functional specification

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Hyper-V management

Article • 01/07/2021

Here we describe how to manage or get info about virtual machines.

In this section

 Expand table

Topic	Description
About Hyper-V management	This section contains info for Hyper-V management.
Hyper-V management reference	This section contains the reference for Hyper-V management.

Hyper-V WMI provider (V2)

Article • 01/07/2021

Purpose

Hyper-V provides a scalable, reliable, and highly available virtualized server computing environment. It enables one or more guest operating systems to run concurrently on a single physical computer. Key uses for virtual machine technology include the following:

- Server consolidation
- Consolidation of development and testing environments
- Simplified disaster recovery

In this section

Topic	Description
About the Hyper-V WMI provider	The WMI provider for Hyper-V enable developers, and scripters, to quickly build custom tools, utilities, and enhancements for the virtualization platform. The WMI interfaces can manage all aspects of the Hyper-V services.
Using the Hyper-V WMI provider	The following topics describe how to use the Hyper-V WMI provider.
Hyper-V WMI classes	The following are the Hyper-V WMI classes.
Hyper-V application health monitoring API	The Hyper-V application health monitoring API is used to monitor the health state of applications running in a virtual machine.
Hyper-V replication API	The Hyper-V replication API is used to control virtual machine replication and failover recovery.
Hyper-V metrics API	The Hyper-V metrics API is used to monitor the health state of applications running in a virtual machine.
Hyper-V networking API	The Hyper-V networking API is used to control networking in Hyper-V.

Topic	Description
Hyper-V migration API	The Hyper-V migration API is used to control storage and live migration in Hyper-V.
Hyper-V virtual Fibre Channel API	The Hyper-V virtual Fibre Channel API is used to control virtual Fibre Channel adapters in Hyper-V.
Hyper-V VM placement API	The Hyper-V virtual machine (VM) placement API contains the compatibility info for the VMs or hosting computer system.
Threshold classes	Contains the classes introduced in Windows 10.
RS2 Classes	Contains the new classes for Windows 10, version 1703.
RS3 Classes	Contains the new classes for Windows 10, version 1709.
Hyper-V glossary	Glossary of terms used in the Windows Virtualization SDK documentation.

Run-time requirements

Hyper-V services require an x64-based system that supports hardware-assisted virtualization running Hyper-V. Programs that interact with the Hyper-V WMI interfaces however, can run remotely on any system that supports WMI.

Related topics

[Hyper-V \(Windows Server 2008 R2 Technical Library\)](#)

[Windows Management Instrumentation](#)

[Windows Server Virtualization](#) [↗](#)

Feedback

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Make your own integration services

08/14/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Starting in Windows 10 Anniversary Update, anyone can make applications that communicate between the Hyper-V host and its virtual machines using Hyper-V sockets -- a Windows Socket with a new address family and specialized endpoint for targeting virtual machines. All communication over Hyper-V sockets runs without using networking and all data stays on the same physical memory. Applications using Hyper-V sockets are similar to Hyper-V's integration services.

This document walks through creating a simple program built on Hyper-V sockets.

Supported Host OS

- Windows 10 and later
- Windows Server 2016 and later

Supported Guest OS

- Windows 10 and later
- Windows Server 2016 and later
- Linux guests with Linux Integration Services. See [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows](#)

Note

A supported Linux guest must have kernel support for:

Bash

```
CONFIG_VSOCKET=y
CONFIG_HYPERV_VSOCKETS=y
```

Capabilities and Limitations

- Supports kernel mode or user mode actions
- Data stream only

- No block memory (not the best for backup/video)

Getting started

Requirements:

- C/C++ compiler. If you don't have one, checkout [Visual Studio Community](#) ↗
- [Windows SDK](#) -- pre-installed in Visual Studio 2015 with Update 3 and later.
- A computer running one of the host operating systems specified with at least one virtual machine. -- this is for testing your application.

Note: The API for Hyper-V sockets became publicly available in Windows 10 Anniversary Update. Applications that use HVSocket will run on any Windows 10 host and guest but can only be developed with a Windows SDK later than build 14290.

Register a new application

In order to use Hyper-V sockets, the application must be registered with the Hyper-V Host's registry.

By registering the service in the registry, you get:

- WMI management for enable, disable, and listing available services
- Permission to communicate with virtual machines directly

The following PowerShell will register a new application named "HV Socket Demo". This must be run as administrator. Manual instructions below.

PowerShell

```
$friendlyName = "HV Socket Demo"

# Create a new random GUID. Add it to the services list
$service = New-Item -Path "HKLM:\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\Virtualization\GuestCommunicationServices" -Name ((New-
Guid).Guid)

# Set a friendly name
$service.SetValue("ElementName", $friendlyName)

# Copy GUID to clipboard for later use
$service.PSChildName | clip.exe
```

Registry location and information:

```
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows  
NT\CurrentVersion\Virtualization\GuestCommunicationServices\
```

In this registry location, you'll see several GUIDs. Those are our in-box services.

Information in the registry per service:

- Service GUID
 - ElementName (REG_SZ) -- this is the service's friendly name

To register your own service, create a new registry key using your own GUID and friendly name.

The friendly name will be associated with your new application. It'll appear in performance counters and other places where a GUID isn't appropriate.

The registry entry looks like this:

Output

```
HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows  
NT\CurrentVersion\Virtualization\GuestCommunicationServices\  
999E53D4-3D5C-4C3E-8779-BED06EC056E1\  
    ElementName    REG_SZ    VM Session Service  
YourGUID\  
    ElementName    REG_SZ    Your Service Friendly Name
```

❗ Note

The Service GUID for a Linux guest uses the VSOCK protocol which addresses via a `svm_cid` and `svm_port` rather than a guid. To bridge this inconsistency with Windows the well-known GUID is used as the service template on the host which translates to a port in the guest. To customize your Service GUID simply change the first "00000000" to the port number desired. Ex: "00000ac9" is port 2761.

C++

```
// Hyper-V Socket Linux guest VSOCK template GUID  
struct __declspec(uuid("00000000-facb-11e6-bd58-64006a7986d3"))  
VsockTemplate{};  
  
/*  
 * GUID example = __uuidof(VsockTemplate);  
 * example.Data1 = 2761; // 0x00000AC9  
 */
```

Tip: To generate a GUID in PowerShell and copy it to the clipboard, run:

```
PowerShell
```

```
(New-Guid).Guid | clip.exe
```

Create a Hyper-V socket

In the most basic case, defining a socket requires an address family, connection type, and protocol.

Here is a simple [socket definition](#)

```
C

// Windows
SOCKET WINAPI socket(
    _In_ int af,
    _In_ int type,
    _In_ int protocol
);

// Linux guest
int socket(int domain, int type, int protocol);
```

For a Hyper-V socket:

- Address family - `AF_HYPERV` (Windows) or `AF_VSOCK` (Linux guest)
- type - `SOCK_STREAM`
- protocol - `HV_PROTOCOL_RAW` (Windows) or `0` (Linux guest)

Here is an example declaration/instantiation:

```
C

// Windows
SOCKET sock = socket(AF_HYPERV, SOCK_STREAM, HV_PROTOCOL_RAW);

// Linux guest
int sock = socket(AF_VSOCK, SOCK_STREAM, 0);
```

Bind to a Hyper-V socket

Bind associates a socket with connection information.

The function definition is copied below for convenience, read more about bind [here](#).

```
C

// Windows
int bind(
    _In_ SOCKET          s,
    _In_ const struct sockaddr *name,
    _In_ int              namelen
);

// Linux guest
int bind(int sockfd, const struct sockaddr *addr,
          socklen_t addrlen);
```

In contrast to the socket address (sockaddr) for a standard Internet Protocol address family (`AF_INET`) which consists of the host machine's IP address and a port number on that host, the socket address for `AF_HYPERV` uses the virtual machine's ID and the application ID defined above to establish a connection. If binding from a Linux guest `AF_VSOCK` uses the `svm_cid` and the `svm_port`.

Since Hyper-V sockets do not depend on a networking stack, TCP/IP, DNS, etc. the socket endpoint needed a non-IP, not hostname, format that still unambiguously describes the connection.

Here is the definition for a Hyper-V socket's socket address:

```
C

// Windows
struct SOCKADDR_HV
{
    ADDRESS_FAMILY Family;
    USHORT Reserved;
    GUID VmId;
    GUID ServiceId;
};

// Linux guest
// See include/uapi/linux/vm_sockets.h for more information.
struct sockaddr_vm {
    __kernel_sa_family_t svm_family;
    unsigned short svm_reserved1;
    unsigned int svm_port;
    unsigned int svm_cid;
    unsigned char svm_zero[sizeof(struct sockaddr) -
                             sizeof(sa_family_t) -
                             sizeof(unsigned short) -
```

```
sizeof(unsigned int) - sizeof(unsigned int)];  
};
```

In lieu of an IP or hostname, AF_HYPERV endpoints rely heavily on two GUIDs:

- VM ID – this is the unique ID assigned per VM. A VM's ID can be found using the following PowerShell snippet.

PowerShell

```
(Get-VM -Name $VMName).Id
```

- Service ID – GUID, [described above](#), with which the application is registered in the Hyper-V host registry.

There is also a set of VMID wildcards available when a connection isn't to a specific virtual machine.

VMID Wildcards

 Expand table

Name	GUID	Description
HV_GUID_ZERO	00000000-0000-0000-0000-000000000000	Listeners should bind to this VmId to accept connection from all partitions.
HV_GUID_WILDCARD	00000000-0000-0000-0000-000000000000	Listeners should bind to this VmId to accept connection from all partitions.
HV_GUID_BROADCAST	FFFFFFFF-FFFF-FFFF-FFFF-FFFFFFFFFFFF	
HV_GUID_CHILDREN	90db8b89-0d35-4f79-8ce9-49ea0ac8b7cd	Wildcard address for children. Listeners should bind to this VmId to accept connection from its children.
HV_GUID_LOOPBACK	e0e16197-dd56-4a10-9195-5ee7a155a838	Loopback address. Using this VmId connects to the same partition as the connector.
HV_GUID_PARENT	a42e7cda-d03f-480c-9cc2-a4de20abb878	Parent address. Using this VmId connects to the parent partition of the connector.*

* **HV_GUID_PARENT** The parent of a virtual machine is its host. The parent of a container is the container's host. Connecting from a container running in a virtual machine will connect to the VM hosting the container. Listening on this VmId accepts connection from: (Inside containers):

Container host. (Inside VM: Container host/ no container): VM host. (Not inside VM: Container host/ no container): Not supported.

Supported socket commands

Socket() Bind() Connect() Send() Listen() Accept()

HvSocket Socket Options

 Expand table

Name	Type	Description
HVSOCKET_CONNECTED_SUSPEND	ULONG	When this socket option is set to a non-zero value sockets do not disconnect when the virtual machine is paused.

Useful links

[Complete WinSock API](#)

[Hyper-V Integration Services reference](#)

Data Exchange: Using key-value pairs to share information between the host and guest on Hyper-V

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Data Exchange is an integration service (also known as key-value pair exchange or KVP) that can be used to share small pieces of information between a virtual machine (guest) and its Hyper-V host. General information about the virtual machine and host is automatically created and stored as key-value pairs. You can also create your own pairs to share custom data.

Key-value pairs consist of a "key" and a "value". Both are strings; no other data types are supported. When a key-value pair is created or changed, it becomes visible to both the guest and the host. KVP data travels over the Hyper-V VMbus and doesn't require any network connectivity between the guest and host.

Once created, key-value pairs remain until they're deleted. Any application that creates key-value pairs should delete them when they're no longer needed. Key-value pairs move with the virtual machine during Live Migration.

Windows guests

On Windows guests, KVP data is stored in the registry under:

text

HKLM\SOFTWARE\Microsoft\Virtual Machine

The data is organized in these subkeys:

- **Virtual Machine\Auto** – Data describing the guest. Created by integration service drivers after they load. Visible to the host as intrinsic data.
- **Virtual Machine\External** – Data pushed to the guest from the host by a user.
- **Virtual Machine\Guest** – Data created on the guest. Visible to the host as non-intrinsic data.
- **Virtual Machine\Guest\Parameter** – Data pushed to the guest from the host that describes the host.

Adding values from within the guest is as simple as creating a new string value under

`HKLM\SOFTWARE\Microsoft\Virtual Machine\Guest`. You must be an administrator in the guest to

modify this location. You can use WMI (PowerShell or other tooling) from the host or a remote machine (with permissions) to retrieve the value.

For information about registry size limits, see the (legacy) article [Registry Element Size Limits](#).

Add a new key-value pair in the guest

In this example, the value of **Status** is set to `Ready`:

PowerShell

```
$regPath = "HKLM:\SOFTWARE\Microsoft\Virtual Machine\Guest"  
Set-ItemProperty -Path $regPath -Name "Status" -Value "Ready" -Type String
```

You can use the same syntax to change the value.

Query key-value pairs in the guest

To query the value of the **External** subkey (data pushed to the guest from the host):

PowerShell

```
$regPath = "HKLM:\SOFTWARE\Microsoft\Virtual Machine\External"  
Get-ItemProperty -Path $regPath -Name "Name"
```

Linux guests

Linux doesn't have a registry, so KVP items are stored in the file system. A daemon process, `hv_kvp_daemon`, must be running to handle processing. For most distributions with Linux Integration Services (LIS) or in-kernel drivers installed, this daemon starts automatically. In some cases, extra steps may be required to install and start the daemon.

The Linux integration services implement data exchange with KVP pools. A KVP pool is a file stored in a specific path. There are four pool files:

text

```
/var/lib/hyperv/.kvp_pool_0  
/var/lib/hyperv/.kvp_pool_1  
/var/lib/hyperv/.kvp_pool_2  
/var/lib/hyperv/.kvp_pool_3
```

These pool files map to the following Windows registry key sets:

- Pool 0: Virtual Machine\External
- Pool 1: Virtual Machine\Guest
- Pool 2: Virtual Machine\Auto
- Pool 3: Virtual Machine\Guest\Parameter

ⓘ Note

For more information about Linux KVP support, see [Linux and FreeBSD Virtual Machines on Hyper-V](#).

The key-value pair infrastructure might not function correctly without a Linux software update. Contact your distribution vendor for an update if you encounter issues.

Pool structure

Each pool file contains records with this structure:

C

```
struct kvp_record
{
    unsigned char key[ HV_KVP_EXCHANGE_MAX_KEY_SIZE ];
    unsigned char value[ HV_KVP_EXCHANGE_MAX_VALUE_SIZE ];
};
```

Those size constants are defined in `hyperv.h` (a kernel header distributed with the Linux kernel sources).

Read and display values from pool 0

This sample reads KVP values from pool 0 and displays them.

C

```
``c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <errno.h>
#include <fcntl.h>
#include "../include/linux/hyperv.h"

typedef struct kvp_record
{
```

```

    unsigned char key [HV_KVP_EXCHANGE_MAX_KEY_SIZE];
    unsigned char value [HV_KVP_EXCHANGE_MAX_VALUE_SIZE];
} KVP_RECORD;

KVP_RECORD myRecords[200];

void KVPAcquireLock(int fd)
{
    struct flock fl = {F_RDLCK, SEEK_SET, 0, 0, 0};
    fl.l_pid = getpid();

    if (-1 == fcntl(fd, F_SETLKW, &fl))
    {
        perror("fcntl lock");
        exit (-10);
    }
}

void KVPReleaseLock(int fd)
{
    struct flock fl = {F_UNLCK, SEEK_SET, 0, 0, 0};
    fl.l_pid = getpid();

    if (-1 == fcntl(fd, F_SETLK, &fl))
    {
        perror("fcntl unlock");
        exit (-20);
    }
}

int main (int argc, char **argv)
{
    char poolName[] = "/var/lib/hyperv/.kvp_pool_0";
    int i;
    int fd;
    int bytesRead;
    int numRecords;

    fd = open(poolName, O_RDONLY);
    if (-1 == fd)
    {
        printf("Error: Unable to open pool file %s\n", poolName);
        exit (-30);
    }

    KVPAcquireLock(fd);
    bytesRead = read(fd, myRecords, sizeof(myRecords));
    KVPReleaseLock(fd);

    numRecords = bytesRead / sizeof(struct kvp_record);
    printf("Number of records : %d\n", numRecords);

    for (i = 0; i < numRecords; i++)
    {
        printf("  Key   : %s\n  Value: %s\n\n", myRecords[i].key,

```

```

myRecords[i].value);
    }

    close(fd);

    return 0;
}

```

Create a KVP item in pool 1

C

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <errno.h>
#include <fcntl.h>
#include "../include/linux/hyperv.h"

typedef struct kvp_record
{
    unsigned char key [HV_KVP_EXCHANGE_MAX_KEY_SIZE];
    unsigned char value [HV_KVP_EXCHANGE_MAX_VALUE_SIZE];
} KVP_RECORD;

void KVPAcquireWriteLock(int fd)
{
    struct flock fl = {F_WRLCK, SEEK_SET, 0, 0, 0};
    fl.l_pid = getpid();

    if (-1 == fcntl(fd, F_SETLKW, &fl))
    {
        perror("fcntl lock");
        exit (-10);
    }
}

void KVPReleaseLock(int fd)
{
    struct flock fl = {F_UNLCK, SEEK_SET, 0, 0, 0};
    fl.l_pid = getpid();

    if (-1 == fcntl(fd, F_SETLK, &fl))
    {
        perror("fcntl unlock");
        exit (-20);
    }
}

int main (int argc, char **argv)
{
    char poolName[] = "/var/lib/hyperv/.kvp_pool_1";

```

```

int    fd;
KVP_RECORD newKvp;

if (3 != argc)
{
    printf("Usage: WritePool keyName valueString\n\n");
    exit (-5);
}

fd = open(poolName, O_WRONLY);
if (-1 == fd)
{
    printf("Error: Unable to open pool file %s\n", poolName);
    exit (-30);
}

memset((void *)&newKvp, 0, sizeof(KVP_RECORD));
memcpy(newKvp.key, argv[1], strlen(argv[1]));
memcpy(newKvp.value, argv[2], strlen(argv[2]));

KVPAcquireWriteLock(fd);
write(fd, (void *)&newKvp, sizeof(KVP_RECORD));
KVPReleaseLock(fd);

close(fd);

return 0;
}

```

Delete a KVP item from pool 1

This sample deletes an item.

C

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <errno.h>
#include <fcntl.h>
#include <uapi/linux/hyperv.h>

typedef struct kvp_record
{
    unsigned char key [HV_KVP_EXCHANGE_MAX_KEY_SIZE];
    unsigned char value [HV_KVP_EXCHANGE_MAX_VALUE_SIZE];
} KVP_RECORD;

void KVPAcquireWriteLock(int fd)
{
    struct flock fl = {F_WRLCK, SEEK_SET, 0, 0, 0};

```

```

fl.l_pid = getpid();

if (-1 == fcntl(fd, F_SETLKW, &fl))
{
    perror("fcntl lock");
    exit (-10);
}
}

void KVPReleaseLock(int fd)
{
    struct flock fl = {F_UNLCK, SEEK_SET, 0, 0, 0};
    fl.l_pid = getpid();

    if (-1 == fcntl(fd, F_SETLK, &fl))
    {
        perror("fcntl unlock");
        exit (-20);
    }
}

int find_record_offset(int fd, char *key)
{
    int bytesRead;
    int offset = 0;
    int retval = -1;

    KVP_RECORD kvpRec;

    while (1)
    {
        lseek(fd, offset, SEEK_SET);
        bytesRead = read(fd, &kvpRec, sizeof(KVP_RECORD));
        if (0 == bytesRead)
        {
            break;
        }

        if (0 == strcmp(key, (const char *) kvpRec.key))
        {
            retval = offset;
            break;
        }

        offset += sizeof(KVP_RECORD);
    }

    return retval;
}

int main (int argc, char **argv)
{
    char poolName[] = "/var/lib/hyperv/.kvp_pool_1";
    int fd;
    int exitVal = -1;

```



```

int    bytesRead;
int    bytesWritten;
int    offset_to_delete;
int    offset_last_record;
KVP_RECORD kvpRec;

if (2 != argc)
{
    printf("Usage: WritePool keyName valueString\n\n");
    exit (-5);
}

fd = open(poolName, O_RDWR, 0644);
if (-1 == fd)
{
    printf("Error: Unable to open pool file %s\n", poolName);
    exit (-10);
}

KVPAcquireWriteLock(fd);
offset_to_delete = find_record_offset(fd, argv[1]);
if (offset_to_delete < 0)
{
    exitVal = -15;
    goto cleanup2;
}

offset_last_record = lseek(fd, -sizeof(KVP_RECORD), SEEK_END);
if (offset_last_record < 0)
{
    exitVal = -20;
    goto cleanup2;
}

if (offset_last_record != offset_to_delete)
{
    lseek(fd, offset_last_record, SEEK_SET);
    bytesRead = read(fd, &kvpRec, sizeof(KVP_RECORD));
    lseek(fd, offset_to_delete, SEEK_SET);
    bytesWritten = write(fd, &kvpRec, sizeof(KVP_RECORD));
}

ftruncate(fd, offset_last_record);

exitVal = 0;

cleanup2:
    KVPReleaseLock(fd);

cleanup1:
    close(fd);

    return exitVal;
}

```

Working with key-value pairs from the host using WMI

The following examples use the WMI v2 namespace. For WMI v1 (older versions), remove the `\v2` segment from the namespace path.

❗ Note

If you're using Windows 8 or Windows 8.1, install Client Hyper-V to get the namespaces.

Read the value from the host

This example gets the value of the key `Status` from a VM named `Vm1`:

PowerShell

```
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_ComputerSystem -Filter {ElementName = 'Vm1'}
$vm.GetRelated("Msvm_KvpExchangeComponent").GuestExchangeItems | % { \
    $GuestExchangeItemXml = ([XML]$_).SelectSingleNode(\
        "/INSTANCE/PROPERTY[@NAME='Name']/VALUE[child::text() = 'Status']")
    if ($GuestExchangeItemXml -ne $null)
    {
        $GuestExchangeItemXml.SelectSingleNode(\
            "/INSTANCE/PROPERTY[@NAME='Data']/VALUE/child::text()" ).Value
    }
}
```

Add or modify key-value pairs from the host

To add a key-value pair from the host, get instances of both the management service and the VM and create a new instance of `Msvm_KvpExchangeDataItem`. When creating the new instance, specify the `Name`, `Data`, and `Source` (must be `0`). Then call `AddKvpItems`.

Querying for host-created key-value pairs is similar to guest queries but requires an additional association hop to `Msvm_KvpExchangeComponentSettingData`. Modifying and deleting values works the same way—specify the same key name and call the appropriate `Modify` or `Remove` method.

❗ Important

The examples below use the v2 namespace. If you're using Windows Server 2008 or Windows Server 2008 R2, remove the `\v2` segment.

Add a new key-value pair

PowerShell

```
$VmMgmt = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_VirtualSystemManagementService
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_ComputerSystem -Filter {ElementName='VM1'}
$kvpDataItem = ([WMIClass][String]::Format("\{0}\{1}:{2}", \
    $VmMgmt.ClassPath.Server, \
    $VmMgmt.ClassPath.NamespacePath, \
    "Msvm_KvpExchangeDataItem")).CreateInstance()

$kvpDataItem.Name = "Name"
$kvpDataItem.Data = "Data"
$kvpDataItem.Source = 0

$VmMgmt.AddKvpItems($vm, $kvpDataItem.PSBase.GetText(1))
```

Query key-value pairs on the host

PowerShell

```
$VmMgmt = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_VirtualSystemManagementService
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_ComputerSystem -Filter {ElementName='VM1'}
($vm.GetRelated("Msvm_KvpExchangeComponent")[0]
).GetRelated("Msvm_KvpExchangeComponentSettingData").HostExchangeItems | % { \
    $GuestExchangeItemXml = ([XML]$_).SelectSingleNode(\
        "/INSTANCE/PROPERTY[@NAME='Name']/VALUE[child::text() = 'Name2']")
    if ($GuestExchangeItemXml -ne $null)
    {
        $GuestExchangeItemXml.SelectSingleNode(\
            "/INSTANCE/PROPERTY[@NAME='Data']/VALUE[child::text() = 'Data2']").Value
    }
}
```

Modify a key-value pair

PowerShell

```
$VmMgmt = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_VirtualSystemManagementService
```

```
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_ComputerSystem -Filter {ElementName='VM1'}
$kvpDataItem = ([WMIClass][String]::Format("\{0}\{1}:{2}", \
    $VmMgmt.ClassPath.Server, \
    $VmMgmt.ClassPath.NamespacePath, \
    "Msvm_KvpExchangeDataItem")).CreateInstance()

$kvpDataItem.Name = "Name"
$kvpDataItem.Data = "Data2"
$kvpDataItem.Source = 0

$VmMgmt.ModifyKvpItems($Vm, $kvpDataItem.PSBase.GetText(1))
```

Remove a key-value pair

PowerShell

```
$VmMgmt = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_VirtualSystemManagementService
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Class \
    Msvm_ComputerSystem -Filter {ElementName='VM1'}
$kvpDataItem = ([WMIClass][String]::Format("\{0}\{1}:{2}", \
    $VmMgmt.ClassPath.Server, \
    $VmMgmt.ClassPath.NamespacePath, \
    "Msvm_KvpExchangeDataItem")).CreateInstance()

$kvpDataItem.Name = "Name"
$kvpDataItem.Data = [String]::Empty
$kvpDataItem.Source = 0

$VmMgmt.RemoveKvpItems($Vm, $kvpDataItem.PSBase.GetText(1))
```

See also

- [Hyper-V integration services](#)
- [Supported Linux and FreeBSD virtual machines for Hyper-V on Windows](#)
- [PowerShell Direct](#)

Move from Hyper-V WMI v1 to WMI v2

08/14/2025

Applies to:  [Windows Server 2025](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#),  [Windows 11](#),  [Windows 10](#),  [Azure Local 2311.2 and later](#)

Windows Management Instrumentation (WMI) is the management interface underlying Hyper-V Manager and Hyper-V's PowerShell cmdlets. While most people use our PowerShell cmdlets or Hyper-V manager, sometimes developers needed WMI directly.

There have been two Hyper-V WMI namespaces (or versions of the Hyper-V WMI API).

- The WMI v1 namespace (root\virtualization) which was introduced in Windows Server 2008 and last available in Windows Server 2012
- The WMI v2 namespace (root\virtualization\v2) which was introduced in Windows Server 2012

This document contains references to resources for converting code that talks to our old WMI namespace to the new one. Initially, this article will serve as a repository for API information and sample code / scripts that can be used to help port any programs or scripts that use Hyper-V WMI APIs from the v1 namespace to the v2 namespace.

MSDN Samples

[Hyper-V virtual machine migration sample](#) 

[Hyper-V virtual Fiber Channel sample](#) 

[Hyper-V planned virtual machines sample](#) 

[Hyper-V application health monitoring sample](#) 

[Virtual hard disk management sample](#) 

[Hyper-V replication sample](#) 

[Hyper-V metrics sample](#) 

[Hyper-V dynamic memory sample](#) 

[Hyper-V Extensible Switch extension filter driver](#) 

[Hyper-V networking sample](#) 

[Hyper-V resource pool management sample](#) 

[Hyper-V recovery snapshot sample](#) 

Samples From Blogs

[Adding a Network Adapter To A VM Using The Hyper-V WMI V2 Namespace](#) 

[Connecting a VM Network Adapter To A Switch Using The Hyper-V WMI V2 Namespace](#) 

[Changing The MAC Address Of NIC Using The Hyper-V WMI V2 Namespace ↗](#)

[Removing a Network Adapter To A VM Using The Hyper-V WMI V2 Namespace ↗](#)

[Attaching a VHD To A VM Using The Hyper-V WMI V2 Namespace ↗](#)

[Removing a VHD From A VM Using The Hyper-V WMI V2 Namespace ↗](#)

[Creating a VM using the Hyper-V WMI V2 Namespace ↗](#)